SHIMADZU

M503-E411
Rev: F

MOBILE X-RAY SYSTEM

MobileDaRt Evolution

MX8 Version / MX9 Version

MobileArt Evolution

MX8 Version

SERVICE MANUAL

Only authorized service personnel is allowed to perform the update.
Perform the update after reading this document carefully.

Preface

The installation, operation, and maintenance should be of the equipment in accordance with the instructions in this manual.

- Safety can be achieved with an individual operator’s attention.
- While operating the equipment, ensure safety of operator and people around one.
- Repair and inspection of the inside of the equipment is dangerous. A person who has received proper training in Shimadzu performs them.

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Revision History

Revision	Date	Changes
ㅡ	2021.7	-
A	2023.1	・chapter 2.5 Added -02 board information to XCONT-2018 ・chapter 2.8 Added check pin to MUX CHARGE 32 KX ・chapter 3.3~3.7 Add board and signal schematic ・chapter 4.1 Added revisions to which measures were taken in software revision ・ chapter 4.2 Revising troubleshooting and adding connection diagrams ・chapter 6 Add factory command to DTOUT command ・Other correction of errors
B	2023.6	・chapter 5.20 Added Replacement FPD-guide right frame ASSY ・ chapter 5.19 Added procedure to Replacement of the H.V.Transformer
C	2024.4	・2.2, 2.5, 2.6, 5.19 Added information about -01/02 of XCONT-2018 ・3.1, 3.2, 3.5, 5.13 Name of relays on MUX POWER have been corrected. (K7->K7 and K8) ・4.2 Added information about trouble shooting for each error code ・4.4 Regarding the adjusted AD value of FSR, the factory default value was added as reference information. ・5.1 Added information of the wire rope replacement procedure ・5.3 Added information of the procedure for replacing the photo sensor ・5.6 Detailed information of slide plate added ・5.7 Added information of the procedure for replacing the arm up/down magnet lock (Replacement procedures differ depending on the time of manufacture). ・5.14 Added information of the procedure for replacing the new type of LED lamp ・5.18 Change the picture of the DC power supply to a new one. ・5.21 Capacitor replacement procedure is added. ・Chapter 7. DR ID-800 Error Code Response Method is added.
D	2024.8	・4.2 Added information about trouble shooting for each error code ・4.7 and 4.8 are added.
E	2025.4	・Added information about MX9 Version ・4.2 Added information about trouble shooting for each error code ・5.22, 5.23 and 7.2 are added.

F	2025.9	・5.24 Added information about replacement of front wheels

1 Introduction

1. Introduction

1.1 Special functions for MobileDaRt/Art Evolution (MX8/MX9 Version)

I. Sound Alarm

The speaker attached to the front cover generates a driving warning sound, X-Ray. Ready up sound, and exposure sound. The sound pattern tone and volume can be set by the customers according to "User Setting" in Reference Guide.

II. Notification Function

If it is time to replace one of the parts on below table, when the key switch is turned on, the replacement request is displayed on the X-ray control panel and the warning indicator lamp lights on.

The kV display indicates the part code for the replacement part. The detailed explanation is written on "Notification Function" in Reference Guide.

As well, refer to "Adjustment Mode" in Installation Manual regarding the relative setting by field service engineers.

KV Display	Dose Display	Parts	Information on Replacement Time Display
U01	L-bat	Battery	Displayed after 2 years from installation.
U02	C-bat	CPU backup battery	Displayed after 2 years from installation.
U03	E-bat	Emergency brake release battery	Displayed after 2 years from installation.
U04	tUbe	X-Ray tube unit	Displayed after 30000 exposures from installation.
U05	Drive	Motor	Displayed after the system travels 7500 km from installation.
U06	rope	Wire rope	Displayed after the arm unit is moved up/down 400 km from installation

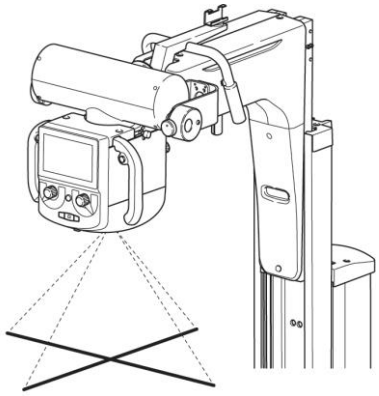
III. Additional Radiography Function

After the battery indicator stops blinking and radiography becomes impossible due to low battery condition, if additional radiography function is used, about 10 additional radiography is possible.

The detailed explanation is written on "Additional Radiography Function" in Reference Guide.

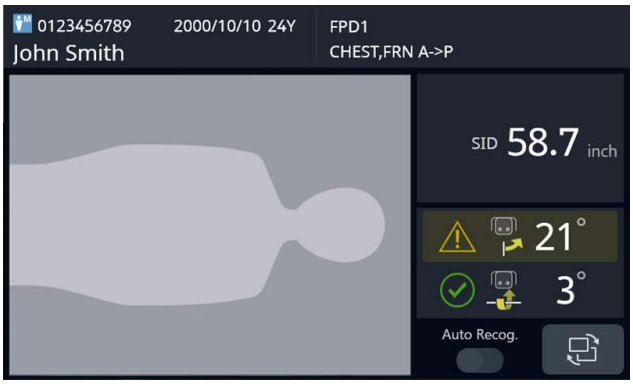
IV. Camera Function and Laser Marker (for MX9 Version only)

The vertical and horizontal laser markers mounted on the collimator provide a clearer view of the center of the X-ray field. (Option for MX9)



Laser Markers

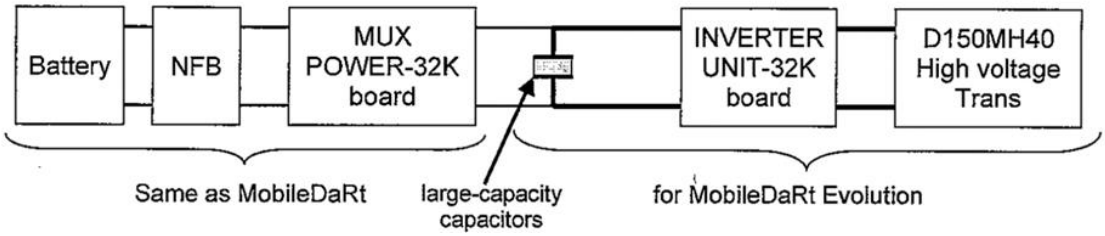
The camera images acquired by the camera mounted on the collimator provide positioning information. (Option for MX9 Premium type only)



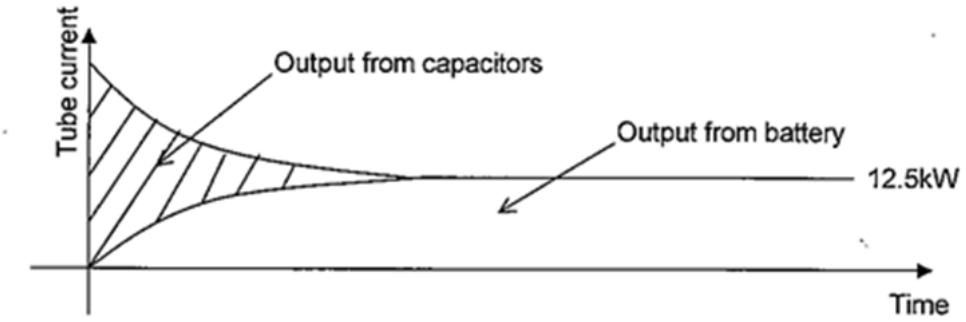
camera image display screen on the collimator

1.2 Outline and attention for the circuit for charging capacitors

1.2.1 High voltage line structure for 32kW type



In case of 32kW type, in order to get 32kW maximum electric power in a short time, the internal large-capacity capacitors are connected to the battery as shown on the above drawing.

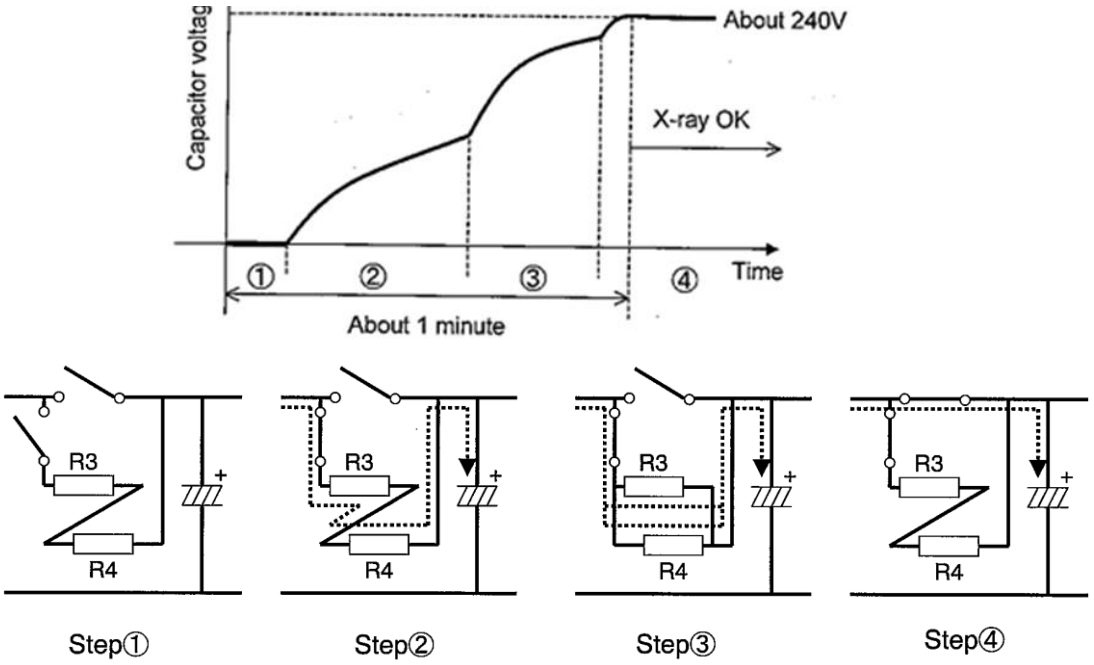


The internal large-capacity capacitors enable to get 32kW maximum electric power in a short time (20msec).

But, in case that radiography condition is a long time and tube current 160mA or less, output electric power is 12.5kW.

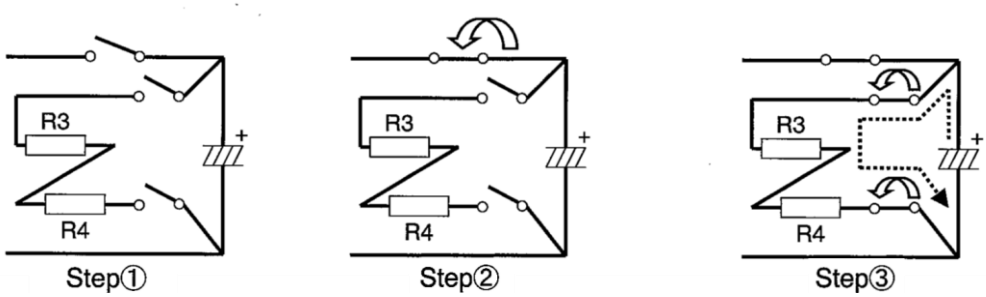
1.2.2 Procedure of charging the internal capacitors

It charges capacitors by switching the connection of resistors R3, R4 in series or parallel.
If capacitors are 0V, it will take about 1 minute to ready-up.
If charge capacitors once and leave in a day, voltage will decrease by about 20%. Therefore, it will take a few second to ready-up.



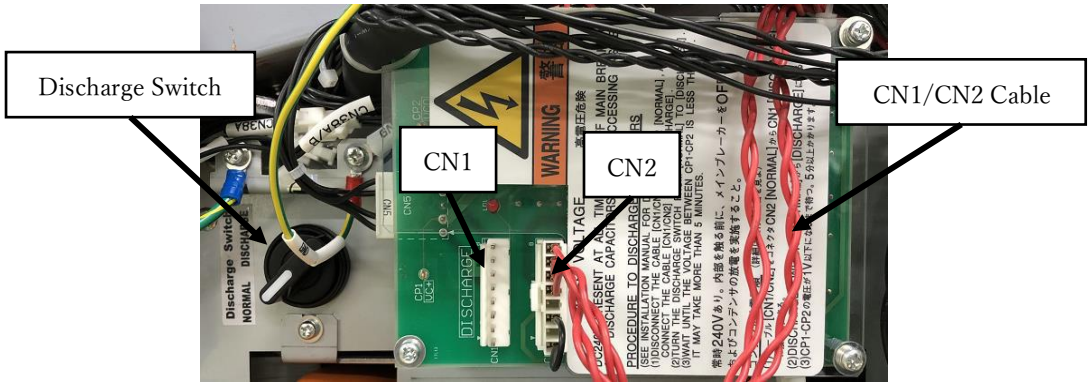
1.2.3 Procedure of discharging the internal capacitors

Even after turning OFF the key switch or the main breaker, electrical charge remains in capacitors. Therefore, it is possible to get an electric shock as it is during maintenance. We must discharge the capacitor when maintenance of this system's power supply.
Capacitors are discharged by connection resistors R3, R4. The resistors are connected by charging connection of the CN1/CN2 cable and turning on discharge switch.



Step1: Normal mode
Step2: The CN1/CN2 cable is connected to connector CN1.
Step③: The resistors are connected by turning on discharge switch. It discharges capacitors.

Disconnect the cable CN1/CN2 from the connector CN2 "NORMAL" and connect it to the connector CN1 "DISCHARGE".
By this connection, LED on MUX CAPACITOR PCB will light on if electricity remains in the capacitors.
Then, discharging will start by turning Discharge switch to the upside.
It normally takes more than 5 minutes until the remaining voltage drops less than 1V.
（The remaining voltage can be measured between CP1 and CP2 on MUX CAPACITOR PCB.）



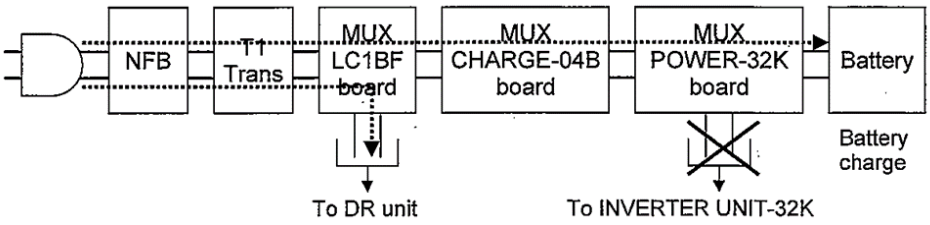
MUX CAPACITOR BOARD

1.3 DR operation and radiography during power plug connection

1.3.1 DR operation during battery charging

When the DR unit is ON and the key switch is OFF, power is supplied to the DR unit and battery when the power plug is connected to a power outlet. Therefore, the DR unit can be operated during charging.

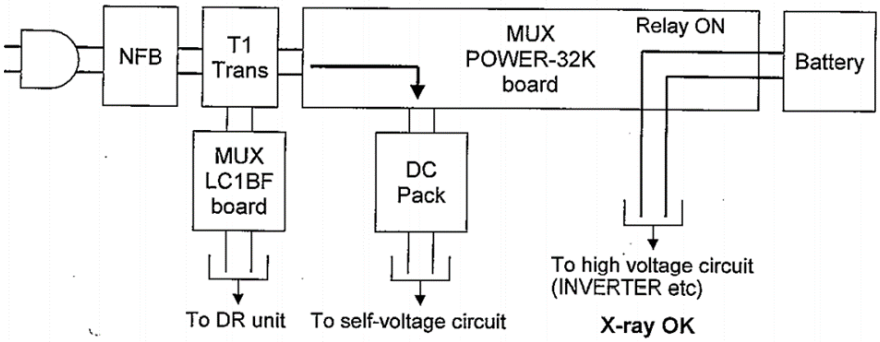
However, the battery does not supply power to the high-voltage line for X-rays. it is impossible to exposure while battery charging.



1.3.2 Radiography during plug connection

When key switch is ON and connect power plug into a power outlet, power is supplied to the internal control circuit, DR unit and except high voltage line for radiography, and battery supplies power to high voltage line for radiography.

It is possible to expose while plug connection.



1.4 Word List

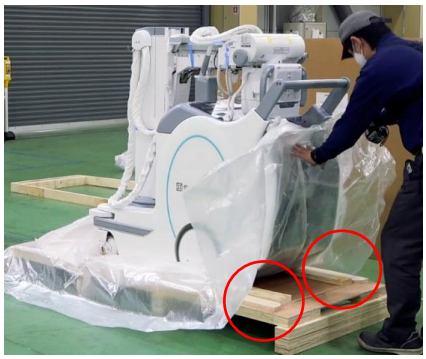
Word	Meaning
IFST	Filament heating voltage signal, which consists of FVR + feedback signal of tube current
IFV	Measured filament current (1V/1A)
VS	Reference inverter working frequency voltage until the tube voltage reaches the preset value
TKV	Measured tube voltage (1V/20 kV)
VN	Reference inverter working frequency voltage after the tube voltage reaches the preset value
TMA	Measured tube current (1V/50 mA)
RKV	Preset tube voltage (1V/20 kV)
RMA	Preset tube current (1V/50 mA)
FVR	Preset filament current (1V/1A)
VP1	+ side of the measured charged voltage of the batteries(3V/100V) / the measured charged voltage of the capacitors (3V/200V) :Changeover by the signal CHK60
VP2	- side of the measured charged voltage of the batteries (3V/100V)
IC	Charging current detection (1V/ 1A)
LV	Power supply confirmation (2V/100%), which detects 10V output from the transformer T1
VIN	Inverter frequency voltage
KV-	Tube voltage feedback signal (-) (1V/20 kV)
KV+	Tube voltage feedback signal (+) (1V/20 kV)
TH1/TH2	Contact signal of the thermal switch of X-ray tube
X/Y/Z	Stator coil for anode rotation of X-ray tube
FBST	Signal of starting kV feedback control, which is generated when TkV reaches 92% of RkV
FSR	Force sensing resistance which is mounted inside driving operation handle to control DC motors

1. Introduction (Overview and Maintenance Notes)

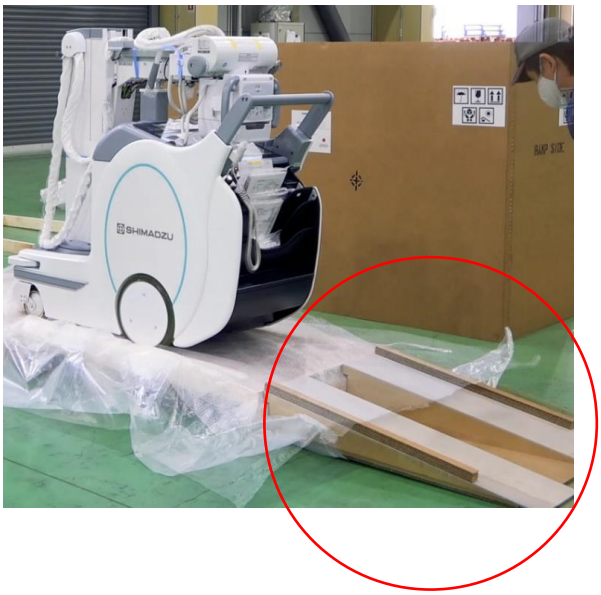
1.5 How to use the slope pallet during unpacking

Slope pallet

I. Remove the screws here and remove the wheel stopper.



II. Set up the slope pallet like this.



III. Set the main breaker to ON(upper position).

IV. Set the key switch here to ON (by turning).

1.6 Differences from the conventional device

1.6.1 Difference between MX8/MX9 and MX7 Version or earlier

Where different	Referential section in the installation manual
<Mechanical> Design concept is the same as that of MX7 Version or earlier except for the shape of parts and the position of mounting screws. Particularly different parts are: Fixed column changed to telescopic column; 15/17 inch monitor changed to 19 inch LCD monitor; Layout of a DR device; Layout of batteries; An irradiation field adjusting control provided also on the reverse side of the collimator; and FPD lock mechanism /docking mechanism are provided.	Installation of an arm cover, Battery Replacement
<Operation panel and adjustment mode> Operation method of the adjustment mode has been changed associated with change of the operation panel (sheet key) design.	Adjustment mode
<Error message> Error messages have been updated so as to gain more detailed information for troubleshooting. As to a motor-related error, for example, a different error code is displayed for the right motor and the left motor respectively.	Error message list
<Main body software> Update procedures for the main body software of XCONT board have been changed. (MX8/MX9 has no software-installed medium attached to the device while MX7 or earlier did. In addition, the software identification procedure for MX8/MX9 also has been changed from MX7 or earlier.) The software name has been changed from SH-MUX200 for MX7 or earlier to SH-MUX300 for MX8/MX9 with no interchangeability between them. The software for MX8 and MX9 also do not have compatibility to each other, however, they will be integrated in future.	Upgrading of software version

1. Introduction (Overview and Maintenance Notes)

Where different	Referential section in the installation manual
<Boards> Boards CL MEAS-17 and DR POWER CONT have been newly installed. The other boards remain the same as before except that DIPSW has some changes incorporated. Additionally, XCONT-2008X, MUX POWER 32K and MU DRIVER-07 are limited in hardware revisions that can be used for MX8/MX9 Version device, and P/N dash number has been changed for easier identification. Except for the initial devices of MX8 Version, XCONT-2008X has been changed to XCONT-2018, and the software for each boards are different. POWER-200D board has been discontinued and switched to the substitute board MUX DR CONT when MX9 Version has been released.	List of switches and jumper wires on the boards
<Option> There are some newly installed options including wireless hand switch, camera function and laser marker (for MX9 Version only) etc. for which, refer to respective installation manuals. The protective screen is almost the same as the conventional screen.	Installation of options

1.6.2 Difference between MX8 and MX9 Version

Where different	Referential section in the installation manual
<Mechanical> In MX9, FPD docking mechanism is also provided for FUJIFILM DR-ID 800.	Installation
<Error message> In case of Premium type of MX9, the error code is also displayed on the 2nd monitor. As well, the error codes related to communication with the 2nd monitor are added.	Error message list
<Main body software> The software for MX8 and MX9 also do not have compatibility to each other, however, they will be integrated in future.	Upgrading of software version
<Boards> Except for the initial devices of MX8 Version, XCONT-2008X has been changed to XCONT-2018, and the software for each boards are different. P/N of XCONT-2018 and CL MEAS-17 are different between MX8 and MX9 Version and they do not have compatibility to each other. POWER-200D board has been discontinued and switched to the substitute board MUX DR CONT when MX9 Version has been released.	List of switches and jumper wires on the boards
<Option> In MX9 Version, camera function and laser marker are added. Refer to respective installation manuals.	Installation of options

2 Required tools and PCB explanation

2. Required tools and PCB explanation

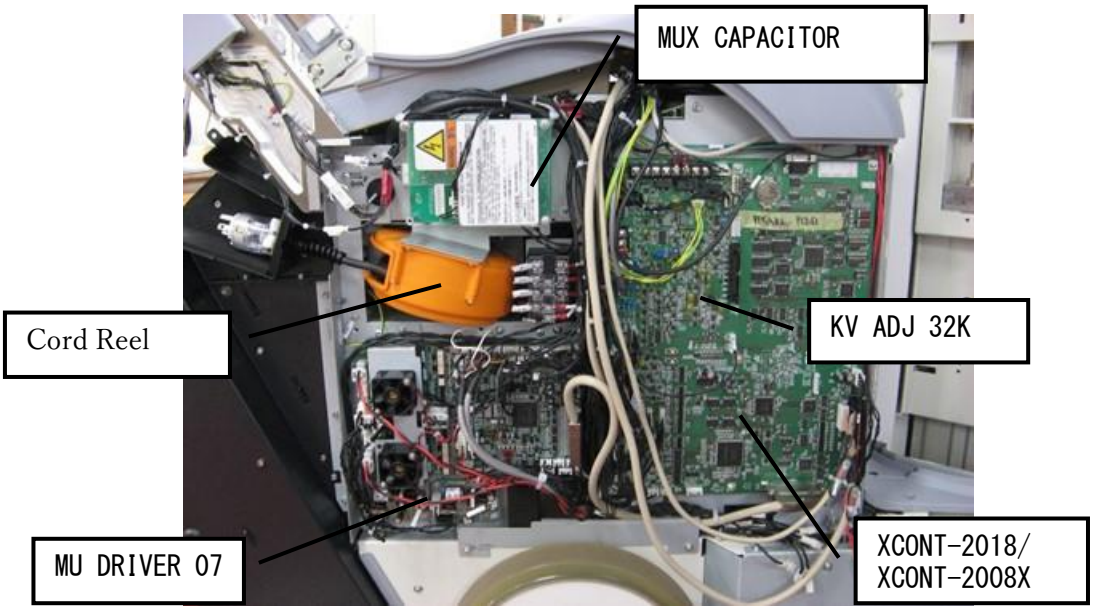
2.1 Tools Required in Maintenance

For installation or maintenance of the system, prepare the following tools:

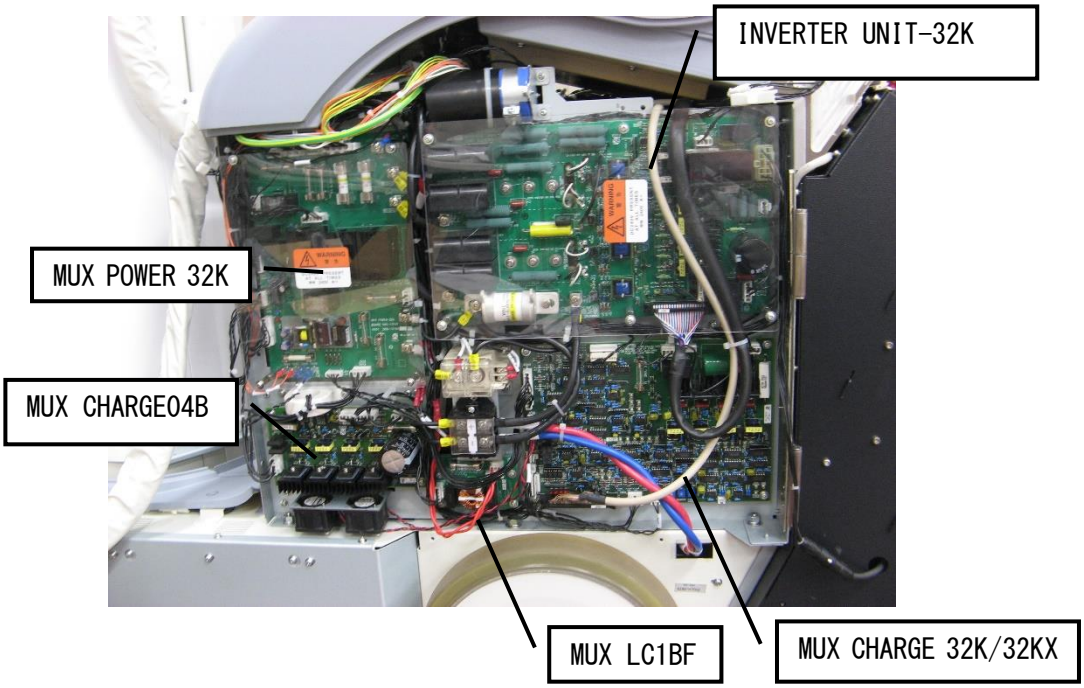
☐ Screwdriver (cross slot type/ straight slot type)
☐ Hexagon wrenches Nos. 2.5 to 10
☐ Nipper
☐ Cutting pliers
☐ Tester (digital)
☐ Alligator clip
☐ Cable tie
☐ Cutter
☐ Grease
☐ Waste
☐ Alcohol or acetone for cleaning
☐ Hexagonal bolt M8×16 (for replacing X-ray tube)
☐ Oscilloscope
☐ Pliers for FU lock nut AW04 (for adjusting X-ray tube rotation)
☐ Push-pull gauge (for adjusting X-ray tube rotation and the handle :0 ~ 20 kgf)
☐ USB port hub :2 ports or more (for setting up the DR system)
☐ Keyboard (USB type)
☐ Mouse (USB type)
☐ DVD Drive (USB type)
☐ External hard disk drive (USB type) or USB thumb drive
☐ PC (Windows10 installed Microsoft EDGE or Google Chrome)
☐ Serial cable for RS-232C (9pin female, cross type for connecting XCONT)

2.2 PCB Layout (32kW)

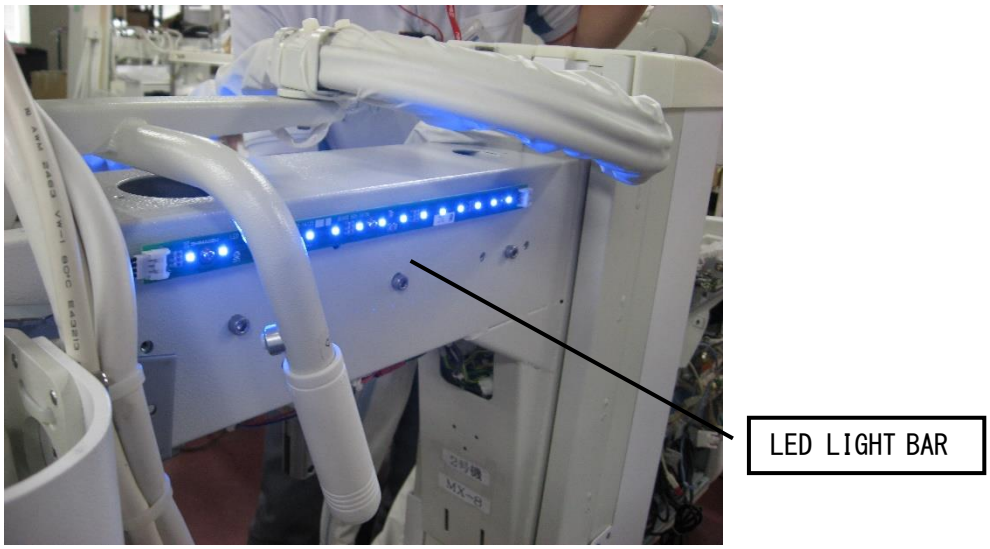
2.2.1 Right side



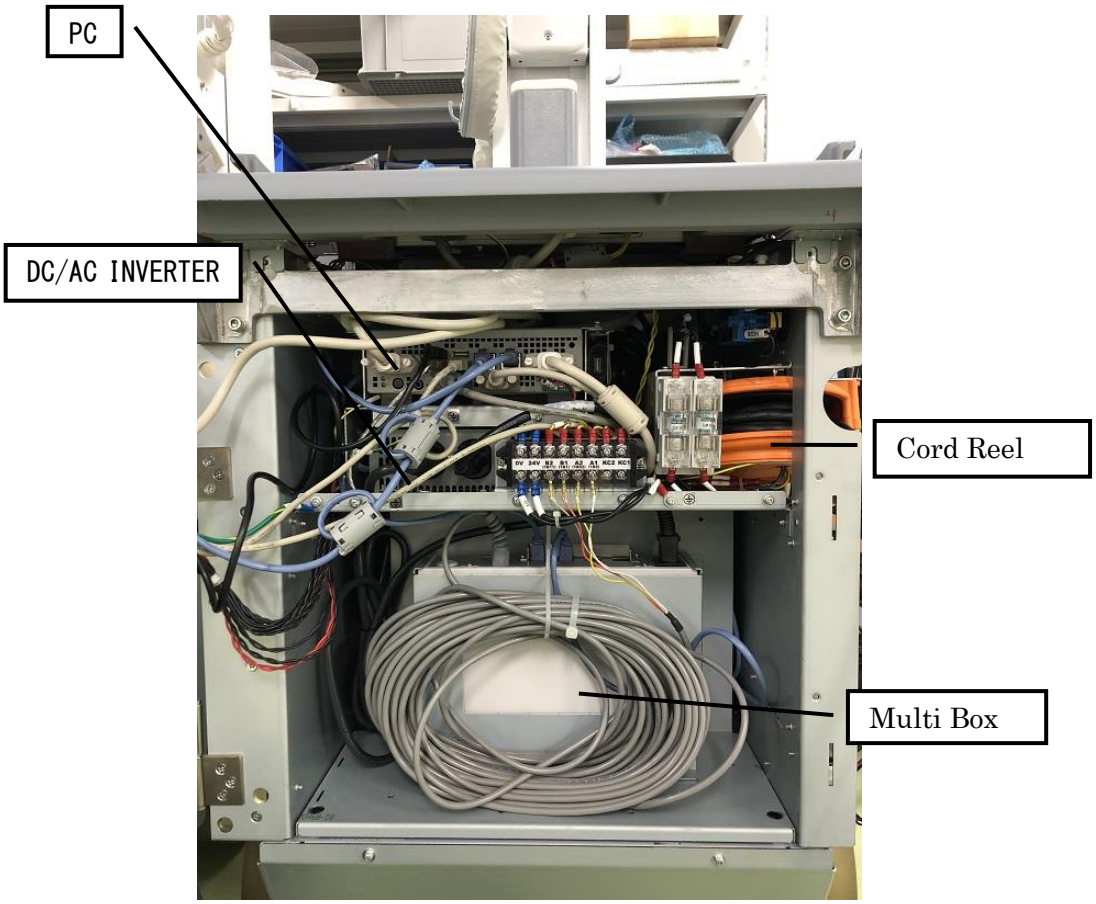
2.2.2 Left side



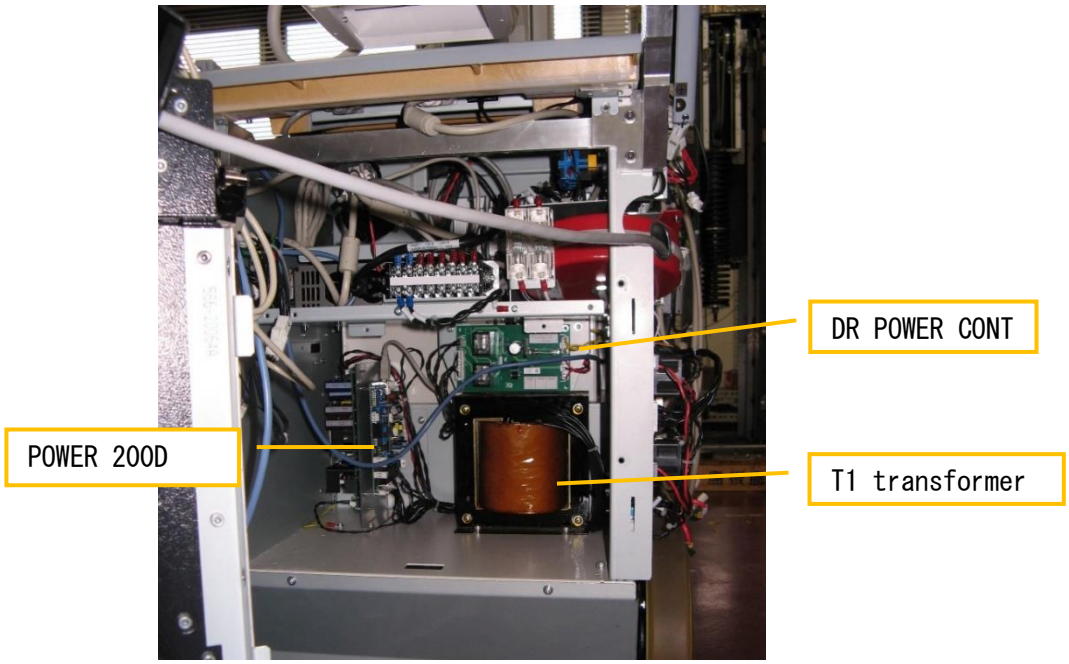
2.2.3 Arm side



2.2.4 Rear side (DR ASSY)



2.2.5 Rear side：Condition after removal of IU BOX



2.2.6 Cautionary Notes for High-Voltage Transformers

There are 2 types of 32 kW type transformers. The names on the labels are the same as the D150MH-40 and should be distinguished by P/N. In this manual, “(S) ” is used for convenience. The D150MH-40(S) is a successor to the D150MH-40, but there are combination conditions for XCONT and INVERTER UNIT-32K boards.

Name	P/N	Combination conditions	
		XCONT	INVETER UNIT-32K
D150MH-40	562-25900	XCONT-2008X: P/N: 565-12406-01, XCONT-2018: Except for 565-15000-03	P/N: 562-26303
D150MH-40(S)	565-16863 or 565-16863-01	XCONT-2018: P/N: 565-15000-01/02 or Rev. K or later of 565-15000 (565-15000-03 in case of MX9)	P/N: 562-26303-01

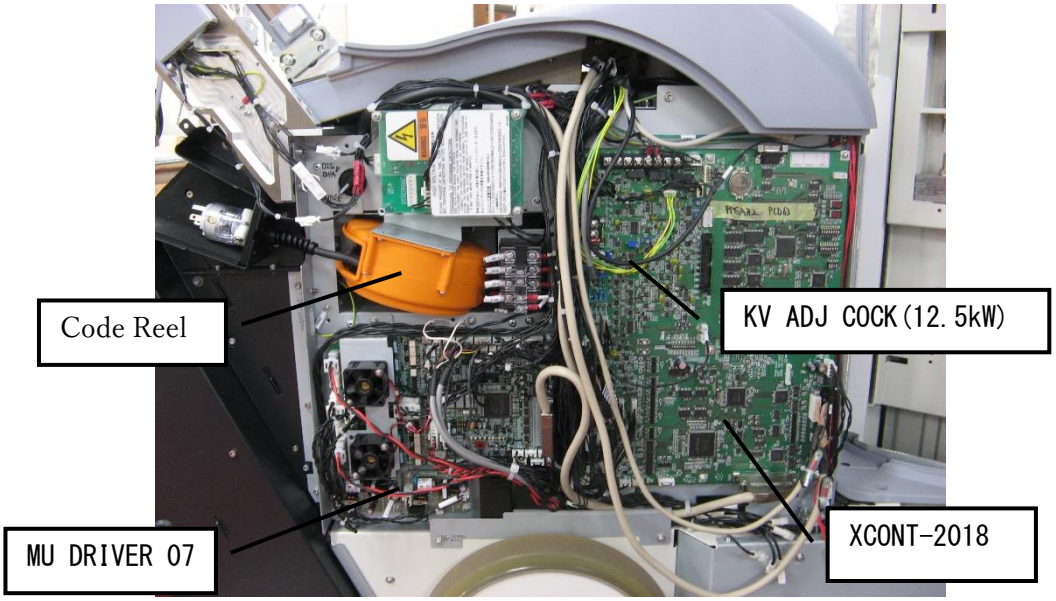
Note:

All MX9 and 32kW type of MobileArt Evolution MX8 Version is equipped with D150MH-40(S).

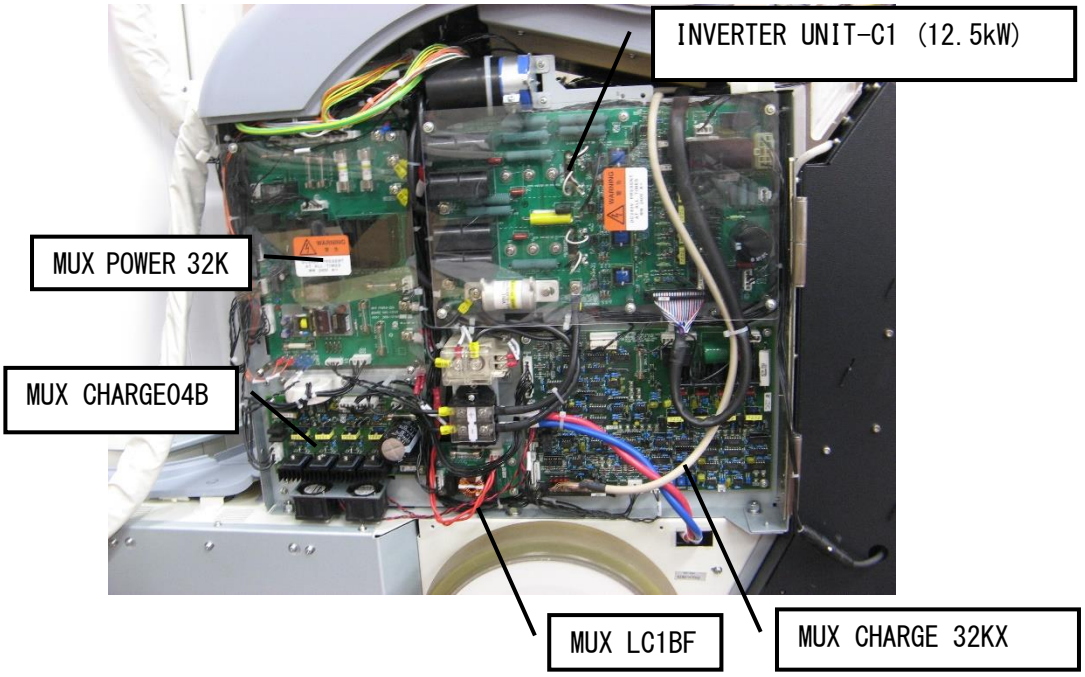
MobileDaRt Evolution MX8 Version is equipped with D150MH-40(S) since around Sep. of 2023.

2.3 PCB Layout(12.5kW)

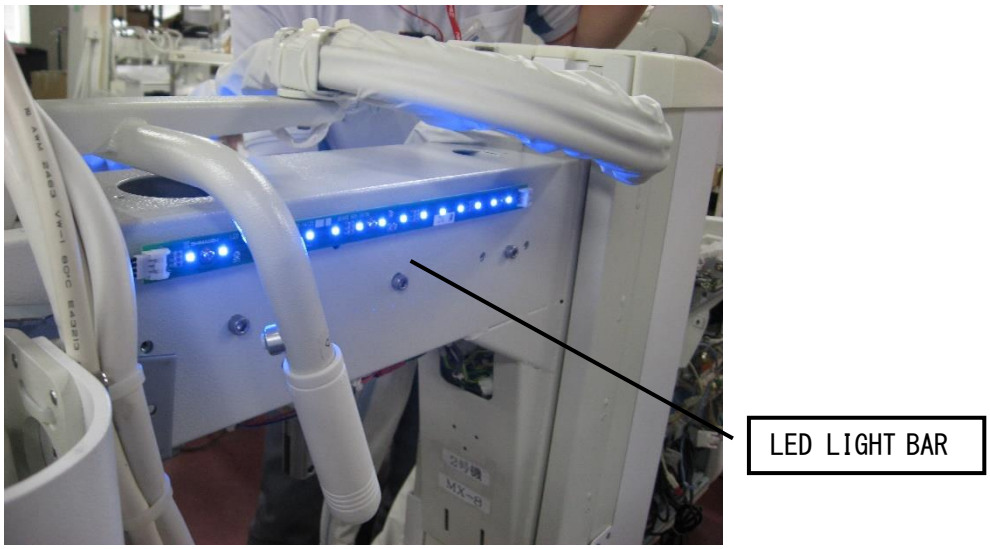
2.3.1 Right side



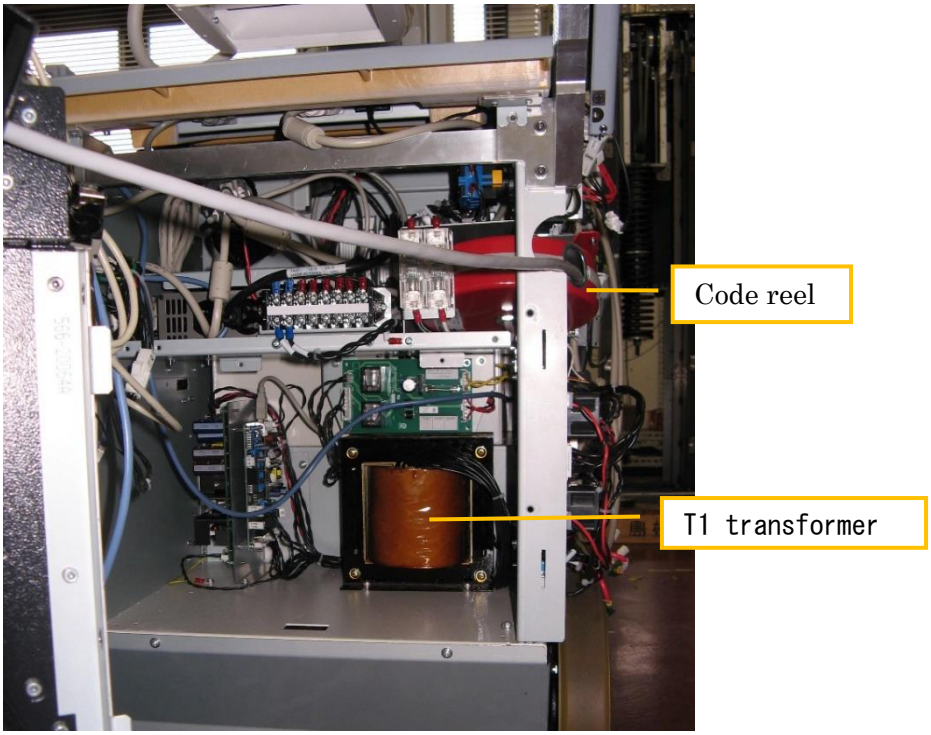
2.3.2 Left side



2.3.3 Arm side



2.3.4 Rear side(DR ASSY)



2.4 Outline of Each PCB

2.4.1 XCONT PCB (XCONT-2008X PCB or XCONT-2018 PCB)

☐ Controls the operation panel.
☐ Offers diversified I/Os.
☐ Performs D/A and A/D conversion.
☐ Detects and controls the tube voltage and detects discharge of the X-ray tube.
☐ Detects and controls the tube current through filament heating.
☐ Detects the tube current.
☐ Interfaces each PCB.
☐ Initializing circuit for FPGA on MU DRIVER-07 PCB.
☐ Pulse count circuit for the dose meter option
☐ Output for the speaker
☐ Circuit for the lighting hand switch
☐ Interface for MUX CAPACITOR PCB, which controls charging of the capacitors
☐ Interfaces for POWER-200D, which controls power supply for DR system
☐ Circuit for 32kW H.V. Transformer
☐ Circuit for 12.5kW H.V. Transformer
☐ Applicable to the FPD lock mechanism
☐ Applicable to the wireless hand switch option.
☐ Control by LED LIGHT BAR
☐ Charge start operation from the standby (Auto power off) condition by plugging into the outlet.
☐ The unit returns automatically from charging and transits to the radiography mode by the operation of various switches.

2.4.2 KV ADJ 32K PCB

☐ KV detection circuit for 32kW H.V. Transformer. This PCB is mounted on XCONT PCB.

2.4.3 KV ADJ COCK PCB

☐ KV detection circuit for 12.5kW H.V. Transformer. This PCB is mounted on XCONT PCB.

2.4.4 MUX CHARGE-32K/32KX PCB

☐ Detects the battery voltage and charged voltage error.
☐ Controls the battery charging current and detects charging current error.
☐ Offers the inverter circuit for the starter.

2.4.5 MUX CHARGE-04B PCB

□ Offers the inverter circuit for battery charging.

□ Detects the battery charging current.

2.4.6 MUX-LC1BF PCB

□ Noise Filter for the power supply of MUX CHARGE-04B PCB.

2.4.7 MUX POWER-32K PCB

□ Turns on and off the power, and offers the automatic power OFF function.

□ Supplies the power from the batteries to each part.

□ Change over AC and DC of the DC pack input.

□ Automatic charge starts when a certain time passed without performing the operation while the product is connected to the outlet.

□ The unit returns from charging and transits to the image shooting mode by the operation from the hand switch or the seat panel during charging.

2.4.8 MUX CAPACITOR PCB

□ Detects the charged voltage of the capacitors for 32kW type.

□ Controls changeover of the resistance for charging of the capacitors.

□ Discharge of the capacitors

2.4.9 INVERTER UNIT-32K PCB

□ Offers the main inverter for 32kW type.

□ Offers the inverter circuit for heating the filament.

□ Controls the starter. Detects the stator coil current.

2.4.10 INVERTER UNIT-C1 基板

□ Offers the main inverter for 12.5kW type.

□ Offers the inverter circuit for heating the filament.

□ Controls the starter. Detects the stator coil current.

2.4.11 MU DRIVER-07 PCB

☐ Controls the DC motors.
☐ Detects the handle operation.
☐ Detects the bumper switch.
☐ Controls the arm catch area.
☐ Controls the electromagnetic lock.
☐ Detects the Stop switch.
☐ Detects the Emergency Brake Release switch
☐ Self-diagnosis circuit of FET for driving DC motors
☐ Travel (Brake bar input) from the standby (Auto power off) condition and the return by the arm operation (All free switch input).

2.4.12 SHEET PANEL MUX300 PCB

☐ Sheet panel for operation of X-ray high voltage generator

2.4.13 POWER-200D PCB

☐ Controls the power supply for DR system.

2.4.14 DR POWER CONT PCB

☐ Control of the power supply for DR unit (The power supply for DR unit is switched between the external power supply and internal battery every time the power is plugged in or off the outlet.)
☐ Rectification and smoothing of the DR device-dedicated AC power supply at the time of connecting the AC power plug (Prevention of arc discharge from the switching relay contacts arising at the time when the DR device-dedicated AC power supply is switched to an external AC power source or internal battery accompanied by disconnection of the AC power plug)

2.4.15 LED POWER-20 PCB

☐ Controls LED lamp

2.4.16 CL MEAS-17 PCB

☐ Leaf opening measurement with a collimator and other items for MobileDaRt evolution

2.4.17 LEAF-MEAS-15 PCB

☐ Leaf opening measurement with a collimator and other items for MobileArt Evolution

2.4.18 LED LIGHT BAR

☐ The condition of the unit is displayed by the emission colors of LEDs mounted on the boards that are installed on both sides of X-ray tube support arm.

2.4.19 RC TRANSMITTER PCB (option)

☐ Controls send for Infrared remote control.

2.4.20 RC RECEIVER PCB (option)

☐ Controls receive for Infrared remote control.

2.5 XCONT (XCONT-2008X / XCONT-2018) PCB

2.5.1 Function

This is the CPU PCB which controls system by using the software.

The following data is stored in this PCB. It is recommended to back up the following data in preparation for PCB replacement.

- FVR data
- Handle data
- Battery data
- User setting data
- Notification Information data
- Error log
- APR data
- Dose Area Product Meter(option) adjustment data
- Dose Area Product Calculation Unit adjustment data
- DR setting mode data
- APR and X-ray condition which was set just before the power was turned off last time
- Time setting data

2. Required tools and PCB explanation

This PCB equipped with the following circuits performs general control.

- I/O circuit, A/D circuit and D/A circuit
- Circuits to detect and feedback TKV and TMA
- TKV OVER detection circuit (X-ray tube discharge detection circuit)
- System timer circuit (X-ray cutoff signal)
- IGBT drive control circuit
- Filament heating FET drive control circuit
- Dynamic display control circuit, sheet key input circuit and sheet key search circuit
- -15V detection circuit
- Power ON reset circuit (master reset signal)
- Charging control
- Interface circuit with each PCBs
- Initializing circuit for FPGA on MU DRIVER-07 PCB.
- Pulse count circuit for the dose meter option
- Output for the speaker
- Circuit for the illuminated hand switch option
- Interface for MUX CAPACITOR PCB, which controls charging of the capacitors
- Interfaces for POWER-200D, which controls power supply for DR unit
- Circuit for 32kW H.V. Transformer
- Applicable to the FPD lock mechanism
- Applicable to the wireless hand switch option
- Control by LED LIGHT BAR
- Charge start operation from the standby (Auto power off) condition by plugging in to the outlet.

  The unit returns automatically from charging and transits to the image shooting mode by the operation of various switches.

2.5.2 PCB replacement

- Remove the KV ADJ 32K/COCK PCB from the old XCONT and attach it to the new XCONT.
- For MobileArt Evolution MX8 Version, be sure to install the XCONT-2018 board (P/N: 565-15000-01/02 or Rev. K or later of 565-15000).
  The Rev.K supports the high-voltage transformer D150MH-40(S) (P/N: 565-16863) and D125PH-C1 (P/N: 562-20700-02). (FPGA revised)
- For MobileDaRt Evolution MX8 Version, when replacing the high voltage transformer from D150 MH-40 (P/N: 562-25900) to D150 MH-40(S) (P/N: 565-16863 or 565-16863-01), be sure to attach the XCONT-2018 board (P/N: 565-15000-01/02 or Rev. K or later of 565-15000).
  As well, INVERTER UNIT-32K must be replaced to P/N:562-26303-01 which corresponds to D150 MH-40(S) together.

Reference:

5.19 Replacement of the XCONT PCB, the INVERTER UNIT PCB and the H.V. Transformer in this manual

- In case of MX9 Version, be sure to install the XCONT-2018 board (P/N: 565-15000-03).
- In MX8 Version, if Dongle is combining the required options (BiAA option), use XCONT-2018 (P/N: 565-15000-02).

2.5.3 Replacing the PCB

After replacing the XCONT, set the DIP switches and jumpers first, then perform initialization of the PCB by setting DIP SW2-8 and SW3-1 to ON. If there is a backup data, upload it. Next, perform FVR adjustment (Confirmation). After that, refer to the installation manual and execute "Adjusting mAs circuit" "Adjustment of the inverter control circuit (Confirmation)" of the XCONT-2018 PCB. Then execute [Adjustment of battery voltage detection circuit].

Note:
Software Rev. 1.50 or later performs initialization according to the settings of DIP SW2-1, 2-2, and 2-6. Set the DIP SW according to the specifications of the device before initialization. In case of less than Rev. 1.50, initialization for D150 MH-40 (P/N: 562-25900) is performed regardless of DIP SW setting.

<table>
<tr><th rowspan="2">Model</th><th colspan="3">DIP SW setting of XCONT</th><th rowspan="2">H. V. Transformer</th><th rowspan="2">XCONT Combination conditions</th></tr>
<tr><th>2-1</th><th>2-2</th><th>2-6</th></tr>
<tr><td rowspan="2">DaRt
MX8 Version</td><td>ON</td><td>OFF</td><td>OFF</td><td>D150MH-40
(P/N: 562-25900)</td><td>XCONT-2008X:
P/N: 565-12406-01,
XCONT-2018:
Except for 565-15000-03</td></tr>
<tr><td>ON</td><td>OFF</td><td>ON</td><td>D150MH-40(S)
(P/N : 565-16863/-01)</td><td rowspan="3">XCONT-2018
P/N: 565-15000-01/02 or
Rev. K or later of 565-15000</td></tr>
<tr><td>Art
32kW type</td><td>OFF</td><td>OFF</td><td>ON</td><td>D150MH-40(S)
(P/N : 565-16863/-01)</td></tr>
<tr><td>Art
12.5kW type</td><td>OFF</td><td>ON</td><td>OFF</td><td>D125PH-C1
(P/N: 562-20700-02)</td></tr>
<tr><td>DaRt
MX9 Version</td><td>ON</td><td>OFF</td><td>ON</td><td>D150MH-40(S)
(P/N : 565-16863/-01)</td><td>XCONT-2018:
P/N: 565-15000-03</td></tr>
</table>

Each adjustment methods are different depending on the model of the high-voltage transformer as follows.

(1)Adjusting the Inverter

Below is an excerpt from the MobileArt/DaRt Evolution Installation Manual.

Note that the method of adjusting the IGBT gate waveform differs between the old and new high-voltage transformers.

Check the inverter pulse width.

1 Connect oscilloscope probes between CP5 [Ga] - CP6 [Ea] and CP10 [Gb] - CP12 [Eb] on INVERTER UNIT board.

2 Perform READY and XRAY operations, and check the gate waveform of IGBT.

3 Confirm that the pulse width at +5 V is as shown in the following table.

Table 9-2

	Models	High Voltage Transformer	pulse width
32kW type	MobileDaRt Evolution	D150MH-40 (P/N : 562-25900)	7.8 - 8.0 μs
		D150MH-40(S) (P/N : 565-16863)	10.9 - 11.1 μs
	MobileArt Evolution high-power type	D150MH-40(S) (P/N : 565-16863)	10.9 - 11.1 μs
12.5kW type	MobileArt Evolution standard type	D125PH-C1	7.8 - 8.0 μs

4 If necessary, readjust the pulse width using the procedure in "9.4.4 Adjusting inverter pulse width of XCONT-2018 board".

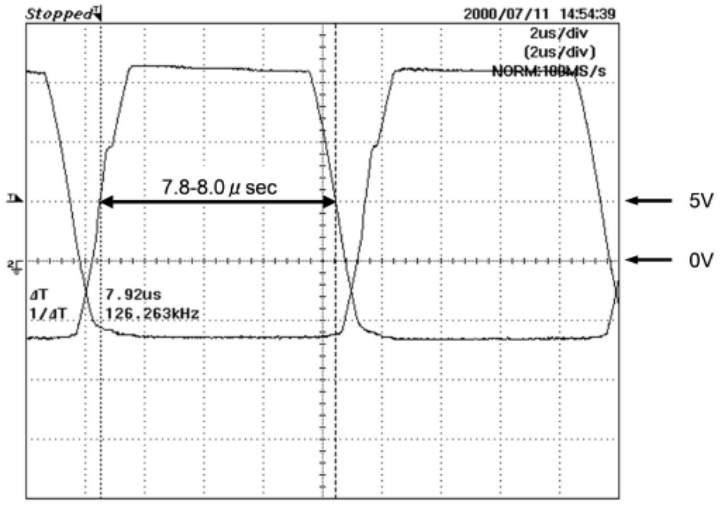


Fig. 9-1 Waveform between CP[Ga] – CP[Ea], CP[Gb] – CP[Eb] on INVERTER UNIT board (for D125PH-C1 and D150MH-40)

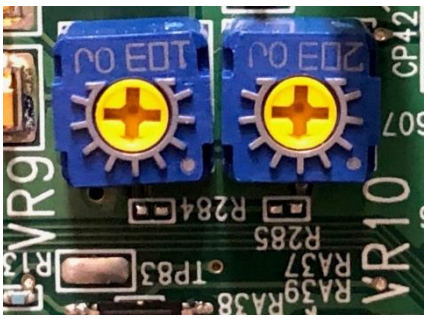
NOTE Check the gate waveform of the time period later than 10 ms after the start of the X-ray exposure. This is to ignore the initial peak value and enable accurate reading.

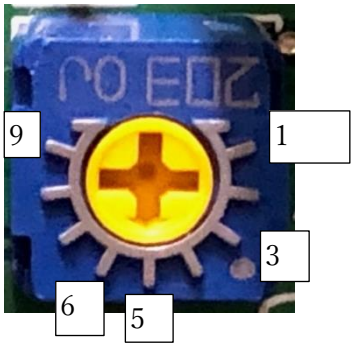
(2)Adjusting VR9, VR10 on XCONT-2018

The way to adjust the trimmer VR9 and VR10 on the XCONT-2018 board is different for the old and new high-voltage transformers.

Only one of the trimmer's crosses is an arrow. Adjust the arrow to point to the memory location in the table below. Regarding VR10, it should be adjusted by measuring the resistance value between CP61［FB RES］ and CP42［VINCHK3］ on XCONT-2018 board. Note that the adjustment value varies depending on the type of transformer.

Model	High voltage transformer	VR9	VR10
Dart MX8	P/N ： 562-25900 D150MH-40	Turn 5 scales in CW	5.8k ohm±5%
	P/N ： 565-16863/-01 D150MH-40(S)	Turn 6 scales in CW	10k ohm (Turn 5 scales in CW)
Art MX8（32kW）	P/N ： 565-16863/-01 D150MH-40(S)	Turn 6 scales in CW	10k ohm (Turn 5 scales in CW)
Art MX8(12.5kW)	P/N ： 562-20700-02 D125PH-C1	Turn 5 scales in CW	10k ohm (Turn 5 scales in CW)
Dart MX9	P/N ： 565-16863/-01 D150MH-40(S)	Turn 6 scales in CW	10k ohm (Turn 5 scales in CW)





(3)Adjusting the tube voltage waveform

In the tube current 2 point adjustment mode, adjust VR4 (kV-ADJ) of the KVADJ board so that the tube voltmeter indicates 80 kV +/- 1 kV, and adjust VR3 (WF-ADJ) so that the tube voltmeter waveform is horizontal. The adjustment point (tube current value: 50mA or 200mA) at this time differs depending on the model of high-voltage transformers as shown on the below table. Anyway, this adjustment is normally not necessary if the high voltage transformer is not replaced.

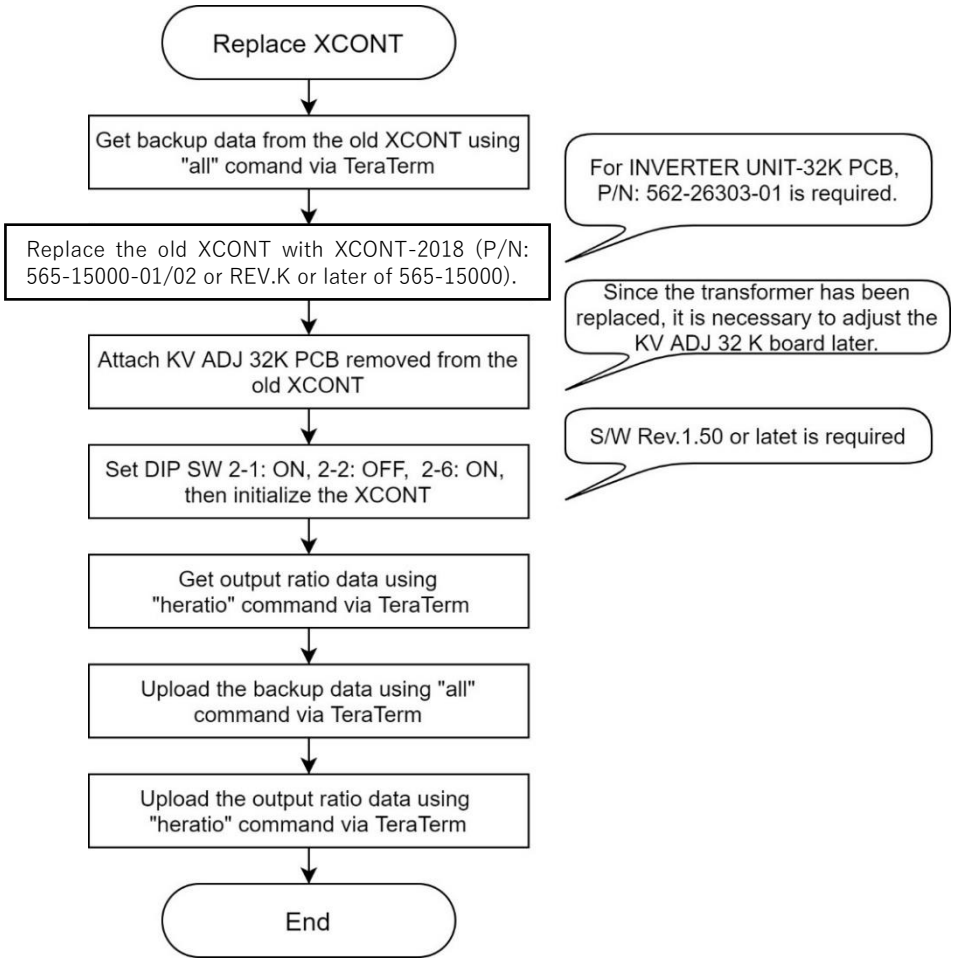
Set the adjustment point as shown below.

Table 9-5

	Models	High Voltage Transformer	Adjustment point
32kW type	MobileDaRt Evolution	D150MH-40 (P/N : 562-25900)	80kV 50mA
		D150MH-40(S) (P/N : 565-16863)	80kV 200mA
	MobileArt Evolution high-power type	D150MH-40(S) (P/N : 565-16863)	80kV 200mA
12.5kW type	MobileArt Evolution standard type	D125PH-C1	80kV 50mA

2.5.4 Flow chart for replacing D150-MH40 with D150-MH40(S)

When the high-voltage transformer type is changed, the XCONT and INVERTER UNIT-32K boards also must be changed. At this time, if the backup data of the original XCONT board before replacement is written to the new XCONT board, the output ratio data for the original high-voltage transformer type will also be written to the new XCONT. Therefore, it is necessary to follow the procedure below to rewrite the output ratio data for the new high-voltage transformer type. (Since the output ratio data in the XCONT board is not rewritten unless initialization is performed with the DIP SW setting corresponding to the type of high-voltage transformer, this procedure is necessary.)



2.5.5 CPU backup battery replacement

When the CPU back up battery is empty, some data (error log and time setting data etc.) stored in the SRAM will be erased. When the message requesting battery replacement is displayed, replace the battery using the following procedure.

- Turn on the key switch to apply the power.
- Replace the CPU backup battery.
- Refer to "Resetting the CPU backup battery usage information" in the installation manual to set the usage date.

2.5.6 LED List

LED No.	Description
LD1	"sh" is normally displayed here since the first operation is done from the console after power on.
LD2	
LD3	Normally, lights off.
LD4	Normally, flickers with very high frequency.
LD5	Normally, flickers with low frequency.
LD6	Normally, lights off.

2.5.7 Check pin list

CP No.	CP Name	Description
CP1	System clock	CLK6M
CP2	System clock	CLK12M
CP3	Vbatt	Lithium battery voltage
CP4	+5V	+5V
CP5	DGND	GND

2.5.8 Jumper list

Jumper	Pin No.	Name	Setting
JP1	1-2	IN	Shorted (Normal setting :The CPU backup battery is connected.)
	2-3	OUT	Open

2.5.9 DIP Switch list

Number	Function	Setting	
		ON	OFF
SW2-1	System setting	DaRt type	Art type
SW2-2	Configuration for 12.5kW type*2	12.5kW type	32kW type
SW2-3	Debug mode	Debug mode	Normal mode
SW2-4	DR Communication 1	DR-200m, Connect maintenance PC	DR-ID800, CXDI, CS-7, Art type
SW2-5	Measured value display mode	Measured value display mode	Normal mode
SW2-6	Configuration for high voltage transformer*2	D150MH-40(S) (32kW)	D150MH-40 (32kW), D125PH-C1 (12.5kW)
SW2-7	DR Communication 2	DR-ID800, DR-200m	CXDI, CS-7, Art type
SW2-8	Initial setting mode	Initial setting mode	Normal mode
SW3-1	EEPROM Initialization	Initialization Mode*1	Normal mode
SW3-2	Keyless Entry option	Option enable	Option disable
SW3-3	Infrared remote controller option	Option enable	Option disable
SW3-4	DR Communication 3, FPD Lock (Art type) *2	DR-200m, Lock Enable (Art type)	DR-ID800, CXDI, CS-7, Lock disable (Art type)
SW3-5	DAP option	Option enable	Option disable
SW3-6	DR Communication 4 *2	DaRt type	Art type or DaRt (CS7 with serial radiography etc.)
SW3-7	CS7 serial radiography*3	With serial radiography kit	Without serial radiography
SW3-8	Adjustment mode	Adjustment mode	Normal mode
SW4-1	RESERVED		Normally OFF
SW4-2	XCONT Software update	Software update mode	Normal mode
SW4-3	RESERVED		Normally OFF
SW4-4	RESERVED		Normally OFF

*1: EEPROM Initialization is performed when SW3-1 is turned ON together with SW2-8.
In Rev. 1.50 and later, initialization is executed according to the setting of SW2-1, SW2-2, and SW2-6.

*2: Added in Rev.1.50.

*3: Added in Rev.1.82.

2.6 KV ADJ 32K

2.6.1 Function

This board is a small board mounted on an XCONT board and has adjustment trimmers for the tube voltage detection circuit for adjustment.

2.6.2 VR list

VR No.	Function	How to adjust
VR3	WF-ADJ	Waveform on KV meter should be flat.
VR4	KV-ADJ	KV meter should display the setting kV±1kV.

2.6.3 PCB replacement

- When XCONT PCB is replaced, the original KV ADJ 32K/COCK PCB can be used without any re-adjustment.
- When the high voltage transformer is replaced, VR3 and VR4 should be re-adjusted by using kV meter.

Note:
When replacing the high-voltage transformer D150MH-40 (P/N: 562-25900) to D150MH-40 (S) (P/N: 565-16863), be sure to install the XCONT-2018 board (P/N: 565-15000-01/02 or Rev. K or later of 565-15000). For the INVERTER UNIT-32K board, install P/N: 562 -26303-01 that supports D150 MH-40(S) (P/N: 565-16863).

2.7 KV ADJ COOK

2.7.1 Function

This board is a small board mounted on an XCONT board and has adjustment trimmers for the tube voltage detection circuit for adjustment.

The constant of the tube voltage detection circuit differs between the 32 kW type and the 12.5 kW type. The XCONT board can be shared by the board. (Error F1B occurs if the correct type of KVADJ board is not installed for the equipment output.)

・When the equipment rating is 32 kW - > KV ADJ 32 K board installed

・When the equipment rating is 12.5 kW - > KV ADJ COCK board installed

2.7.2 VR list

VR No.	Function	How to adjust
VR3	WF-ADJ	Waveform on KV meter should be flat.
VR4	KV-ADJ	KV meter should display the setting kV±1kV.

2.7.3 PCB Replacement

- When XCONT PCB is replaced, the original KV ADJ 32K/COCK PCB can be used without any re-adjustment.
- When the high voltage transformer is replaced, VR3 and VR4 should be re-adjusted by using kV meter.

Note:

When replacing the high-voltage transformer with D150MH-40 (P/N: 562-25900) to D150MH-40 (S) (P/N: 565-16863), be sure to install the XCONT-2018 board of Rev. J or later. For the INVERTER UNIT-32K board, install P/N: 562 -26303-01 that supports D150 MH-40(S) (P/N: 565-16863).

2.8 MUX CHARGE 32K/32KX

After May 2020, MUX CHARGE-32K board was switched to the successor MUX CHARGE-32KX board. MUX CHARGE-32KX board can be used as a substitute in case of the failure of MUX CHARGE-32K.

Note: Basically, the new board MUX CHARGE-32KX has compatibility with the old board MUX CHARGE-32K except for 12.5kW type of the initial version of MobileArt Evolution which is provided with halogen lamp. The brightness of the collimator lamp will be not enough if MUX CHARGE-32KX is combined with the halogen lamp type of the collimator lamp for 12.5kW type of MobileArt Evolution.
On the other hand, MUX CHARGE-32K cannot be combined with MX8 Version of MobileArt Evolution.

2.8.1 Function

- Detection the battery voltage and charged voltage error.
- Controls the battery charging current, and detects charging current error. The inverter itself is included in MUX CHARGE-04B PCB.
- Inverter circuit for the starter and the collimator lamp. (However, the collimator lamp is driven by XCONT board instead of MUX CHARGE-32K/32KX in case of the newer models MX7, MX8 and MX9.)

2.8.2 PCB replacement

<In case of MUX CHARGE-32K>
The VR1 (Lamp), the VR2 (Starter) and the VR3 (Reference frequency) should be adjusted. Refer to “Adjusting MUX CHARGE-32K board” in Installation Manual.
As well, confirm the detected battery charged voltage referring to “Adjusting the battery voltage detection circuit” in Installation Manual.

<In case of MUX CHARGE-32KX>
Since the potentiometers VR1 (lamp), VR2 (Starter), and VR3 (reference frequency) do not exist, only confirmation of the dip switch setting is required.
Confirmation of “Adjusting the battery voltage detection circuit” in Chapter 7 of Installation Manual is necessary.

2.8.3 LED list

LED No.	Signal Name	Description
LD1	CHON	Lights ON during charging the battery.
LD2	LAMP ON	Lights ON when 70/60Hz inverter is working ON.
LD3	STON	Lights ON when collimator lamp is selected for 70/60Hz inverter.
LD4	VCOK	Lights ON when the charged voltage of capacitors reaches about 200V and the signal VCOK generates.

2.8.4 VR list (Only for MUX CHARGE-32K)

Name	Function	How to adjust
VR1	Inverter pulse for Lamp	Refer to "Adjusting MUX CHARGE 32K" in Installation Manual.
VR2	Inverter pulse for Starter	
VR3	Reference frequency (140Hz)	

2.8.5 DIP switch list (Only for MUX CHARGE-32KX)

Switch No.	Function	Switch setting	Default setting
S1-1	Depends on the models	OFF : MobileDaRt Evolution ON : Others	OFF
S1-2	No use	OFF	OFF
S1-3	No use	OFF	OFF
S1-4	No use	OFF	OFF

2.8.6 Check pin list

CP No.		Signal Name	Description
32K	32KX		
CP1	CP52	-15V	-15V
CP2	CP53	GND	GND
CP3	CP58	+15V	+15V
CP4	CP72	VP1	+ side of the measured charged voltage of the batteries (3V/100V) / the measured charged voltage of the capacitors (3V/200V) : Changeover by the signal CHK60
CP5	CP63	VP2	- side of the measured charged voltage of the batteries: 3V/ 100V
CP6	CP13	Q2	Gate pulse for Starter and Lamp
CP7	CP15	Q1	Gate pulse for Starter and Lamp
CP8	CP27	GND	GND
CP9	CP21	IC1	Measured charging current of the batteries : 1V/1A
CP10	CP60	120+	+ side of the measured charged voltage of the batteries: 3V/ 100V
CP11	CP59	120-	- side of the measured charged voltage of the batteries: 3V/ 100V
CP12	CP54	VC	Measured charged voltage of the capacitors: 3V/ 200V
CP13			No use
CP14		50kHz	
CP15	CP38	GND	GND
CP16		120Hz/140Hz	Reference frequency for Starter and Lamp
CP17		Lamp/Starter	Lamp/Starter change over
CP18			No use
CP19	CP26	LV	Line Voltage (2V/100%)
CP20	CP16	Q14C	Gate pulse for charging the battery
CP21		Q14C	Gate pulse for charging the battery
CP22	CP9	Q23C	Gate pulse for charging the battery
CP23		Q23C	Gate pulse for charging the battery
CP24	CP20	LF	4kHz pulse for 0.4A or 0.18A of setting charging current
CP25	CP22	PIC	Setting value of charging current: 1V/1A
CP26	CP23	HF	26kHz pulse for 2.0A or 1.0A of setting charging current
CP27	CP17	CLK1	26kHz clock pulse for 2.0A or 1.0A of setting charging current
CP28	CP19	CLK2	4kHz clock pulse for 0.4A or 0.18A of setting charging current
CP29	CP14	Q4	Gate pulse for Starter and Lamp
CP30	CP12	Q3	Gate pulse for Starter and Lamp
CP31	CP56	GND	GND

2.8.7 Fuse list

Fuse No.	Description
F1	250V,0.5A fuse for line voltage detection circuit

2.9 MUX CHARGE-04B

2.9.1 Function

Generation of battery charging voltage (inverter circuit)

2.9.2 PCB replacement

No adjustment is necessary.

2.9.3 Check pint list

CP No.	CP Name	Description
CP1	IC2	Output for charging the battery
CP2	IC1	
CP3	Q14C	Gate pulse for charging the battery
CP4	Q23C	
CP5	IC	Measured charging current. 1V/1A
CP6	+15V	+15V
CP7	GND	GND

2.9.4 Fuse

Fuse Name	Description
F1	250V, 5.0A fuse for AC125V input power supply for charging the battery

2.10 MUX-LC1BF

2.10.1 Function

Noise Filter for MUX CHARGE-04B Substrate

A noise filter for the output of a T1 transformer, in which the output of MUX-LC1BF becomes the input of MUX-CHARGE-04B.

2.10.2 Replacement PCB

No adjustment is necessary.

2.10.3 Fuse

Fuse Name	Description
F1	250V, 6.25A fuse for AC125V input power supply for charging the battery

2.11 MUX POWER-32K

2.11.1 Function

- Power ON/OFF, Auto Power Off Circuit
- Battery to supply power to all parts
- AC/DC input switching circuit to DC power source
- Auto charging starts after a certain period of time without any operation when plugged in.
- During charging, the camera returns from charging with a hand switch or seat panel operation and enters shooting mode.

2.11.2 PCB replacement

Confirm the detected battery charged voltage.
“Adjusting the battery voltage detection circuit” in Installation Manual.

2.11.3 Fuse list

Fuse No.	Description
F1	250V, 2.0A fuse for AC100V input power supply from the power plug for +5V, ±15V and +24V DC pack
F2	250V, 2.0A fuse for±120V power supply from the battery for +5V, ±15V and +24V DC pack
F3	250V, 6.25A fuse for±120V power supply from the battery for filament heating (INVERTER UNIT) and starter/collimator lamp (MUX CHARGE-32K)
F4	660V, 10A fuse for +120V power supply from the battery for DC motors (MU DRIVER-07)
F5	660V, 10A fuse for −120V power supply from the battery for DC motors (MU DRIVER-07)
F6	250V, 1A fuse for 12VDC power supply PS1(VAF512)
F7	250V, 5.0A fuse for charging the battery by using SSR
F8	250V, 5.0A fuse for charging the battery by using SSR
F9	250V, 10A fuse for power supply L, L12 from T3 transformer for collimator lamp
F10	250V, 2A fuse for +120V power supply from the battery for DR system

2.11.4 LED

LED No.	Function
LD1	Lights on when the key switch is turned on and DC12V output is available from the DC power supply PS1 on this PCB.

2.12 MUX CAPACITOR

This PCB controls charging and discharging of the capacitors, which are provided to generate 32kW output of X-ray.

2.12.1 Function

- Detects the charged voltage of the capacitors.
- Controls changeover of the resistance for charging of the capacitors.
- Discharge of the capacitors

2.12.2 LED list

LED No.	Description
LD1	When the capacitor is discharged (when the cable "CN1/CN2" is connected to the connector CN1), it lights up if there is any charge remaining.

2.12.3 Check pin list

CP No.	Name	Description
CP1	VC+	+ side of the measured charged voltage of the capacitors
CP2	VC0	- side of the measured charged voltage of the capacitors

2.12.4 PCB replacement

No adjustment is necessaty.

2.13 INVERTER UNIT-32K

Note:

The available P/N vary depending on the high voltage transformer installed in the device.

- P/N: 562-26303 for D150MH-40 (P/N: 562-25900) installed
- P/N: 562-26303-01 for D150MH-40(S) (P/N：565-16863) installed

When replacing the high-voltage transformer from D150 MH-40 (P/N: 562-25900) to D150MH-40(S) (P/N: 565-16863), arrangement of the XCONT-2018 board (P/N: 565-15000-01/02 or Rev. K or later of 565-15000) is also necessary.

2.13.1 Function

This PCB equipped with the following circuits controls the power system.

- IGBT circuit for 32kW inverter
- Filament heating circuit
- Starter circuit

2.13.2 PCB replacement

After replacing, adjust FVR and confirm TkV.
As well, take care of the DIP switch setting.

2.13.3 Check pin list

CP No.	CP Name	Description
CP1	CL	Filament heating voltage
CP2	CO	
CP3	Gc	IGBT gate signal
CP4	Ec	
CP5	Ga	IGBT gate signal
CP6	Ea	
CP7	YC	Stator coil current detection (1V or more is normal.)
CP8	Gd	IGBT gate signal
CP9	Ed	
CP10	Gb	IGBT gate signal
CP11	ZC	Stator coil current detection (1V or more is normal.)
CP12	Eb	IGBT gate signal
CP13	GND	GND
CP14	CL1	Filament heating voltage for the large focus
CP15	CS1	Filament heating voltage for the small focus

2.13.4 Fuse

Fuse Name	Description
F1	Provided for protection (600V,150A or 660V,160A)

2.13.5 DIP Switvh

Number	Function	Setting	Default
SW1-1	Unit Setting	Normally OFF	OFF
SW1-2	Not used	Normally OFF	OFF

SW1 is not mounted with ASSY Rev. G or later.

2.14 INVERTER UNIT-C1

2.14.1 Function

This PCB equipped with the following circuits controls the power system.

- IGBT circuit for 12.5kW inverter
- Filament heating circuit
- Starter circuit

2.14.2 PCB replacement

After replacing, adjust FVR and confirm TkV.

As well, take care of the DIP switch setting.

2.14.3 Check pin list

CP No.	CP Name	Description
CP1	CL	Filament heating voltage
CP2	CO	
CP3	Gc	IGBT gate signal
CP4	Ec	
CP5	Ga	IGBT gate signal
CP6	Ea	
CP7	YC	Stator coil current detection (1V or more is normal.)
CP8	Gd	IGBT gate signal
CP9	Ed	
CP10	Gb	IGBT gate signal
CP11	ZC	Stator coil current detection (1V or more is normal.)
CP12	Eb	IGBT gate signal
CP13	GND	GND

2.14.4 Fuse

Fuse Name	Description
F1	Provided for protection (600V,100A)

2.14.5 DIP Switch

Number	Function	Setting	Default
SW1-1	Unit Setting	Normally ON	ON
SW1-2	Not used	Normally OFF	OFF

2.15 MU DRIVER-07

2.15.1 Function

This PCB has CPU and controls bellows.

- Controls the DC motors.
- Detects the handle operation.
- Detects the bumper switch.
- Controls the arm catch area.
- Controls the electromagnetic lock.
- Detects the Stop switch.
- Detects the Emergency Brake Release switch.
- Self Diagnosis of FET for driving DC motors
- Travel (Brake bar input) from the standby (Auto power off) condition and the return by
- the arm operation (All free switch input).

The software program of FPGA on this PCB is written by XCONT PCB during power on.

Note:
The right DC motor uses ＋120V an the left DC motor uses --120V for driving.
The machine moves in reverse direction when the right DC motor rotates in forward direction.
The machine moves in forward direction when the left DC motor rotates in forward direction.

2.15.2 PCB replacement

No adjustment is necessary.

2.15.3 LED list

LED No.	Description
LD1	+24V power supply
LD2	Lights on when FET thermal switches for Q11, Q14 (for driving the right DC motor) are ON
LD3	Lights on when FET thermal switches for Q1, Q8 (for driving the left DC motor) are ON
LD4	Lights on when the contact of K11 is CLOSE.
LD5	Lights on when Arm Catch Sensor SW4 is ON
LD6	Lights on when Arm Release Switch SW1 is ON
LD7	Lights on when Bumper Switches are ON
LD8	Lights on when Forward Fine Driving Operation Switch is ON
LD9	Lights on when Backward Fine Driving Operation Switch is ON
LD10	Lights on when Arm Lock Release Switch is ON
LD11	+5V power supply
LD12	＋15V power supply for the left DC motor
LD13	Lights on when 3.3V output is OK
LD14	7 segment display (Refer to the below table about the meaning.)
LD15	7 segment display (Refer to the below table about the meaning.)
LD16	7 segment display (Refer to the below table about the meaning.)
LD17	＋15V power supply for the right DC motor
LD18	Lights off when Emergency Switch is ON
LD19	Lights on when Driving Brake Release Bar is ON

2.15.4 The meaning of 7 segment display of LD14, LD15, LD16 according to the setting of rotary switch SW2

Setting of SW2	Meaning of 7 segment display	Value
0	The latest error	Error code *1
1	Input signal from the left FSR	0～1023 *2
2	Input signal from the right FSR	0～1023 *2
3	Measured current of the left DC motor	0～1023 ***2**
4	Measured current of the right DC motor	0～1023 ***2**
5	Measured driving speed of the left DC motor	-511～511 ***3**
6	Output command for the left DC motor	-127～127 ***4**
7	Measured driving speed of the right DC motor	-511～511 ***3**
8	Output command for the right DC motor	-127～127 ***4**
9	AD value VOUT1	0～1023 ***2**
A	AD value VOUT2	0～1023 ***2**
B	AD value VOUT3	0～1023 ***2**
C	AD value VOUT4	0～1023 ***2**
D	Not used	
E	Not used	
F	Version Number	0～999 ***5**

*** 1** [E00] is displayed if no error was recorded.
If some errors were recorded, [E] and sum of the available error codes on the following table is displayed.

Error code (Hexadecimal)	Content of the error
01	Short circuit of FET for driving the left DC motor
02	Short circuit of FET for driving the right DC motor
04	Not used
08	Abnormal condition of FET
10	FET Thermal switches are ON
20	Dry cell battery error
40	Melted contact of K11
80	Communication error with XCONT

*2 Displayed in unsigned 10 bit Hexadecimal

Display		Value
000	→	0
001	→	1
:		
00A	→	10
:		
3FF	→	1023

*3 Displayed in signed 10 bit Hexadecimal

Display		Value
000	→	0
001	→	1
:		
1FF	→	511
200	→	Not defined
201	→	-511
:		
:		
3FE	→	-2
3FF	→	-1

*4 Displayed in signed 8 bit Hexadecimal

Display		Value
000	→	0
001	→	1
:		
07F	→	127
080	→	Not defined
081	→	-127
:		
:		

0FE → -2
0FF → -1

*5 Displayed in unsigned decimal

Display		Value
000	→	0
001	→	1
:		
999	→	999

2.15.5 Check pin list

CP No.	Signal Name	Description
CP1	Q1 DRIVE	Driving Signal for Q1 (for driving the right DC motor)
CP2	Q2 DRIVE	Driving Signal for Q2 (for driving the right DC motor)
CP3	LEFT_MCOM	Measured current of the left DC motor
CP4	M120	-120V power supply after passing the contactor K11
CP5	LC2	Measured current of the left DC motor (after passing the filter):0V at normal condition, 0.2V/1A CP4 is the ground CP during measuring the voltage at CP5.
CP6	Q9 DRIVE	Driving Signal for Q9 (for driving the right DC motor)
CP7	Q8DRIVE	Driving Signal for Q8 (for driving the right DC motor)
CP8	+3.3V	
CP9	+5V	
CP10	GND	
CP11	120G	
CP12	-120V	
CP13	LC1	Measured current of the left DC motor (before passing the filter)
CP14	5L	5V power supply to control the left DC motor
CP15	M120	-120V power supply after passing the contactor K11
CP16	+1.2V	
CP17	+120V	

CP No.	Signal Name	Description
CP18	120G	
CP19	RC2	Measured current of the right DC motor (after passing the filter):0V at normal condition, 0.2V/1A CP18 is the ground CP during measuring the voltage at CP19.
CP20	5R	5V power supply to control the right DC motor
CP21	P120	+120V power supply after passing the contactor K11
CP22	FOUT_R	Measured current of the right DC motor (after VF conversion)
CP23	VFCLK_R	Clock signal for VF conversion for the right DC motor
CP24	VFCLK_L	Clock signal for VF conversion for the left DC motor
CP25	FOUT_L	Measured current of the left DC motor (after VF conversion)
CP26	RIGHT_MCOM	Measured current of the right DC motor
CP27	RC1	Measured current of the right DC motor (before passing the filter)
CP28	Q11 DRIVE	Driving Signal for Q11 (for driving the left DC motor)
CP29	Q12 DRIVE	Driving Signal for Q12 (for driving the left DC motor)
CP30	RFSR	Input signal from the right FSR :2.5V at neutral position
CP31	LFSR	Input signal from the left FSR :2.5V at neutral position
CP32	BTT	Voltage of the brake release battery Typical value: 4V (battery voltage 18V) EMPTY :3.27V (battery voltage 14.4V)
CP33	AD4	Analog Input (Spare)
CP34	DAG	Analog GND
CP35	DA2	Analog Input (Spare)
CP36	DAG	Analog GND
CP37	DA1	Analog Input (Spare)
CP38	Q15 DRIVE	Driving Signal for Q15 (for driving the left DC motor)
CP39	Q14 DRIVE	Driving Signal for Q14 (for driving the left DC motor)

2.15.6 DIP Switch list

SW No.	Function	Setting	Factory setting
SW1-1	No use	Always OFF	OFF
SW1-2	No use	Always OFF	OFF
SW1-3	No use	Always OFF	OFF
SW1-4	No use	Always OFF	OFF
SW3	FSR type	“NEW FSR ASSY”：New type. “OLD FSR ASSY”：Old type	Set to “NEW FSR ASSY”

2.15.7 Jumper list

JP No.	Name	Setting	Description
JP2	「 FAN24V」	○	Normal
	「 FAN5V 」	×	Don’t set.
JP3	「 PS 」	○	Normal
	「 AS 」	×	Don’t set.
JP4	「 PS 」	○	Normal
	「 AS 」	×	Don’t set.
JP5	「 PS 」	○	Normal
	「 AS 」	×	Don’t set.

2.16 SHEET PANEL

2.16.1 Function

This PCB is the sheet panel for X-ray high voltage generator.

2.16.2 PCB replacement

No adjustment is necessary.

2.17 POWER 200D/MUX DR CONT

2.17.1 Function

This PCB is included in DR ASSY and controls the power ON/OFF circuit for DR system.
With the release of the MX9 version, the POWER 200D was replaced by the MUX DR CONT.
MUX DR CONT can also be used as a replacement for the POWER 200D.
However, POWER 200D cannot be used for MX9 version.

2.17.2 PCB Replacement

Set VR1 and VR2 at the middle position. No need to adjust. (Only for POWER 200D)

2.17.3 LED list

In case of POWER 200D

LED No.	Description
LD1	Lights ON when the switch “DR” is pressed.
LD2	Lights ON when PC for DR system is ON.
LD3	Lights ON when the signal “PC_PERMIT” from XCONT-2018 PCB is available. (“PC_PERMIT” is available when the key switch is turned on and the password is entered in case with the keyless entry option.)
LD4	Lights ON when the input power from the battery is supplied and DC24V output is available from the DC power supply M8 on this PCB.
LD5	Lights ON when the power of PC is ON.
LD6	Lights ON when the power plug is connected.

2.17.4 VR list (Only for POWER 200D)

Name	Function	How to adjust
VR1	Output Pulse width from M2A	Pulse width should be about 6 sec. after the switch “DR” is pressed. Normally, set at the middle position.
VR2	Output Pulse width from M1B	Pulse width should be about 2 sec. after the power plug is connected. Normally, set at the middle position.

2.17.5 Check pin list

In case of POWER 200D

CP No.	Signal Name	Description
CP1	+24V	+24V
CP2	+5V	+5V
CP3	GND	GND

2.17.6 Fuse(On DR ASSY)

Fuse Name	Description
F21	250V, 2A fuse for the input power circuit from AC125V output of T1 transformer for DR system

2.17.7 DIP switch list

In case of POWER 200D

SW No.	Function	Setting	Default
SW1	For checking	When SW1 is ON, power for DR system can be turned ON without permission signal “PC_PERMIT” from XCONT PCB. (This switch is just for service use. If you forget returning to OFF, DR system can be turned on by connecting the power plug and turning on the switch “DR” without turning on the key switch or without entering the password.)	OFF
SW2	Not used	Normally OFF	OFF

In case of MUX DR CONT, refer to Installation Manual.

2.18 DR POWER CONT

2.18.1 Function

Control of power supply for DR unit

- Detects the insertion and removal of the power plug and switches the power supply for the DR unit with external AC power or internal battery.
- A circuit for rectifying and smoothing the AC power supply to reduce the potential difference is provided in order to prevent the occurrence of arc discharge at the contact of the switching relay due to the potential difference at the time of switching the AC/DC power supply and the occurrence of contact welding and fuse breakage.

2.18.2 PCB replacement

No adjustment is necessary.

2.18.3 Relay

Relay No.	Function
K1	Turns ON when the power plug is connected, and supplies power from the power plug to the DR system. (When the power plug is not connected, K1 is turned off to supply power from the battery to the DR system.)
K2	ON when DR system is powered ON

2.18.4 Fuse

Fuse No.	Function
F1	250 V, 2.0 A fuse in the input circuit from the 125 V output from the DR system-dedicated T1 transformer

2.19 LED POWER-20

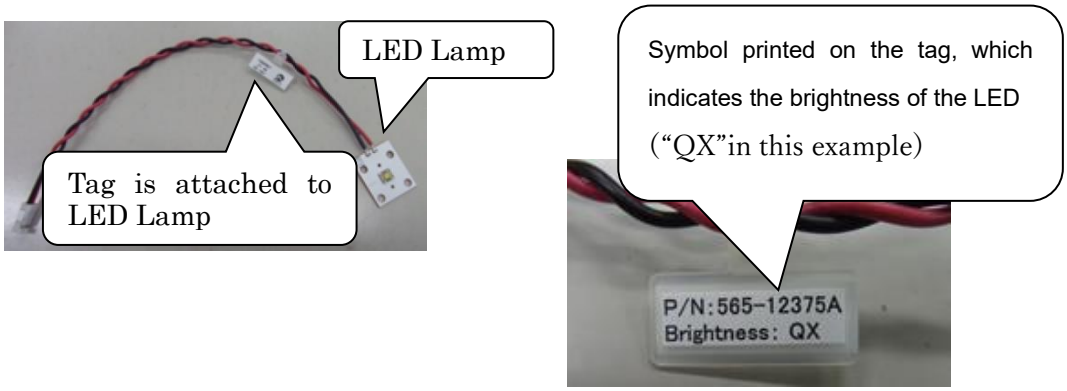
2.19.1 Function

This PCB is placed in the collimator. The LED lamp control circuit, DIP switch for brightness adjustment, etc. are mounted on this PCB.

2.19.2 DIP switch list

<In case of LED LIGHT ASSY: P/N 565-12375-02> (shipped before around April 2022)
Even if the same voltage is applied to the LED lamp, the brightness varies slightly.
Therefore, the SW1 on the LED POWER- 20 PCB is set to supply an appropriate voltage to each LED to be connected, and a constant luminance is obtained.
Each LED lamp (LED LIGHT ASSY) has the following tag. Sets SW1 based on the symbol printed on this tag.

Switch No.	Tag			
	PX	PY	PZ	QX
SW1-1	OFF	OFF	ON	ON
SW1-2	ON	ON	OFF	ON
SW1-3	OFF	ON	ON	OFF

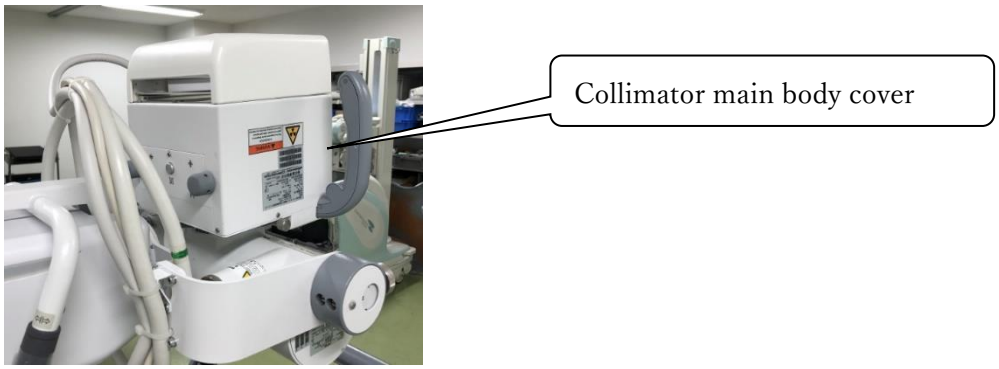


<In case of LED LIGHT ASSY: P/N 565-14690-02> (shipped after around April 2022)
The LED rank symbol is not available on the tag attached to the LED LIGHT ASSY.
In this case, set the DIP switch SW1 on the LED POWER-20 board as shown below.

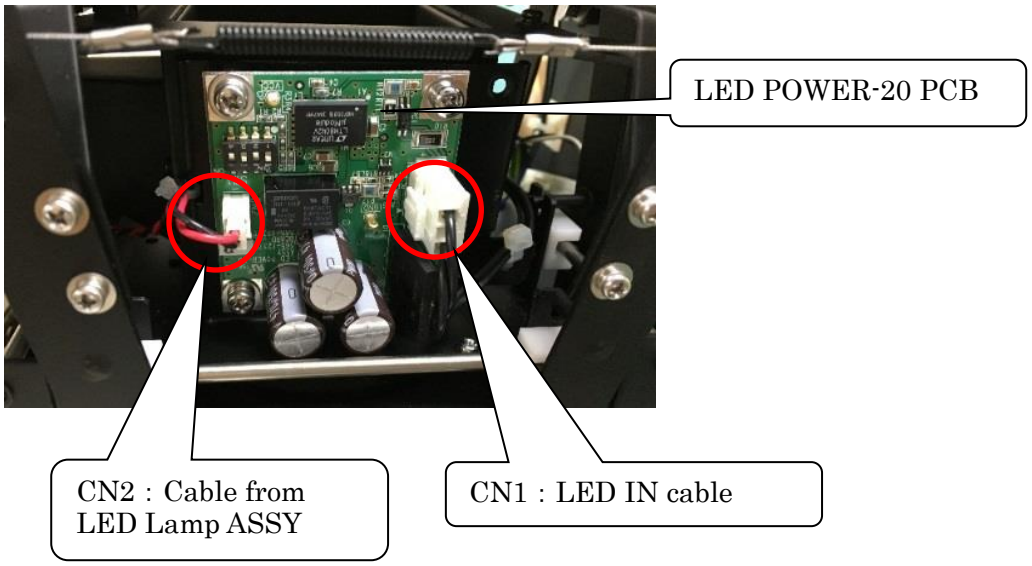
Switch No.	
SW1-1	ON
SW1-2	ON
SW1-3	OFF

2.19.3 Replacement Procedure

Remove the collimator main body cover.



Disconnect the two cables (shown in the figure below) connected to the LED POWER-20 PCB, replace the PCB. In case of LED lamps shipped before around April 2022, set SW1 according to the table in Chapter 2.19.2. while referring to the symbol printed on the tag attached to the LED lamp.



On the other hand, for LED lamps shipped after around April 2022, the LED rank symbol is not available on the tag attached to the LED LIGHT ASSY. In this case, set the DIP switch SW1 on the LED POWER-20 board according to Chapter 2.19.2. (SW1-1 ON、SW1-2 ON、SW1-3 OFF、SW1-4 OFF)

When MX9 version does not have the 3D camera option, the angle of the collimator detected by the angle sensor of this CL MEAS-17 board is displayed.
Since the attitude of the CL MEAS board may be shifted when the collimator cover is removed, it is necessary to check whether the angle display is not shifted after replacing LED POWER-20 board. Follow the procedure "7.26.3 Checking the collimator angle" in the Installation Manual to check the angle, and if necessary, adjust the collimator angle detection again.

2.20 CL MEAS-17

Note:
This board is mounted in the collimator for MobileDaRt Evolution.
The MobileArt Evolution comes with a LEAF MEAS-15 board.

Note that the MX8 and MX9 have different P/N and are not compatible with each other.

Model	CL MEAS-17 ASSY
DaRt MX8 Version	P/N：565-13320
DaRt MX9 Version	P/N：565-13320-01

2.20.1 Function

Collection of sensor information around the collimator such as the opening measurement of H/V leaf or collimator angle (only for MX9 Version).

2.20.2 DIP switch list

The operation is different in the normal mode and self check mode by OFF/ON of SW1-4, accordingly, the following section describes the settings in both modes.

#	SW1-4: OFF (Normal mode)	SW1-4: ON (Self check mode)
SW1-1	Not used	OFF: AD_CH1 is fixed to V leaf. ON: AD_CH1 is fixed to H leaf. (*)
SW1-2	Not used	Not used
SW1-3	Not used	Not used
SW1-4	Operation as the normal mode	Operation as the self check mode

(*)The purpose to be fixed to either one of H/V leaf is because it is impossible to read the value correctly if it is not fixed to either one when it is intended to measure the voltage value of leaf opening with a circuit tester at the check pin which is described later.
The timing to be fixed shall be after the judgment of self check completion (7 seconds after the start of checking).
Until that moment, the system has been switched to either one in the same way as the normal mode.

2.20.3 Check pin

This board has no check pins but the through hole is provided as a J10 connector instead.
The following table shows the assignment of J10 through-hole.

Pin	Signal name	Description
1	GND	GND supply
2	+15V	+15V potential check
3	+3.3V	+3.3V potential check
4	+5.5V	+5.5V potential check
5	REF_5V	REF5V potential check
6	AD_CH1	V/H leaf potential check (*)
7	AD_CH2	Analog 2 potential check
8	AD_CH3	Analog 3 potential check
9	AD_CH4	Analog 4 potential check
10	GND	GND supply

Note:
When you want to measure the potential of AD_CH1 (No. 6 pin) with a circuit tester, it is required to set the SW1-1 to the side you want to measure after setting the SW1-4 to ON position (Self check mode) and then start the measurement after fixing the potential to either one of H/V leaf. When the potential is measured with an oscilloscope, it is possible to measure in the normal mode as SW1-4 OFF, but it is necessary to pay attention to ensure that the potential has been switched to H/V on a periodic basis.

2.20.4 LED

Since the display mode of LD1 and 2 varies when the operation mode of board is in the normal mode (DipSW1-4: OFF) and self check mode (DipSW1-4: ON). This section describes those cases respectively.

(1) Display in the normal mode (DipSW1-4: OFF)

#	Display content	Display mode
LD1	Board status	Normal: Blinking at intervals of 1 second Disconnection with serial communications with the host: Lights off Activation of the watchdog timer: Lights on
LD2	CN4 input condition	No input: Lights off Input of either one of 2ch. inputs (close): Blinking once a second Input of both 2ch. inputs (close): Blinking twice a second
LD3	Analog power supply condition	Lights on when the analog power supply is fed (when ANALOG_ON is asserted)
LD4	Power supply feed condition	Power supplied: Lights on always

(2) Display in the self check mode (DipSW1-4: ON)

#	Display content	Display mode
LD1	Check completed	Checking: Blinking in 1 second period alternately with LD2 At check completion: Lights on
LD2	Check result	Checking: Blinking in 1 second period alternately with LD1 Check OK: Lights on Check NG: Lights off
LD3	Analog power supply status	Lights on when the analog power supply is fed (when ANALOG_ON is asserted) [ditto]
LD4	Power feeding status	Power supplied: Lights on always [ditto]

2.20.5 Self-check functions

The self check function is mounted on CL MEAS-17, which checks that there is no failure in board components by a H/W single without connecting to the host (Check is completed only in 7 seconds).

The execution procedure of the self check is only to connect the load of each connector and turn on the DipSW1-4.

The execution result will be displayed by using LED (LD1 and 2) on the board and 7 segment LED x 3 digits connected to the J9 connector.

The timing to turn on the DipSW1-4 could be any time, accordingly, it is allowed to feed power in the condition of ON or there is no problem even if the switch is turned ON during power feeding.

However, the photo sensor of CN4 (or the micro switch) is required to be in the close condition during checking (i.e. checking that "close" can be detected normally).

The following table shows the circuit list that are checked by self checking.

Check location	7 segments display	Check content
Serial communication	[J 1]	Check whether the serial communication is normal
Universal SPI port	[J 3]	To be expanded in future. (Always do PAS in the first version)
Analog 1	[J 4]	Check that A/D value is in the normal range.
Analog 2	[J 5]	Check that A/D value is in the normal range.
Analog 3	[J 6]	Check that A/D value is in the normal range.
Analog 4	[J 7]	Check that A/D value is in the normal range.
V leaf opening rate	[Cn2]	Check that A/D value of V leaf is in the normal range.
H leaf opening rate	[Cn3]	Check that A/D value of H leaf is in the normal range.
Digital SW	[Cn4]	Check that the switch state is "close" in both 2 channels.
Acceleration sensor	[ACC]	Execute the self test function of the acceleration sensor itself and check the acceptance (not used).
Analog power supply	[AnA]	Check that the analog power supply can be turned on and stable.

The following table shows the display content to LED while checking and when the check is completed.

Check condition	LD1	LD2	7 seg.LED
Checking (7 seconds)	Blinking alternately with LD2	Blinking alternately with LD1	[-]Hyphen circulation
Check completed : OK	Lights on (Display of completion)	Lights on (Display of OK)	[PA.S] display
Check completed : NG	Lights on (Display of completion)	Lights off (Display of NG)	Circulates the check points after blinking of [FAL]. Location name lights on at the OK point. Location name blinks at the NG point.

2.20.6 Replacement Procedure

When the MX9 version does not have the 3D camera option, the angle of the collimator detected by the angle sensor of this CL MEAS-17 board is displayed.
Therefore, it is necessary to calibrate the angle sensor when replacing the CL MEAS board.
In addition, the attitude of the CL MEAS board may be shifted when the collimator cover is removed. After removing this cover, it is necessary to check whether the angle display is not shifted. If the cover is removed, follow the procedure "7.26.3 Checking the collimator angle" in the Installation Manual to check the angle, and if necessary, adjust the collimator angle detection again.

2.21 LEAF MEAS-15

Note:

This board is mounted in the collimator R-20C for MobileArt Evolution.

The MobileDaRt Evolution comes with a CL MEAS-17 board.

2.21.1 Function

Collection of sensor information around the collimator such as the opening measurement of H/V leaf.

2.21.2 Check pin

Pin	Signal name	Description
1	5V CHECK	+5V supply check
2	GND	GND supply
3	SSD	
4	VREF 10V	REF+10V supply check
5	LEAF SEL	V/H select signal
6	V LEAF	V leaf potential check
7	H LEAF	H leaf potential check

2.21.3 LED

#	Display content	Display mode
LD1	LEAF HL	Normal: Blinking
LD2	5V ON	Power supplied: Lights on always

2.22 RC RECEIVER and RC TRANSMITTER PCB (option)

2.22.1 Function

These PCBs are the receiving and sending side of the Infrared remote control unit option.

2.22.2 PCB replacement

No adjustment is necessary.

3 Circuit Explanation

3.1 Power ON/OFF operation

Related PCB: MUX POWER-32K BOARD

3.1.1 Power ON operation

I. When the key SW is turned ON, all three contacts of the key SW close.
(All 3 contacts of connector JKEY on MUX POWER -32 K board are closed.)
II. Battery voltage DC 240 V is input to PS 1 and 12 V is output.
III. Latch relay K9 turns on and K6 relay turns on.
IV. As a result, the current suppression resistors R1 and R2 (150 ohms) of the control unit are passed through (connector JRS) to supply power to each part.
V. When power is supplied to the DC packs PS1 and PS2 through the connector JDC, 5 V, 15 V and 24 V are output and the control board starts operating.
VI. After checking the CPU board XCONT and completing FPGA writing (about 5 seconds), the MGRS signal is sent to the MUX POWER -32 K board via the MUX CHARGE -32 KX board to turn on the relay K4. As a result, relays K7 and K8 are turned on, and the path through the suppression resistors R1 and R2 is shorted to allow sufficient load current to flow.
VII. As for the device, it can be operated (running and arm unlock) when the MGRS signal comes out.

Problem:
The status indicator light does not light up properly and shooting operation is not possible.

Cause:
Under normal conditions, when the key SW is turned ON, it means that the lines of KEYSW1 and KEYSW2 are short-circuited, and the CPU of the XCONT board recognizes it as signal KEYSW through the MUX CHARGE -32 KX board. When the CPU is unable to recognize this due to a contact failure of the key SW, etc., a failure case occurs.

3.1.2 Power OFF operation

I. When the key switch is set to OFF, the relay K7 and K8 turn OFF first. Though the input to the PS1 in the DC pack stops, the output remains for a while. Accordingly, the relay K6 turns OFF after a while.
   By this operation, the power supplied through the resistors R1 and R2 remains for a while because excessive load is applied on contacts if all contacts are opened at the same time.

3.1.3 Auto power OFF operation

I. In the power ON status, the CPU outputs the AUTOOFF signal through XCONT and MUX CHARGE-32KX PCB to set the relay K5 to ON. As a result, the latch relay K9 turns OFF, the relays K6 and K7, K8 turn OFF, the power supplied to the DC pack stops, then the system stops.
II. In this status, PS1 on MUX POWER-32K PCB is continuing operation. Accordingly, if the system is left in the auto power OFF status, the batteries are discharged in a long period due to the fine discharge current. When the power supply cable is connected to the outlet in the condition of auto power off, the equipment returns from auto power off and starts charging.

3.2 Operation when power plug is connected (Changeover between AC and DC of DC pack input)

Related PCB: MUX POWER-32K BOARD

3.2.1 Power plug connection

I. When the power plug is connected, the 100 V output of the transformer T1 (100 E-0 E) is input to the MUX POWER -32 K board by the connector JT1C.
II. Relay K1 turns on and the power supply line to the DC pack switches to AC100V instead of from the battery.
III. 100 V AC is supplied to the DC pack and 5 V, 15 V and 24 V are output from the DC pack and the control board starts operation.
IV. At the same time as (2), relay K3 is turned on.
V. When relay K3 is ON, ACON1, ACON2 are short-circuited, go through the MUX CHARGE -32 KX board to the XCONT board (as signal LVON), and the CPU recognizes the power plug connection.
VI. Since the other contact of relay K3 (two contacts) is short-circuited by the pattern, the connection to the coils of relays K6, K7 and K8 remain even when the power plug is connected. For this reason, K6, K7 and K8 can be turned on by 12 V output from the PS1 when the key switch is turned on even while the power plug is connected, and shooting becomes possible. When the key switch is turned OFF, the K6, K7 and K8 are turned OFF and the load is separated from the battery, allowing battery charging.

Other function:

I. If the device is left untouched when it is plugged in, auto charging starts after a certain period.
II. Even during charging, the camera returns to shooting mode by hand switch or seat panel operation.
III. When the power cable is connected to an outlet while Auto Power Off is on, it will resume Auto Power Off and start charging.

3.3 Detection of battery voltage

Related PCBs: MUX POWER-32K, MUX CHARGE-32KX, XCONT-2018

3.3.1 Detecting Battery Voltage

I. Twenty batteries are connected in series, 10 batteries on the plus side and 10 batteries on the minus side.

The middle point is connected to the chassis via resistances of several M ohm (on the MUX POWER-32K and MUX CHARGE-32KX PCB).

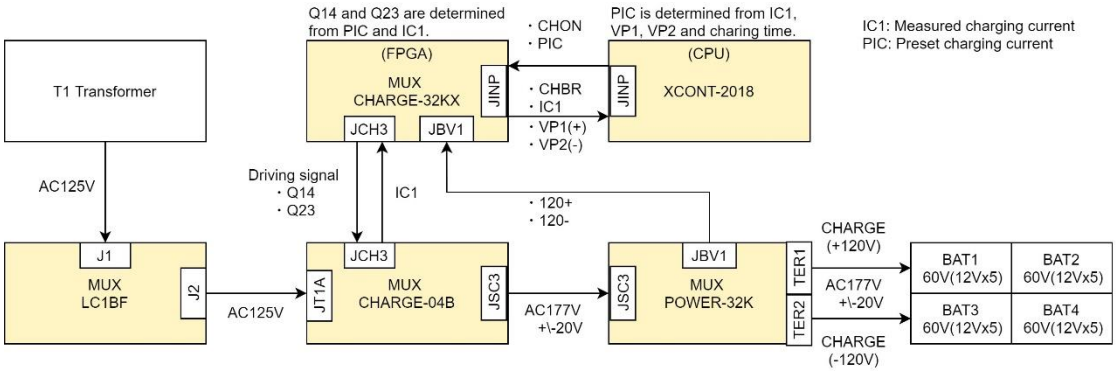
II. The plus, the minus and the middle point of the batteries are offered at the following terminals on the MUX POWER-32K PCB.

Battery plus side : TER1 [B120+]
Battery middle point : TER5 [B0]
Battery minus side : TER2 [B120-]

III. The values at the terminals above are converted into the battery voltage detection value ("120+" signal at CP60 and "120-" signal at CP59) on the MUX CHARGE-32KX PCB via the resistors R8 to R11 (10 M ohm) and the connector JBV1.

The relationship between the actual battery voltage and the detection value ("120+" and "120-" signals) is approximately "3V/100V". For example, when the actual battery voltage is 120V, the detection value at CP60 or CP59 is about 3.6V.

IV. Finally, the "120+" and "120-" signals are sent as the VP1 and the VP2 to the XCONT PCB.



3.3.2 Battery Charged Voltage Over

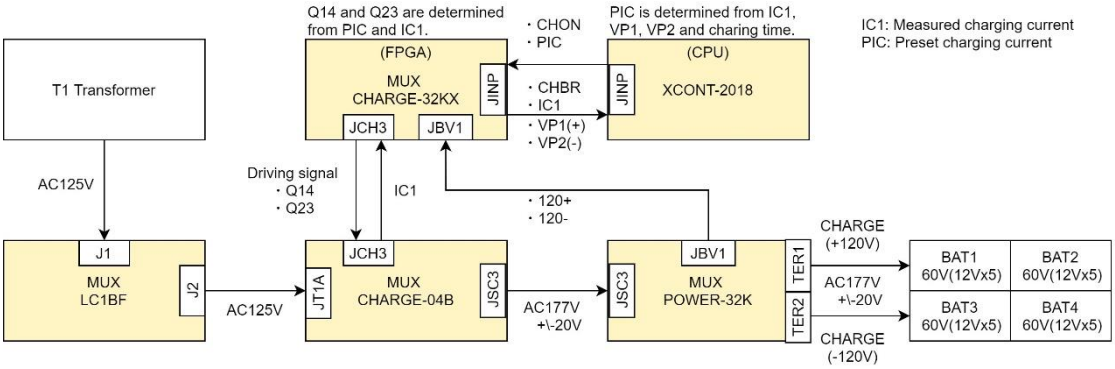
I. The operational amplifiers A8, 120+ and 120- are used to detect overvoltage. The threshold is 4.67V. Converted to battery voltage, 4.67 x 100/3 = 156 V (Each positive side and negative side)

II. As a result, it is sent to the XCONT board as an error signal CHBR2. CHON signal turns OFF and battery charging is aborted. Error F5b is displayed. As mentioned above, error F5 b is hardware detection, but separately there are F 51 and F 53 which have similar meaning in software detection.

3.4 Charging circuit for batteries

Related PCB: MUX POWER-32K, MUX CHARGE-32KX, MUX CHARGE-04B

3.4.1 Charging



I. When the power plug is connected, the 125 V (125 F-0 F) output of the transformer T1 is input through the MUX-LC1BF board to the connector JT1A on the MUX-CHARGE-04B board.

II. The 125 V output of (1) is rectified and charged to a capacitor C6 (approximately 177 +/- 20 V) and connected to the inverter circuit of the MUX CHARGE-04B board. The output of this inverter circuit is connected to the battery through connector JSC 3 on the MUX POWER -32 K board.

III. The inverter circuit of the MUX CHARGE-04B board is controlled by drive signals Q 14 C/Q 23 C from the MUX CHARGE -32 KX board through connector JCH 3. The MUX CHARGE -32 KX substrate PWM controls the inverter so that the measured charging current value IC1 (CP 21: 1V/1A) is the same as the set charging current value PIC (CP 22: 1V/1A). The measured charging current value is detected by the current transformer CT1 on the MUX-CHARGE-04B substrate.

IV. The set charge current value PIC is determined according to the table below according to the current battery charge voltage value.

STEP	Contents
START	Pre-charging starts with PIC 0.5A up to about 10 minutes. If the charged voltage doesn't reach to 280V within 5 min., proceeds to STEP 1. If the charged voltage reaches to 280V within 5 min., proceeds to STEP 3. If battery is already charged more than about 85% of full charged condition under "Battery charging restriction setting mode", battery charging stops even within 5 min without proceeding to STEP1 or STEP3.
STEP1	PIC is 2.0A. (1.3A during charging under ON condition of DR system.) Proceeds to STEP 2 when the charged voltage reaches to 280V.
STEP2	PIC is 1.0A. Proceeds to STEP 3 when the charged voltage reaches to 285V.
STEP3	PIC is 0.4A. Proceeds to STEP 4 when the charged voltage reaches to 290V.
FINISH	PIC is 0.18A. Charging finishes when the charged voltage reaches to 290V.

In addition, the switching frequency of the battery charging inverter is switched to two levels according to the set charging current value PIC as shown in the table below.

PIC	Frequency of the inverter
2.0A	13kHz
1.0A	
0.5A (Beginning of charging)	2kHz
0.4A	
0.18A	

V. A CH-ON signal from XCONT is required to initiate control of the inverter for battery charging. When the key SW is in the OFF state, the CH-ON signal and the set charge current value PIC occur approximately 10 seconds after the power plug is connected, regardless of the current battery charge voltage value.

3. Circuit Explanation

Reference:

The power for DR system is supplied from 125V output of T1 transformer when the power plug is connected. Therefore, battery charging is possible even under ON condition of DR system.

By the way, during battery charging under ON condition of DR system, power for both DR system and battery charging is supplied from 125V output of T1 transformer.

Therefore, the preset charging current PIC at STEP1 is decreased to about 1.3A though it is normally 2.0A.

The green LED LD7 on MUX CHARGE-32KX will light on when CH-ON is available.

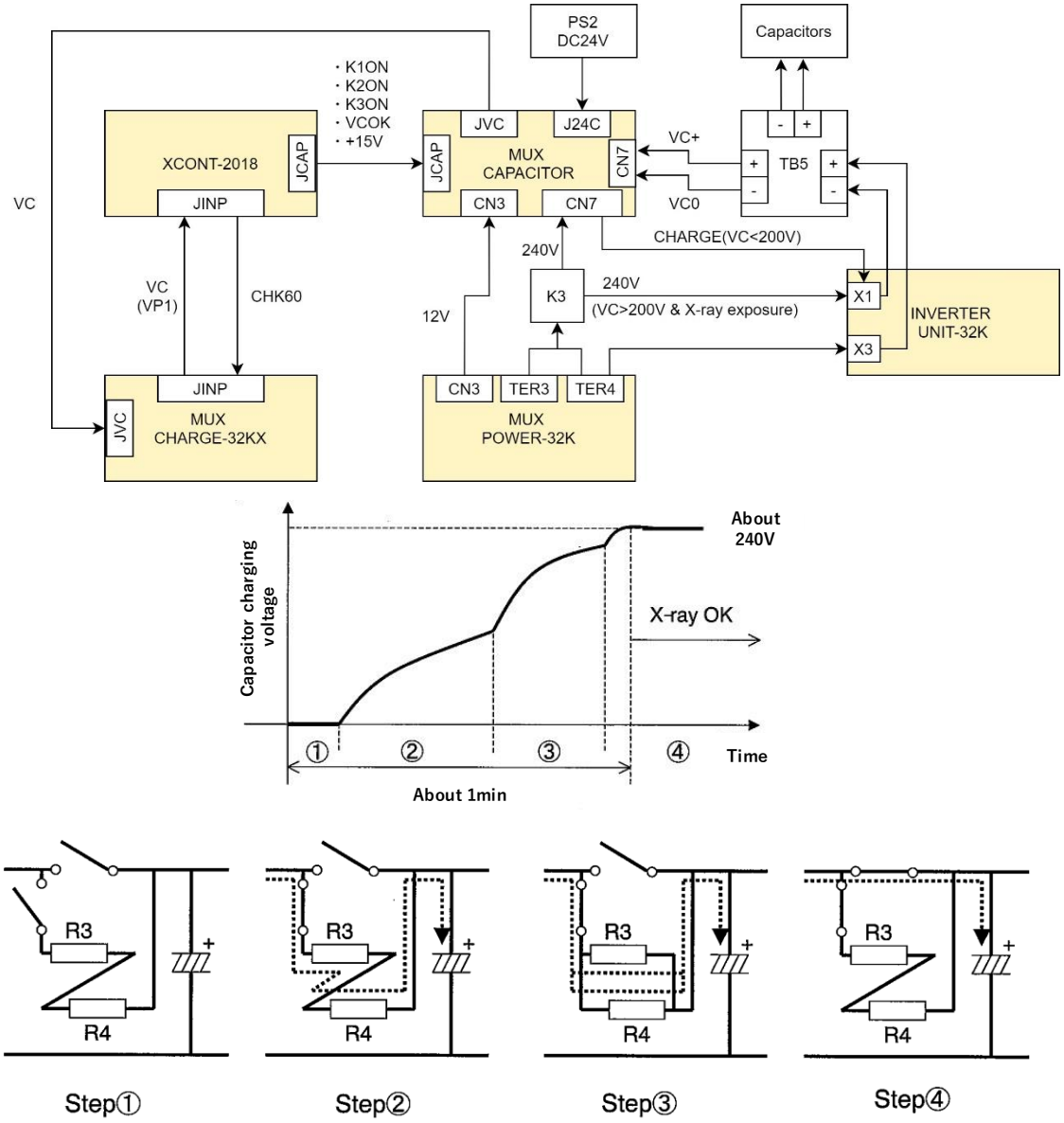
Other function:

I. If the device is left untouched when it is plugged in, auto charging starts after a certain period of time.
II. Even during charging, the device returns to radiography mode by hand switch, control panel, key switch, driving handle or arm brake release switches operation.
III. Auto Power Off status
IV. Battery charging automatically stops 10 hours after it starts.

3.5 Charging circuit for capacitors

Related PCB: MUX CAPCITOR, XCONT-2018, MUX CHARGE-32KX, MUX POWER-32K

3.5.1 Charging



I. When the key SW is turned on and the power is turned on (To be precise, when the MGRS signal is generated and relay K7 and K8 above are turned on,), DC 12 V is sent from the MUX POWER -32 K board through connector CN3 to the MUX CAPACITOR board, relay K7 and K8 are turned on, and contact K7B, K7C and K8B, K8C are closed. Also, DC 24 V power supply is input to the MUX CAPACITOR board through connector J 24 C. This 24 V is the power source to operate this board and the external relay K3.

II. About 10 seconds after the MGRS signal is turned on in the control section, each signal for starting capacitor charging is input from the XCONT board through the connector JCAP. First, a "K1 ON" signal is input from the JCAP (5) pin, and relay K5 is turned on, resulting in relay K1 being turned on. Thus, 240 V from the battery is charged to the capacitor through a total of 300 ohms of R3 (150 ohms) and R4 (150 ohms) connected in series to the MUX CAPACITOR substrate through CN2.

III. The voltage of the capacitor enters the board from the CN 7 (1) (3) pin and is sent from the connector JVC to the MUX CHARGE -32 KX board through a 10 megohm resistance. On the MUX CHARGE -32 KX board, the capacitor voltage from the JVC and the positive side voltage of the battery are switched by the signal "CHK 60" from the XCONT board and sent to the XCONT board through the analog signal check pin VP1.

IV. When the capacitor voltage reaches about 170 V (Usually about 30 ~ 60 seconds after charging starts), the XCONT substrate sends a "K2 ON" signal through pin (4) of JCAP to this substrate, and relay K6 is turned on, resulting in K2 being turned on. As a result, the connection between R3 (150 ohms) and R4 (150 ohms) switches from serial to parallel, allowing the battery to charge through 75 ohms to the capacitor.

V. When the capacitor voltage reaches about 200 V (Usually about 10 ~ 30 seconds after K2 is turned on), the "VCOK" signal is generated by the A 12 B and M 8 B on the MUX CHARGE -32 KX substrate and sent to the XCONT and MUXCAPACITOR substrates. In the MUX CAPACITOR board, this "VCOK" is received through the connector JCAP, and the relay K3 on this board is turned on. As a result, the last protection to turn on the external relay K3, which is connected to the MUX CAPACITOR board through CN7, has been released.

VI. When the READY operation is performed, the "K3ON" signal is input from the XCONT board through the connector JCAP to the main board, and the relay K4 on the main board is turned on. With this, the external relay K3 connected to the main board through CN7 is turned on, and the battery and capacitor are connected without resistance, so that shooting is possible. This "K3 ON" signal turns off about 5 seconds after shooting or about 30 seconds after stopping the READY operation without shooting.

3.5.2 Discharge

I. When discharging the capacitor, change the connection of the cable "CN1/CN2" "connected to CN2 to CN1. While the charge remains, LD1 on the MUX CAPACITOR substrate lights up. When the discharge switch connected to CN5 is turned on, R3 (150 ohms) and R4 (150 ohms) connected to CN1 are discharged through a total of 300 ohms connected in series.
II. On the other hand, since CN2 is OPEN, even when the power of the device is ON, charging resistors R3 and R4 are not connected to the charging circuit, and there is no connection for supplying voltage to the coil of the external relay K3 connected to CN7, so charging to the capacitor is not possible.

3.6 Inverter circuit for starter and collimator lamp circuit

Related PCB:

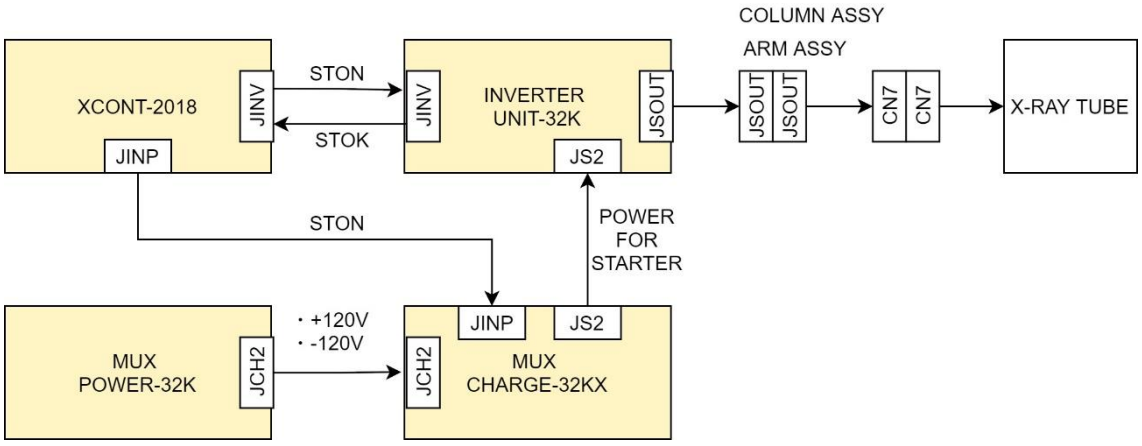
MUX CHARGE-32KX, INVERTER UNIT-32K, XCONT-2018, LED POWER-200 PCB

3.6.1 Explanation

I. Starter circuit

I. For the starter drive, the inverter is mounted on the MUX CHARGE -32 KX board.

II. The input voltage is approximately 240 V of battery voltage. Dip switch S1 on the MUX CHARGE -32 KX board sets the inverter waveform to a pulse width appropriate for the starter drive.



II. Collimator lamp circuit

I. The collimator lamp is driven by DC24V which is supplied to LED POWER -20 PCB inside the collimator via TER6 and TER7 on MUX POWER -32 K PCB when the FET Q 13 on the XCONT PCB is turned ON.

II. The circuit for making the output voltage for driving the LED constant is included on LED POWER-20 PCB.

3.7 Main inverter unit, filament heating inverter unit and starter circuit

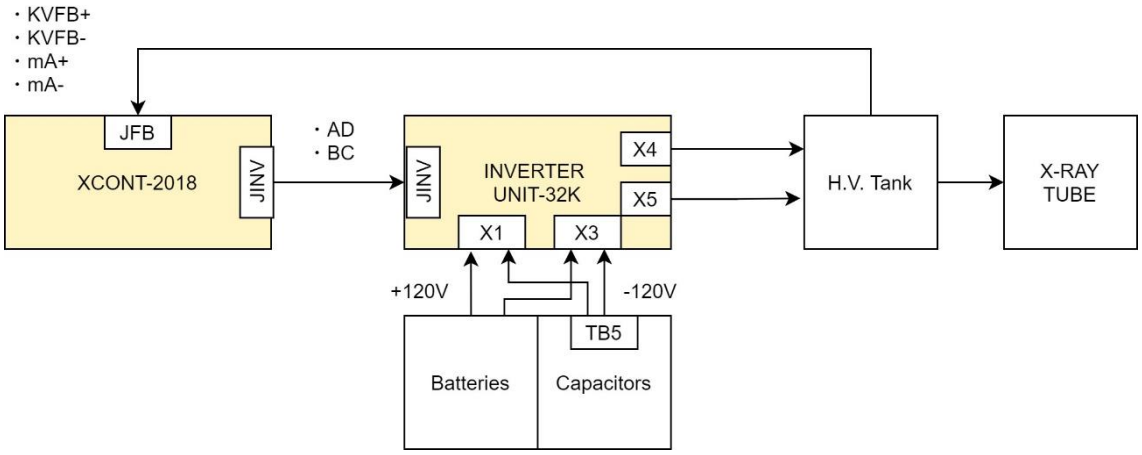
Related PCB:

INVERTER UNIT 32KX, INVERTER UNIT C1, XCONT-2018

3.7.1 Main inverter unit

Inverter for generating high pressure. It has IGBT and gate drive circuitry.

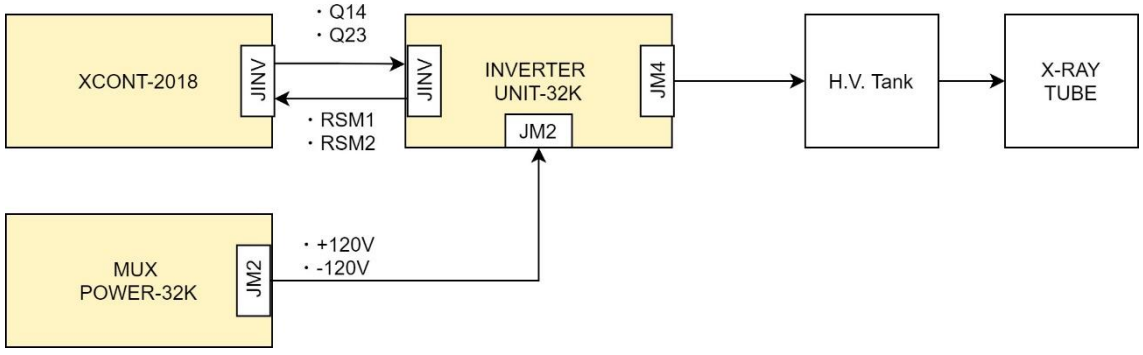
The gate drive signal comes from the XCONT substrate.



3.7.2 Filament heating inverter unit

I. This inverter unit heats the filament and is equipped with the FET and the gate drive circuit. The gate drive signal comes from the XCONT PCB.
Besides the inverter, a part of the primary current detection unit is mounted also.
(The current transformer CT1 etc.)

II. Focus size Large/Small is selected by the relay K1.

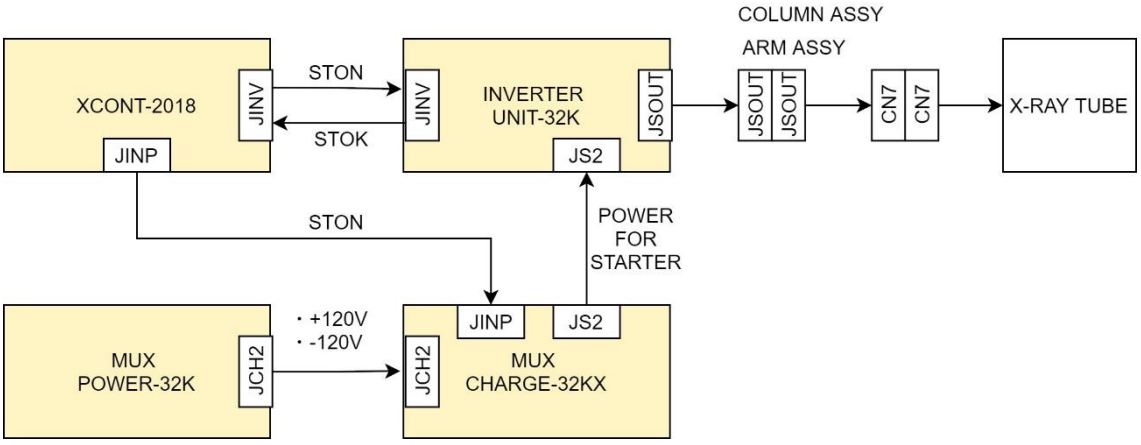


Reference:

Because a part of the current detection unit is mounted, it is required to confirm adjustment of the FVR value (preset value of filament current) if the entire INVERTER UNIT-32K PCB is replaced for replacement of the IGBT.

3.7.3 Starter circuit

The power for starter circuit is turned on/off by the inverter on the MUX CHARGE-32KX PCB. The starter circuit on INVERTER UNIT-32K PCB detects the stator coil current and inserts a phase advancing capacitor.



Reference:

If the dip switch SW1-1 on INVERTER UNIT-32K PCB is ON, the STON signal may not be able to turn ON due to the interference with the SSR Q10.

Therefore, SW1-1 should be set to OFF (“MUX” side).

3.8 Power ON/OFF circuit for DR system

Related PCB:

Power 200D PCB, DR ASSY, XCONT PCB, DR POWER CONT PCB

3.8.1 Power Supply for Power 200D PCB

I. When XCONT is operating, 24 V from XCONT enters the POWER -200 D board connector JPWD. As a result, 24 V passes through CP1 and is input to the three-terminal regulator of M9, and 5 V is output. This 5 V is the main power supply for the POWER -200 D substrate. As a result of the 5V power supply of the board being obtained, the board is in an operational state. Also, 24 V is used for relay operating power supplies, etc.

3.8.2 ON operation of DR system

I. When the SW of the DR system is pressed, the signal is input to the photocoupler PC1C from the connector JSW of the POWER -200 D board, and an H-level pulse of about 2 seconds is generated in M1A. The output of M1A goes to M5A (1). At this time, since the PC is not yet on, the connector JUSB connected to the PC's USB port is in a no-input state, and as a result, M5A (2) is at H level. As a result, M5A (3) becomes H-level for about 2 seconds.

II. M5A (3) goes to H level, and M5B (4) goes to H level. Also, M5B (5) is at H level if the PC _ PERMIT signal comes from the XCONT board through the connector JPWD, and the output M5B (6) is at H level. (The PC _ PERMIT signal is sent from the XCONT board by turning on the key SW and, if a keyless option is provided, by inputting a password.) And M2A's output goes to H-level for about six seconds. The output of M2A finally enters M7 (1) - (4). On input to the M7, all four outputs (K1, K2, K3, LED5) work.

III. When K1 operates, relay K2 of the DR POWER CONT board connected to the connector JDCP turns ON. As a result, battery voltage is applied to the PS 12 (24 V power supply for LCD and PC, for DC-AC converter input) to power the DR system. It is also input to DC power supply M8 through connector J 240 on the POWER -200 D board to generate 24 V. In addition, by operating the relay K4, the fan near the PC connected to the JFN will operate.

IV. When K2 operates, the contacts of K2 are closed and input to the XCONT board as a PC _ RUN signal through the connector JPWD, allowing the CPU to recognize that the DR system is operating.

V. When the K3 is activated, the LED (LED 5) in the power supply switch of the DR system lights up.

VI. When the PC is powered on while M6B (5) is at H level (About 6 seconds after M2A output is at H level, i.e. pressing the DR system power switch), 5V is output from the USB port of the PC and 5V is input to the connector JUSB. As a result, even if M6B (6) goes to H level and the output of M2A (13) (M6B (5)) returns to L level, the inputs of M7 (1) - (4) are kept at H level. That is, the power of the DR system is kept on.

Even if the supply of 24 V from XCONT is lost, for example by turning off the key switch, the POWER -200 D board can continue operating because the output of the DC power supply of the M8 is present and 24 V is present because the output of the 3-terminal regulator remains as well, since the J 240 is connected to the battery. This allows the DR system to continue operating even when the key switch is turned off while the DR system is operating.

3.8.3 OFF operation of DR system

When PC is shut down by SHUT DOWN operation from DR system, DC5V from USB port of PC disappears, then M6B⑥ becomes low level and the input signal to M7①～④ becomes low level.
As a result, all of K1, K2, K3 on Power 200D PCB and “K2” on DR POWER CONT PCB turn off, then power supply for DR system is shut down.
At the same time, the power supply to J240 also disappears, then Power 200D PCB stops working.

3.8.4 Function when the power plug is connected

When the power plug is connected, the signal “LVON” is sent to XCONT PCB via MUX POWER-32K and MUX CHARGE-32KX PCB.
As well, this signal is sent to POWER-200D via the connector JPWD. When the power plug is connected, M7⑤ becomes high level and the relay K6 turns on.
Finally, the relay “K1” on DR POWER CONT PCB, which is connected to POWER-200D via the connector JAC, turns on.
Then, the power supply for DR system is changed from the battery to AC180V, which is transformed by T4 transformer from AC125V output of T1 transformer.
“K1” on DR POWER CONT PCB turns on 2 seconds after connection of the power plug, because there is the delay circuit by M1B and M5D on POWER-200D PCB.
That is to avoid the unstable action of the relay “K1” due to any chattering when the power plug is connected. On the other hand, “K1” turns off immediately after the power plug is disconnected.

4 Trouble Shooting

4.1 Error Code List

4.1.1 Alarm List

Code	Error condition	Treatment method
F64 F65	Invalid hand switch input	Release the hand switch button and this error disappears.
F68	The shooting operation was performed before the FPD device was ready.	Wait until the DR system has completed READY before shooting. Select the shooting menu (communicate APR) again and this error disappears.
F6d	EMG signals are being input.	Remove any malfunction of the equipment. Turn off the emergency stop switch and this error disappears.
F76	Bad value sent from DR (bad APR communication).	Correct the shooting protocol in the DR console. Once the DR system sends the correct value, the error disappears. Update F/W to Rev. 1.37 or later
H1	Calculated, the amount of heat stored in the X-ray tube device after the image was taken is over. Can't shoot until the warning light disappears.	Reduce shooting conditions or stop shooting until the warning light goes off. When the X-ray tube machine cools and the temperature drops, this error disappears.
H2	In actual measurement, the amount of heat stored in the X-ray tube device after imaging is over. You can't shoot until the warning light disappears.	Stop shooting until the warning light goes off. When the X-ray tube device cools and the temperature drops, this error disappears.
H4	No thermal signal input. You can't shoot until the warning light disappears.	
E	CPU battery voltage dropped.	Normal running and shooting is possible, but the CPU backup battery needs to be replaced.
EEE	No CPU battery.	
E5	The emergency brake release battery voltage dropped.	You can drive and shoot normally, but you can't release the brakes in an emergency. The battery needs to be replaced for emergency brake release.

4. Trouble Shooting

Code	Error condition	Treatment method
bbb	The main battery voltage dropped to a level that made it impossible to shoot.	If the battery is low, charge it.
b1	Due to the low battery mode, the set shooting conditions exceed the time limit of the image accumulation time setting.	If the battery is low, charge it. The warning is cleared when battery power is restored.
bb0	Main battery voltage does not shoot It dropped to the level and the device auto-powered off. The device may have been left unattended, forgetting to turn off the key.	If the battery is low, charge it.
bb1	Main battery voltage does not shoot It dropped to the level and the device auto-powered off. The equipment may have been left unattended with the DR powered on.	If the battery is low, charge it. If charging does not restore battery capacity, the battery may be degraded.
A2	The cable from the handle short connector (CN 38 A, CN 38 B) to the MU DRIVER PCB is broken.	Check that the cable from the short connector on the handle (CN 38 A, 38 B) to the MU DRIVER board is not broken. Make sure the handle short connector (CN 38 A, 38 B) is securely connected.
A4	The brake release switch is on when the power is turned on.	Release the brake release bar. You can take pictures, but you can't move the device until the warning is lifted.
A5 A6	The microtremor switch (forward: A5 or backward: A6) is on when the power is turned on.	Release the microtremor switch. You can take pictures, but you can't move the device until the warning is lifted.
A7 A8	Driving handle input (left: A7 or right: A8) was detected at power-on.	Release the running wheel. You can take pictures, but you can't move the device until the warning is lifted.
A9	The obstacle detector is on when the power is turned on.	Eliminate the input to the obstacle detector. You can take pictures, but you can't move the device until the warning is lifted.

Code	Error condition	Treatment method
A10 A11	The arm lock release switch (A10) and release lever (A11) are on when the power is turned on.	Release the arm lock release switch (A10) and release lever (A11). You can take pictures, but you can't move the device until the warning is lifted.
d62	An anomaly in the arm movement detection encoder was detected.	Power cycle the device to flush the arm Post echo when operating with a stroke Check if da is idle. Update F/W to Rev. 1.50 or later.

4.1.2 Failure error message list (generator control)

Code	Error condition	Treatment method
F01	Power supply voltage 20% greater than the standard value	Checking the outlet power voltage Checking XCONT board signal "LV"
F02	The power supply voltage has fallen more than 30% below the standard value.	
F03	No "-15 V" check signal entered	Check DC Power Connections Checking the XCONT board signal -15 V
F04	Main power relay or anti-entry relay welding condition	Checking Voltage Between [TER3] and [TER4] When Key Switch Is Off Contact welding of power relays K6, K7 and K8 on MUX Power -32K Update F/W to Rev. 1.50 or later
F11	The tube voltage at the time of imaging was detected to be more than 30 kV higher than the set value.	Confirmation of the signals of XCONT at the time of radiation exposure: (TKV), (KV +), (KV-), (VP1), (VP2), (VIN)
F12	Measured tube voltage at the time of shooting is more than 15% lower than the set value	
F13	Measured tube voltage at the time of shooting is 10 kV or less	
F14	Microdischarge of the tube ball occurred three times and the radiation was stopped.	Aging of X-ray tubules Replacement of X-ray tube if not improved

Code	Error condition	Treatment method
F15	The measured tube voltage at the time of shooting is more than 10% lower than the set value. Your battery may be performing poorly.	Check battery voltage Verifying FVR Adjustment Confirmation of the signals of XCONT at the time of radiation exposure: (TKV), (KV +), (KV-), (VP1), (VP2), (VIN)
F16	The kV feedback switching signal is not detected during shooting. It is also possible that the battery performance has decreased.	Check battery voltage Verifying FVR Adjustment Confirmation of the signals of XCONT at the time of radiation exposure: (TKV), (KV +), (KV-), (VP1), (VP2), (VIN)
F17	A microdischarge of the bulb occurred once.	Aging of X-ray tubules
F18	Microdischarges of the tubules occurred twice.	Aging of X-ray tubules
F19	The anode or cathode tube voltage exceeds 90 kV.	Aging of X-ray tubules Replacement of X-ray tube if not improved
F1a	The kVADJ board is not installed.	Check the kVADJ PCB on the XCON.
F1b	A kVADJ substrate that is not compatible with the high-voltage transformer installed in the device is installed.	
F1c	The FPGA revision on the XCONT board does not support the high voltage transformer D150MH-40 (P/N: 565 -16863).	Check the revision seal of the XCONT board FPGA, M35.
F21	The measured tube current at the time of shooting is more than 100 mA larger than the set value, it was detected SW.	Check the FVR adjustment. Check the signal of the XCONT at the time of radiation exposure (TKV), (TMA), (IFV), (VIN). Update F/W to Rev. 1.50 or later (F21, F24, F32)
F22	The measured tube current value at the time of shooting is more than 50% smaller than the set value.	
F23	Measured tube current value at the time of shooting is 10mA or less	

4. Trouble Shooting

Code	Error condition	Treatment method
F24	The measured tube current at the time of shooting is more than 100 mA larger than the set value, it was detected HW.	Check the FVR adjustment. Check the signal of the XCONT at the time of radiation exposure (TKV), (TMA), (IFV), (VIN). Update F/W to Rev. 1.50 or later.
F31	The measured value is more than 50% larger than the set value of the filament heating, it was detected SW.	
F32	The measured value is more than 50% larger than the set value of the filament heating, it was detected HW.	
F33	Measured value of filament heating is 1V or less.	Check the connection of the filament heating line. Check the FVR adjustment. Check the signal of the XCONT at the time of radiation exposure (TKV), (TMA), (IFV), (VIN). Update F/W to Rev. 1.52 or later.
F34	The measured value is more than 50% larger than the set value of filament heating, it was detected HW.	
F35	The measured value is more than 50% larger than the set value of filament heating, it was detected SW.	
F41	Capacitor voltage 300V or more	・MUX CAPACITOR 基板の[VC+]-[VC0]間の電圧確認
F42	60 seconds after capacitor charging starts, the capacitor voltage becomes 170 V or less.	
F43	コンデンサ充電開始して 10 秒後にコンデンサ電圧が 30V 以下	Check voltage between VC + and VC0 on MUX CAPACITOR board
F44	Capacitor charge normal, but no signal VCOK	Check signal of MUX CHARGE -32 KX board, VC, VP1
F45	Condenser charging relay K2 does not become standby after 30 seconds of operation	Check the voltage between "VC +" and "VC0" on the MUX CAPACITOR board. Update F/W to Rev. 1.61 or later.
F46	The capacitor charging relay K3 has not been turned on even though it was just before the start of radiation.	

Code	Error condition	Treatment method
F47	After reaching standby, the capacitor voltage fell below the standby level for more than 30 seconds.	Check the voltage between "VC +" and "VC0" on the MUX CAPACITOR board. Update F/W to Rev. 1.61 or later.
F51 F53	Charging battery + side (F51), - side (F53) voltage value is 155 V or more.	Checking the signals VP1 and VP2 on the charging XCONT board Checking the signals of the charging MUX CHARGE -32 KX board: [120 +] [120 -] [VP1] [VP2]
F52 F54	Charging battery + side (F 52), - side (F 54) voltage value becomes 50 V or less.	
F55	Voltage on the + side of the battery is greater than 12 V or more on the - side while the battery is running	Checking the XCONT board signals P1 and VP2 during battery operation Checking the signals of the MUX CHARGE -32 KX board when it is running on a battery: [120 +] [120 -] [VP1] [VP2]
F56	When the battery is running, the voltage on the battery-side is more than 12 V and greater than the voltage on the + side.	
F57	The voltage on the + side of the battery being charged is greater than 12 V or more than the - side	Checking the signals VP1 and VP2 on the charging XCONT board Checking the signals of the charging MUX CHARGE -32 KX board: [120 +] [120 -] [VP1] [VP2]
F58	The voltage on the battery-side during charging is greater than 12 V or more than the + side	
F5a	Charging current exceeded 3.8 A. “CHBR1 signal”	Checking the signal IC of the charging XCONT board Checking the signal IC1 of the charging MUX CHARGE -32 KX board Checking the signal of the charging MUX CHARGE-04B board, IC-
F5b	+ or - side battery voltage of 155 V or more “CHBR2 signal”	Confirmation of MUX CHARGE -32 KX board signal "120 +," "120 -"
F5c	The outlet was connected with the main breaker turned off.	Check the main breaker Check battery voltage

4. Trouble Shooting

Code	Error condition	Treatment method
F5E	During charging, the measured value exceeded the command current by more than 0% for 10 seconds.	Checking the signal “IC” of the charging XCONT board Checking the signal “IC1” of the charging MUX CHARGE -32 KX board Checking the signal “IC-" of the charging MUX CHARGE-04B board
F5F	During charging, the measured value was less than 30% of the command current for 10 seconds.	Checking the signal "IC" of the charging XCONT board Checking the signal "IC 1" on the charging MUX CHARGE -32 KX board Checking the signal "IC-" on the MUX CHARGE-04B board during charging Checking the fuse "F1" on the MUX CHARGE-04B board
F61	The starter drive signal does not return even after the starter start time has ended.	Check the signal "YC" "ZC" on the INVERTER UNIT-32K. Grounding should be done with "GND" on the INVERTER UNIT-32K.
F6E	Shooting does not end until backup time/mAs after shooting	Confirmation of XCONT signals "TKV," "TMA" and "KVT" during radiation exposure Update F/W to 1.53 or later.
F6F	No KVT input after 250 msec of EXR output	
F71	Area Dosimeter Communication Timeout Error	Check area dosimeter connections and settings
F72	Area Dosimeter Communication Error	Rebooting the equipment and checking the shooting operation
F73	Area Dosimeter Failure Error	Replacing an Area Dosimeter
F81	The setting of DIP switch 2-2 on the XCONT board does not match the maximum power output of the device (32 kW/12.5 kW)	Check the DIP switch settings on the XCONT board.
F85	The setting of DIP switch 2-6 on the XCONT board does not match the type of high voltage transformer for the 32kW type installed in the device	

Code	Error condition	Treatment method
F86	The setting of DIP switch 2-1 on the XCONT board does not match the type of device (MobileDaRtEvolution/MobileArt Evolution).	Check the DIP switch settings on the XCONT board.

4.1.3 Failure error message list (moving control)

If you see errors from “d15” to d 27 to “d60”, take the following steps in common:

I. Move the device to a safe location.

II. Do not turn the power back on and use the emergency brake release function to move it.

Code	Error condition	Treatment method
d15	Welding of the relay in the motor drive was detected.	Check the continuity between Pin 1-2 of “CN17” with connected.
d16	A short-mode failure of the right FSR was detected.	Check the cable between FSR L ASSY and MUDRIVER for damage to the film caused by pinching. Check the FSR L ASSY for any ground contact.
d17	A short-mode failure of the left FSR was detected.	Check the cable between FSR R ASSY and MUDRIVER for damage to the film caused by pinching. Check the FSR R ASSY for any ground contact.
d20	A malfunction was detected in the motor FET of the right motor drive circuit.	With the unit powered down, check that the diodes on the MUDRIVER are conducting on the right “D11, 12, 14, 15” and left “D 1, 2, 8, 9”. Update F/W to Rev. 1.62 or later.
d21	A malfunction was detected in the motor FET of the left motor drive circuit.	
d22	A short circuit was detected in the right motor drive circuit.	
d23	A short circuit was detected in the left motor drive circuit.	

Code	Error condition	Treatment method
d24	A rotation error of the right motor was detected.	Check if “CN7” are connected incorrectly on the MU DRIVER board. Make sure that “CN7” on the MU DRIVER board are securely connected. Check that the GND cable of the right motor's encoder cable is securely screwed to the device housing.
d25	A rotation error of the left motor was detected.	Check if “CN6” are connected incorrectly on the MU DRIVER board. Make sure that “CN6” on the MU DRIVER board are securely connected. Check that the GND cable of the left motor's encoder cable is securely screwed to the device housing.
d26	The overcurrent of the right motor was detected.	Check that the wheels on the right side of the device are not locked or deformed and in contact with the device.
d27	The overcurrent of the left motor was detected.	Check that the wheels on the left side of the device are not locked or deformed and in contact with the device.
d28	A malfunction was detected in the right motor encoder.	Check if “CN7” on MU DRIVER board is connected. Check that the GND cable of the right motor's encoder cable is securely screwed to the device housing. In the speed display mode, check that the value of the right motor is appropriate (no abnormal value such as zero is displayed).

Code	Error condition	Treatment method
d29	A malfunction was detected in the left motor encoder.	Check if “CN6” on MU DRIVER board is connected. Check that the GND cable of the left motor's encoder cable is securely screwed to the device housing. In the speed display mode, check that the value of the left motor is appropriate (no abnormal value such as zero is displayed).
d30	The temperature rise of the motor FET of the right motor drive circuit was detected.	Check that “CN19” on the MU DRIVER board are securely connected. After dropping the main breaker of the device, check the temperature of the "d30": FET “Q11, 12, 14, 15” on the MUDRIVER.
d31	The temperature rise of the motor FET of the left motor drive circuit was detected.	Check that “CN18” on the MU DRIVER board are securely connected. After dropping the main breaker of the device, check the temperature of the “d31”: FET “Q1, 2, 8, 9” on the MUDRIVER.
d40	Fuse blown was detected.	Check the continuity of the fuses “F4” and “F5” installed on the left side of the device.
d42	No 24V power.	Check the connection of “CN9” on the MUDRIVER. Check that the connector on the DC power “PS2” mounted at the front of the unit is securely connected.
d44	No + 5/+ 15 V power supply for right motor	Check that the DC power “PS22” connectors on the right side of the unit are securely connected. Check that the “CN22 and 23” on the MUDRIVER board are securely connected.
d45	No + 5/+ 15 V power supply for left motor	Check that the DC power “PS21” connectors on the right side of the unit are securely connected. Check that the “CN22 and 23” on the MUDRIVER board are securely connected.
d60	A broken wire was detected.	Check the strut wires for breakage.

Code	Error condition	Treatment method
d61	Connection error with MU DRIVER board	Make sure that the “CN21” on the MU DRIVER board and the “CNUM” on the XCONT board are securely connected.

4.2 What to do if an error message appears

4.2.1 H1 and H2

The CPU calculates the accumulated heat capacity from past exposures and a warning message appears. If the number of exposures is small but this message appears or does not disappear after a period of time, check the voltage of the XCONT battery and initialize the SRAM. Unplug the battery and leave it for about 10 minutes with the power off to clear the SRAM. If that doesn't work, try initializing the CPU.

4.2.2 H4

This message is issued when the thermal switch of the tube bulb is activated. If the bulb is not hot at all but this message appears, check the thermal switch signal. Since this H4 and F6 d are also issued when the 24 V power supply is not present, check the output of the DC 24 V power supply (PS2) to the XCONT, cable wiring, etc.

A similar phenomenon occurs when the CPU fails to recognize "key switch ON" due to a faulty contact of the key switch, the MGRS signal is not generated, and the DC 24 V power drops because relays K7 and K8 of the MUX POWER -32 KX are not turned on and charge suppression resistors are not shorted. See also “4.2.26 F6d”.

4.2.3 F01, F02

Software detection

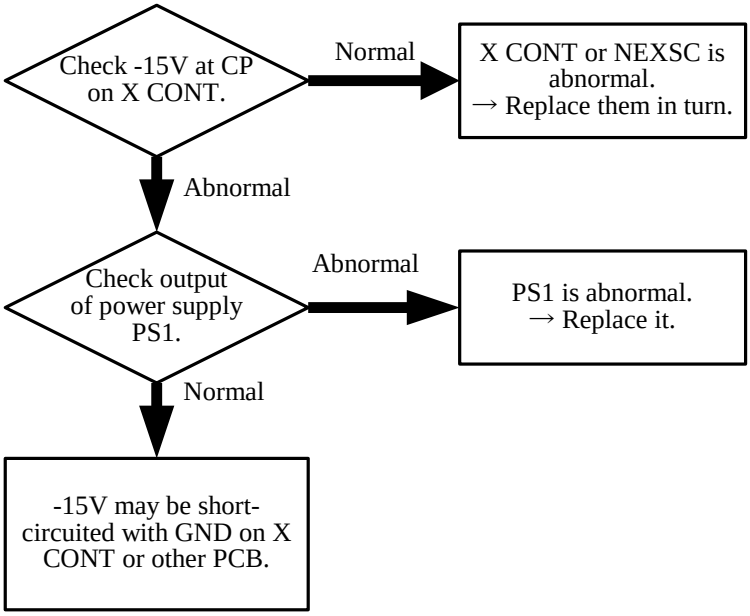
- F01: Power supply voltage is about 20% or more higher than normal (installed value)
- F02: Power supply voltage is about 30% or more lower than normal (installed value)

I. Check that the voltage of the CP 23 (LV) on the XCONT is within 2 V +/- 20% to -30%.
II. If this error occurs even though it is within the above range, the XCONT is faulty.
III. If not, check fuse F1 on MUX CHARGE -32 KX.
IV. If the fuse F1 is blown, replace the fuse. If the F1 breaks repeatedly, replace the MUX CHARGE -32 KX.
V. If the fuse F1 is not blown, check between pins 7 and -8 of connector JT1B for the presence of AC10V input from T1 transformer.
VI. If AC10V input is coming, replace the MUX CHARGE -32 KX.
VII. If there is no AC10V input, check the input voltage of the T1 transformer and the cabling from the AC10V output. If they are fine, replace the T1 transformer.

4.2.4 F03

Hardware detection

Abnormal -15 V power supply for control circuit.



4.2.5 F04

If the power supply remains energized for 10 seconds even though the key switch is off when the power plug is not plugged in, it is judged that "Contact welding of main power relay "K7, K8" or inrush current prevention relay "K6" on MUX POWER -32 K has occurred" and error F04 is displayed.

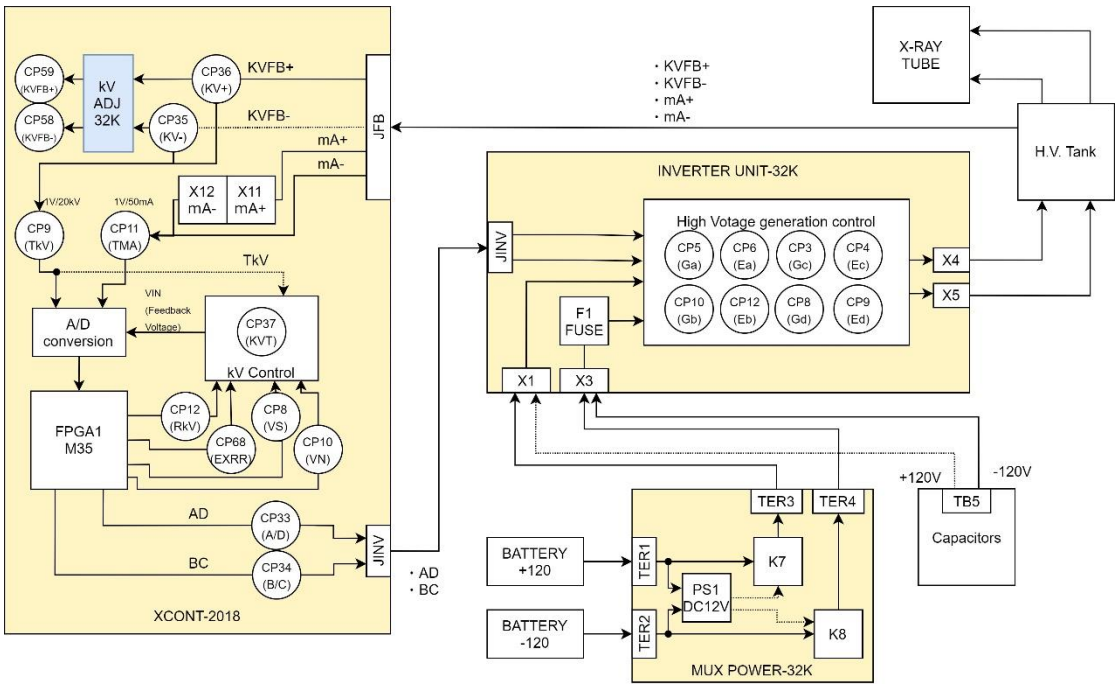
Check the voltage between terminals TER3 and TER4 on the MUX POWER-32 K when the key switch is off, and if it is not 0 V, the contacts of the above relays may be welded. (However, in case of 32 kW type, since the large-capacity capacitor is connected between TER3 and TER4, even after the key switch is turned off, the voltage does not immediately go to 0 V and gradually decreases.)

On the other hand, if the power supply from the outlet is unstable and the CPU cannot detect the outlet connection status for more than 10 seconds even though the power plug is connected to the outlet, F04 will also be detected, so it is necessary to make sure that the outlet power supply is stable.

In order to avoid false positive error, software Rev. 150 of XCONT-2018 has been released. You may be able to improve it by upgrading the software to the newest revision. In this device, there are few cases where contact welding of power relays K6, K7, K8, etc. has actually occurred.

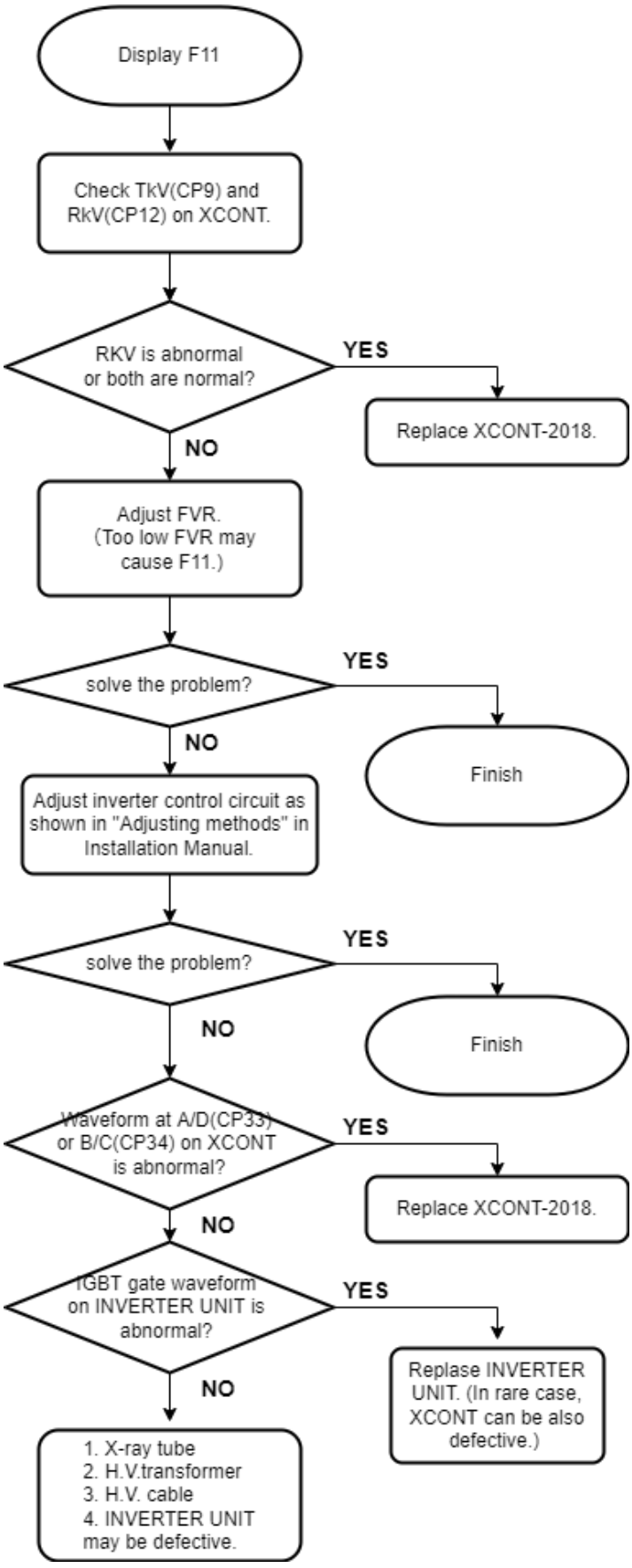
4.2.6 F11 (KV OVER)

Software to detect. Measurements were taken 10 ms after the start of exposure and at 100 ms intervals thereafter. However, it is not detected when the shooting time is 20 ms or less.
This error code means that the measured tube voltage is about 30 kV or more higher than the set value.



Investigate according to the flowchart below and on the next page.
Note:
When checking the gate waveform of the IGBT, follow the instructions in "Adjusting the Inverter Control Circuit on the XCONT -2018" in the installation manual and take measures such as short-circuiting to prevent the generation of high pressure.

4. Trouble Shooting



4.2.7 F12 (KV UNDER)

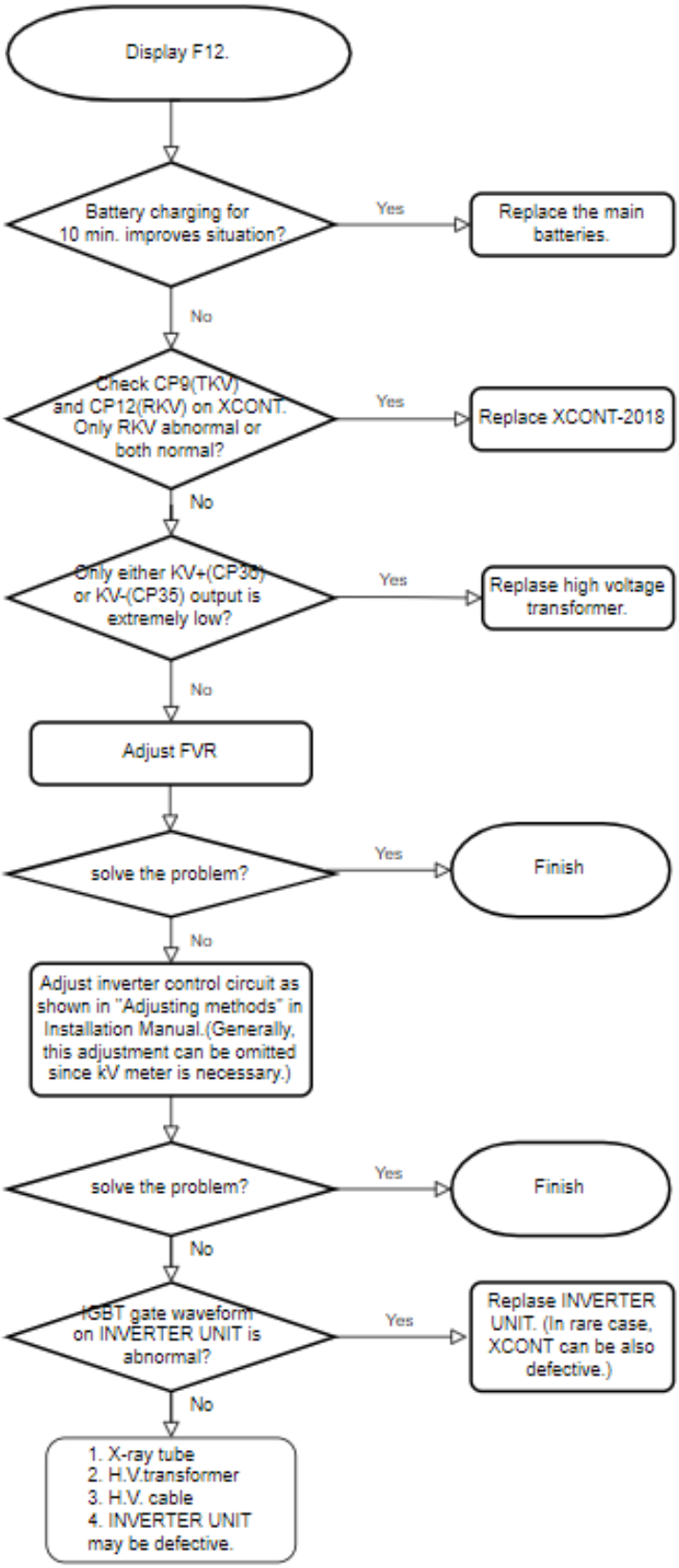
Software detection. Measurements were taken 10ms after exposure and at 100ms intervals thereafter. However, it is not detected when the shooting time is 20ms or less.
The measured tube voltage is more than 15% lower than the set value - > If this error occurs, stop X-ray irradiation.
These errors are complementary to F15 and F16.

Follow the instructions below and the flowchart on the next page to investigate.

Note:
When checking the gate waveform of the IGBT, follow the instructions in "Adjusting the Inverter Control Circuit on the XCONT -2018" in the installation manual and take measures such as short-circuiting to prevent the generation of high pressure.

Note:
As the battery deteriorates, the tube voltage may drop when the battery is low, especially in high shooting conditions. In this case, recharging the battery reduces the frequency of malfunctions.
In some cases, F 12, F 15, and F 16 may occur due to lack of capacity even when all 5 of the battery level indicator LEDs are on. In that case, too, if an additional charge clearly improves the battery, there's a big chance of a bad battery.

4. Trouble Shooting



4.2.8 F13 (NO KV)

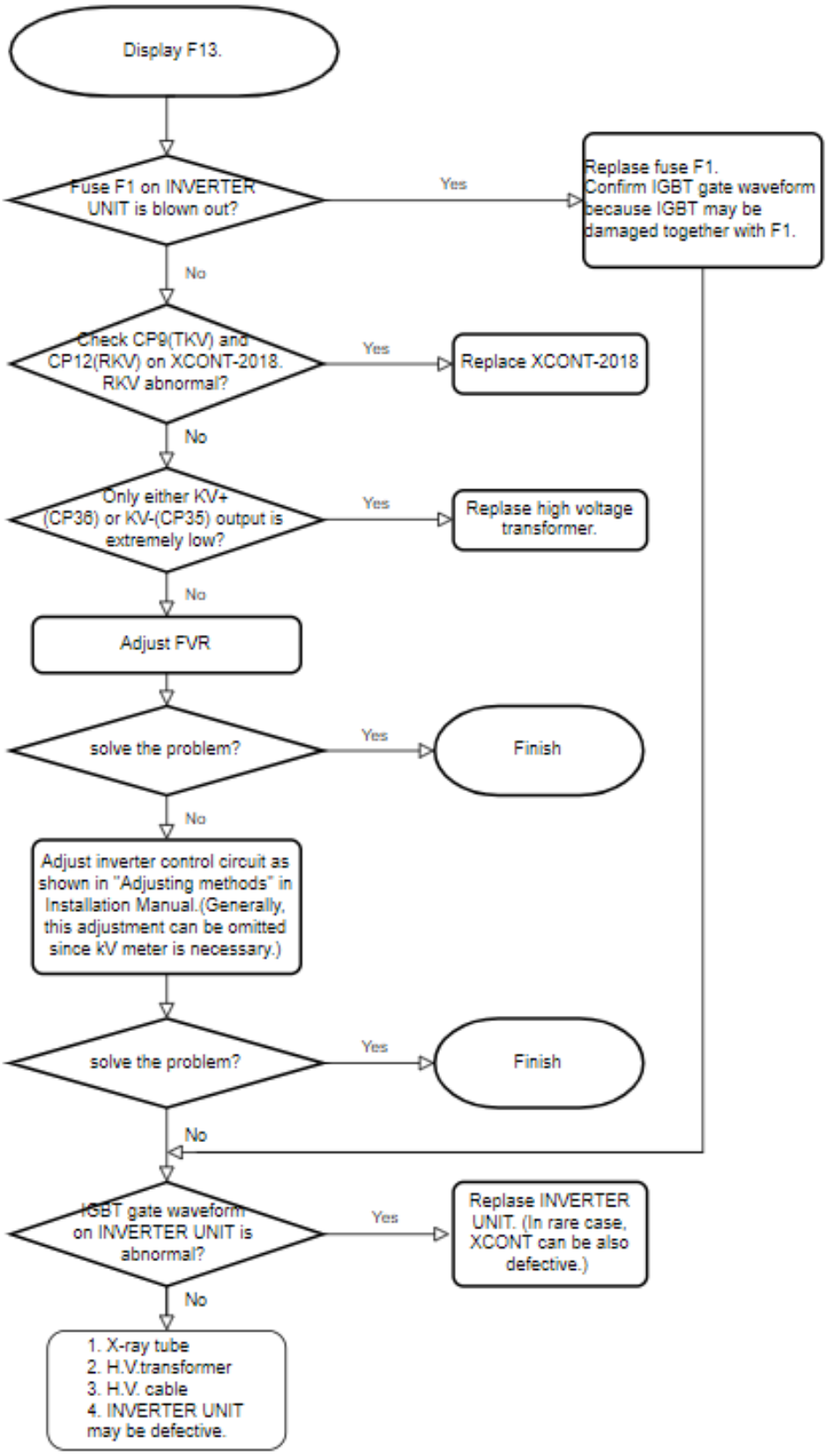
The measured tube voltage by the software detects about 10 KV or less.
Measurements were taken 10ms after exposure and at 100ms intervals thereafter. However, it is not detected when the shooting time is 20ms or less.

Follow the instructions below and the flowchart on the next page to investigate.

Note:
When checking the gate waveform of the IGBT, follow the instructions in "Adjusting the Inverter Control Circuit on the XCONT-2018" in the installation manual and take measures such as short-circuiting to prevent the generation of high pressure.

Note:
If the resonant capacitors C9 and C 18 on the INVERTER UNIT are damaged, the same as when the fuse F1 is blown, the IGBT has a normal gate waveform but no measured tube voltage (TkV is 0). Again, replace the INVERTER UNIT.

4. Trouble Shooting



4.2.9 F14 (KV BRAKE)

Measured tube voltage abnormality or bulb discharge due to hardware was detected.

- When 3 micro-discharges occur during 1 shot, F 14 occurs and shooting operation is stopped.
- When the first micro-discharge occurs, F 17 is recorded in the log, and when the second micro-discharge occurs, F 18 is recorded (hidden), but shooting resumes after a short (15 msec) pause.

Follow the instructions below and the flowchart on the next page to investigate.

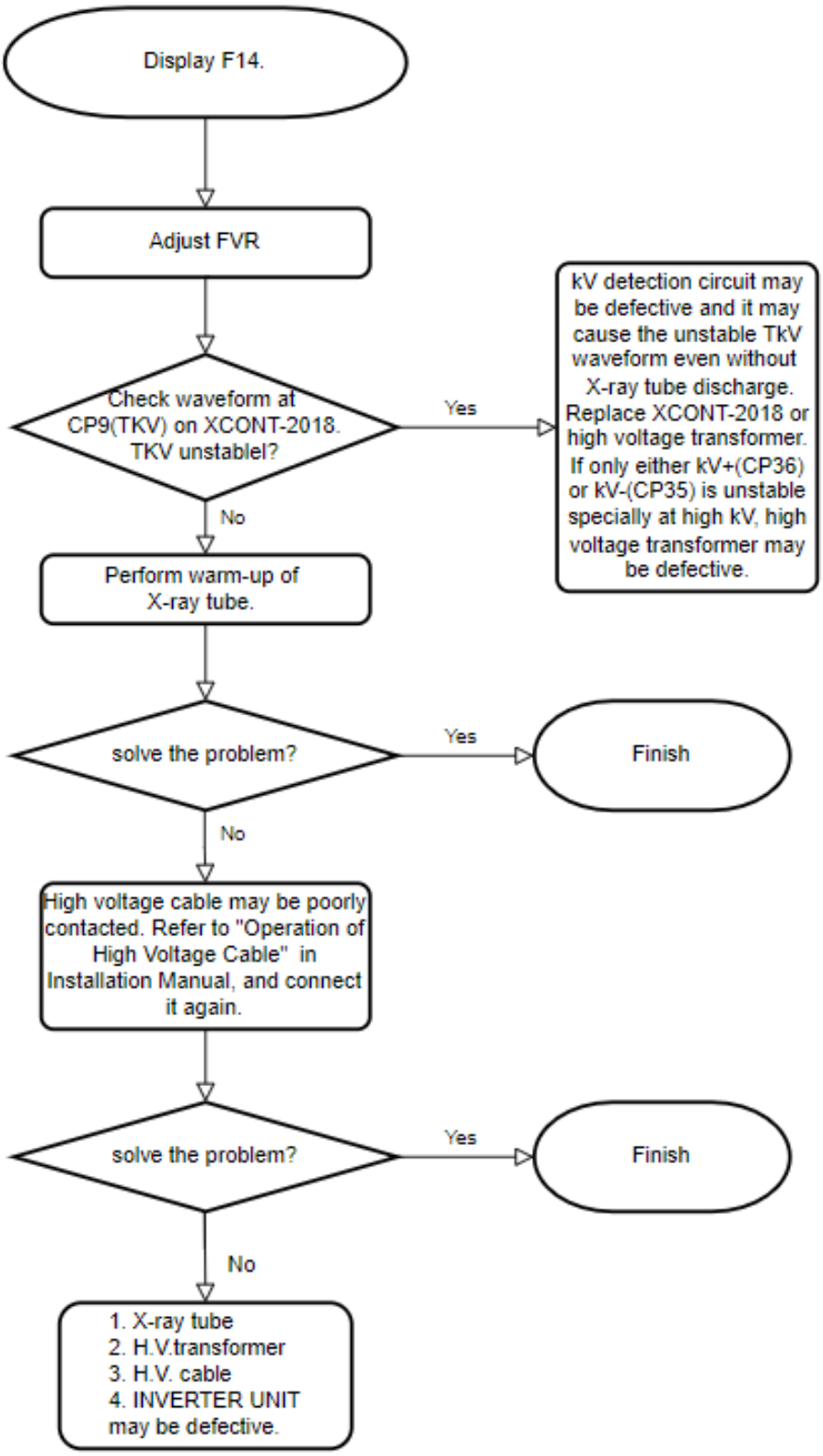
This error message assumes a high-pressure system (especially a tube bulb) discharge. In many cases, this can be resolved by performing aging, so first perform FVR adjustment to ensure that the TkV and TmA waveforms are normal, and then perform aging.

If we can confirm the following phenomenon, we can clearly conclude that it is a high-pressure system discharge.

I. It makes a pan-like discharge sound during exposure.
II. It does not occur at low KV, but an error message occurs when increasing KV.
III. The glass is visibly clouded as you look through the bulb radiator. (by deposition of target material)
IV. A clear roughness can be seen in the target when viewed through the bulb radiating orifice.
V. It can be seen peeking through the tube radiating orifice to confirm the spark discharge like a thick lightning when exposed.

Note:
Be sure to protect against radiation with lead glass.
Since an aluminum filter is installed in the tube radiating port, remove the lead cone first and also remove the aluminum filter when looking inside the glass tube.

4. Trouble Shooting



4.2.10 F15 (KV UNDER)

Software detection. Measurements were taken 10ms after the start of exposure and at 100ms intervals thereafter. However, it is not detected when the shooting time is 20ms or less.
The measured tube voltage is more than 10% lower than the set value. However, X-ray irradiation does not stop when this error occurs (because the degree of tube voltage drop is relatively minor). These errors are complementary to F12 and F16.
See Section F12 for how to deal with it.

4.2.11 F16 (KV UNDER)

Software detection
The KV feedback switching signal FBST (a signal that occurs when the measured tube voltage reaches a value greater than 92% of the set value) was not detected during the irradiation: However, X-ray irradiation is not stopped even if this error occurs, and F16 is displayed after the shooting is finished. These errors are complementary to F12 and F15.
No matter how short the shot, F 16 can still be detected.
See Section F12 for how to handle this.

4.2.12 F17 および F18 (KV BRAKE)

Measured tube voltage abnormality or bulb discharge due to hardware was detected.
When a total of 3 micro-discharges occur during a single shooting, F 14 occurs and the shooting operation stops.
When the first micro-discharge occurs, F17 is recorded in the log, and when the second micro-discharge occurs, F 8 is recorded (hidden), but shooting resumes after a short (15 msec) pause.

See Section F 14 for how to deal with it.

4.2.13 F19

Hardware detection of anode or cathode tube voltage exceeding 90 kV.
In this case, too, it is highly likely that a bulb discharge occurred.
See Section F 14 for how to deal with it.

4.2.14 F1a

KVADJ confirmation signal * KVADJ CON is not detected. When this error is detected, X-ray irradiation is prohibited. Check that the KVADJ is mounted correctly on the XCONT and that the connection connector JFB2 between the two boards has a poor contact.

4.2.15 F1b

The signal *COCK indicating the type of KVADJ is not compatible with the system. For a 32kW type, a KVADJ 32 K must be installed, and the * COCK signal must be High. If the * COCK signal is Low, this error is detected, and X-rays are prohibited.

- 32 kW type: KVADJ32 K mounted - > * COCK signal is high
- 12.5 kW Type: Equipped with KV ADJ COCK - > * COCK signal is low

Check that the KVADJ (KVADJ32K) of the type suitable for the system is mounted correctly on the XCONT and that the connection connector JFB2 between the two boards has a poor contact.
It can also occur if software from another device is accidentally inserted when upgrading software.

4.2.16 F1c

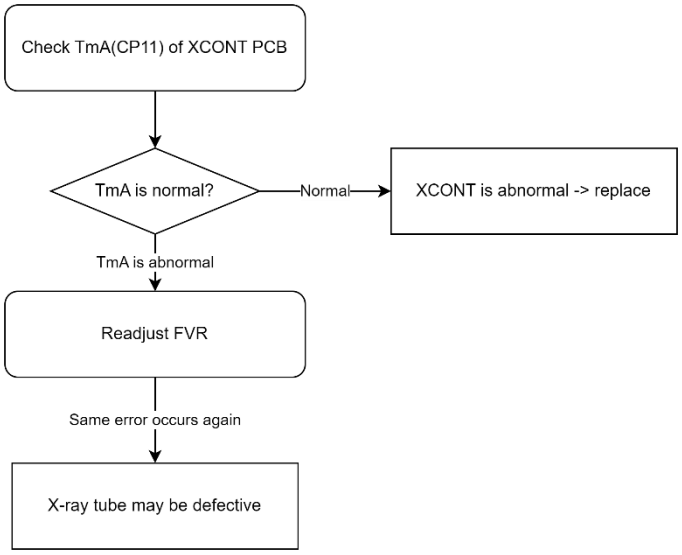
The FPGA revision on the XCONT-2018 does not support the high-voltage transformer D 150 MH -40 (S) (P/N: 565 -16863). Use a J- or later XCONT-2018.

4.2.17 F21,F24 (MA OVER) 及び F22 (MA UNDER)

F 21 and F 22 are detected by software at 10 ms after the start of exposure and at 100 ms intervals thereafter.
However, it is not detected when the shooting time is 20 ms or less.
F 24 is hardware detection.

F 21, F 24: The measured tube current is about 100 mA more than the set value.
F 22: Measured tube current is about 50% or less of the set value

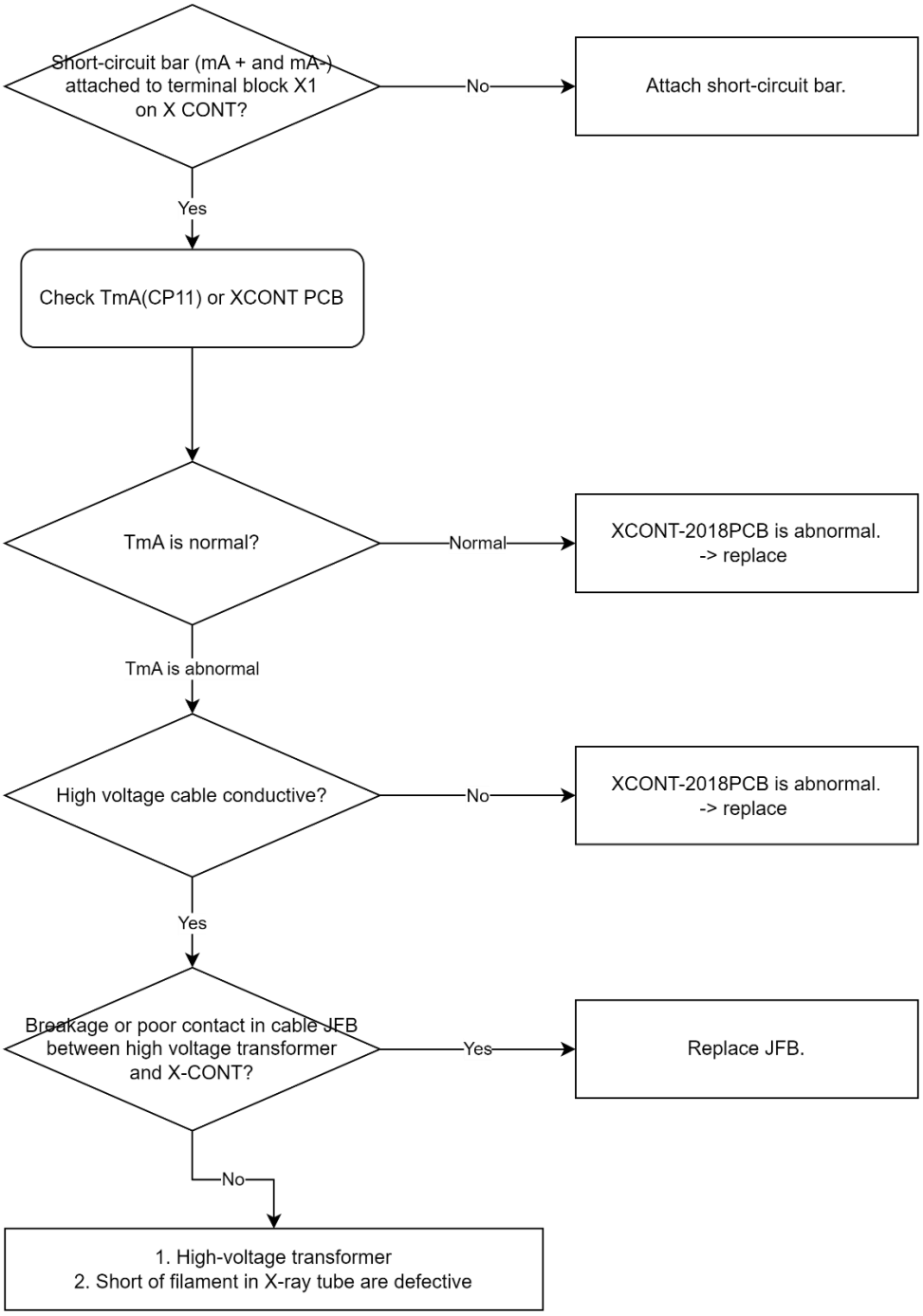
If this message appears, also check the flow chart of the heating system of F31 and F32.



4.2.18 F23 (NO MA)

Measured tube current of less than about 10 mA was detected 10 ms after the start of exposure by software, and measured at 100 ms intervals.

However, it is not detected when the shooting time is 20 ms or less.



How to check the bulb filaments

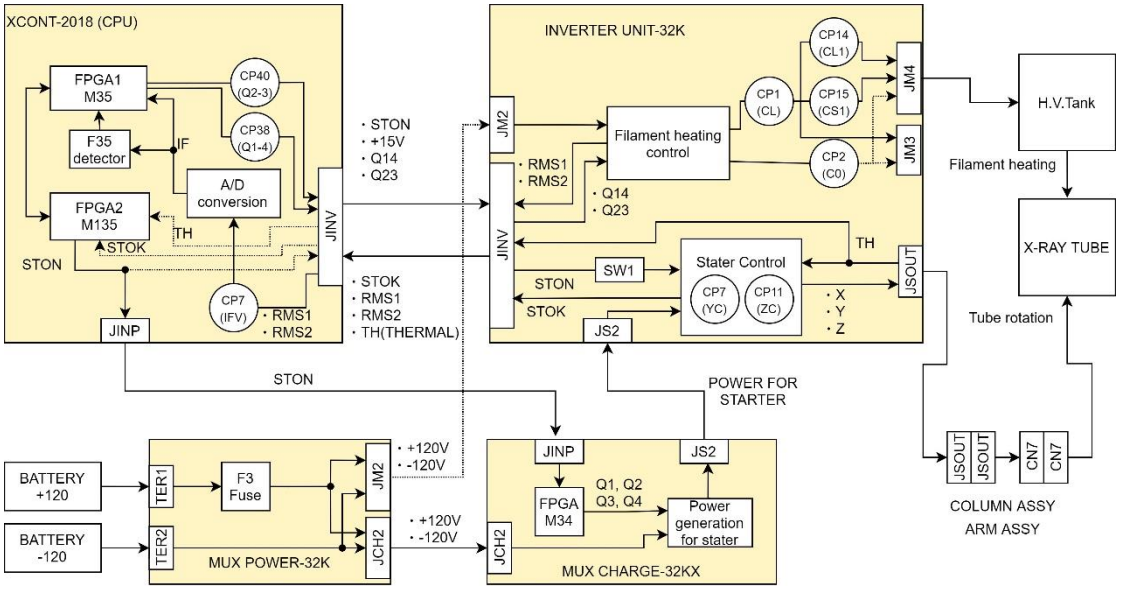
Remove the collimator and visually observe the inside of the tube from the tube radiation port to check the filament lighting during READY. Be sure to use lead glass for radiation protection.

Note:
An abnormal F3 fuse on the MUX POWER can also cause this error because the power supply for heating the filament is not properly supplied between pins 1-3 of connector JM2 of the INVERTER UNIT.
In particular, if F3 fuse repeatedly blows, it is possible that a short circuit may have occurred somewhere on the output side. Therefore, it is necessary to investigate the cause of the blown fuse by referring to paragraph 2 of "4.2.20 F32, F33 (IF UNDER)."

4.2.19 F31, F34, F35 (IF OVER)

Filament heating current occurs when over

- F31 is S/W detection
- F34, F35 for H/W detection (Added from XCONT-2018, F34 in FPGA, F35 independent circuit)

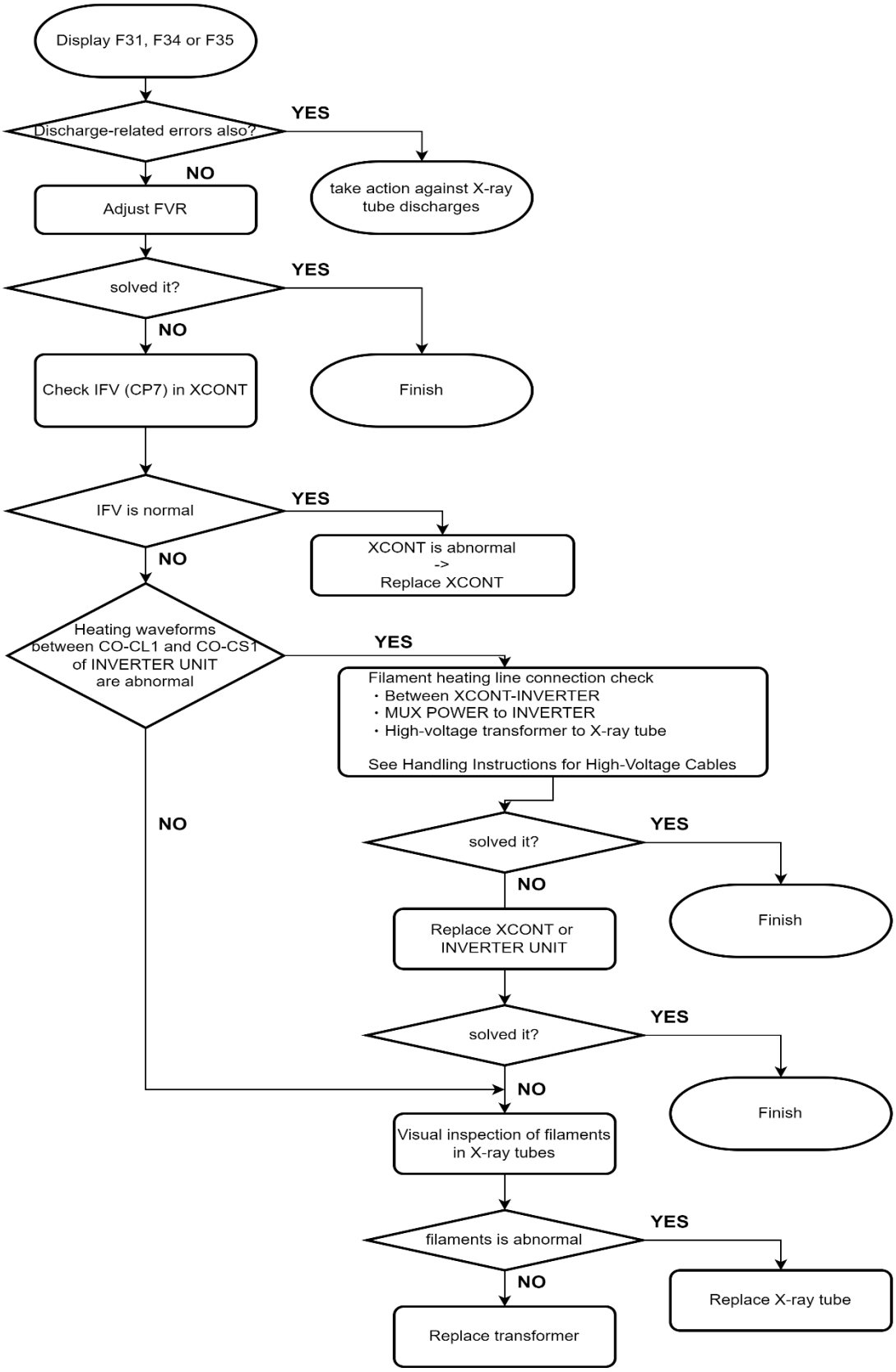


There are only a few cases in which a normal breakdown causes an over.
Furthermore, it is less likely to occur for anomalies on the high pressure side, and relatively more likely to occur on the low pressure side (Boards, cables, etc.).

- If the IFV (CP7) of the XCONT at the time of the error is confirmed to be normal, it becomes a problem after the A/D conversion in the XCONT.
- If the IFV at the time of the error is high (note that there is a period of about 2 seconds after the Ready starts that is higher than the FVR), it is likely to be an INVERTER or XCONT related to control. Another possibility is a cable between the XCONT and the INVERTER.
- If the problem does not always occur, check the connection and replace the boards as necessary.

In X-ray tube discharges, even minor discharges may generate F34 and F35 for H/W detection due to noise, so when these errors occur, check whether there are any other errors generated in the discharges. If, at the same time, an error occurs in the discharge, then X-ray tube discharge correspondence is required. As for the F35, there have been many reports of its occurrence, but it was presumed to be a false positive immediately after the shooting ended, and a correction was made in Rev 1.52.

Note that the heating waveform confirmation may be omitted because the waveform disappears the moment an error occurs, making measurement rather difficult.


Display F31, F34 or F35
Discharge-related errors also?
YES
take action against X-ray tube discharges
NO
Adjust FVR
solved it?
YES
Finish
NO
Check IFV (CP7) in XCONT
IFV is normal
YES
XCONT is abnormal
->
Replace XCONT
NO
Heating waveforms between CO-CL1 and CO-CS1 of INVERTER UNIT are abnormal
YES
Filament heating line connection check
・Between XCONT-INVERTER
・MUX POWER to INVERTER
・High-voltage transformer to X-ray tube
See Handling Instructions for High-Voltage Cables
NO
solved it?
YES
Finish
NO
Replace XCONT or INVERTER UNIT
solved it?
YES
Finish
NO
Visual inspection of filaments in X-ray tubes
filaments is abnormal
YES
Replace X-ray tube
NO
Replace transformer

4.2.20 F32, F33 (IF UNDER)

- F32

  Filament current occurs below 50% of set value. S/W detection.

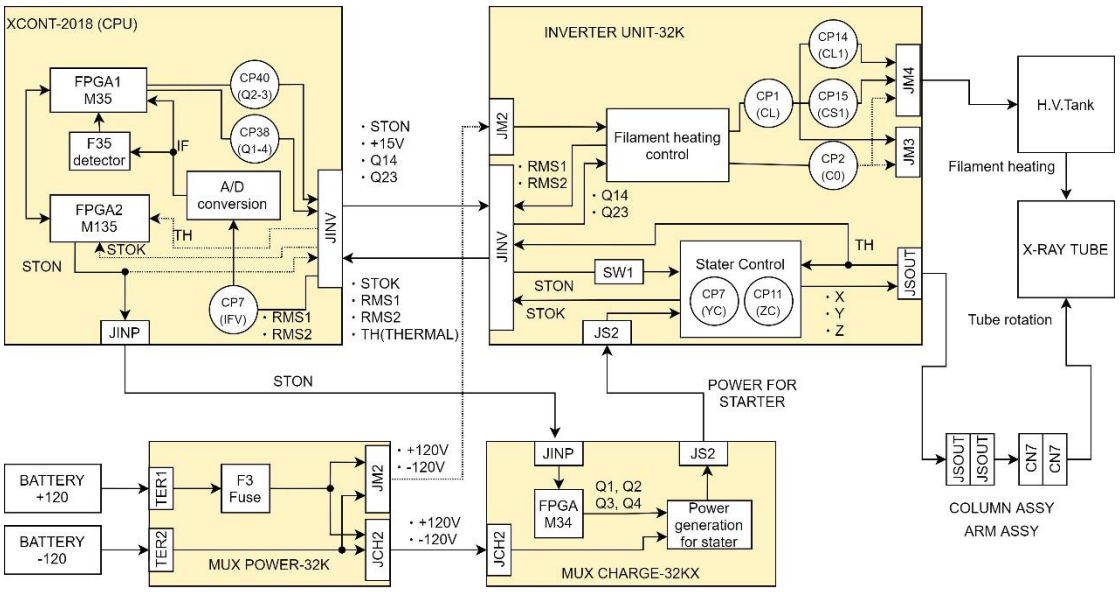
  Even in the case of broken filaments or broken wires or poor contact on the high pressure side, the result may be F 32 instead of F 33.

- F33

  It occurs when the filament current is less than 1 A (IFV less than 1 V). S/W detection.

  It is mainly for detecting filament breakage, but it also occurs due to breakage or contact failure on the high-pressure side, blown fuse, etc.

If the fuse (F3 on the MUXPOWER-32K) is blown, the fuse will be F33. Note that this fuse is common to the power supply to the starter circuit (the MUXCHARGE-32K) in addition to the filament heating, and therefore, if there is trouble in the starter circuit, the fuse may blow, resulting in F33.



Follow the instructions below and the flowchart on the next page to investigate.

I. Check fuse F3 on MUX POWER -32 K.

II. If fuse F3 is blown, replace the fuse.

If the fuse always blows even after replacing the fuse, remove connectors JM2 (supply to INVERTER) and JSCH2 (supply to MUX CHARGE) on the MUX POWER-32K alternately and check which connection causes the fuse to blow.

4. Trouble Shooting

・ When the fuse blows on the INVERTER side
The most likely cause is component failure of the filament heating inverter circuit, of which short-circuit failure of the FET is the most likely. Check if the FET is short-circuited. Alternatively, check the path from the output JM3 of the INVERTER to the high-voltage transformer for a short circuit. Check the resistance value of the filament coils C-CL, C-CS.

・ When the fuse blows on the MUX CHARGE-32 K/32 KX side
The most likely cause is component failure of the inverter circuit for the starter, of which short-circuit failure of the FET is the most likely. Check if the FET is short-circuited. Alternatively, check if there is a short circuit in the path to the INVERTER and X-ray tube device after output JS2 of the MUX CHARGE. Check the resistance value of the stator coils X,Y and Z.

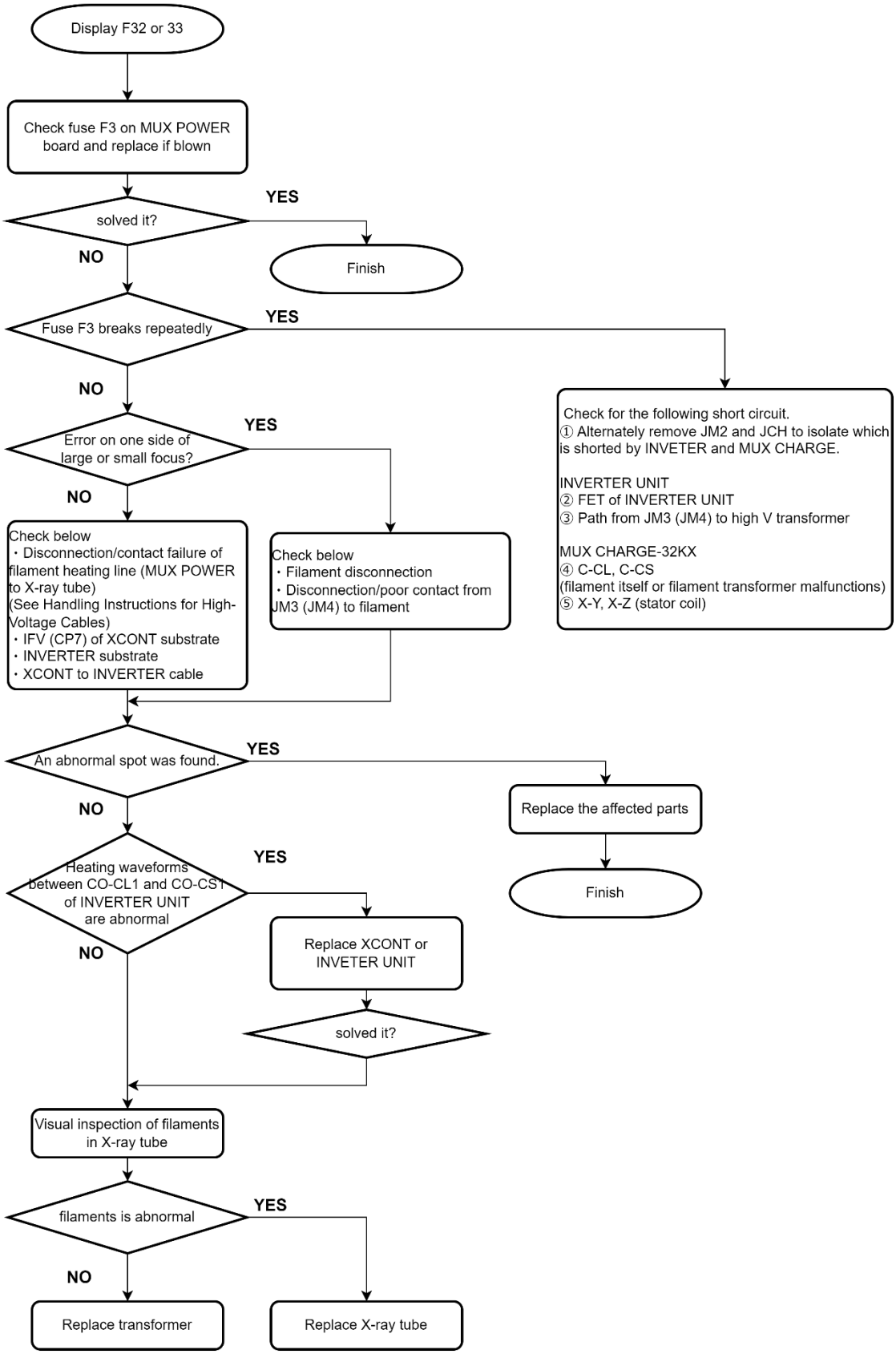
III. If the fuse F3 is not blown, check to see if the error occurs in both the large and small focus or only one focus.
- Case of constant occurrence only at one focus
Since the filament heating control is normal, there is a high possibility of a filament break in the X-ray tube device or a break or contact failure in the path from JM3 (or JM4 if two foci) to the filament.
- Case of constant occurrence only at both focus
Disconnection of both filaments (unlikely), disconnection and contact failure of the common line (MUX POWER-32K to X-ray tube), and XCONT/INVERTER for control in common with small and large focus remain possible.
- -- Detailed confirmation of disconnection and contact failure --
  - Unplug the high-voltage cable at the cathode side of the high-voltage transformer and measure the resistance between the pins to check for broken "filament" + "high-voltage cable". If a new cable is available, the measurement can be made by unplugging it at the X-ray tube side and connecting a new high-voltage cable.
  - Disconnect the high-voltage cable at the X-ray tube side and measure the resistance between the pins to check for disconnection on the high-voltage side of the "high-voltage cable" + "inside the high-voltage transformer."
  - By removing JM3 (JM4 if two foci) connected to the INVERTER and measuring the resistance between the pins, it is possible to check whether there is a disconnection on the low-pressure side of the high-voltage transformer.

IV. IV. If the cause cannot be identified above, the cause may be the XCONT, INVERTER, or cable between XCONT and INVERTER.

- If the IFV (CP7) of the XCONT at the time of the error is confirmed to be normal, it becomes a problem after the A/D conversion in the XCONT. -- > Try replacing it.
- If the cable connection is faulty, try replacing the cable.
- If there is nothing wrong with the cable connection, try replacing the INVERTER UNIT.

V. If it recurs despite the above, visually check the filament of the X-ray tube and replace the X-ray tube if damage is found.

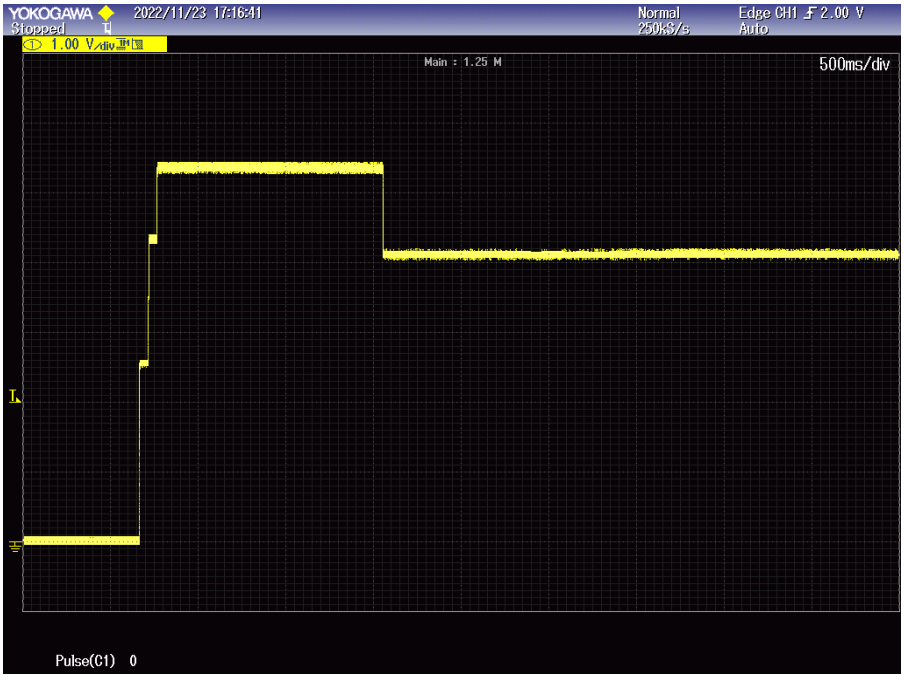
VI. If there is no abnormality in the X-ray tube, it is considered to be an abnormality in the high-pressure transformer (the filament transformer inside it).
- > Replace the high-pressure transformer.


Display F32 or 33
Check fuse F3 on MUX POWER board and replace if blown
solved it?
YES
Finish
NO
Fuse F3 breaks repeatedly
YES
NO
Error on one side of large or small focus?
YES
NO
Check for the following short circuit.
① Alternately remove JM2 and JCH to isolate which is shorted by INVETER and MUX CHARGE.
INVERTER UNIT
② FET of INVERTER UNIT
③ Path from JM3 (JM4) to high V transformer
MUX CHARGE-32KX
④ C-CL, C-CS
(filament itself or filament transformer malfunctions)
⑤ X-Y, X-Z (stator coil)
Check below
・Disconnection/contact failure of filament heating line (MUX POWER to X-ray tube)
(See Handling Instructions for High-Voltage Cables)
・IFV (CP7) of XCONT substrate
・INVERTER substrate
・XCONT to INVERTER cable
Check below
・Filament disconnection
・Disconnection/poor contact from JM3 (JM4) to filament
An abnormal spot was found.
YES
NO
Replace the affected parts
Finish
Heating waveforms between CO-CL1 and CO-CS1 of INVERTER UNIT are abnormal
YES
NO
Replace XCONT or INVETER UNIT
solved it?
Visual inspection of filaments in X-ray tube
filaments is abnormal
YES
NO
Replace transformer
Replace X-ray tube

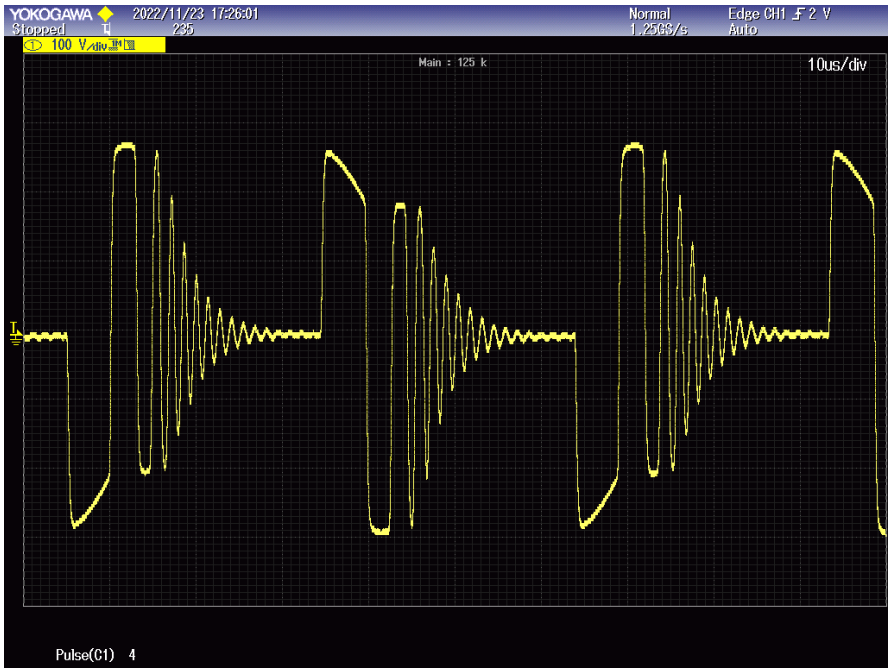
4. Trouble Shooting

Examples of IFV and C-CL waveforms at normal, high pressure disconnection, and low pressure disconnection are shown.
Note that when an error occurs, filament heating stops approximately 0.3 seconds after the Ready starts, so it is not easy to obtain the C-CL waveform.

Normal (FVR: about 4.00)



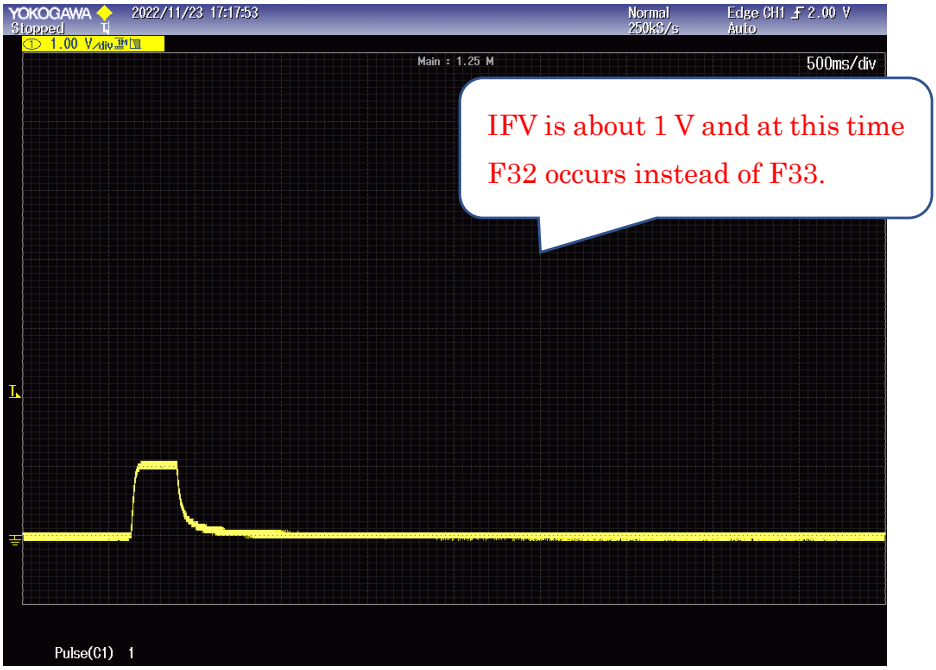
IFVwave （1 V/div、0.5sec/div）



C0-CLwave（50V/div、10usec/div）

4. Trouble Shooting

For high pressure side open (= filament break)

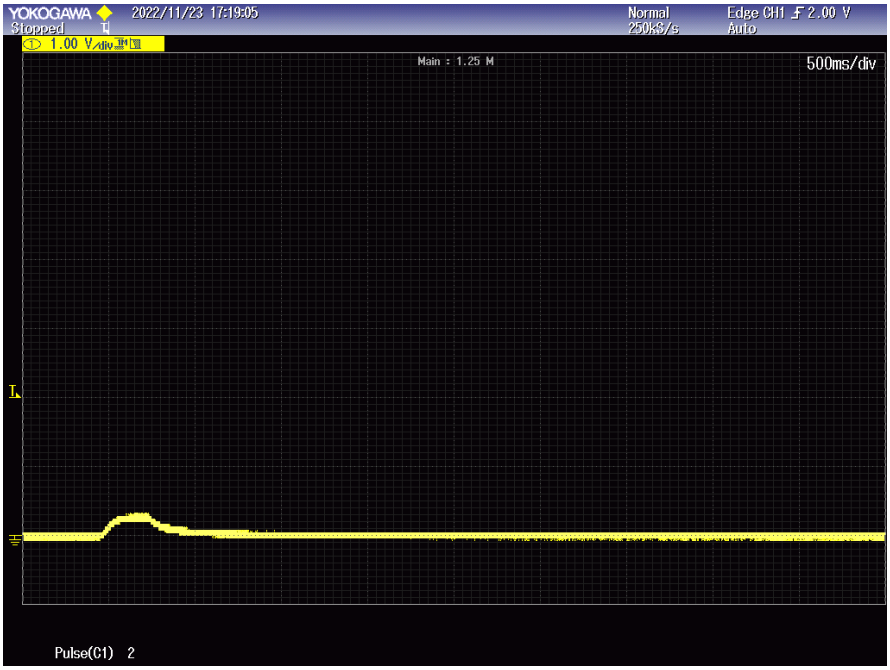


IFVwave　（1 V/div、0.5sec/div）

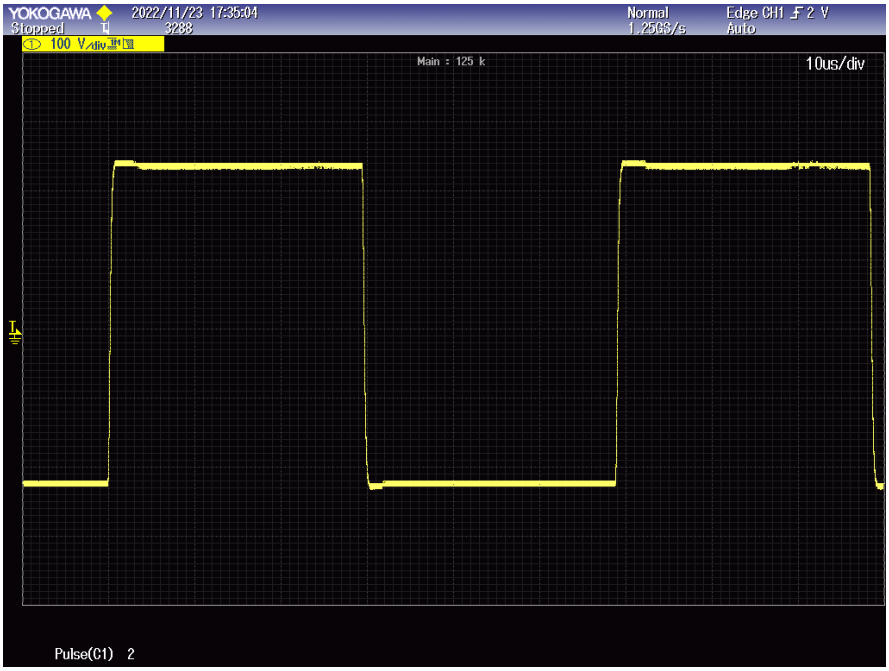
C0-CLwave（50V/div、10usec/div）

4. Trouble Shooting

Low pressure side open case



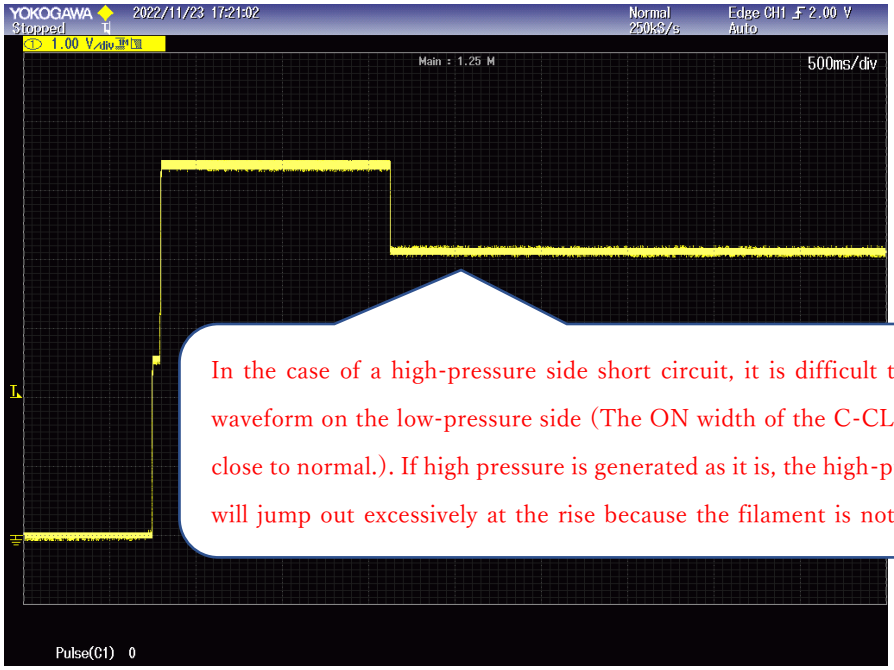
IFVwave（1 V/div、0.5sec/div）



C0-CLwave（50V/div、10usec/div）

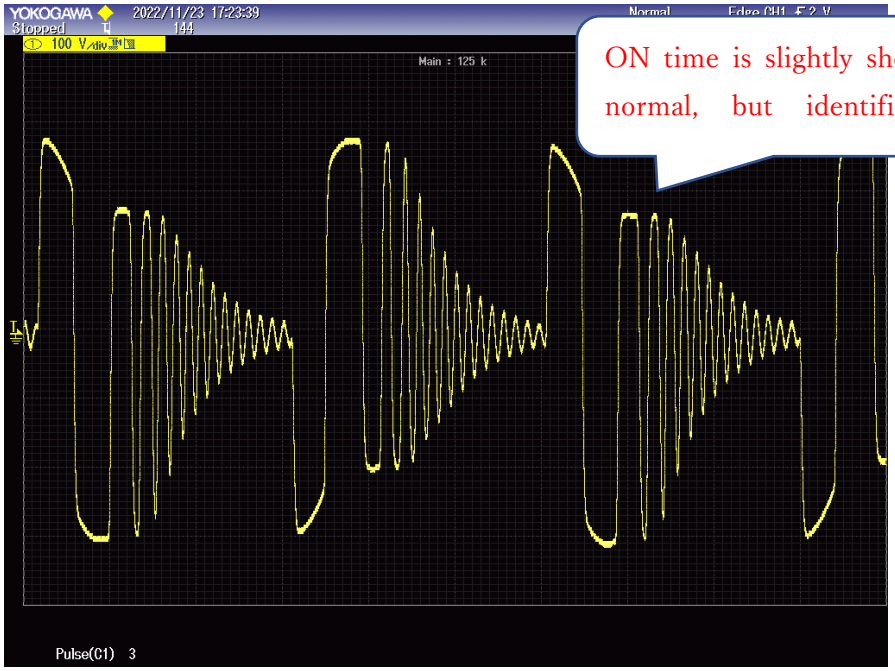
4. Trouble Shooting

< Reference > In case of high pressure side C-L short circuit



In the case of a high-pressure side short circuit, it is difficult to distinguish the waveform on the low-pressure side (The ON width of the C-CL waveform is also close to normal.). If high pressure is generated as it is, the high-pressure waveform will jump out excessively at the rise because the filament is not lit, TOV will be

IFVwave （1 V/div、0.5sec/div）



ON time is slightly shorter than normal, but identification is

C0-CLwave（50V/div、10usec/div）

4.2.21 F41, F42, F43, F44, F45, F46, F47

This is an error that detects a large capacitor charging malfunction.

F41: The charged voltage of capacitors exceeds 300V.

F42: The charged voltage of capacitors is less than 170V even 60 seconds after starting charging. (The charged voltage of the capacitors normally reaches about 170V about 30-60 seconds after charging starts. Then, the relay K2 turns on to change the connection of the charging resistance R3 and R4.)

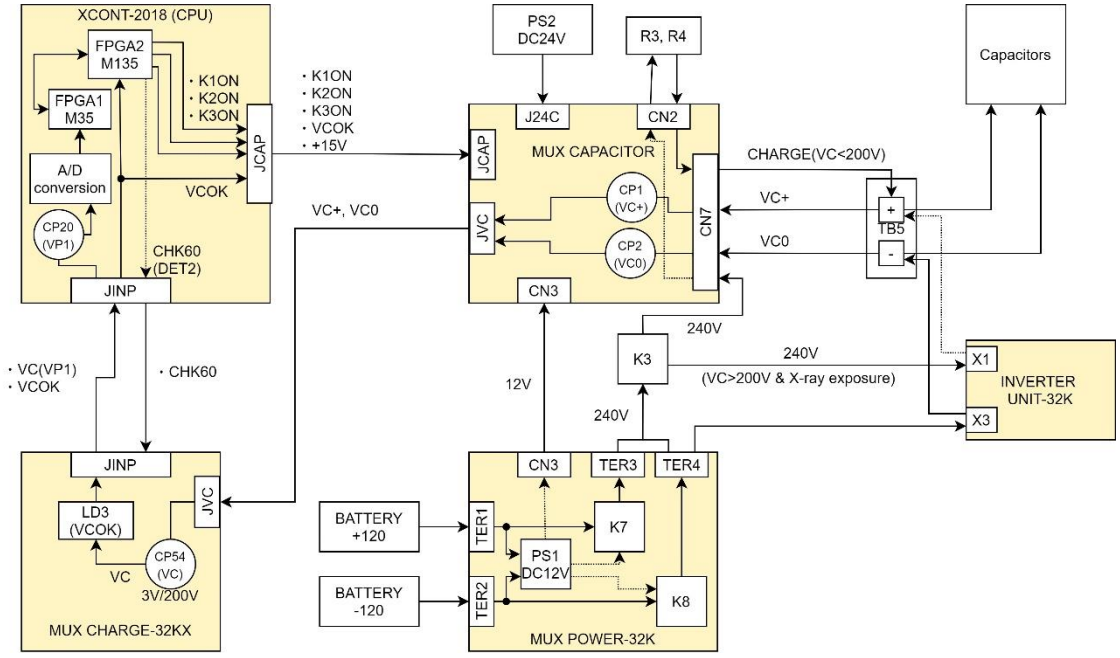
F43: The charged voltage of capacitors is less than 30V even 10 seconds after starting charging.

F44: The signal VCOK is not available though charging of capacitors is normal.

F45: The machine does not become Stand-by condition within 30 seconds after the relay K2 on MUX CAPACITOR PCB turns on. (The charged voltage of the capacitors normally reaches 220-240V about 10-30 seconds after the relay K2 turns on. Then, Stand-by indicator on the control panel lights on.)

F46: Relay K3 for charging capacitor is not turned on immediately before the start of exposure.

F47: The charged voltage of the capacitors becomes under Stand-by level for more than 30 seconds after the machine becomes Stand-by condition.



4. Trouble Shooting

Follow the instructions below and the flowchart on the next page to investigate.

I. The capacitor voltage can be measured with the CP below, so investigate where the fault occurs in the path.
   - Between CP1 (VC+) and CP2 (VC0) on MUX CAPACITOR
   - CP54 (VC)/CP72(VP1) on MUX CHARGE-32 KX : 3V/200V
   - CP 20 (VP1) on XCONT : 3V/200V

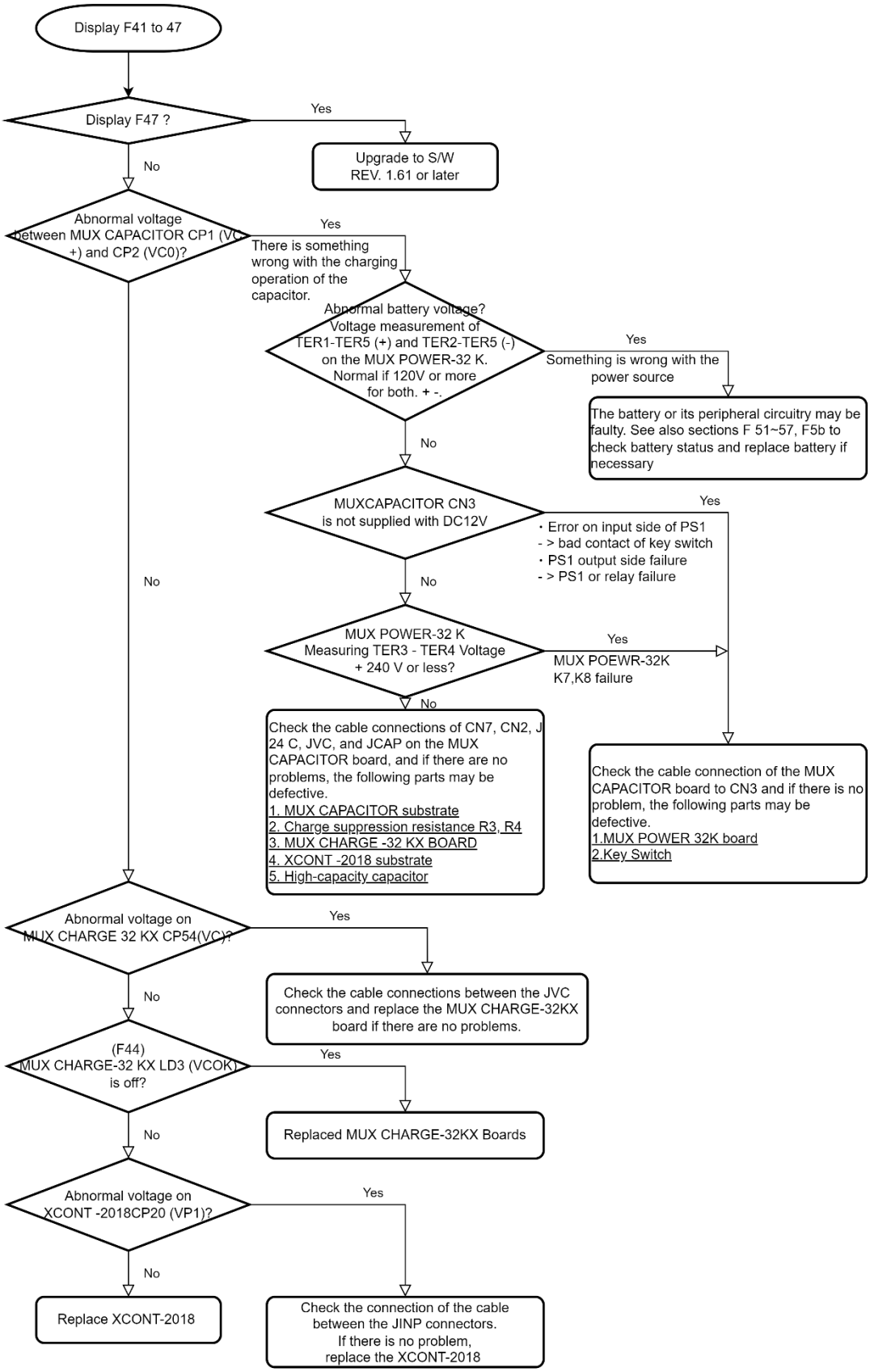
   The voltage at CP72 on MUX CHARGE-32KX and CP20 on XCONT alternates with the battery + side voltage (:3V/100V) every few seconds, but the waveform is flat when charging from the battery to the capacitor is performed correctly.

II. The VCOK signal related to the F 44 is generated by detecting that the voltage of the CP 54 (VC: 3 V/200 V) has exceeded about 200 V on the MUX CHARGE -32 KX (LD 3 is lit) and is sent to the XCONT via the connector JINP. The fault is identified from the actual capacitor voltage, the measurement of the CP 54 voltage, the lighting status of the LD 3, etc.

III. If the actual capacitor voltage is abnormal, measure the battery voltage, check the connection of connector CN2 (Charge suppression resistance R3, R4) on the MUX CAPACITOR, operate each relay on the MUX CAPACITOR (see "Capacitor Charging Circuit" in Chapter 3 of this service manual), and check the wiring around the capacitor and the MUX CAPACITOR.

IV. There was a case where DC 12V did not come to connector CN3 of MUX POWER to MUX CAPACITOR due to a contact failure of one of the contacts of the key switch and the capacitor charging circuit did not operate at all. In this case, the power relays K7 and K8 on the MUXPOWER are also not turned on, so the battery voltage is not supplied to the inverter or the running motor.

4. Trouble Shooting



4.2.22 F51, F52, F53, F54, F55, F56, F57, F58, F5b

Error detecting abnormal battery charge voltage. Software detection except for F5b

F51: Battery + side voltage during charging is 155 V or more.

F52: Battery + side voltage during charging is 50 V or less.

F53: The battery-side voltage during charging is 155 V or more.

F54: The battery-side voltage during charging is less than or equal to 50 V.

F55: The + side voltage during battery operation is more than 12 V larger than the - side.

F56: The -side voltage during battery operation is more than 12 V higher than the + side.

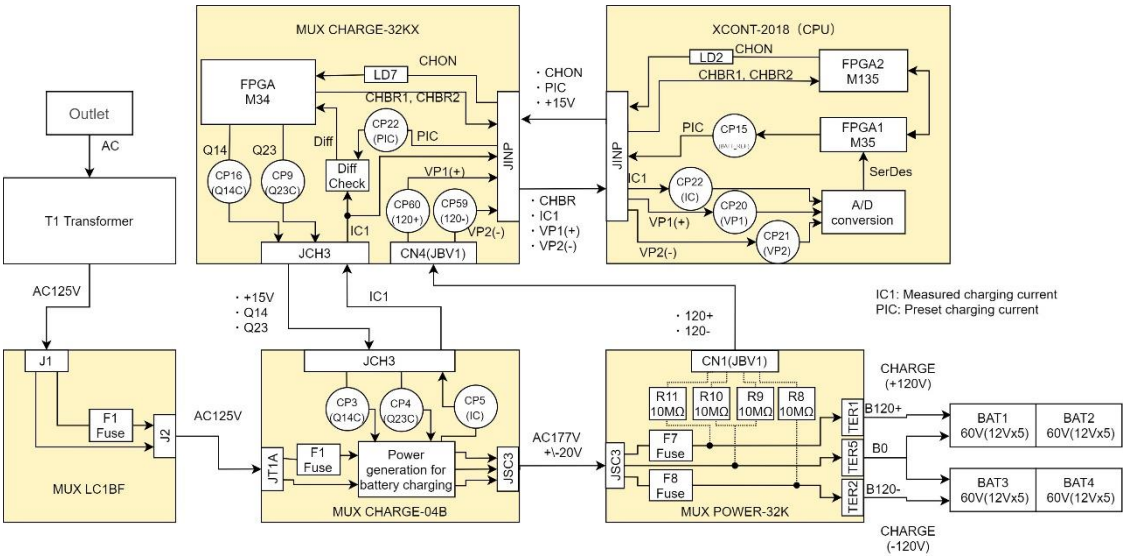
Battery running: Running, shooting, DR device running medium

F57: The + side voltage during battery charging is more than 12 V higher than the - side.

F58: The -side voltage during battery charging is more than 12 V higher than the + side.

F5b: Battery voltage on the + or - side is 155 V or more.

(Hardware detection. CHBR2 signal occurs)



Follow the instructions below and the flowchart on the next page to investigate.

I. Make sure the main breaker is turned on. Also, make sure that the appropriate battery code is set according to the type of battery as described below.
   In case of PXL12090F2V0/Z -> Battery cord 6
   In case of PXL12072H FR -> Battery cord 5

II. The charged voltage values on +/- both sides of the battery are measured between terminals TER1 to TER5 and TER2 to TER5 of the MUX POWER-32 K, respectively. Note, however, that the difference between +/- sides (unbalance) may increase during charging and radiography, even if there is only an unbalance of about 1 V in the standby condition without any operation such as radiography, driving on the floor or charging.

III. If the voltage is abnormal at this terminal (+/- 155 V or less than or equal to 50 V on one side, respectively. Or the voltage difference between the + side and the - side is 12 V or more.), the battery itself or its peripheral circuit may be faulty because the battery voltage is actually abnormal. It is also advisable to check for poor contact around the battery, the presence or absence of the AC125V output of the T1 transformer, which is the power supply for charging the battery, and the actual charging current waveform (see "Checking charging operation" in the installation instructions).
If this charging current waveform is abnormal, it is difficult to determine whether it is due to the battery itself or to the MUX CHARGE-32 KX/-04 B that drives and controls the charging, and you have to replace these in order.

IV. When the battery is degraded, even if there is no unbalance in the standby condition, there is a case in which the voltage rises rapidly on the side of +/- whose capacity is degraded after charging starts, and conversely, when the device is used during running on the floor or radiography, the voltage drops rapidly and the unbalance expands. (In this case, if the degradation proceeds further, it is considered that an unbalance occurs even in the standby condition.)
In particular, if a large number of bbb, bb0, bb1 etc. remain in the log, it is highly likely that the battery has deteriorated. (See paragraph (4) of 4.7)

<Note>
When "Check charging operation" in the installation instructions is performed with the battery fully charged, F 51, F 53, F 5 b, etc. may occur.
In the adjustment mode, the battery is forcibly charged for one minute at a constant value without switching the charging current value, so the charging voltage is likely to be over.

Due to a contact failure of the power relays K7 and K8 (which operate at DC12V given when the key switch is on) on the MUX POWER-32K, even when the key switch is off, a half-way voltage was supplied from the battery to the control circuit, causing an abnormal noise from inside the device, or an imbalance or over discharge of battery voltage.

V. If the voltage is normal between terminals TER1-TER5 and TER2-TER5, check the battery voltage detection adjustment (see "Battery voltage detection adjustment" in the installation instructions). If this adjustment is not performed correctly, the CPU (XCONT) will misrecognize the battery charged voltage.
If the problem recurs after performing the above, check the voltage at CP 60 (120 +) and CP 59 (120 -) of the MUX CHARGE -32 KX. If the voltage is abnormal here, it is possible that the MUX CHARGE -32 KX or MUX POWER-32 K (Are the 10 megohm battery voltage detection resistors R8, R9, R 10 and R 11 normal?) is faulty, or that the contact is faulty at its connector.
If the voltage at above CP60(120+)/CP59(CP120-) are normal, but the voltage at CP72(VP1)/CP63(VP2) on MUX CHARGE-32KX are abnormal, it means that the circuit between CP60-CP72 or CP59-CP63 on MUX CHARGE-32KX is faulty.
The voltage at CP72 alternates with the capacitor voltage every few seconds, but the waveform is flat when charging from the battery to the capacitor is performed correctly.

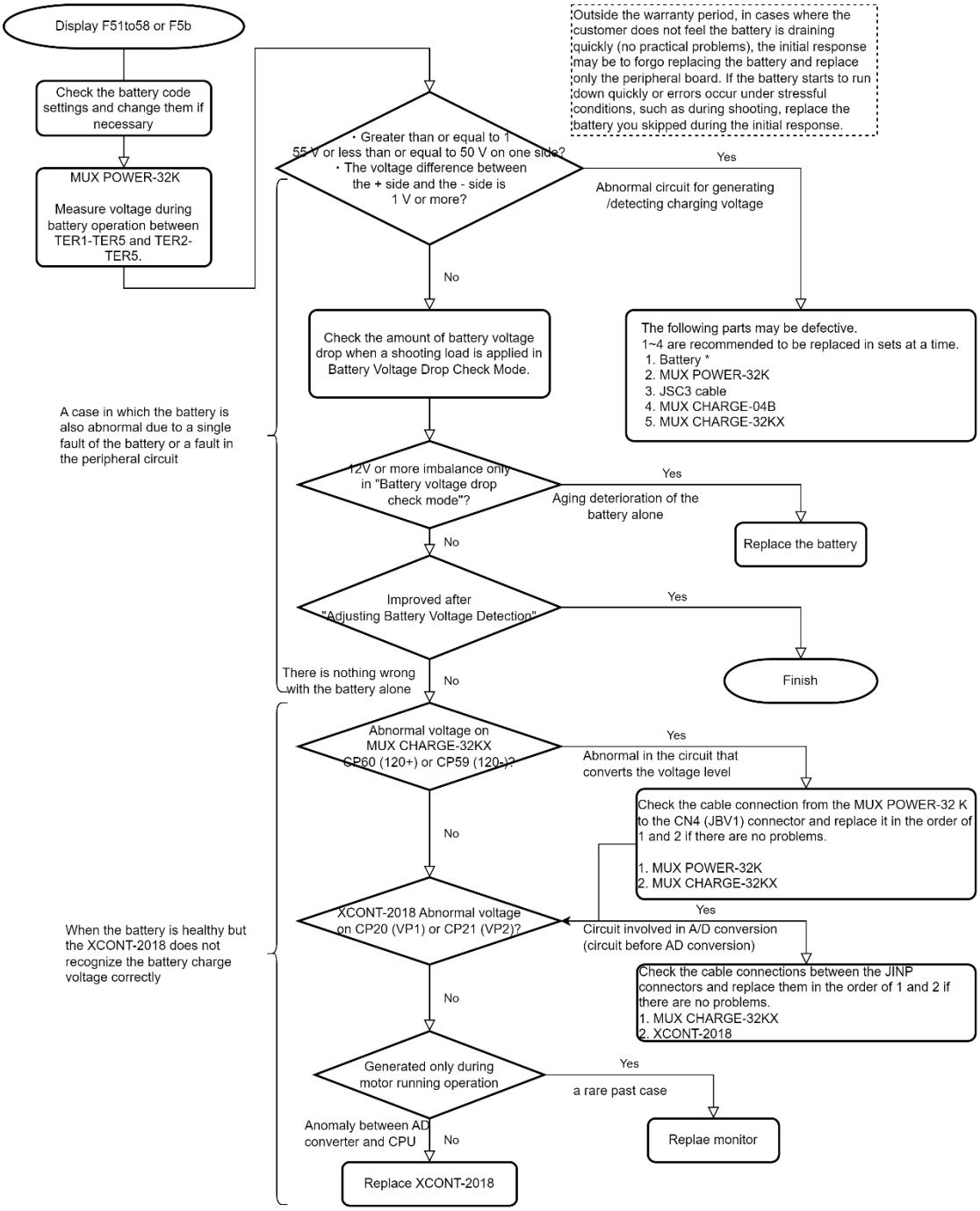
VI. If the voltage on the MUX CHARGE-32 KX is also normal, the XCONT is likely faulty. We measured the voltage at CP 20 (VP1) and CP 21 (VP2) of the XCONT, and if it is normal here, the circuit up to the MUX CHARGE-32 KX is definitely normal.
->Replace the XCONT.

However, there was a special case in which the 5V, 15VDC power supply (PS1) output did not rise up normally at the time of device startup, which caused malfunction in the error detection circuit, causing error F5b to be erroneously detected at startup. If a battery-related error occurs even though the voltages on CP20 (VP1) and CP21 (VP2) are normal, use an oscilloscope to check the rise waveform of the output voltages of the 5V and 15VDC power supplies, and check that both 5V and 15VDC power supply voltage rise to the specified voltage instantly.
Note that the voltage at XCONT CP20 alternates with the capacitor voltage every few seconds as same as CP72(VP1) on MUX CHARGE-32KX, but the waveform is flat when charging from the battery to the capacitor is performed correctly.

VII. In some rare cases, charging related errors F 51, F 52 to F 5 b occurred the moment the travel brake was released due to a motor failure. In that case, you can determine which motor has the fault by disconnecting the power cables (CN 14, C 16) between the left and right motors and the MU DRIVER one by one and testing the cables by connecting them in reverse.

VIII. If any of the three wires of the JSC3 cable between the MUX CHARGE-04B board and the MUX POWER-32K board have contact failure, + or - side may not be charged properly, causing unbalance. In such a case, replacement of the JSC3 cable only sometimes can solve unbalance. However, replacement of the set of 3 parts including both boards together with JSC3 cable is sometimes necessary in order to solve unbalance completely.

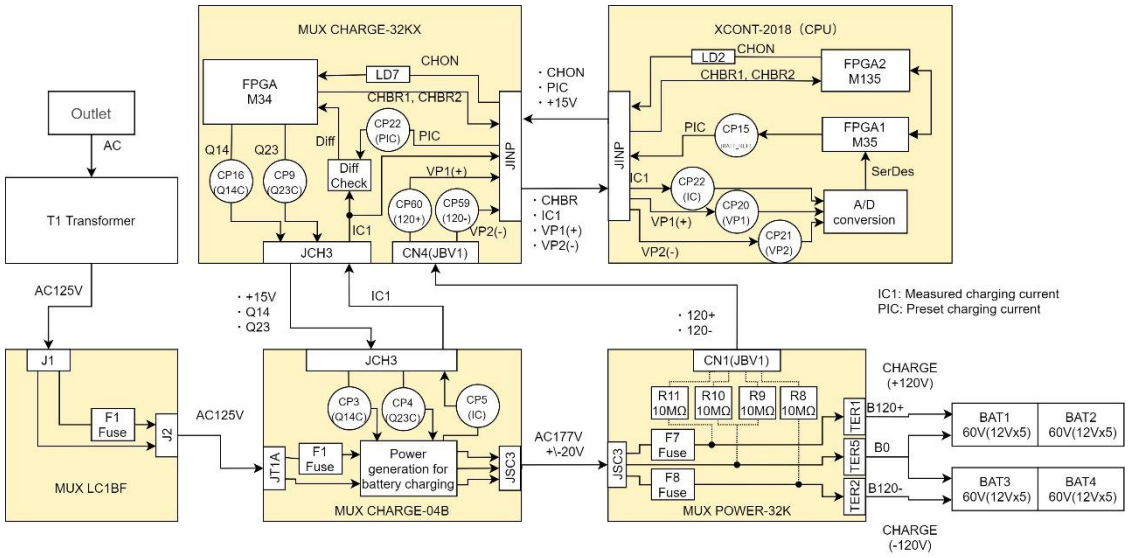
4. Trouble Shooting



4.2.23 F5a, F5E, F5F

Error detecting abnormal battery charging current. F5a is detected by hardware. F5E, F5F are detected by software.

- F5a :Battery charging current is 3.8A or more. CHBR1 signal is generated.
- F5E : During charging, the measured charge current value was more than 50% higher than the set value for 10 seconds.
- F5F : During charging, the measured charge current was less than 30% of the set value for 10 seconds.



The measured charging current value can be measured at CP22 (IC) on XCONT-2018 board, and the setting value can be measured at CP22 (PIC) on MUX CHARGE-32KX board in 1V/1A relationship. For example, if both are approximately 0.5V during pre-charging of which charging current value is 0.5A, we can judge that the charging current value is controlled correctly. If the measured charging current value is abnormal, check the following.

Check the battery charging function according to "Checking charging operation" in the installation instructions.
The actual charging current waveform can be checked with the check pin below.
MUX CHARGE-04B CP 5 (IC) --> XCONT CP 22 (IC)

Note:
Note that F51, F53, F5b, etc. may occur when "Check charging operation" in the installation instructions is performed with the battery fully charged. (In the adjustment mode, the battery is forcibly charged for one minute at a constant value without switching the charging current value, which can easily result in overcharging voltage.)

If either of the check pins above shows a normal charging waveform, the actual charging operation itself is considered normal. In particular, if normal charging waveforms can be seen on all the check pins above, there is a possibility that the detection circuit on the XCONT is faulty.
If the waveform of the MUX CHARGE-04B at CP5 (IC) is abnormal, there is a high possibility that the charging operation is actually abnormal, so check the following before the error occurred.

I. AC 125V input from T1 transformer to MUX CHARGE-04B including checking of F1 fuse on MUX-LC1BF.
   By the way, the rating of this fuse has been changed from 4 A to 6.25 A since the end of 2023. If this fuse is blown, it can be replaced with 6.25 A regardless of the shipping time.
II. Fuse F1 on MUX CHARGE-04B
III. Gate waveform of inverter for battery charging at CP3 (Q14C), CP4 (Q23C) on MUX CHARGE-04B.
IV. Gate waveform of inverter for battery charging at CP16 (Q14C), CP9 (Q23C) on MUX CHARGE -32 KX
V. CHON (charge on signal) on MUX CHARGE-32 KX: Green LED LD 7 lights up while charging.
VI. CP 22 (PIC) on MUX CHARGE -32 KX: Set charge current value sent from CPU

If the above check does not identify the defective part, replace the following parts in order.
- Batteries (replace all 20 at the same time)
- MUX CHARGE -32 KX
- MUX CHARGE-04B

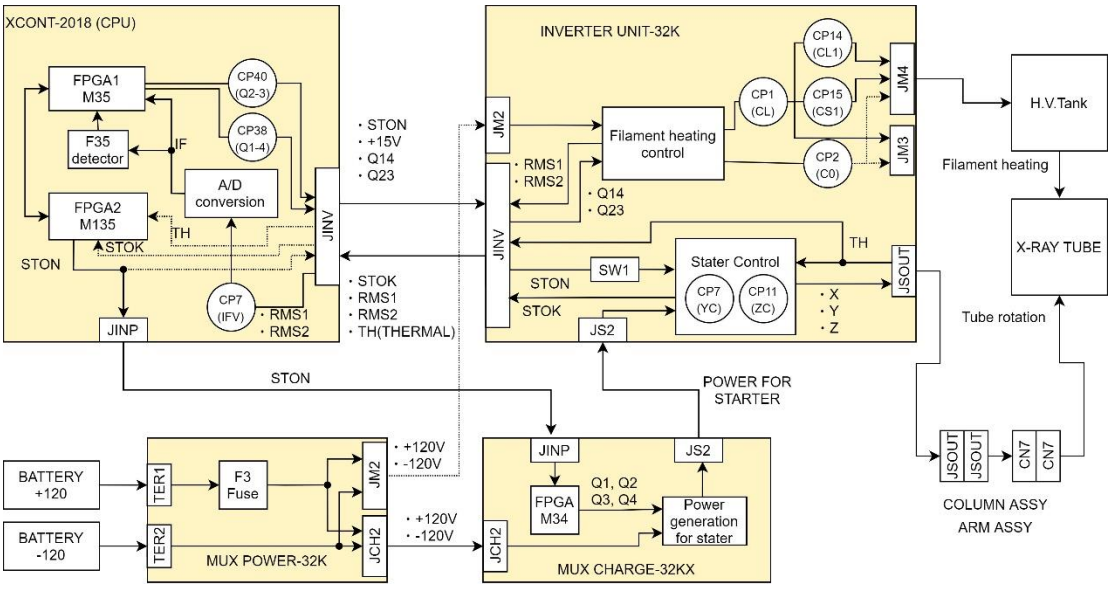
4.2.24 F5c

This error message means "The power plug is connected when the main breaker is off.". Detected when the voltage on either side of the battery is less than or equal to 50 V, preventing battery charging from starting.

4.2.25 F61

Starter failure (no current detection confirmation signal STOK from starter)

No current flows through the stator coils and the rotor does not rotate.



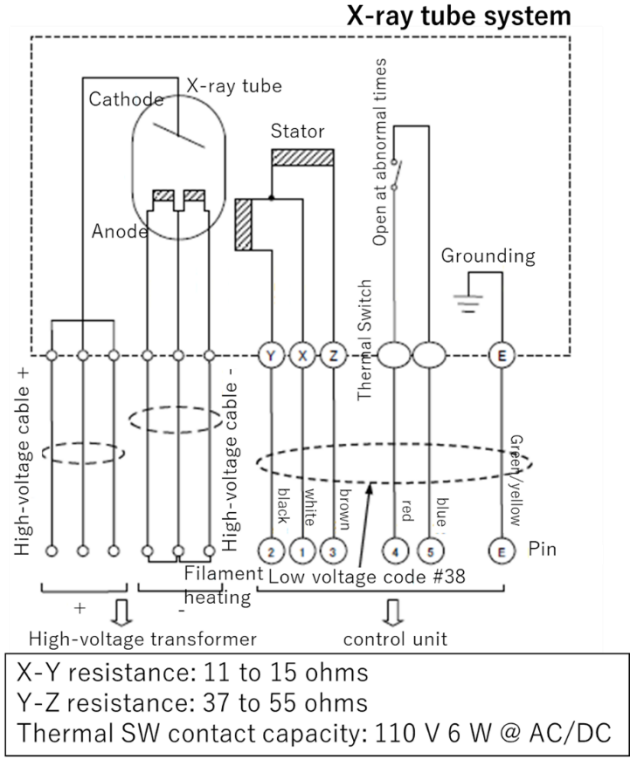
I. Make sure that the INVERTER UNIT Dip switch SW1 -1 is set to OFF ("MUX" side). (However, the SW1 is not available in some cases depending on system model and manufacturing date.)

II. Check that fuse F3 on the MUX POWER-32K is not blown, and if so, replace it.
If F3 blows repeatedly, refer to paragraph 2 of "4.2.20 F32,F33 (IF UNDER)" to investigate the cause.

III. If the fuse F3 is not blown, measure the current values of the X-Y and X-Z coils with CP7 (YC) and CP11 (ZC) of the INVERTER UNIT, respectively. (A voltage of 1V or more is normal and produces a STOK signal on the INVERTER UNIT.) If this is normal, replace the XCONT.

IV. If the current values of the normal X-Y and X-Z coils are not measured on the INVERTER UNIT, measure the resistance values of the X-Y and X-Z coils of the X-ray tube. The standard resistance value is the same, although the tube bulb varies depending on the rating (32 kW, 12.5 kW).

Measuring point	PIN	Standard Resistance
X-Y	1 - 2	11 - 15
X-Z	1 - 3	37 - 55



V. If the resistance value of the X-Y and X-Z coils of the X-ray tube is abnormal, replace the X-ray tube.

If the resistance value of the coil is normal and there is no contact failure of the low voltage cable (X, Y and Z cables), it is possible that the INVERTER UNIT or MUX CHARGE -32 KX has failed, so try replacing it.

4.2.26 F62

After entering HX1, XOK does not return even after 1 second. Usually this message doesn't come up. Check if the DIPSW setting on the XCONT is wrong.

4.2.27 F64

HX1 signal input was present when the power was turned on. (The 2nd step of the hand switch was pressed.)
If the operator did not press the hand switch by mistake, it is possible that the hand switch is defective or the XCONT board is defective.

4.2.28 F65

AR1 signal input was present when the power was turned on. (The 1st step of the hand switch was pressed.)
If the operator did not press the hand switch by mistake, it is possible that the hand switch is defective or the XCONT board is defective.

4.2.29 F68

The radiography operation was performed before the ready preparation of the FPD was completed. If the operator did not press the hand switch by mistake, it is possible that the hand switch is defective or the XCONT board is defective.
If the above F64 or F65 also remains in the log, it is possible that the wrong input from the hand switch also caused F68.

4.2.30 F6d

EMG (emergency stop switch) signals are being input. (Detected on MU DRIVER)
Check the emergency stop switch circuit. However, note that F6 d can also be caused by a malfunction of the 24 V DC power supply used in the circuit of the emergency stop switch. For example, F6 d can occur when the fixing screw of the brake part for pivoting on the support is loosened, the cable of the magnet is broken, and the DC 24 V drops due to ground fault.
Also at this time, due to the malfunction of the circuit, which also uses 24 VDC, H4 (meaning "thermal relay has operated") may be displayed or the arm-related brake may not be released.
A similar phenomenon occurs when the CPU fails to recognize "key switch ON" due to a defective contact of the key switch, the MGRS signal is not generated, the charge suppression resistance is not shorted because relays K7 and K8 of the MUX POWER-32 K are not turned on, and the DC 24 V power supply drops during brake release operation.

4.2.31 F6E

Exposure does not end beyond backup time/mAs.
This is considered to be an X CONT malfunction (X RAY CUT OFF CIRCUIT malfunction).
However, since measures to prevent false detection of F6E have been implemented in REV. 1.53 of the XCONT -2018, make sure that it is REV. 1.53 or later.

4.2.32 F6F

Abnormal exposure control circuit
The kV rise signal KVT did not return 250ms after EXR was output.
There is a high probability of a misconfiguration or an XCONT malfunction.
In some cases, failure of high-voltage cables was the cause. (The meaning of the error is similar to F 12, F 13, F 15, F 16, etc., so please refer to that as well.)

4.2.33 F76

Transmission data error from the DR device (APR number out of range, unknown model code or operative code sent, etc.) Correct the shooting protocol in the DR console.
This error also occurs when conditions are sent from the DR even though the camera is in the shooting mode, so make sure that no hand switch operation is performed when sending conditions from the DR. It could be a faulty hand switch, hand switch I/F board, wireless hand switch and its cables, or it could be a faulty XCONT if it still happens when you remove them.

4.2.34 F81

The combination of the rated output stored in the EEPROM and the Dip SW 2 -2 settings on the CPU (XCONT) is incorrect.

I. Check the CPU for setting errors in Dip SW 2 -2
II. After checking the settings of Dip SW2 -1, SW2 -2, and SW2 -6, initialize XCONT.
III. Hardware failure in the Dip SW itself
IV. Make sure the correct software is installed.

4.2.35 F85

The combination of the high voltage transformer settings stored in the EEPROM and the Dip SW 2 -6 settings on the CPU board (XCONT) is incorrect.

I. Check the CPU board for setting errors in Dip SW 2 -6
II. After checking the settings of Dip SW2 -1, SW2 -2, and SW2 -6, initialize XCONT.
III. Hardware failure in the Dip SW itself
IV. Make sure the correct software is installed.

4.2.36 F86

The combination of the DaRt/Art settings stored in the EEPROM and the Dip SW2 -1 settings on the CPU board (XCONT) is incorrect.

I. Check the CPU board for setting errors in Dip SW 2 -1
II. After checking the settings of Dip SW2 -1, SW2 -2, and SW2 -6, initialize XCONT.
III. Hardware failure in the Dip SW itself
IV. Make sure the correct software is installed.

4.2.37 A2

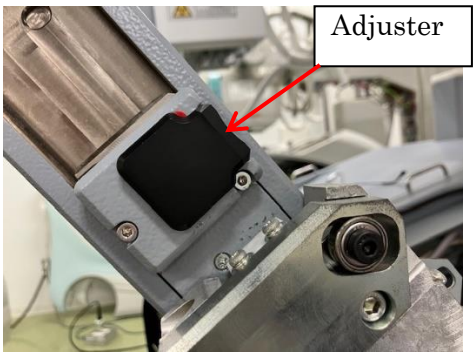
This error is basically an error caused by user error.

A device with the option to change the height of the handle, with the handle in an intermediate position, not at the top or bottom.

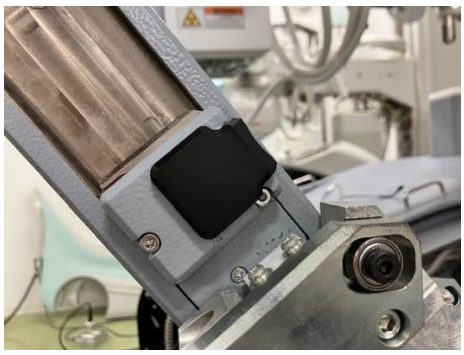
->If this error is detected, the brake is not released when the brake release lever is operated during a stop. Also, when this error is detected while driving, the brakes are applied.

Make sure the handle position is firmly fixed to either the top or bottom edge.

In the picture below, the red markings in the upper left of the black adjuster indicate that the handle height is not in the proper position, or the adjuster is caught mechanically.



Abnormal



Normal

Also, check that the micro switch (Left and right, 2 in total) for detecting the position of the handle is connected and operating normally.

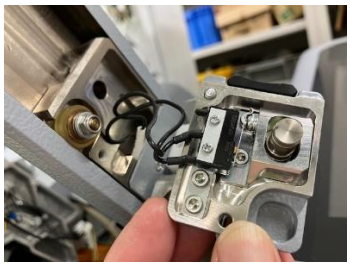
Remove the two screws and turn the adjuster over to see the micro switch. Check that the left and right switches are operating correctly as described below. If necessary, adjust fixing position or replace the switch.

・When normal, i.e., the position of the handle is firmly fixed to either the top or the bottom.

→ The micro switch is ON.

・When abnormal, i.e., the position of the handle is halfway between the top and the bottom, or the adjuster is pressed by hand to change the height of the handle.

→ The micro switch is OFF.



Switch is OFF
when abnormal.

Switch is ON
When normal.

4.2.38 A4~A11

These errors are basically errors caused by user error.
At power-on, input from each operating switch was detected.
Check each of the following operating switches and their cable connections, and if there is no abnormality, replace the MU DRIVER.
However, if it happens every time the A7 or A8 (related to the left and right FSRs) is turned on, perform the handle adjustment first according to the installation instructions, replace the FSR if it does not improve, and if it still fails, replace the MU DRIVER.
Also, if these errors appear randomly when the power is turned on, it could be an XCONT failure.

- A4: Brake release switch
- A5: Microtremor switch (forward)
- A6: Microtremor switch (backward)
- A7: Driving steering wheel input (input from left FSR)
- A8: Driving steering wheel input (input from right FSR)
- A9: Obstacle detector
- A 10: Arm unlock switch
- A 11: Arm unlock lever

4.2.39 d15

Contact welding of relay K 11 (relay that turns OFF when EMG SW is pressed) in motor drive section is detected.
If LD 4 of the MU DRIVER 07 is lit while EMG SW is pressed, it indicates that the contact of K 11 is welded.

4.2.40 d16, d17

Short circuit of left FSR (d17) or right FSR (d16) was detected. (The left and right meanings are assigned to the even and odd error codes so that "Odd is left, even is right".)
When the AD value of the FSR becomes close to 0, which is usually an improbable value, the FSR is judged to be damaged and short-circuited with the ground and D16 or D17 is displayed, and the device is gently stopped to prevent running malfunction. The FSR should be replaced if this error occurs.

4.2.41 d20, d21

An abnormality (open or short) in the FET of the right motor drive circuit (d20) or the left motor drive circuit (d21) on the MU DRIVER 07 was detected.
Basically, it is considered necessary to replace the MU DRIVER 07, but other than that, check the following items.

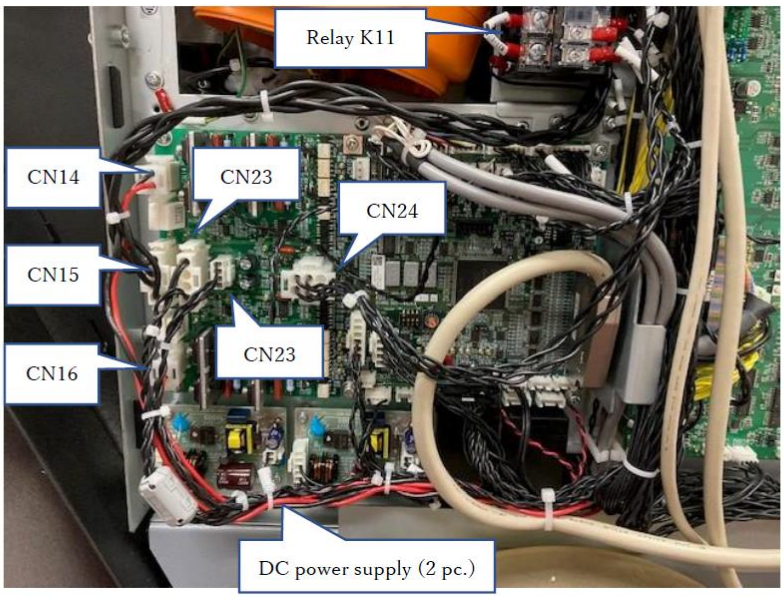
I. This error also occurs when the supply of 240 V power from the battery to the MU DRIVER 07 has stopped, for example, when the fuse F4 or F5 on the MUX POWER 32 K has blown out or the battery has over discharged.
   The state of 240 V power supply from the battery can be checked by the following check pin on the MU DRIVER-07 board.
   F4: Fuse of +120 V power supply (right motor drive power supply)
   → Can be checked with CP17 (+120 V).
   F5: Fuse of -120 V power supply (left motor drive power supply)
   → Can be checked with CP12 (-120 V).
   It should be noted that the error was not solved even after replacing the fuses F4 and F5 and finally replacing the MU DRIVER-07 board solved the error in several cases. In that case, for preventing recurrence of the error in future, please perform revision up of XCONT software to REV. 1.85 or later as well as "**Procedure for improving the wiring route around the MU DRIVER-07 board**" which is explained on the next page.
II. Note that this error may also occur if the EMG SW is released without turning off the key SW. Check the error log and if F6d was recorded immediately before this error, it is most likely due to the miss operation described above.
III. Poor contact between XCONT CNUM to MU DRIVER-07 CN 21. - In this case, error d 20 or d 21 is generally considered to occur intermittently.
IV. DC 15 V power supply for control MU DRIVER 07 board or relay K 11 (See also d 15.) for driving the motors is defective.
   The normal operation of this relay K11 can be confirmed by the following check pin on the MU DRIVER-07 board.
   CP21 (P120): +120 V power supply (power supply for driving the right motor)
   CP15 (M120): -120 V power supply (power supply for driving the left motor)

Procedure for improving the wiring route around the MU DRIVER-07 board

1. Outline

Since the propagation of noise between the wires around the MU DRIVER-07 board may affect the stable operation, the procedure for changing the wiring route to reduce noise propagation is described here.

Although there are some differences depending on the time of manufacture of the equipment, the conditions of wiring route and fixing are as follows.



Changes (1)

The cables from the motor to the MUDRIVER board CN14/CN16 and the input/output cables of the small DC power supply under the MU DRIVER-07 board are held together by the wiring bands. Separate these cables without being held together. As well, separate the input line of the DC power supply from the output line.

Changes (2)

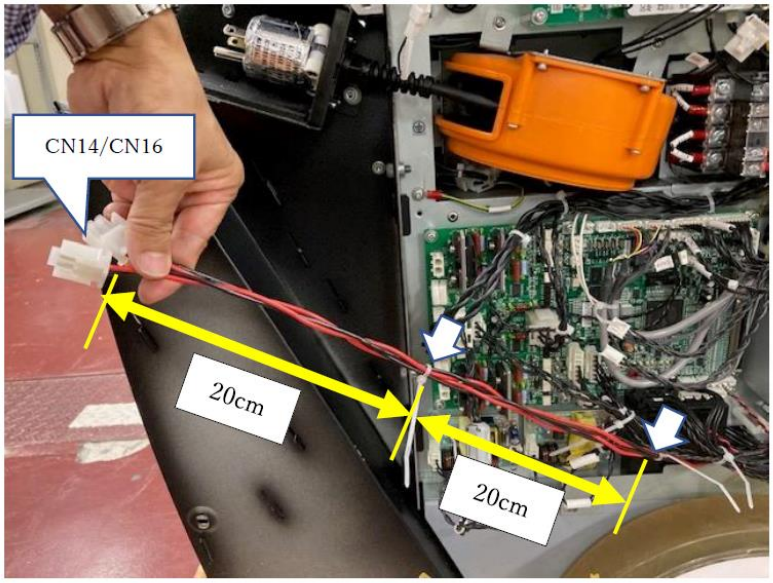
Fix the cable from the relay K11 to the MUDRIVER board CN24 by the wiring band together with the cable to the CN15.

2.Detailed procedure

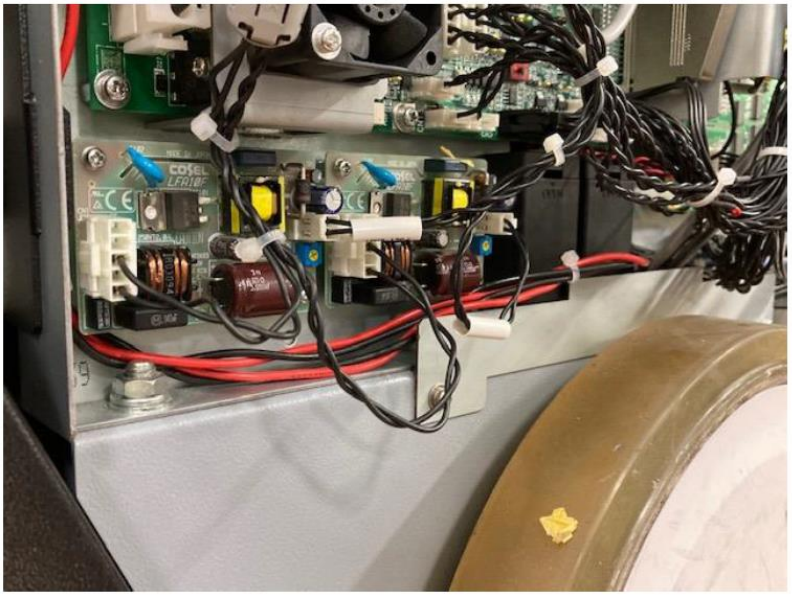
< Change 1>

(1)Remove the wiring bands which are fixing the cables (red-black line) from the motor to the CN14/CN16 and the input/output cables of the small DC power supply under the MU DRIVER-07 board.

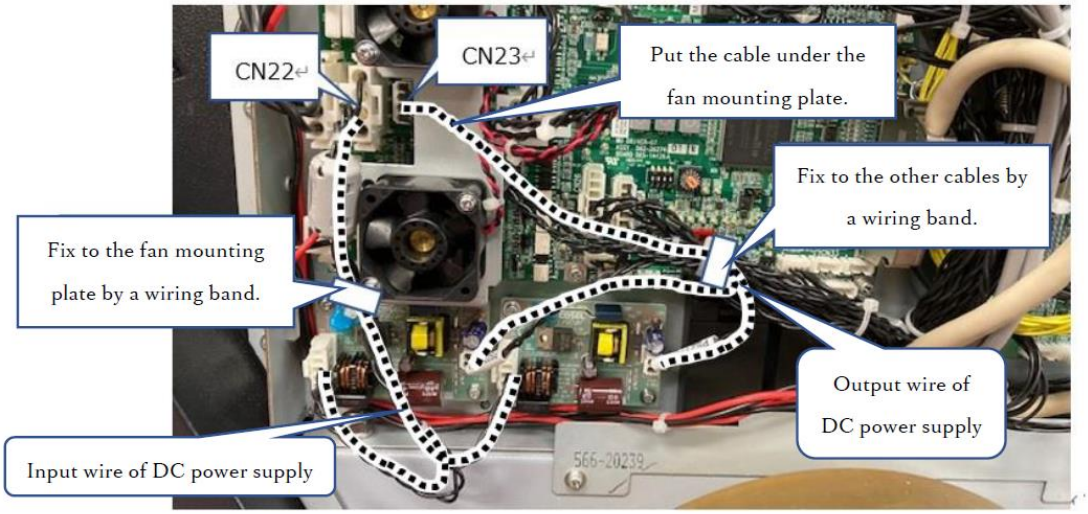
(2)Tie the CN14 and CN16 cables with wiring bands at the locations indicated by the arrows in the picture below (2 locations).



(3)Reconnect the CN14 and CN16 cables (red-black line) to the MU DRIVER-07 board along the chassis as shown in the photo below. (Do not fix the CN14 and CN16 cables together with the input and output lines of the DC power supply.)

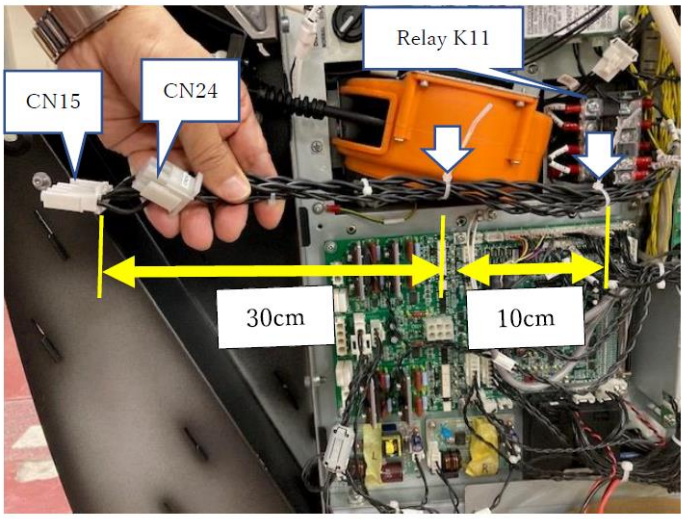


(4)The input and output wires of the DC power supply should be separated from each other. The wiring route is as shown in the photo below.
In the case of the device with a fan, fix the input wire to the fan mounting plate by a wiring band to prevent winding into the fan.

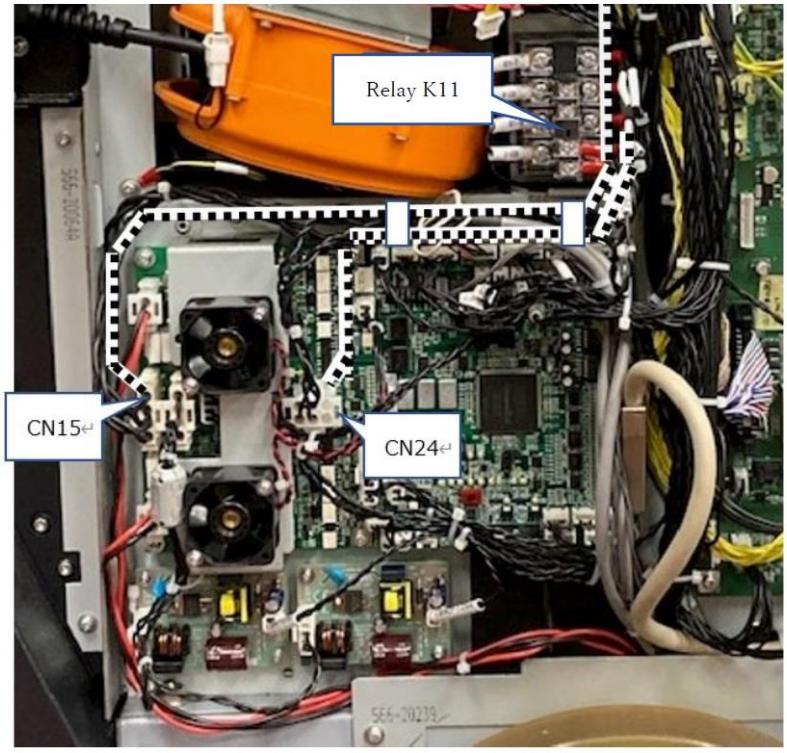


< Changes 2>

(1)Remove two cables from the relay K11 to the MU DRIVER-07 board CN15/CN24 from the MU DRIVER-07 board once, and tie only these two cables with wiring bands at the positions (2 locations) indicated by the arrows below.



(2)Reconnect the cables to CN15/CN24. The below picture shows the wiring route.



4.2.42 d22, d23

A short in the FET for driving the right motor (d22) or the left motor (d23) was detected.

4.2.43 d24, d25

Abnormal rotation of right (d 24) or left (d 25) motor
Feedback signal input from the encoder of the running motor was detected even though there was no input from the left or right running steering wheel (FSR). Or a feedback signal input in the opposite direction to the input from the driving steering wheel was detected.
Check the cable connections between the encoders and the connectors CN6 (left motor) and CN7 (right motor) on the MU DRIVER, including grounding the GND cable.

4.2.44 d26, d27

Overcurrent in the right (d 26) or left (d 27) motor was detected.
(For example, it is detected when a current value of 6 A or more is detected for more than 10 seconds during low-speed running (0.8 km/h or less) and when a current value of 8 A or more is detected for more than 10 seconds during high-speed running.)

It is highly likely that either the left or right running motor is faulty.
The current value of the motor can be measured with CP 5 (left side running motor) and CP 19 (right side running motor) on the MU DRIVER-07 with CP 4 and CP 18 as GND, respectively.

Note:
The CP18 is "120G" (the midpoint of the battery), while the CP4 is "M120" (-120 V), and there is a potential difference of about 120 V between them, so the current values of the left and right motors cannot be measured simultaneously. Be sure to measure each one separately with an oscilloscope.

When the running steering wheel is in the neutral position (non-operating), no voltage is generated on either check pin (0V). The correlation between the current value flowing through the actual motor and the amount of voltage displacement on the check pin is 0.2V/1 A, so if the check pin voltage exceeds approximately 1.2V (at low speed), error D26 or D27 will occur. (To be precise, error D26 or D27 occurs when the voltage of the check pin above is integrated into the frequency converted by the V/F converter to reach a value equivalent to 6A (at low speed) or 8A (at high speed) flowing for 10 seconds.)
Voltage measurements at both check pins during running may determine which motor is defective, left or right.

If the voltages of both check pins are in the normal range, there is a possibility of an overcurrent detection circuit failure on the MU DRIVER or XCONT, but even in this case, the motor is not necessarily normal.
It should be noted that a defective MU DRIVER may cause the motor to drive unsteadily, resulting in error d 26 or d 27.

Other possibilities:.

I. Older equipment can leave track marks on the floor and also cause errors D26 or D27 due to age-related deterioration of the front wheel casters, especially due to increased mechanical load during turning.
II. Poor contact between XCONT CNUM to MU DRIVER-07 CN21. In particular, care must be taken with the connector CNUM on the XCONT because the gap between it and the side cover is small.

4.2.45 d28, d29

Abnormal rotation of the right (d28) or left (d29) motor was detected.
(During running operation, there is no feedback signal input from the encoder.)
The operation of the encoders of the left and right running motors can be checked according to the "display of running speed" in the installation manual, so check whether an abnormal value such as zero is displayed.
Check the cable connections between the encoders and the connectors CN6 (left motor) and CN7 (right motor) on the MU DRIVER, including grounding the GND cable, and replace the motor if there are no problems. If the motor is normal, it is likely that the MU DRIVER is faulty.

4.2.46 d30, d31

An increase in the temperature of the FET of the right (d30) or left (d31) motor drive circuit was detected. (The thermal sensor on the MU DRIVER was activated to the OPEN state).
The thermal sensor will be restored when the temperature drops, but power ON/OFF reset is required to use the device properly.
Check the operation of the FAN on the side of the MU DRIVER 07. (However, this FAN is not always working. It may work after running about 50m.)
Also, make sure that JP2 on the MU DRIVER is set to the "FAN 24V" side, not the "FAN 5V" side.

4.2.47 d40

Fused fuse for motor.
Check the fast-break fuses F4 and F5 on the MUX POWER-32 K.

4.2.48 d42

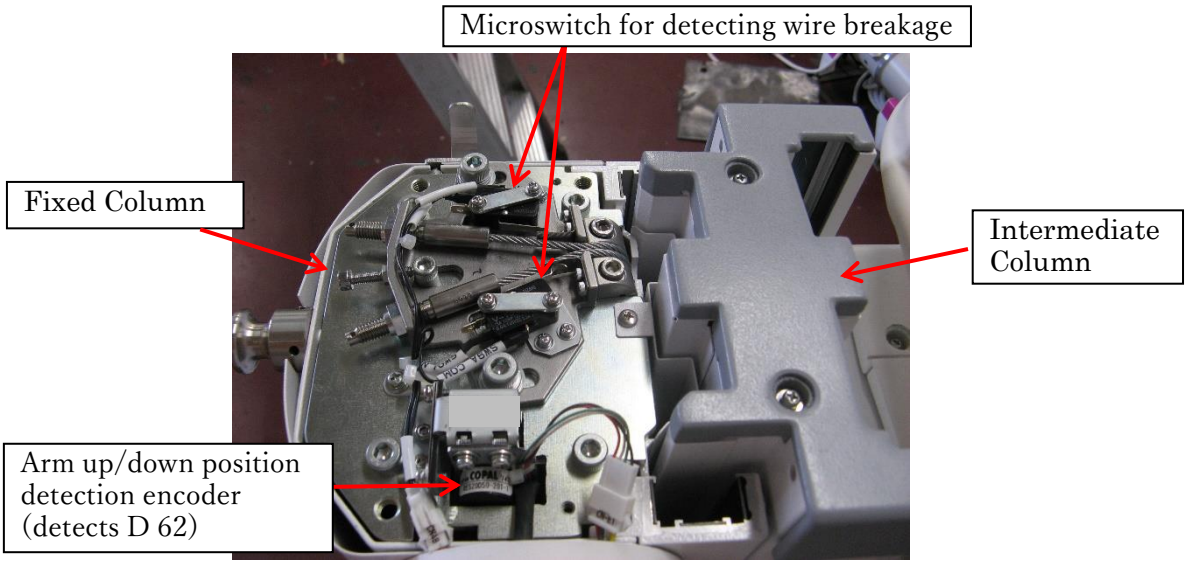
No 24 V power supply is supplied to the MU DRIVER.
Check connection and input/output of PS2 with 24 VDC power supply. In addition, there has been a case where the wire cover of the magnet for the brake of the arm front/rear movement was broken and caused a ground fault, and the MU DRIVER board was damaged, resulting in d42. Therefore, check for any abnormalities around the magnets.

4.2.49 d44, d45

A + 5 V/+ 15 V power supply for the right (d 44) or left (d 45) motor is not supplied to the MU DRIVER-07. Check the cable connections between the DC power supply PS 21 (for the left motor) and PS 22 (for the right motor) located under the MU DRIVER and the connectors CN 22 and CN 23 on the MU DRIVER.

4.2.50 d60

Error due to broken wire.
Check the connection and normal operation of the wire disconnect detection microswitch (below) at the upper end of the fixed post.



4.2.51 d61

MU DRIVER connection error
Check the cable connection between CN 21 on the MU DRIVER and CNUM on the XCONT.

4.2.52 d62

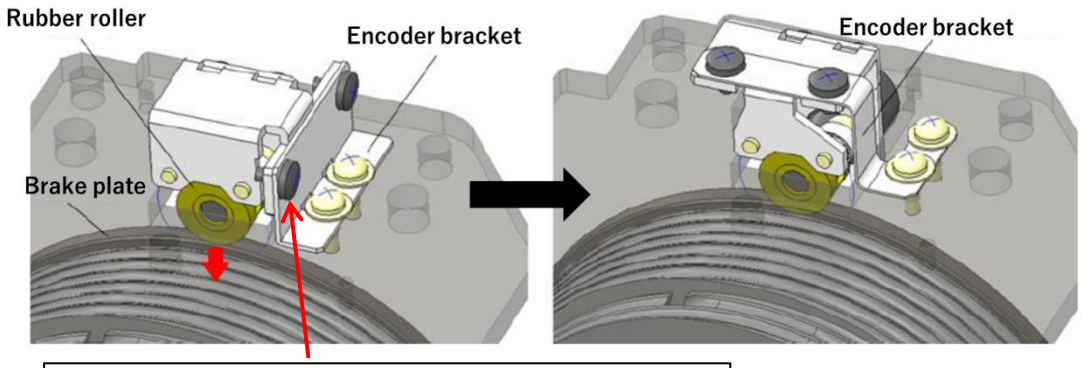
Abnormality was detected in the encoder (below) for detecting the arm vertical position.

Specifically, in a device whose encoder use setting is set to "ON," an error is detected when the encoder value is stored in the catch without changing even once while the arm is unlocked and released from the catch for more than one second.

When this error is displayed, check that the rubber roller of the encoder input shaft rotates smoothly over the entire stroke of the arm up and down, and if necessary, adjust the position of the encoder attachment. Also, check that the connection between the encoder and the MU DRIVER is faulty or broken.

However, in order to simplify the above adjustment, from the equipment shipped in the first half of 2022, the following “UP/DOWN POSITION DETECTOR ASSY" has been changed to a new model and no fitting adjustments are required.
If you perform an adjustment of the position of the encoder and the rubber roller rotates smoothly over the entire stroke of the arm's up and down movement but still gets error D62, replace the encoder with the new UP/DOWN POSITION DETECTOR ASSY because it is likely to be damaged.
Note that REV. 1.52 on the XCONT-2008X and REV.1.51 on the XCONT-2018 correct the malfunction of erroneously detecting D62, so check the software REV. first before considering replacing the ASSY.

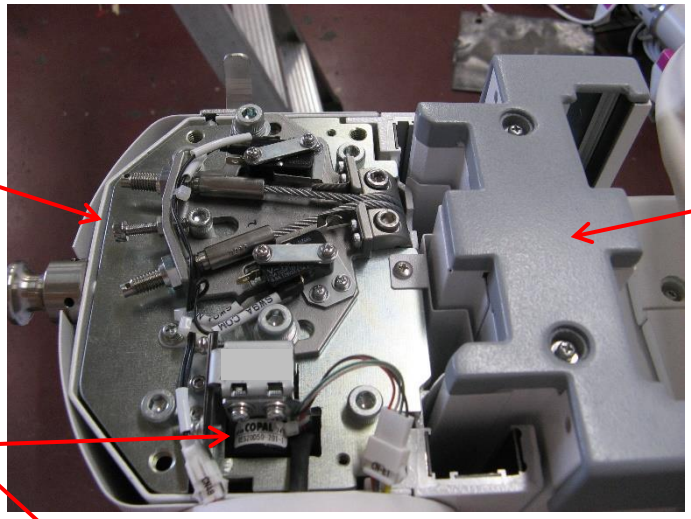
	Conventional model (old model) →	New model (D 62 countermeasure)
P/N	SSU566-22220 UP/DOWN POSITION DETECTOR ASSY	SSU566-27770 UP/DOWN POSITION DETECTOR ASSY
Features	It is necessary to adjust the contact between the brake plate and the rubber roller by shifting the position of the encoder up and down.	No adjustment is required. With a leaf spring, the brake plate and rubber roller automatically contact each other with the optimum strength.

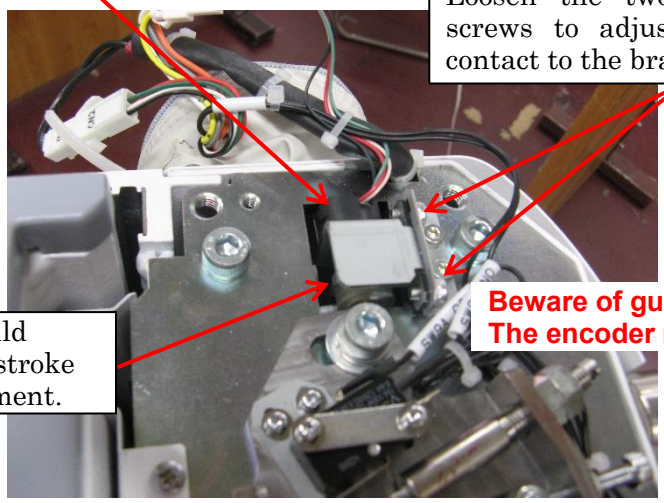


Encoder mounting screws (2 pieces)
: For adjusting the contact of the rubber roller

How to adjust for the old model of UP/DOWN POSITION DETECTOR ASSY

If the input axis of the UP/DOWN POSITION DETECTOR ASSY encoder does not rotate correctly in conjunction with the arm up/down movement, it will lead to D62. To prevent this, it is necessary to check and adjust that the rubber roller of the encoder input shaft rotates smoothly over the entire stroke of arm up/down movement.
If there is a position where the roller does not rotate during the arm moving up and down, loosen the two encoder mounting screws (see the figure on the previous page and the photo below) and adjust the mounting position to ensure that the roller always touches the brake plate for arm up/down movement. However, be careful not to apply too hard, as it may damage the encoder and lead to D62.





4.2.53 bbb

Battery voltage dropped below 237V in case of Battery Code “6” or below 240V in case of Battery Code “5”. In this condition, all of 5 remaining battery level indicator LEDs light off and exposure is disabled. (However, you can perform some additional exposure if you use additional exposure function.)

Note:
Repeated instances of extremely low battery voltage will reduce battery life. To prevent this, we recommend the following.

① When the DR unit is on all the time, the battery voltage drops rapidly. Therefore, it is highly recommended that the power plug be connected when the DR unit is left on. That way, the DR unit is powered from an AC power outlet and the battery doesn't drain.
② By turning off the key switch in the state of (1) above, battery charging can be performed simultaneously.
③ When the last one of the battery indicator LEDs is flashing, you can avoid the situation where error bbb occurs by charging the battery early. Charge the battery immediately when all the remaining battery indicator LEDs are off.

Repeated charging when the battery voltage is high enough also reduces battery life. To prevent this, it is recommended to charge the battery after two or more of the battery level indicator LEDs are turned off. Avoid carrying out further charging from a fully charged state of the battery without consuming the battery at all.
If the battery charge limit function is set to ON, when the remaining battery capacity is approximately 85-90% or more, full charging will not start even if the power plug is connected, so there is no need to worry about overcharging.

4.2.54 bb0

The battery voltage drops to a level that makes it impossible to perform radiography, then Auto Power OFF function has worked.
When the phenomena suggesting the defective battery (F12, F15, F16 and others) occurs , and if this error bb1 remains many times in the past error log, there is the possibility of the harsh use leading to battery over discharge in the past.
This error code bb1 has been set to determine whether the battery is good or bad and to estimate the cause of the battery failure.

4.2.55 bb1

With the DR device on, the battery voltage drops to a level that makes it impossible to perform radiography, then Auto Power OFF function has worked.
When the phenomena suggesting the defective battery (F12, F15, F16 and others) occurs , and if this error bb1 remains many times in the past error log, there is the possibility of the harsh use leading to battery over discharge in the past.
In particular, there is a possibility that the DR device was left ON without connecting the power plug to the wall consent for a long time.
This error code bb1 has been set to determine whether the battery is good or bad and to estimate the cause of the battery failure.

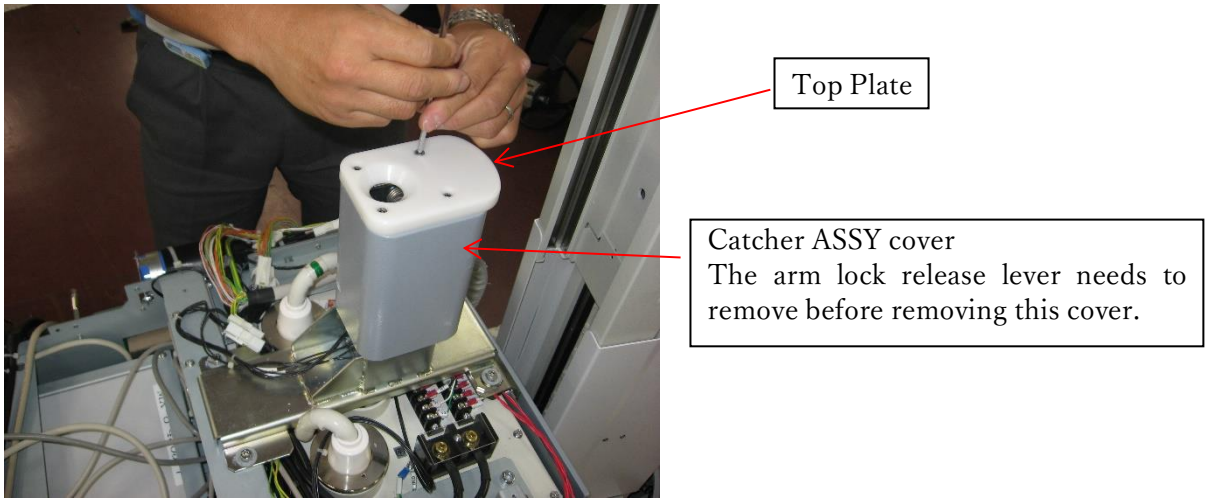
4.3 Sudden slow down of driving speed

Adjust as follows when the driving speed becomes slow because the micro switch for arm detection becomes off during driving on the bumps etc. on the floor. As well, if arm locking procedure is not correct, the arm cannot be locked by the catcher tightly and the arm detection switch easily becomes off during driving.

Explain to the customers lock the arm with the correct procedure according to Operation Manual. Specially, it is important to push X-ray tube toward the column until it hits the mechanical stopper, then, move it down toward Catcher ASSY.

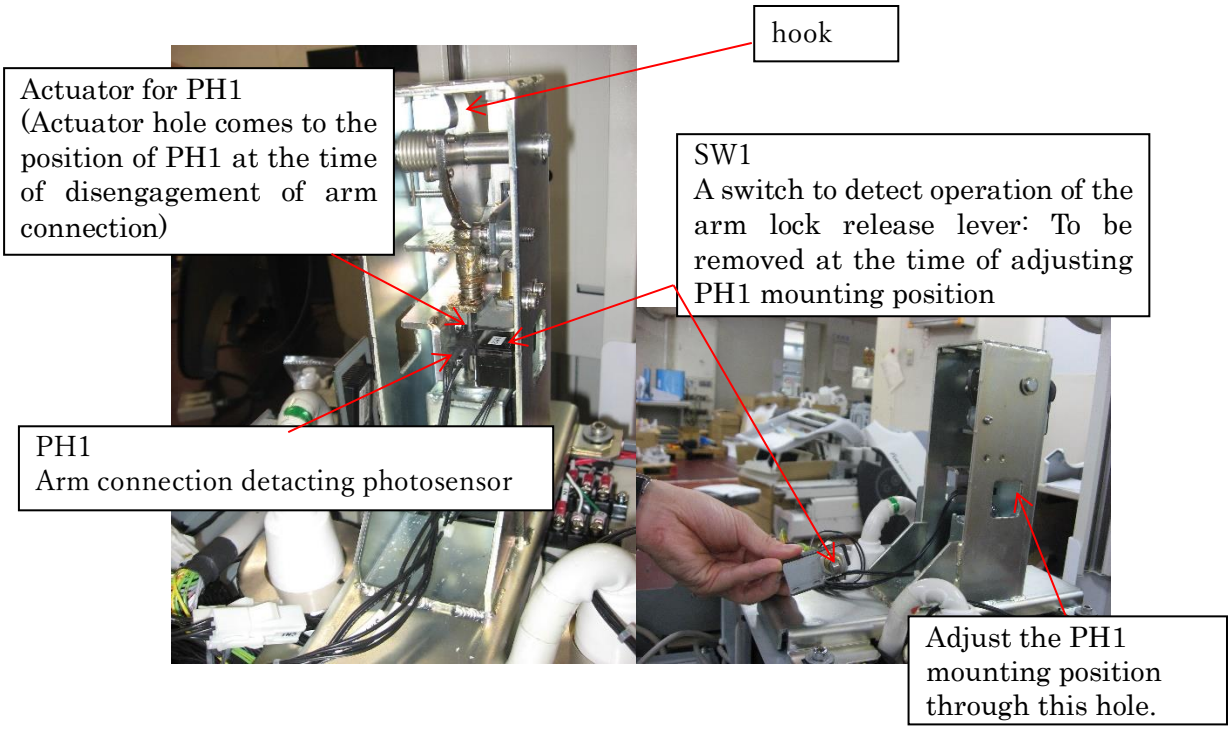
Catcher ASSY arm connection checking sensor adjusting procedure

Turn off the power, then, remove the top plate and Catcher ASSY cover.



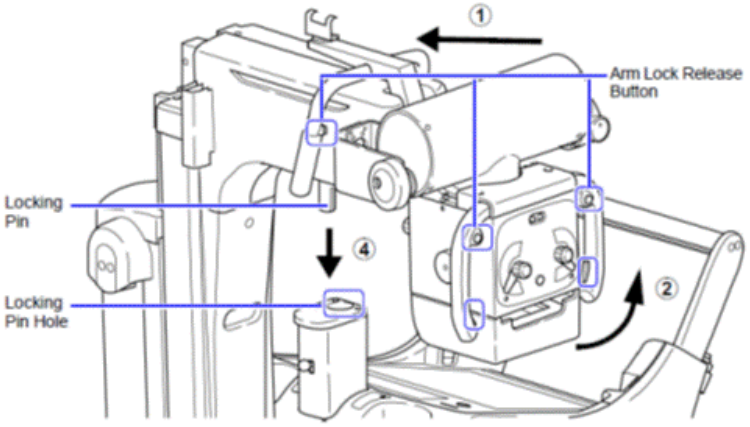
When the arm connection is disengaged (as shown in the photo below), an arm connection detecting hole that is provided on the actuator is situated at the position of photosensor PH1 with PH1 turned off. At the time of arm connection, a claw hooks on the arm catching pin, which elevates the actuator and the actuator hole comes off the photosensor PH1, turning on the PH1. → PH1 mounting position need be adjusted so as to turn on the PH1 at the time of arm connection.

4. Trouble Shooting



4. Trouble Shooting

Reference: Correct Arm Locking Procedure



1 Push the X-ray tube unit as far as possible toward the column until it makes contact with the stopper, while pressing the arm lock release button.

The unit may feel heavy just before reaching the stopper. Be sure to push it all the way in.

2 Return the tilt of the X-ray tube unit and collimator to the center position.

3 Slowly push the X-ray tube unit down while pressing the arm lock release button.

4 Insert the locking pin, located on the lower surface of the arm, into the locking pin hole in the arm joint section.

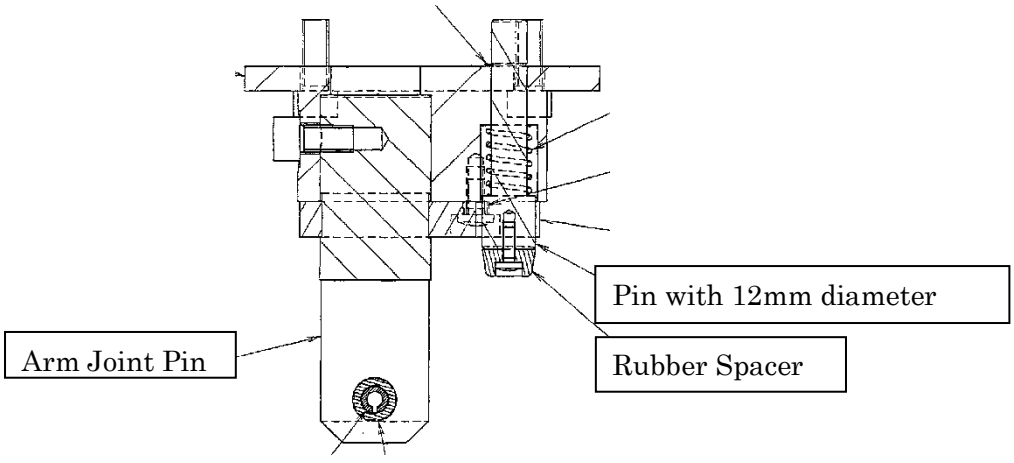
After locking the arm, a sound is emitted, the status indicator lamp switches OFF.

Note:

If you move the arm down to Catcher ASSY without pushing X-ray tube toward the column until it hits the mechanical stopper, the arm normally cannot be locked and you will not hear the buzzer sound.

If the arm can be locked without pushing X-ray tube toward the column, check if the rubber spacer on the end of the pin with 12mm diameter beside Arm Joint Pin is damaged or not.

Replace the rubber spacer if it is damaged. Otherwise, you can insert the washer with about 1mm thickness as the temporary solution.



4.4 Driving has become duller than before

Adjust the handle according to “Adjustment Mode” in the installation manual to see whether the condition is improved. If the condition is improved by adjusting the handle, the performance of the FSR may have been degrading with age.
In the case of recurrence, replace the FSR.

<Standard calibration data (normal case).>

Calibration position		AD value (Handle correction value)		
L	R	L	R	
3	9	799	807	3.0kg
2	8	744	756	1.5kg
1	7	663	687	1.0kg
4	10	362	352	1.0kg
5	11	276	282	1.5kg
6	12	219	220	3.0kg

For example, if the left FSR is degraded and the left motor runs duly, the amount of change from the neutral position (around 500) of the handle correction value to each position of the left FSR may become small as shown below after readjustment.
There will be no problem if readjustment brings about improvement. In the case of recurrence (the handle correction value continues to deviate with age), it is necessary to replace the FSR. In some extreme cases, it has been reported that the handle correction value does not change at all when applying force to the FSR.

<Abnormal case>

Calibration position		AD value (Handle correction value)		
L	R	L	R	
3	9	643	807	3.0kg
2	8	593	756	1.5kg
1	7	563	687	1.0kg
4	10	471	352	1.0kg
5	11	457	282	1.5kg
6	12	390	220	3.0kg

< Reference: Factory Default for Handle Correction Values >

The difference between the handle correction value in neutral position and 1.0 kg must be 50 or more.

The difference between the handle correction value at 1.0 kg and 1.5 kg must be 20 or more.

The handle correction value does not reverse when 1.5 kg and 3.0 kg is applied.

< Failure causes other than FSR >

(1) MU DRIVER-07 is defective.

Since this board controls the motor drive, it is the component to be suspected after the FSR failure. In some cases, the left and right FSRs repeatedly failed due to the defective MU DRIVER.

(2) Fuse F4 or F5 on MUX POWER-32K board is defective. (However, if these fuses are completely blown, error d20 or d21 should occur.)

If these fuses are faulty, the power supply to the MU DRIVER-07 board and motor will not be supplied properly, resulting in "very heavy driving". The state of these fuses can be checked by the following check pin on the MU DRIVER-07 board.

F4: Fuse of +120 V power supply (right motor drive power supply)

→ Can be checked with CP17 (+120 V).

F5: Fuse of -120 V power supply (left motor drive power supply)

→ Can be checked with CP12 (-120 V).

(3) Relay K11 is defective.

This relay turns OFF when the EMG switch is pressed, and cuts off the supply of the ±120 V power supply to the motor. (However, if the relay causes a malfunction or contact failure and completely blocks the ±120 V power supply, error d20 or d21 should occur.)

The normal operation of this relay can be confirmed by the following check pin on the MU DRIVER-07 board.

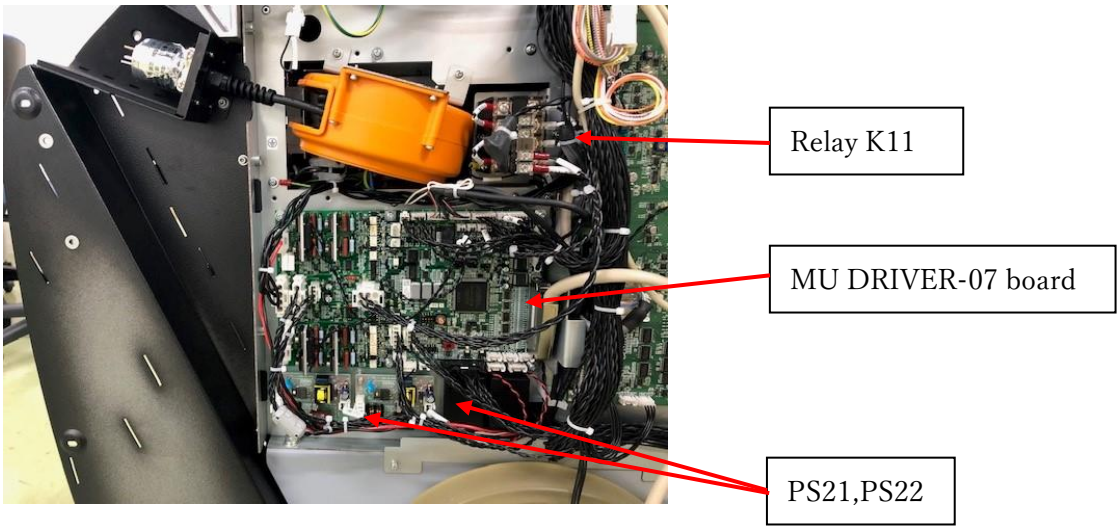
CP21 (P120): +120 V power supply (power supply for driving the right motor)

CP15 (M120): -120 V power supply (power supply for driving the left motor)

(4) DC15V power supply for motor control PS21 or PS22 is defective.

These are two small power supplies for controlling the left and right motors with about 5 cm square which are attached to the underside of the MU DRIVER-07 board. (However, if this power supply is completely defective and 15 VDC is not supplied, some error should be displayed.)

(5) The cable between FSR and MU DRIVER-07 board is defective.

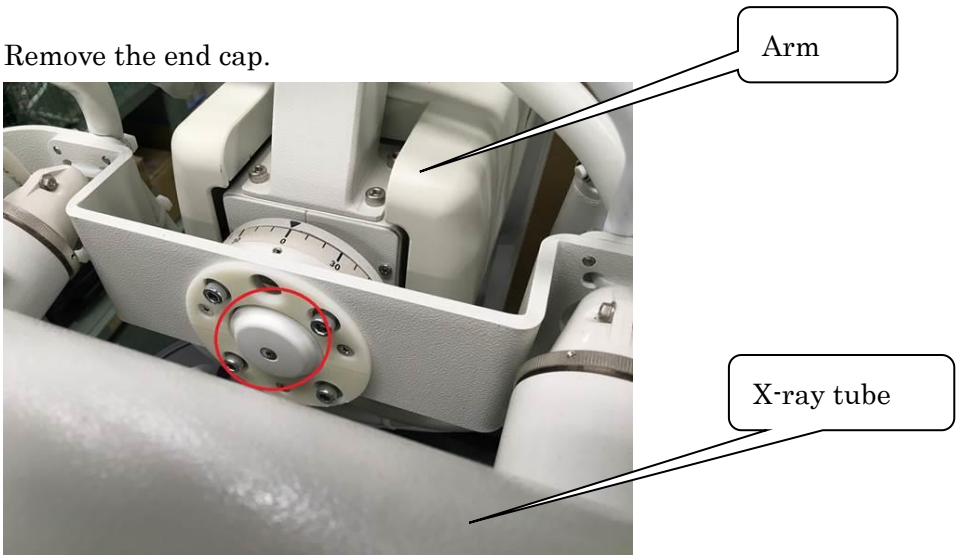


4.5 X ray tube droops under its own weight when tilted around the horizontal axis

There was a report of a case where X-ray tube + collimator droops due to its own weight when tilted around the horizontal axis.
In that case, adjust click feeling and tube rotation operating force for tube horizontal-shaft rotation according to “Maintenance of rotary X-ray tube section” in the installation manual.

Tube rotation operating force can only be adjusted by tightening the lock nut. (Tightening the plunger to adjust rotation operating force is incorrect procedure.)
Tightening the plunger will certainly increase tube rotation operating force (heavy), but the plunger is designed to give a click feeling every 90 degrees, not to adjust rotation operating force.
Before adjusting tube rotation operating force, the plunger performs a uniform "tighten to full and then loosen 45 degrees" adjustment. Then, adjust the rotation operating force by tightening the lock nut. Please refer to the installation manual for details.

I. Remove the end cap.



II. Though the adjustment of the click stop plunger has already been performed at the time of shipment from the factory, check the plunger adjustment according to the following procedure.
Tilt X-ray tube + collimator approximately 45 degrees.

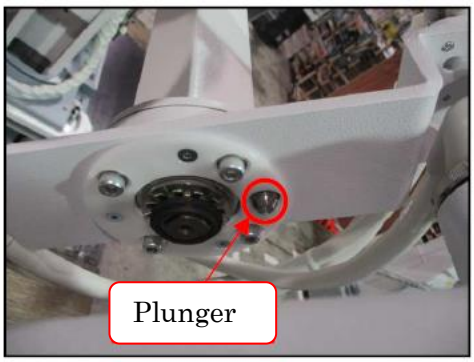
Tilt X-ray tube + collimator approximately 45 degrees during adjustment of the plunger.

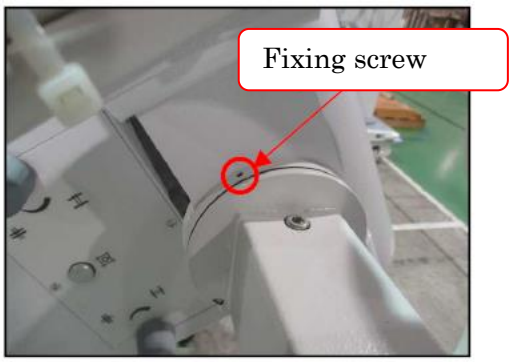
III. Loosen the plunger completely first. Before loosening the plunger, first loosen the fixing screw which holds the shaft of the plunger from the side, and then loosen the plunger. (The plunger cannot be adjusted until the fixing screw is loosened.)

IV. After the plunger is fully tightened, loosen it by 45 degrees. At this time, please adjust carefully because the contact condition of the plunger cannot be visually observed.

Then, tighten the fixing screw to prevent the plunger from turning.

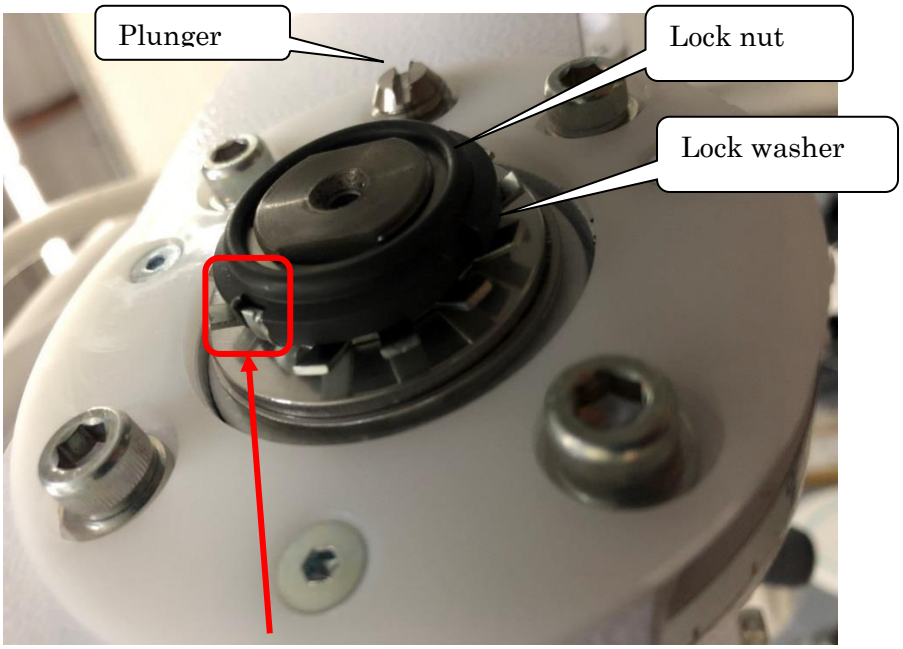
(If the fixing screw is not tightened, the plunger will become loose after X-ray tube is rotated several times.)





V. Pull down the claws of the lock washer to adjust the tightening of the lock nut.

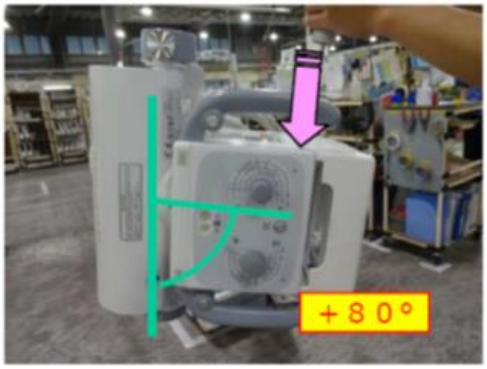
4. Trouble Shooting



The claw of the lock washer is inserted into the groove of the lock nut by bending at one place. The lock nut can be rotated by tilting the claw with a flat screwdriver. Then, only by tightening the lock nut, the rotation torque is adjusted to be about 20 N at the measurement position shown in the figure below. Before measurement,
X-ray tube should be fully rotated about 10 times to fit the contact surface.

VI. Tilt X-ray tube + collimator to +80 deg. position around the horizontal axis as shown below.

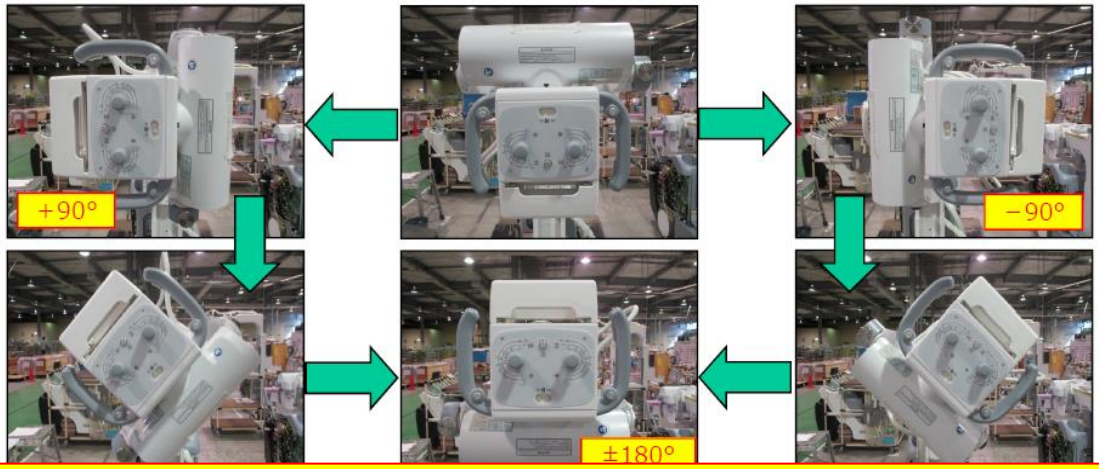
VII. When a push-pull gauge is applied to the arrow tip of the collimator cover, a force is gradually applied in the direction of the arrow, then X-ray tube starts moving. Read the value of the gauge when X-ray tube starts moving.



20N±3N

If it is out of the above range, adjust the degree of tightening of the lock nut again.

VIII. Rotate X-ray tube + collimator and confirm that the click works at ±90deg. and ±180deg. Position.Also, make sure that the collimator does not drop due to its own weight at angles where clicking does not work (at ±80 degrees, ±45 degrees, etc.).



Don't rotate X-ray tube + collimator exceeding 180deg. to avoid the damage upon High Tension cables and their holders etc.

IX. If the collimator drops down due to its own weight, tighten the lock nut a little more to adjust.

X. When the lock nut is tightened or loosened for adjustment, make full rotation about 10 times to familiarize the contact surface each time, then, measure the rotating torque and check if X-ray tube + collimator does not drop.

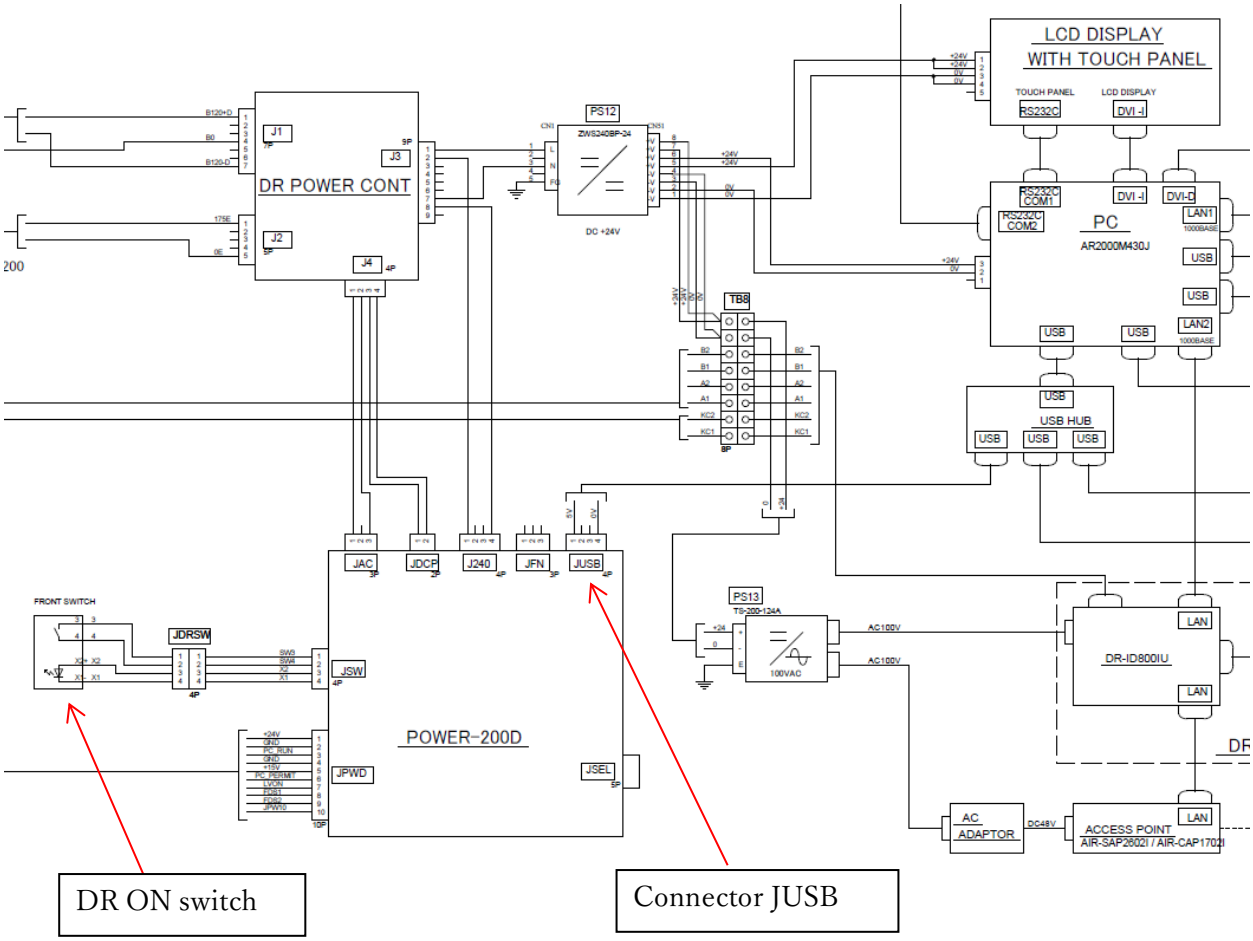
XI. After the adjustment is completed, align the claw of the lock washer with the groove of the lock nut and bend it so that it does not become loose.

If the claw and groove are not aligned, turn the lock nut clockwise (direction of tightening) and fold it at the position where it first fits.

4.6 DR system turns off several seconds after booting up

When the key switch is turned on and XCONT board is in operation, pressing the DR ON switch turns on the PC and starts the DR system. The DC5V output from the USB port of the PC is input to the connector JUSB on the POWER-200D PCB, then, it establishes a self-closing circuit. Even if you release your hand from the DR ON switch, the PC will continue working by this self-closing circuit.
(Refer to "3.8 Power ON/OFF circuit for DR system" of this Service Manual.)

If the above self-closing circuit does not work for some reason, the power supply of the PC will be turned off a few seconds after the hand is released from the DR ON switch, resulting in the phenomenon of "DR system boots up once but turns off after several seconds".



If above phenomenon happens, the following trouble cause is maybe possible.

(1) The PC is not outputting 5 VDC correctly.

(2) Defective USB HUB or USB cable connected between PC and POWER-200D PCB.

(3) Defective POWER-200D PCB

Even if the above parts are not necessarily replaced, the quality of each part can be judged to some extent by the following method, so please try it.

<Trouble shooting>

The USB cable from the POWER-200D PCB is connected to the USB port of the USB HUB. Whether the hub is bad or not can be determined by temporarily connecting the USB cable of the POWER-200D to another USB port on the back of the control PC.

It can also be tested by plugging it into a USB port on a service laptop. If the problem disappears after connecting to the USB port of the service PC, it can be determined that 5V is not coming out of the USB port of the control PC.

In addition, the USB cable for the bar code reader may be disconnected or short-circuited, or the USB port of the PC may not emit 5V due to overload.

You can also test it by connecting the USB cable to the USB port of your laptop for service. If you connect it into the USB port of the service PC and the phenomenon disappears, you can prove that 5V is not coming out of the USB port of the control PC.

In addition, PC may not output 5V from the USB port due to an overload or the damaged USB cable from the bar code reader etc.
In this case, disconnect all USB cables connected to the PC and connect only the USB cable from the POWER-200D PCB.

Proceed checking as follows.

(1) If DC5V is not coming out from the control PC due to the influence of other USB cables, it will be complicated. Therefore, disconnect all cables except the USB cable to the POWER -200 D from the control PC and check if the phenomenon does not change.
If the phenomenon disappears, the control PC is normal.

(2) Connect the USB cable from POWER-200 D to the other port on the USB hub and check for symptoms. If the problem disappears, the USB hub port is faulty. Replace USB HUB.

(3) If connecting the USB hub to another USB port on the control PC eliminates the problem, the USB hub is normal. The USB port on the control PC is wrong. Alternatively, you can check by connecting only the USB cable of the POWER-200D directly to a different USB port on the control PC.

(4) If the phenomenon does not change in each of the above tests, there is a possibility that

5VDC is not really emitted from the control PC.

Finally, it is confirmed by connecting it to the USB port of the service PC, and if the phenomenon disappears, it can be judged that the USB port of the control PC is defective.

If the problem persists, the POWER-200D PCB may be defective. If the LED 2 (USBON) on the POWER-200D PCB does not turn on even when connected to the USB port of the service PC, there is a high possibility that the DC5V input circuit on the POWER-200D PCB is defective.

4.7 Battery life is poor

If battery life is poor, the battery charge control circuit board or the main battery itself may be defective. However, if the customer tells you that the battery is draining faster than before relatively, you should carefully judge whether the main battery is critically defective and needs immediate replacement, or whether it has only begun to deteriorate somewhat relative to when the unit was originally delivered and the main battery does not need replacement.

In the event of a failure or claim including “Battery life is poor” mentioned in this chapter that the main battery is suspected to be defective, obtain the following data for trouble shooting, and send it to Service Management Department together with the trouble report to determine whether the battery needs to be replaced or not, if necessary.

1. Procedure for determining the quality of the main battery

(1) Check that the battery code is set correctly.

As shown in the table below, the battery type has been changed from the equipment shipped since around September 2022.

If the battery code is incorrect, the charging operation and the display of battery indicator may become abnormal, leading to a "fast battery drain" phenomenon.

If the incorrect battery code was set, change it to the correct battery code and check if the phenomenon is improved or not.

	Previous model →	Current model
P/N, Model name	SSU074-71014-11 PXL12090F2V0/Z	SSU074-74017-01 PXL12072H FR
Battery code	6	5

(2) Check whether the voltage on the +/- side of the battery is unbalanced.

(The MUX POWER-32K board terminal board can be measured between TER1-TER5 (+ side) and TER2-TER5 (- side).)

If the unbalance between + and - side is large, the battery may not be charged enough and the display of battery indicator may be abnormal, leading to the phenomenon “Battery life is poor.” or "Battery drain fast."

Note, however, that even if there is only an unbalance of about 1V in the stand-by condition without any operation such as exposure, driving on the floor or battery charging, the unbalance may increase during charging and exposure.

< Unbalance during charging >

About 5 or 10 minutes after the charging starts, use a tester to measure the voltage on the terminal board of the MUX POWER-32K board, and check whether the unbalance spreads. Note that the voltage drops immediately after charging is stopped, so keep charging and measure the voltage.

	Battery voltage at + side	Battery voltage at - side
Stand-by condition		
5 min. after battery charging starts		
10 min. after battery charging starts		

If the unbalance spreads during charging, there is the possibility of the capacity decrease of the battery itself and another possibility of failure on the charging control circuit side. ->Try to find the trouble cause of the unbalance according to Chapter 4.2.22.

In case that the battery is degraded, the voltage of the degraded side of +/- increases too quickly during charging and it will make charging end too early. On the other hand, it may decrease rapidly during radiography or driving on the floor, then it may lead to the phenomenon “Battery life is poor.” or "Battery drain fast."

In particular, if a large number of bbb, bb0 or bb1 remain in the log as described in (4), battery degradation is most likely the cause of the unbalance.

As well, if the voltage of only one block of BAT1,2,3 and 4 is much different from other 3 blocks, the battery itself should be abnormal.

< Unbalance during exposure >

Perform "Battery Voltage Drop Check Mode" to check the drop status of the +/- side battery voltage during exposure. If there is almost no unbalance in the stand-by condition, but the unbalance increases to 5-10 V or more during exposure, the main battery itself is surely defective.

	Battery voltage at + side	Battery voltage at - side
Stand-by condition		
During exposure		

(3) Take exposure under the following X-ray condition under “Low Battery” condition with only 1 or 2 bars of the battery indicator light, and measure the output waveform [TkV] on the XCONT board with an oscilloscope. Be sure to age the X-ray tube before exposure.

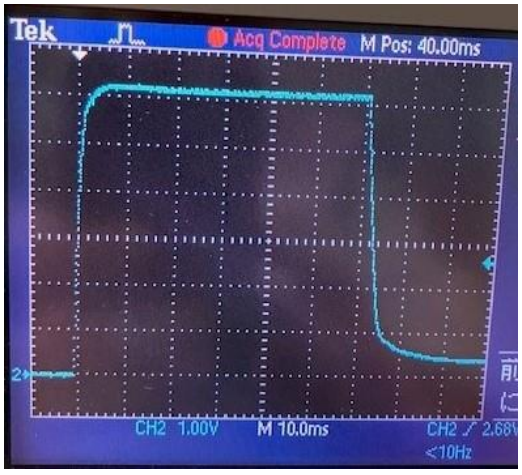
If the TkV waveform rises to the setting kV value and then gradually decreases, the main battery may have deteriorated to a low capacity.

In case that you take exposure without “Low Battery” condition with 3 or more bars of the battery indicator light, even if the battery is actually degraded, it may be difficult to find apparent abnormality in the TkV waveform, and you may overlook the battery failure. Therefore, it is easier to determine whether the battery is defective or not when the battery indicator status is as low as possible. Anyway, it is not absolutely necessary to measure TkV waveform under “Low Battery” condition in case that it is difficult.

・Large focus,125kV,5mAs(250mA,20msec)

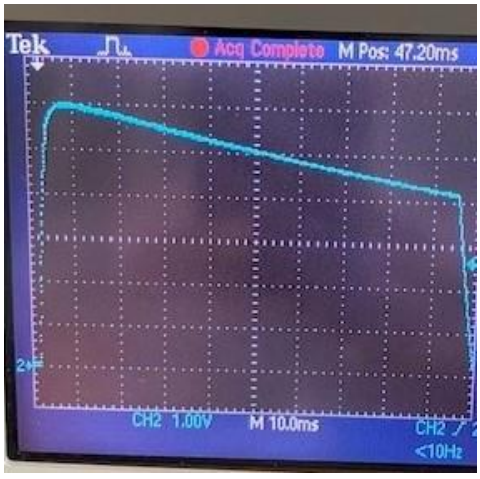
The settings for the oscilloscope are as follows:

[TkV]：1V/div、5msec/div



Example：Normal TkV waveform

(Measured when 3 bars light.)



Abnormal TkV waveform

(Measured when 2 bars light.)

(4) By entering the dtout→factry command of PC maintenance, all data of the device including the error log can be obtained in a text readable form as shown on Chapter 6 “PC Maintenance”, so please send the file to Service Management Department together with information of the type of XCONT board and the software REV.
(Under REV. 1.52 of XCONT-2008X, there was the software bug “The number of occurrences of bbb, bb0 and bb1 cannot be counted correctly. ")

With the above log file, we can find out what kinds of battery related errors have occurred in the past, battery usage, and more.
For example, if a large number of bbb, bb0 or bb1 remain in the log, you know that the battery has repeatedly overdischarged, and you can determine that the battery is likely to be degraded.

If “DR Auto Shutdown” is not set, it can cause overdischarge when the device is left with DR ON.
The function “DR Auto Shutdown” is supported by Canon, Fuji Film and KONICA (since July 2023), however the setting of “features enabled/disabled” and “the time until the Shutdown” etc. must be set correctly.

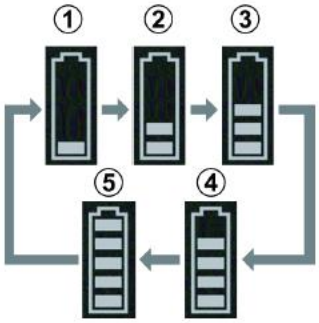
4.8 Battery Charging does not start even though only 3 bars on Battery Indicator light

There is no strict correspondence between the determination of battery charging requirement and the display of battery indicator.

Therefore, the phenomenon such as "Charging does not start even though only 3 or 4 bars of the battery indicator light." or "Charging starts even though all of 5 bars of the battery indicator light." does not directly indicate a malfunction of the device. However, if the phenomenon "Charging does not start even though only 3 bars of battery indicator light." continues many times, it is likely that something is wrong with the device.

[Conditions for starting battery charging]

If the Battery Charge Restriction (see “Battery Code Setting Mode” in the Installation Manual) is set to ON, the device checks the current battery level as the pre-charging to determine if the actual charging is required or not. During this time of pre-charging, the battery indicator lights up repeatedly in sequence as follows: 1st bar light → 2nd bar light → 3rd bar light → 4th bar light → 5th bar light → 1st bar light →・・・.



The battery is continuously charged at a lower current than the actual charging, even during the pre-charging for up to 10 minutes.

The condition under which the actual charging is started is that the voltage on either + or - side of the battery does not reach 133 V during pre-charging. In other words, when the voltage on either + or - side reaches 133 V, the pre-charging ends and the actual charging does not start. Therefore, the time required for checking the necessity for the actual charging is not fixed.

On the other hand, the battery indicator is not simply based on the current battery voltage, but is based on various complex factors. Therefore, it does not uniformly correspond to the battery charging requirement determination.

The condition that determines "No charging required" is generally described as "Battery level is 90% or more" or "4 or 5 bars of the battery indicator light." Therefore, this can lead to misunderstandings. However, it actually does not correspond to the number of the bars which light on the battery indicator. Therefore, "Charging does not start when only 3 or 4 bars on the battery indicator light" or "Charging starts when 5 bars on the battery indicator light" does not directly indicate a malfunction of the device.

[Case of actual malfunction of the device]

(1)Check whether the voltage on the +/- side of the battery is unbalanced.
(The MUX POWER-32K board terminal board can be measured between TER1-TER5 (+ side) and TER2-TER5 (- side).)

If there is an unbalance, when the higher voltage of either + or – side reaches 133 V during pre-charging, it will be judged as "No charge required." However, since the battery indicator is displayed taking into account the lower voltage, there are cases where less than 3 bars light. Note, however, that even if there is only an unbalance of about 1V in the standby state (the standby state without any operation such as radiography, running or charging), the unbalance may increase during charging and radiography.

If there is an unbalance, also see Section 4.7 and Section 4.2.22 to investigate the cause of the unbalance. In many cases, the battery degrades and causes an unbalance. By reviewing the logs, it is somewhat possible to determine if there was a situation that caused battery degradation. (See paragraph (4) of 4.7.) As well, if the voltage of only one block of BAT1,2,3 and 4 is much different from other 3 blocks, the battery itself should be abnormal.

(2)By temporarily turning off the setting of Battery Charge Restriction to allow the device to fully charge, the device may be able to correctly determine whether charging is required.
For example, if you repeatedly stop charging without waiting for the battery to reach full charge, it sometimes causes the phenomenon “The charging judgment may not be performed correctly.”

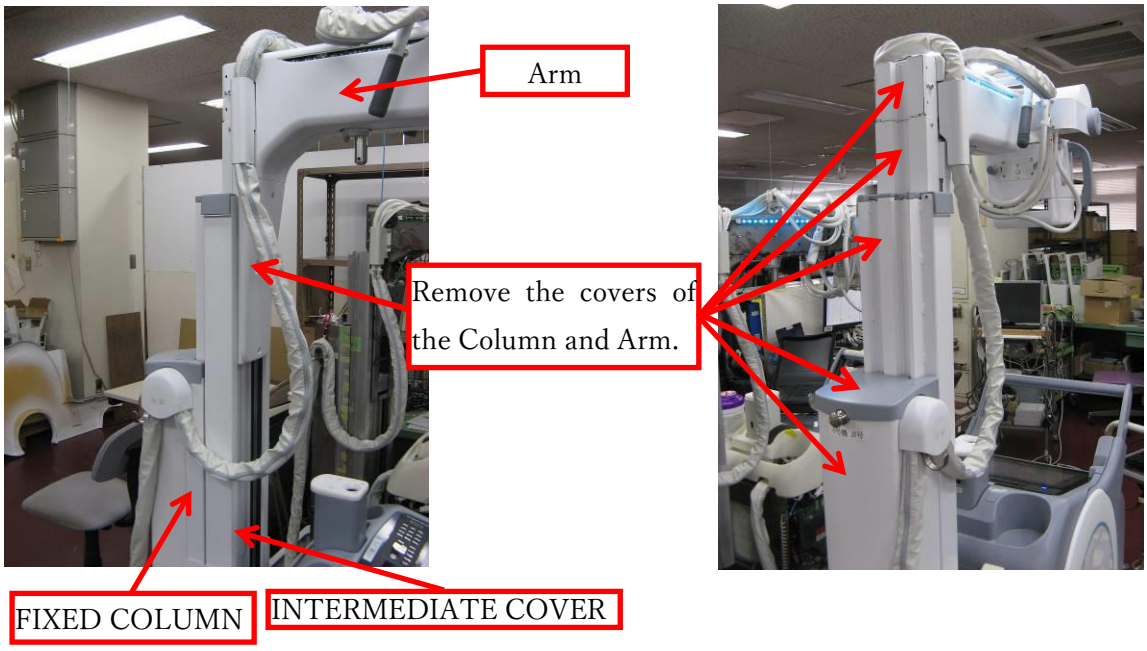
(3)If the above trouble shooting cannot solve the phenomenon, you may be able to correctly determine the necessity of the actual charging by initializing the XCONT board with turning on SW2-8 and SW3-1 according to Installation Manual.

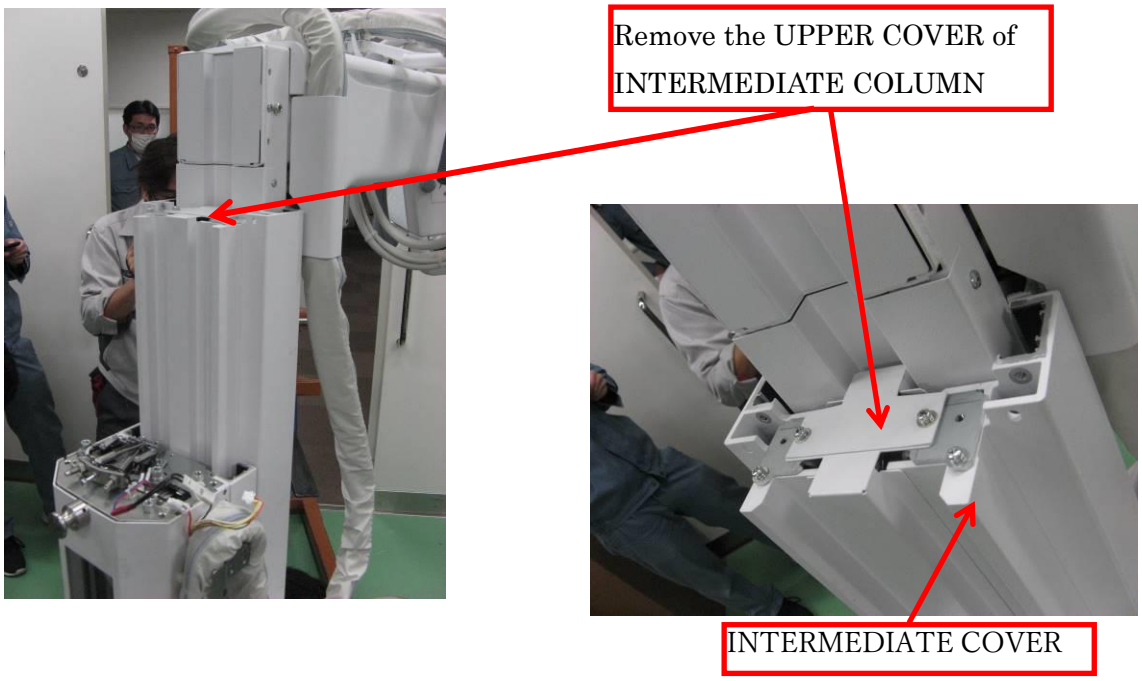
5 Parts replacement

5.1 **Wire rope replacement** (Reference work time 2 persons x 2.5 hours)

5.1.1 Preparation

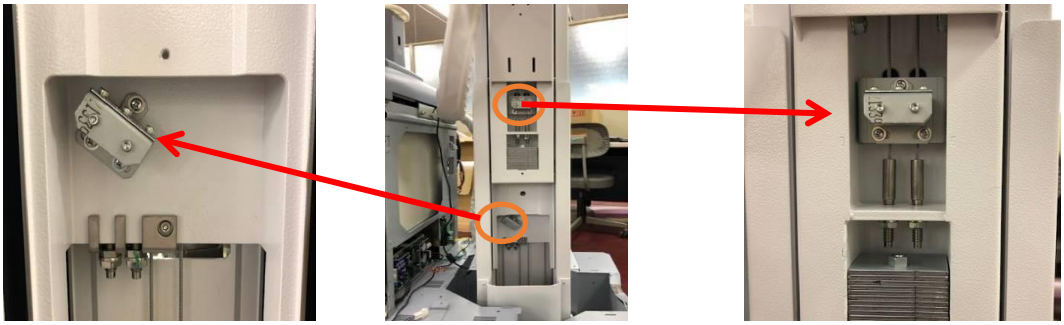
I. Remove the covers of the column and the arm.



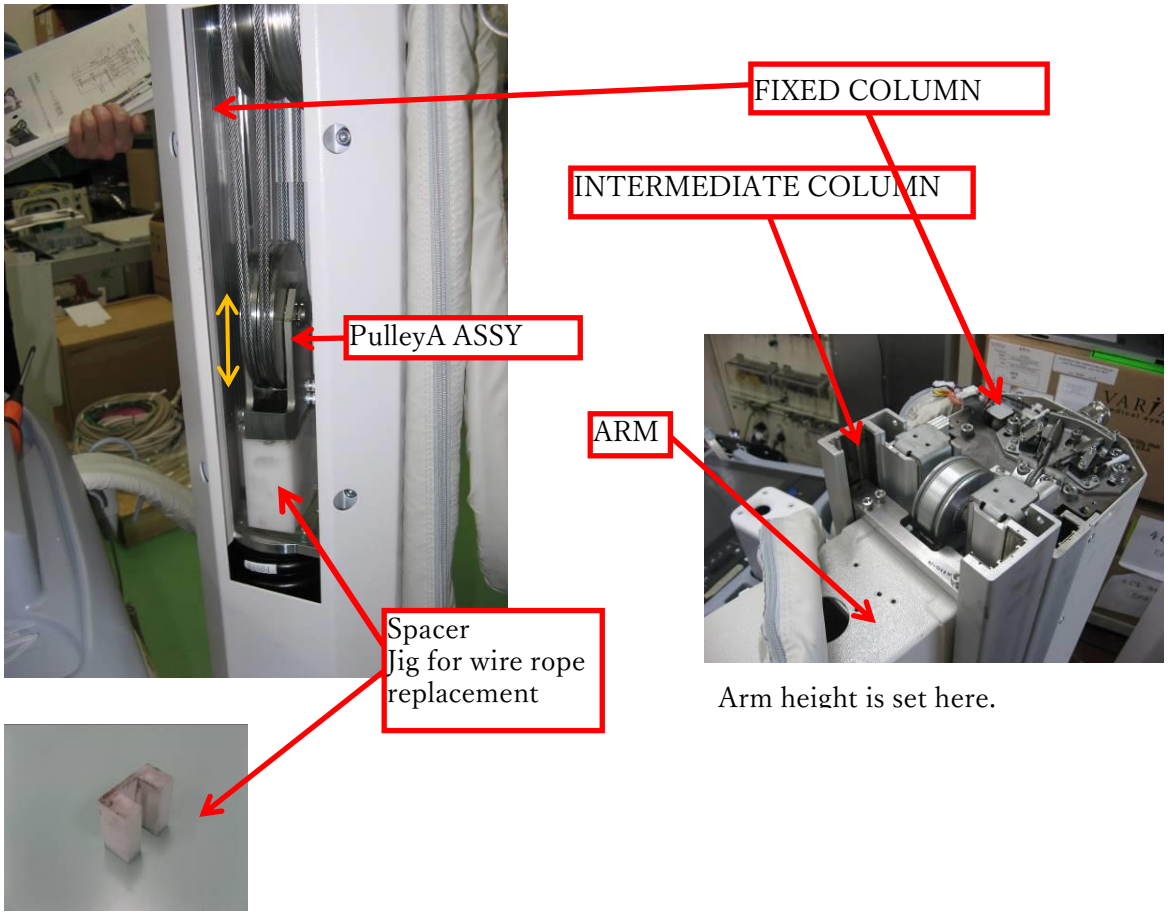


5. Parts replacement

II. Remove the breakage detection lock on the INTERMEDIATE COLUMN and ARM.



III. First, insert a spacer from a wire rope replacement jig into the gap of the spring balance ASSY built into the FIXED COLUMN and lift the arm until the bottom of the pulley A ASSY touches the top end of the spacer.



Arm height is set here.

Note:

This spacer is included in both below mentioned jigs for wire rope replacement.

P/N :566-21942 “Wire Replacement Jig, Column”

(For replacing 566-21399-01 “Wire rope set 1”)

P/N :566-21946 “Wire Replacement Jig, Arm”

(For replacing 566-21399-02 “Wire rope set 2”)

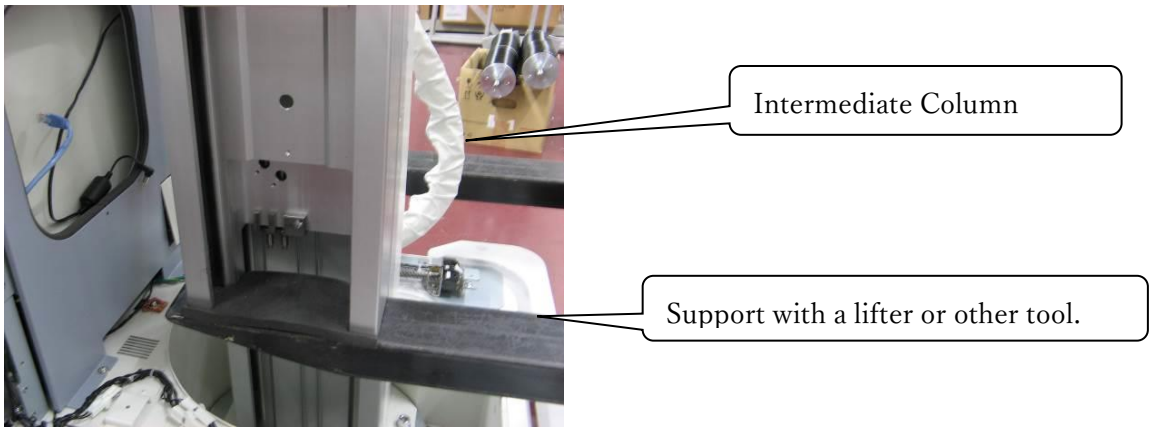
5.1.2 Replacement of Wire Rope for Spring Balance ASSY

The procedure of replacing the Wire Rope Set for Spring Balance ASSY (P/N:566-21399-01 “Wire Rope SET 1”) is explained below. This set includes 2 wire ropes for suspending INTERMEDIATE COLUMN and 2 wire ropes for spring balance.

I. Support the Intermediate Column so that no load is applied to the wire ropes.

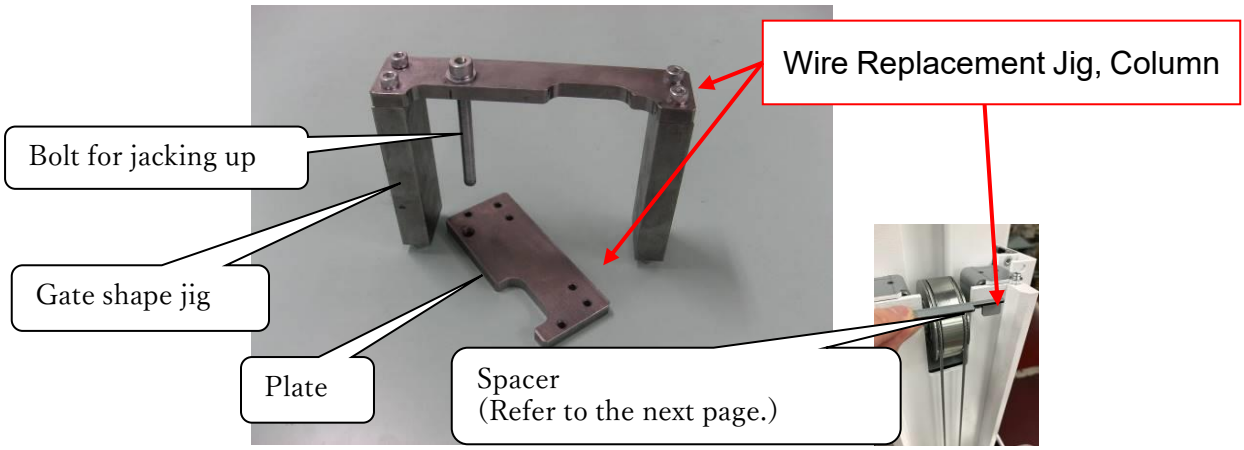
< How to support the Intermediate Column using a lifter, etc >

Support the underside of the Intermediate Column with a lifter, etc.

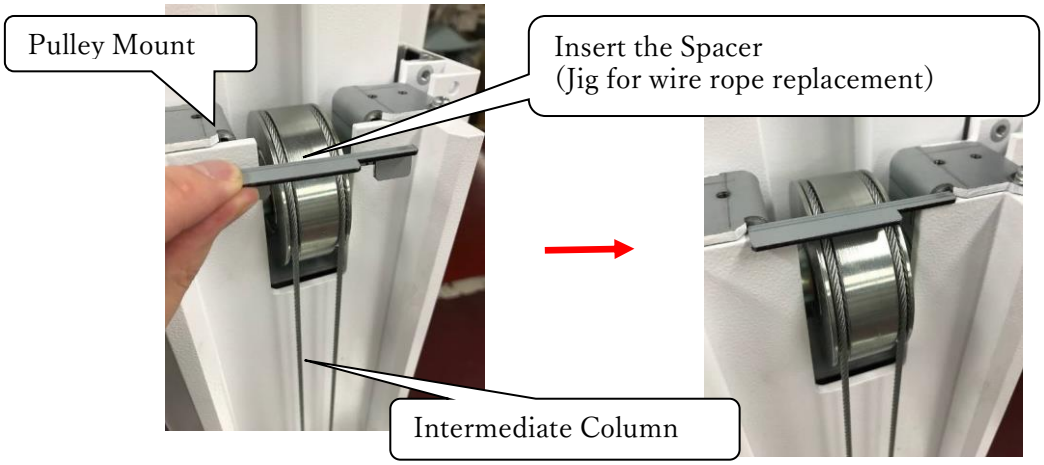


<How to support the Intermediate Column by using the Jig>

Wire replacement jig for Spring Balance ASSY shown below : "P/N:566-21942 Wire Replacement Jig, Column" use to support.

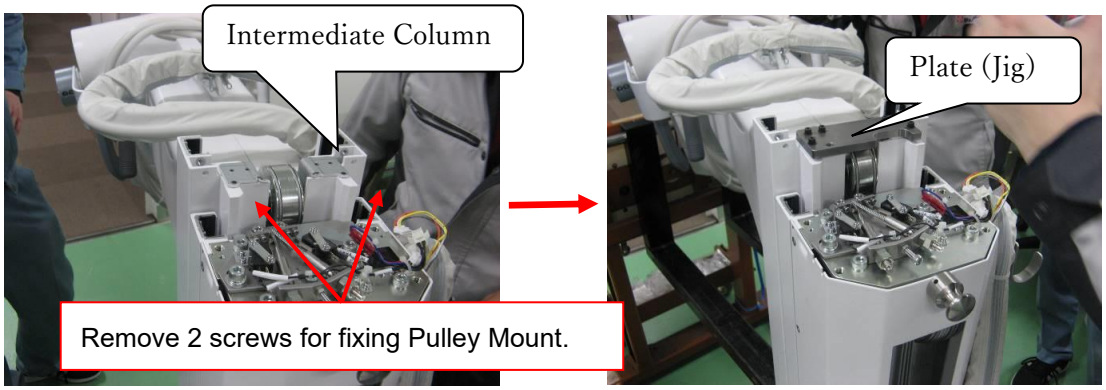


Insert the Spacer (Jig for wire rope replacement) into the gap between the upper end of the Intermediate Column and the Pulley Mount to prevent the position of the Pulley Mount from shifting due to the tension of the wire rope when the fixing screws for the Pulley Mount are removed to fix the Plate (Jig) in the next step.

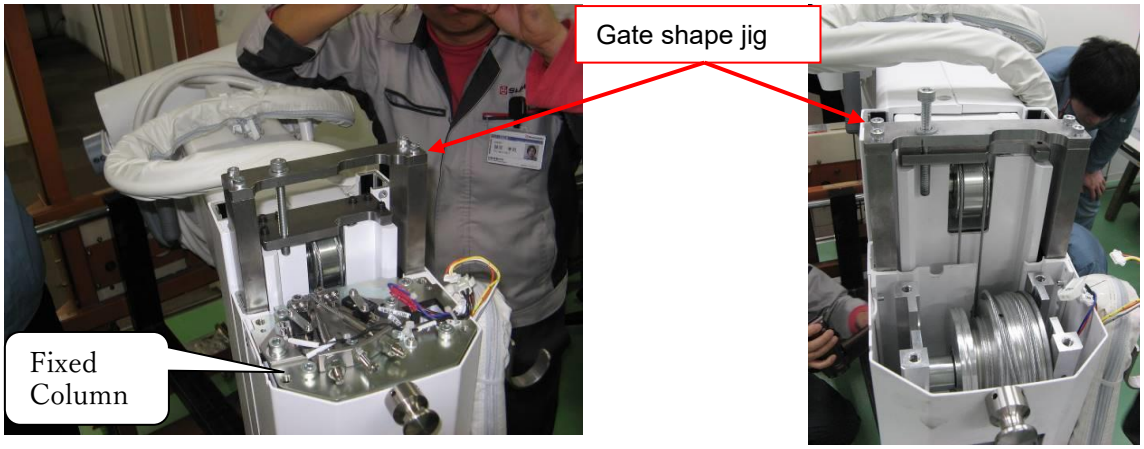


5. Parts replacement

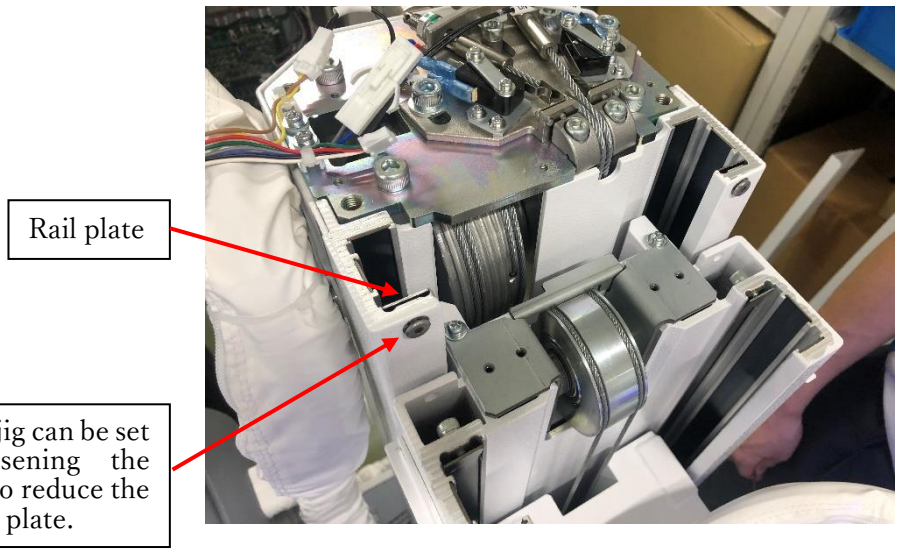
Remove 2 screws for fixing the Pulley Mount on the upper end of Intermediate Column, then, put the Plate (Jig) on the Pulley Mount and fix them together with 6 bolts.



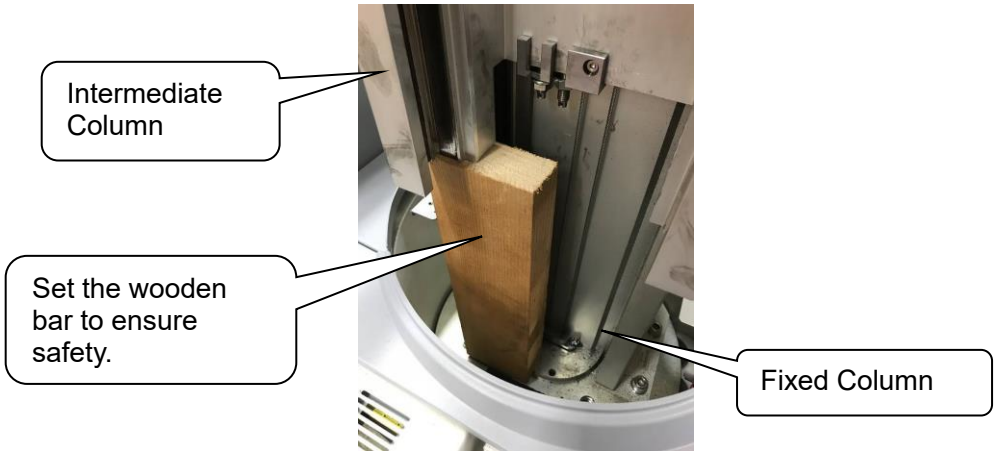
Set the Gate Shape (Jig for wire rope replacement) on the top end of Fixed Column, then, lift up the Intermediate Column by using the bolt for jacking up (Jig) to around the upper limit position as shown on the lower right picture.



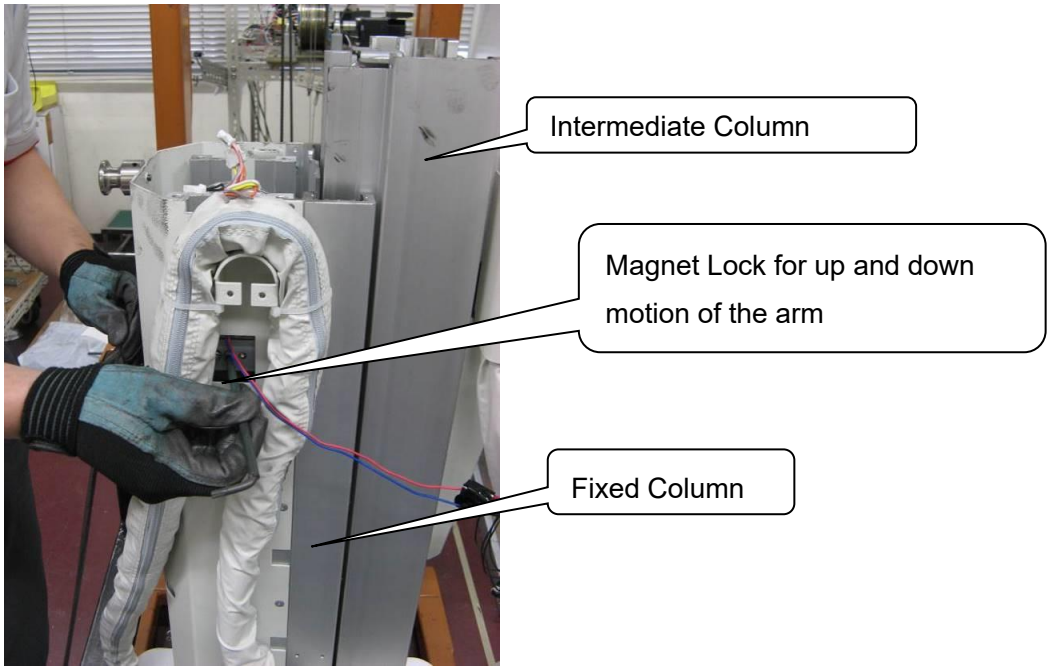
At this time, if the bulge of the Rail Plate at the top end of the Fixed Column prevents you from setting the Gate Shape jig, loosen the screw slightly. If you loosen it too much, the head of the screw will interfere with the Intermediate Column.



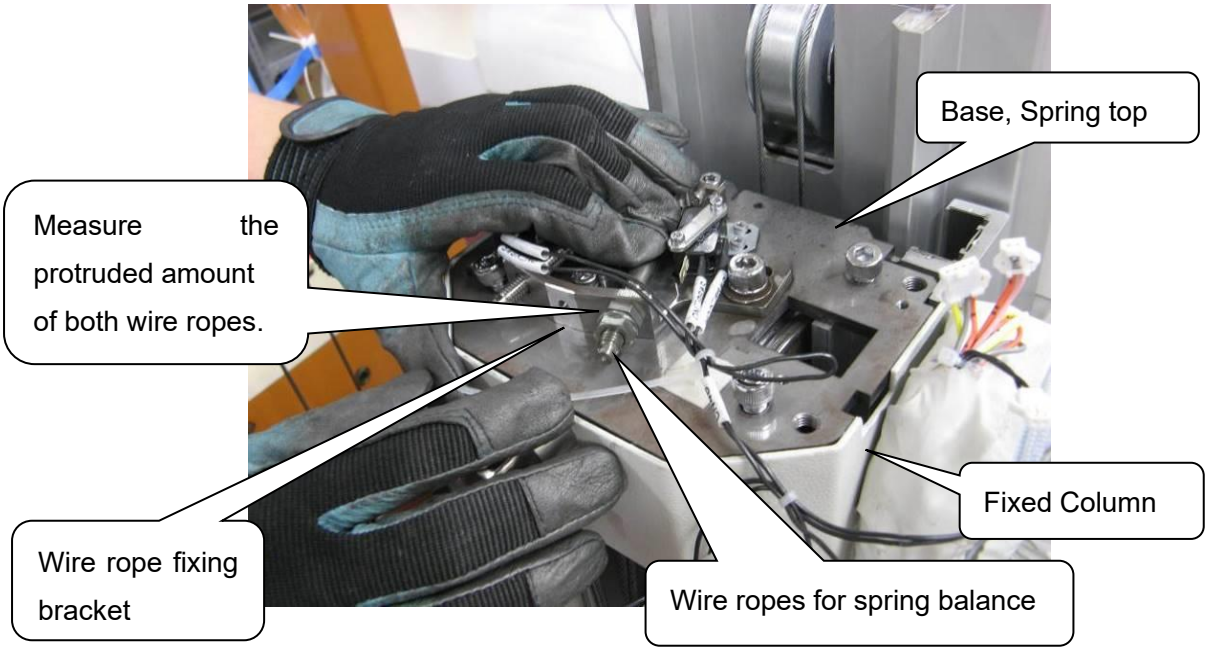
To ensure safety, set the wooden bar under Intermediate Column as shown in the below picture so that it will not fall even if the jig becomes damaged. The size of the wooden bar should be as follows. Height:290-300mm, Width:40-110mm, Depth:40-80mm. (The below picture shows the wooden bar with 300mm height, 80mm width and 40mm depth.)



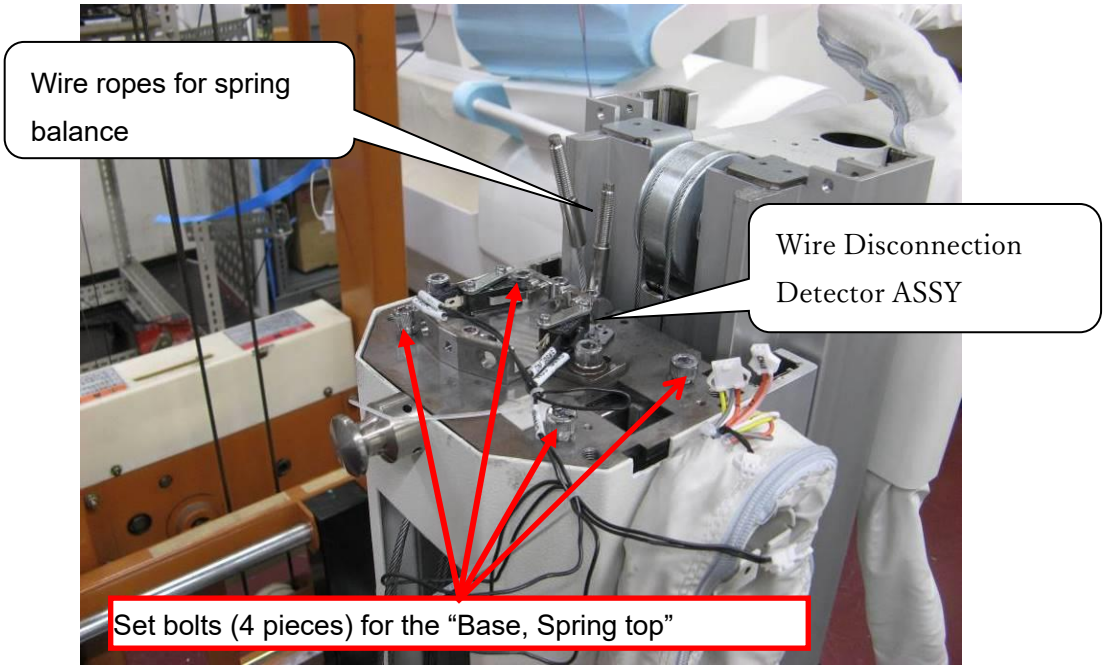
II. Jack up the Magnet Lock for up and down motion of the arm by screwing in the socket head bolt M5 × 6mm in order to disengage locking to facilitate to remove the wire rope from the Helical Cam (pulley).



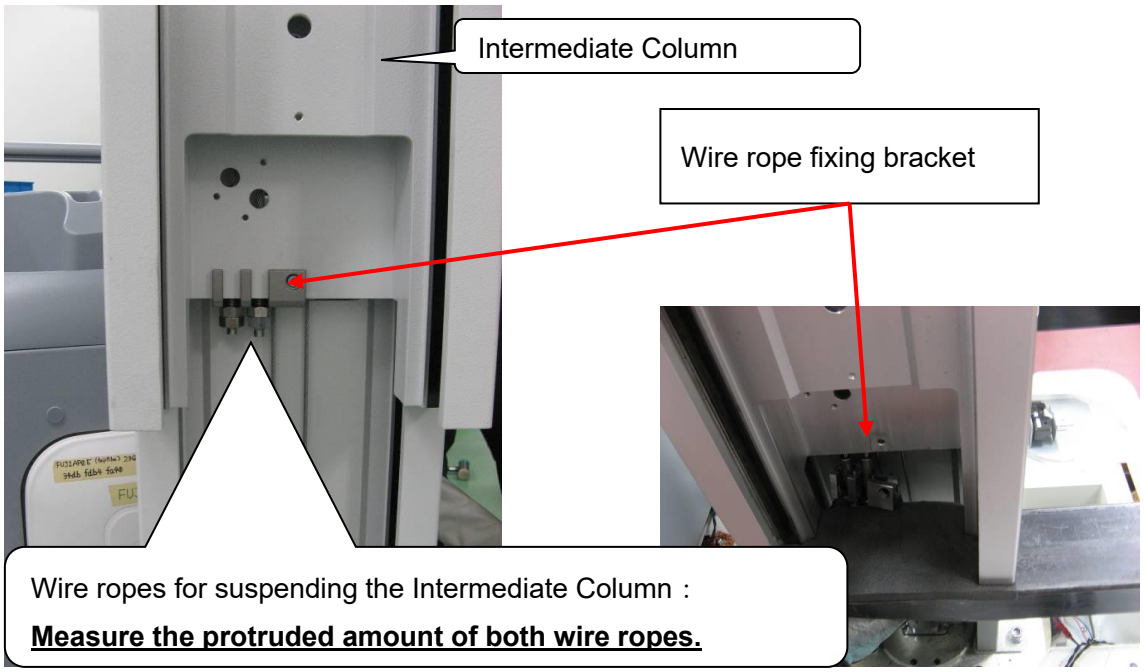
III. Remove the nut for fixing the wire ropes for spring balance (2 pieces). **However, measure the protruded amount of the wire rope end against the fixing bracket to match the tension adjustment after the replacement of wire ropes.** (In the drawing, the amount of protrusion is 22.5 mm, but in practice there are variations, so be sure to record the current value and adjust the value so that it is the same after replacement.)



IV. Remove two wire ropes from the fixing bracket, disconnect the connector of the Wire Disconnection Detector ASSY, and remove four set bolts to remove the “Base, Spring top”.

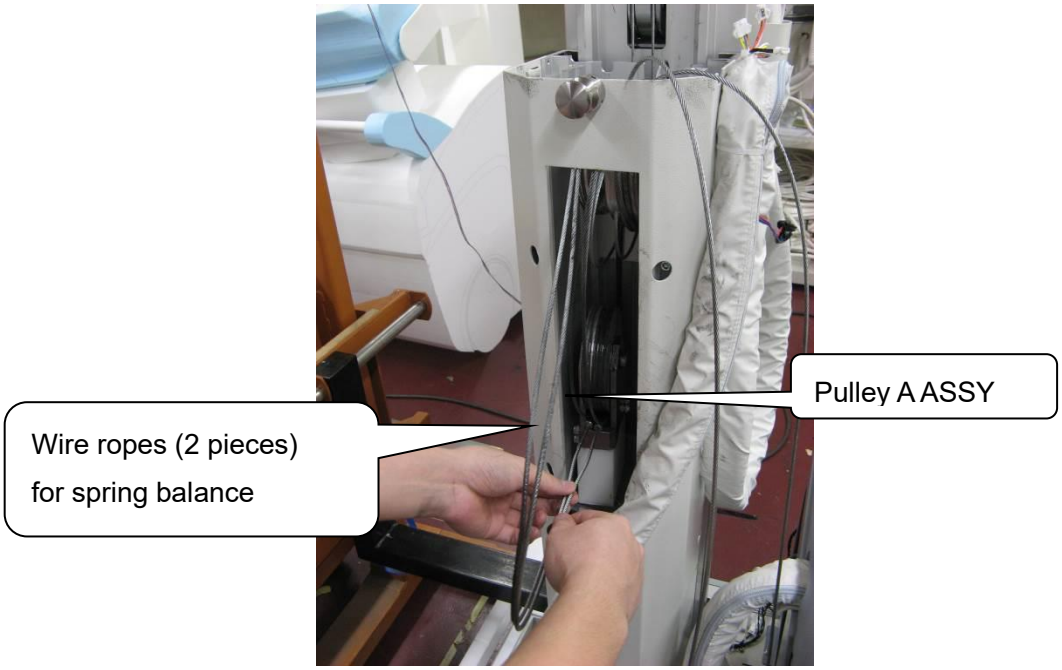


V. Remove the wire rope fixing bracket below the Intermediate Column so that the wire ropes (2 pieces) for suspending the Intermediate Column can be free to the Intermediate Column. However, measure the protruded amount of the wire rope end against the fixing bracket to match the tension adjustment after the replacement of wire ropes.

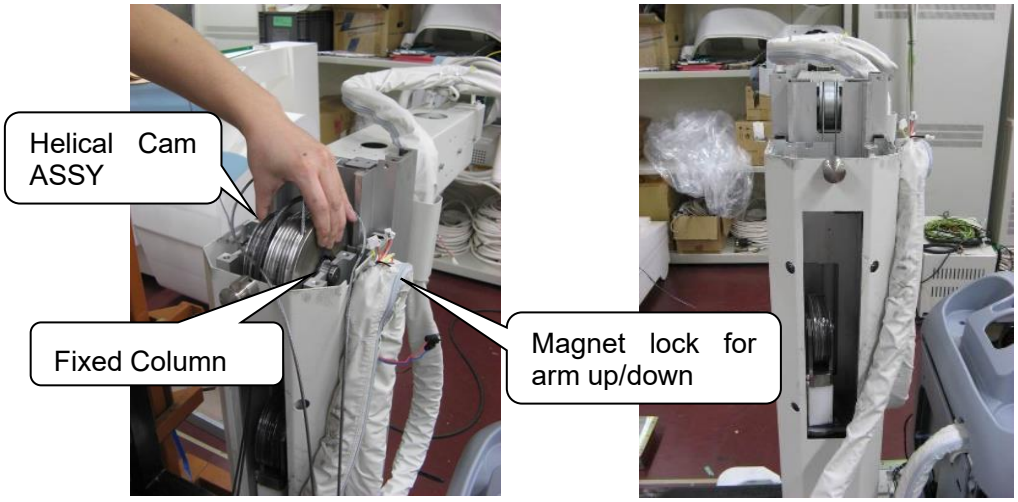


VI. Then, pull out the wire ropes (2 pieces) for suspending the Intermediate Column upward.

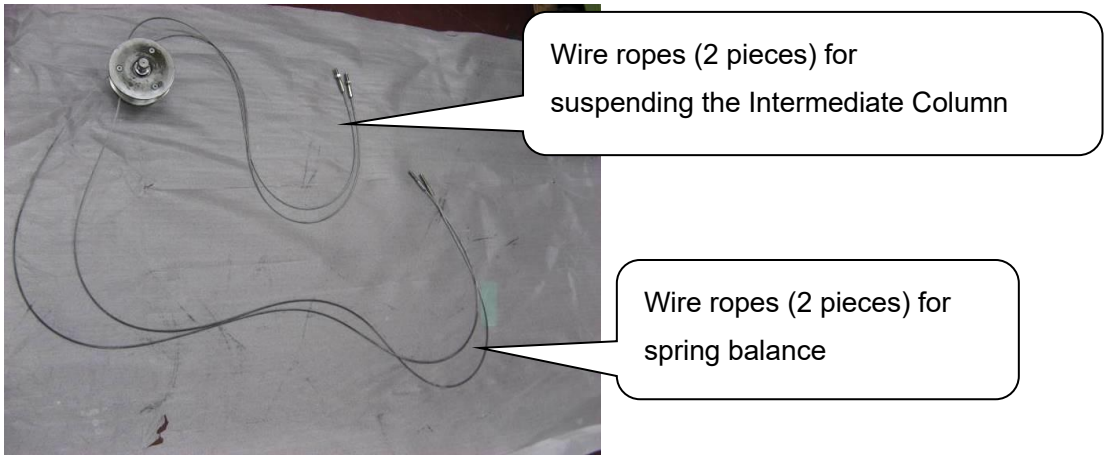
VII. Wind the wire ropes (2 pieces) for spring balance from the Pulley A ASSY to remove.

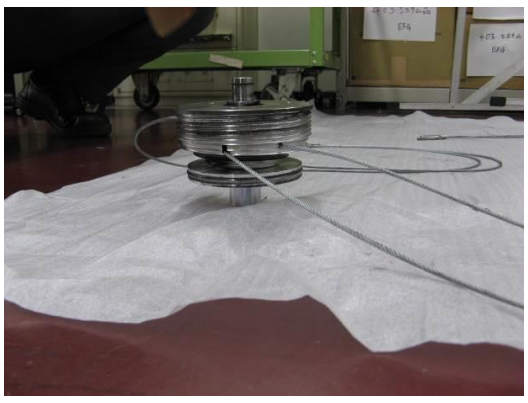


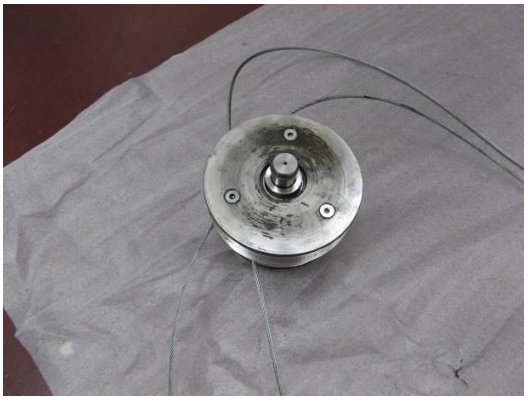
VIII. Remove the Helical Cam ASSY (including 4 pieces of wire ropes) from the Fixed Column. When Helical Cam ASSY is removed, the suction surface of the magnet lock for arm up/down movement appears inside the Fixed Column. Perform degreasing of the suction surface.



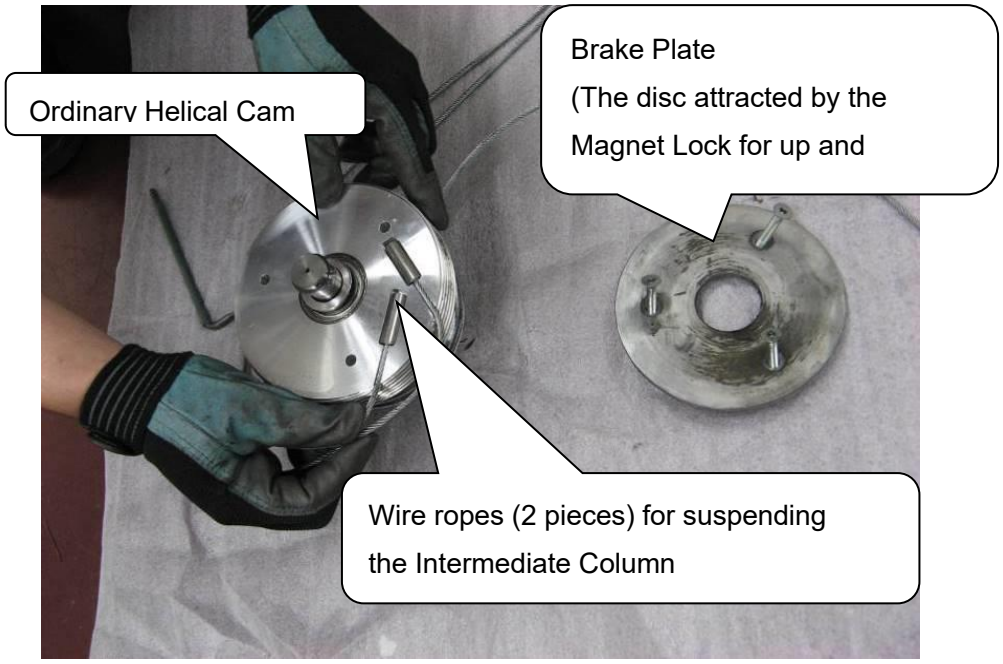
Helical Cam ASSY



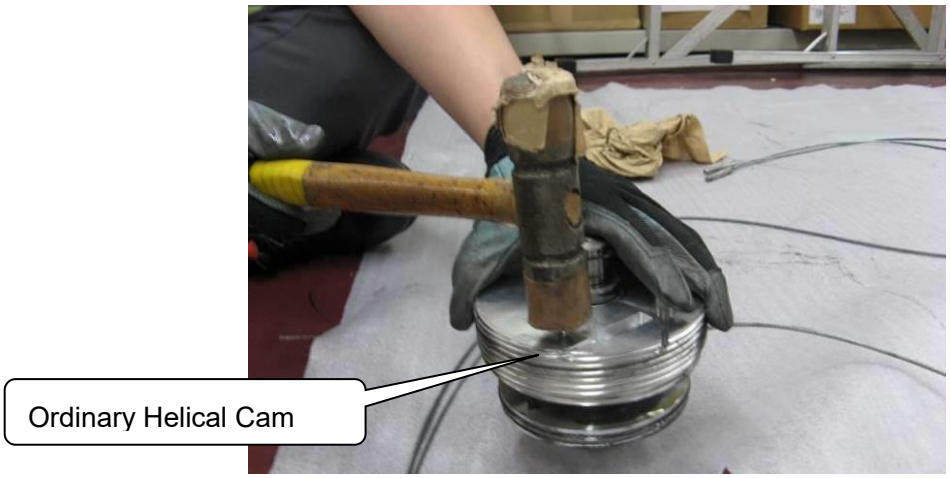




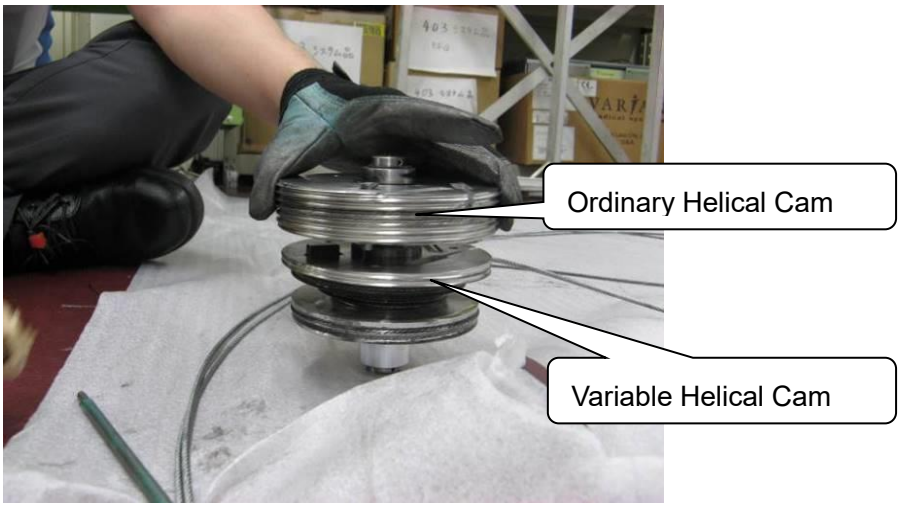
IX. Remove the Brake Plate by removing 3 countersunk screws and then remove two wire ropes for suspending the Intermediate Column, which have been fit in the hole of the Ordinary Helical Cam (whose radius is constant pulley). Pay attention not to make a mistake to replace because the length of ropes is different by approximately 5 cm.

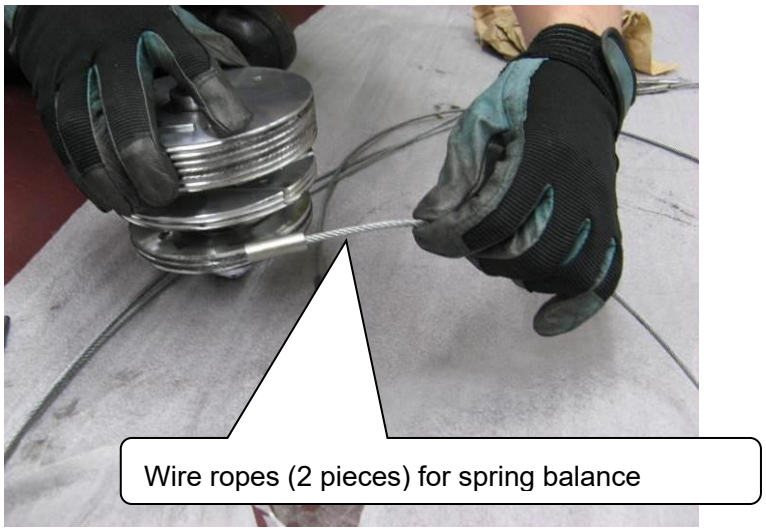


X. Next, displace the Ordinary Helical Cam in the axial direction in order to remove the wire ropes for spring balance. If the key is too hard and the Ordinary Helical Cam cannot be displaced, screw in the countersunk screw for fixing the Brake Plate again and strike with a hammer. (The thread hole for fixing the brake plate has been machined as to be the Variable Helical Cam (i.e. the pulley whose radius changes in helical form) that is located beneath the Ordinary Helical Cam.)

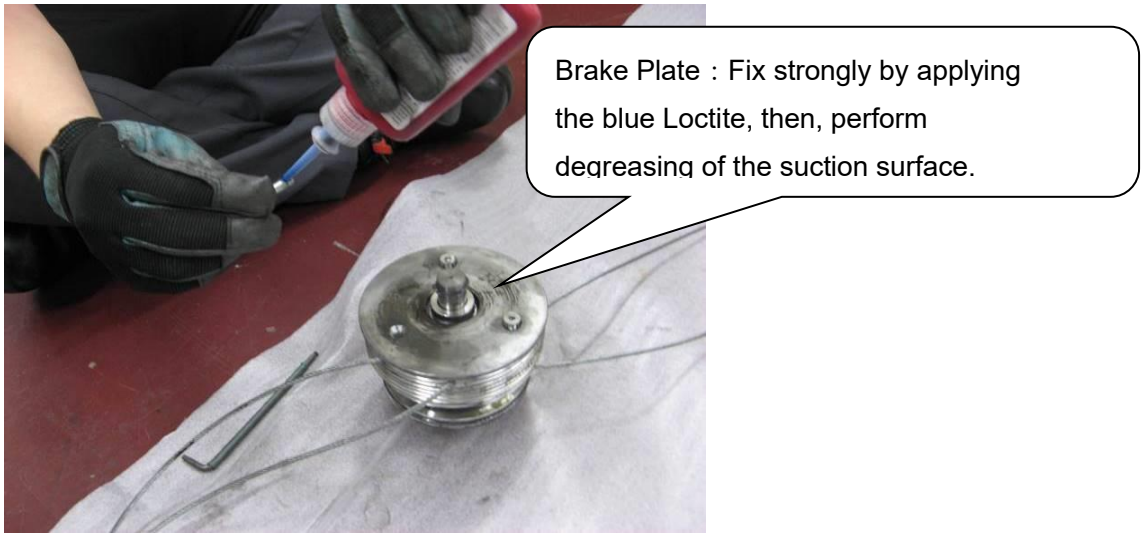


XI. Displace the Ordinary Helical Cam upward and then remove two wire ropes for spring balance which have been fit in the hole of Variable Helical Cam. Pay attention not to make a mistake to replace because the length of ropes is different by approximately 5 cm.

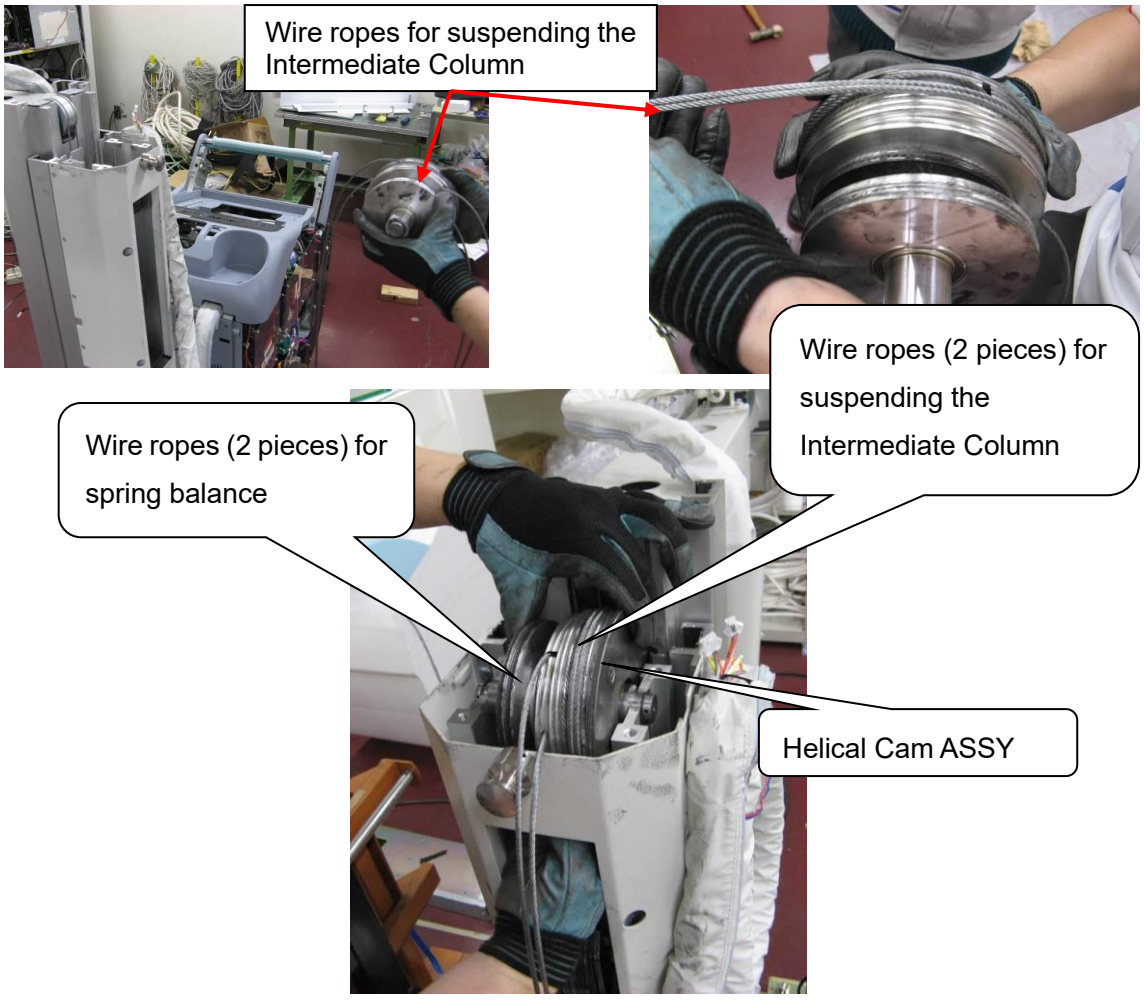




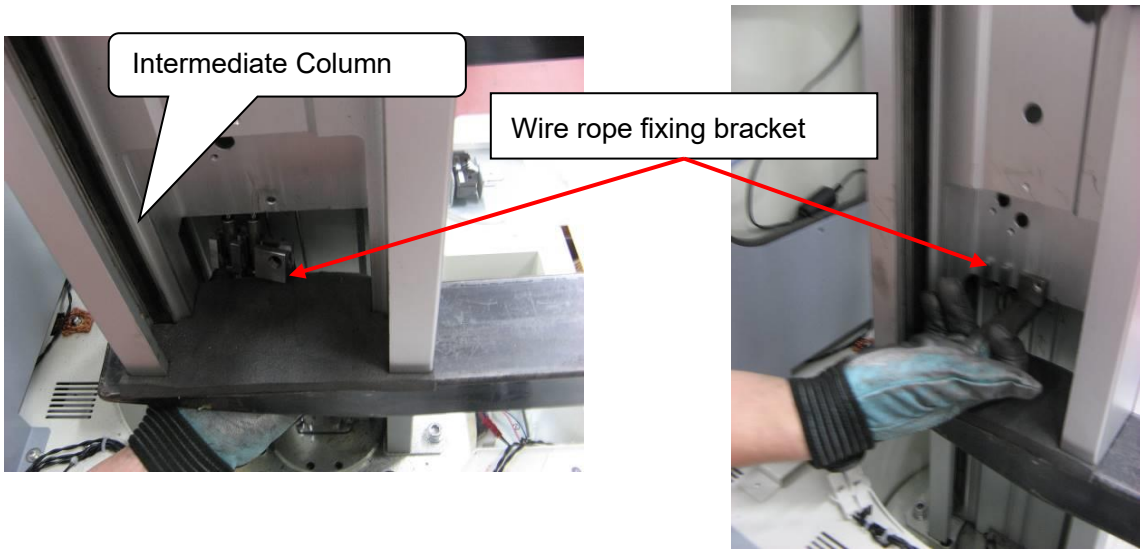
XII. Replace the wire ropes with new ones and assemble in the reverse procedure. To install the Brake Plate, make sure to adjust the phase in rotation direction because there is a back-facing hole on the rear surface of Brake Plate to avoid the interference with the terminal fittings of the wire rope. As well, apply the blue Loctite to the fixing bolts to fix Brake Plate strongly. Be careful, because if this part is loosened, the wire rope may come off from the Helical Cam ASSY. Also, perform degreasing of the suction surface of the Brake Plate. Replacement of the Brake Plate together with the wire ropes is recommended. (Generally, when the wire ropes are replaced, the machine is maybe used for many years. Therefore, the suction surface of the Brake Plate is maybe already damaged.)



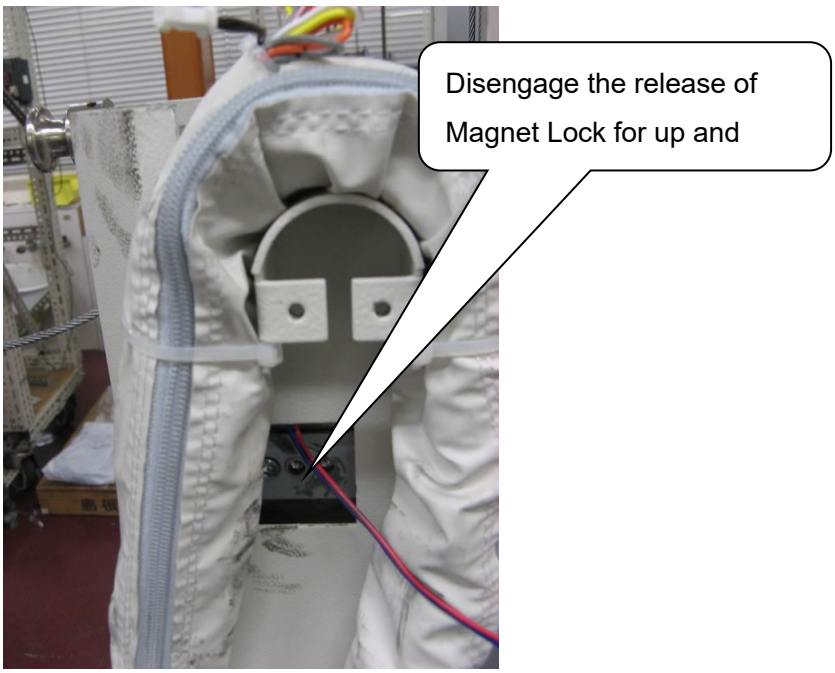
XIII. Attach the Helical Cam ASSY to the Fixed Column. At this time, wind the wire ropes for suspending the Intermediate Column around the Ordinary Helical Cam one time in advance.



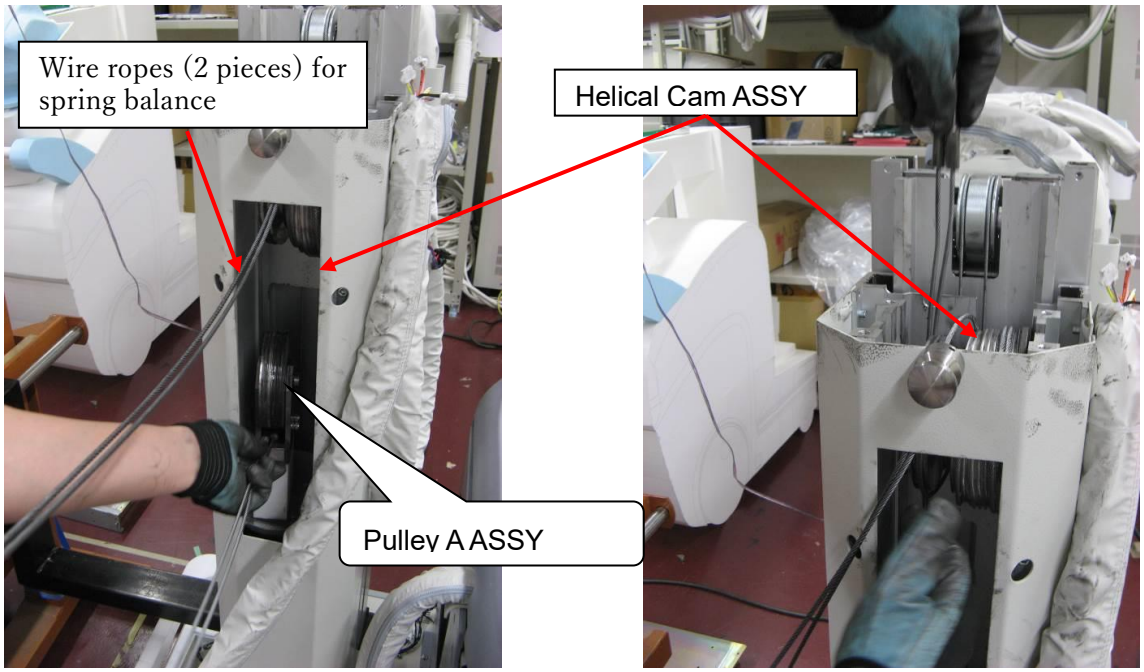
XIV. Fix the wire ropes for suspending the Intermediate Column again to the bottom end of Intermediate Column. Adjust so that the protruding rate of the wire rope end against the fixing bracket becomes the same dimension as that before the replacement. (Both two ropes)

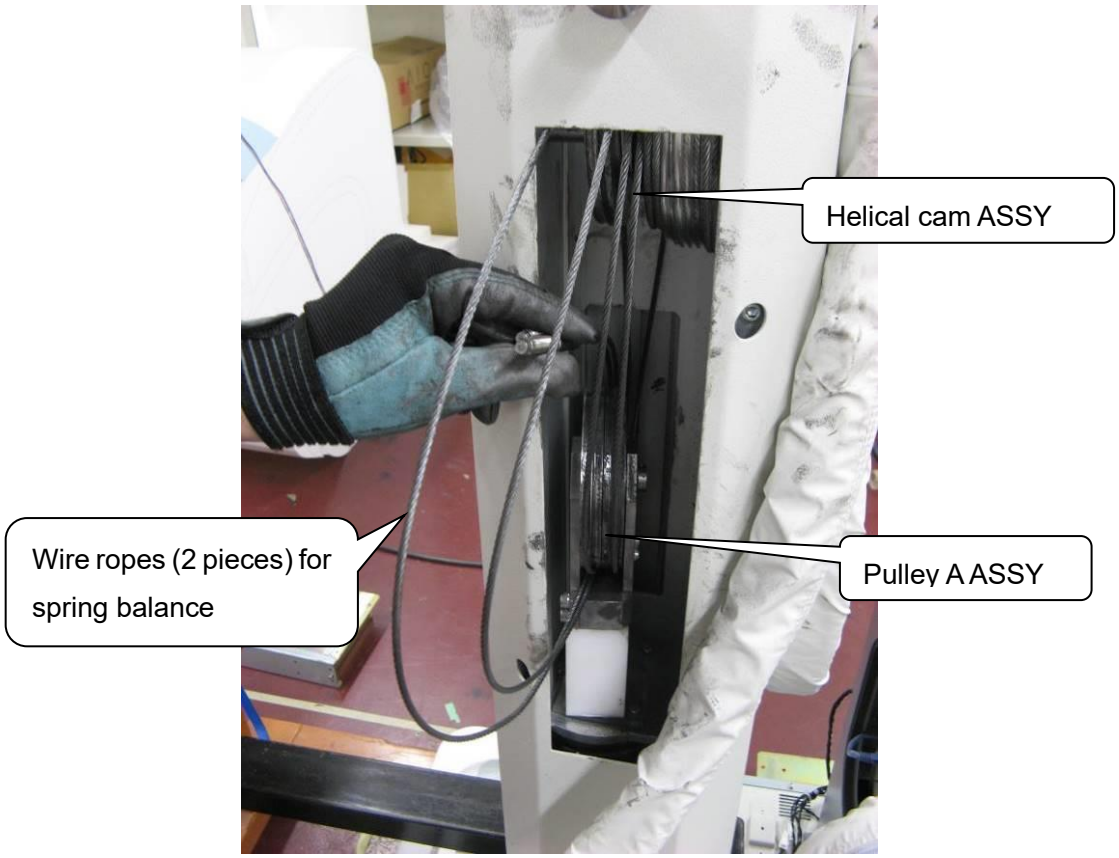


XV. Release the jack up of the Magnet Lock for up and down motion of the Arm to make the brake engaged so that the wire rope can be wound on the Helical Cam ASSY easily.

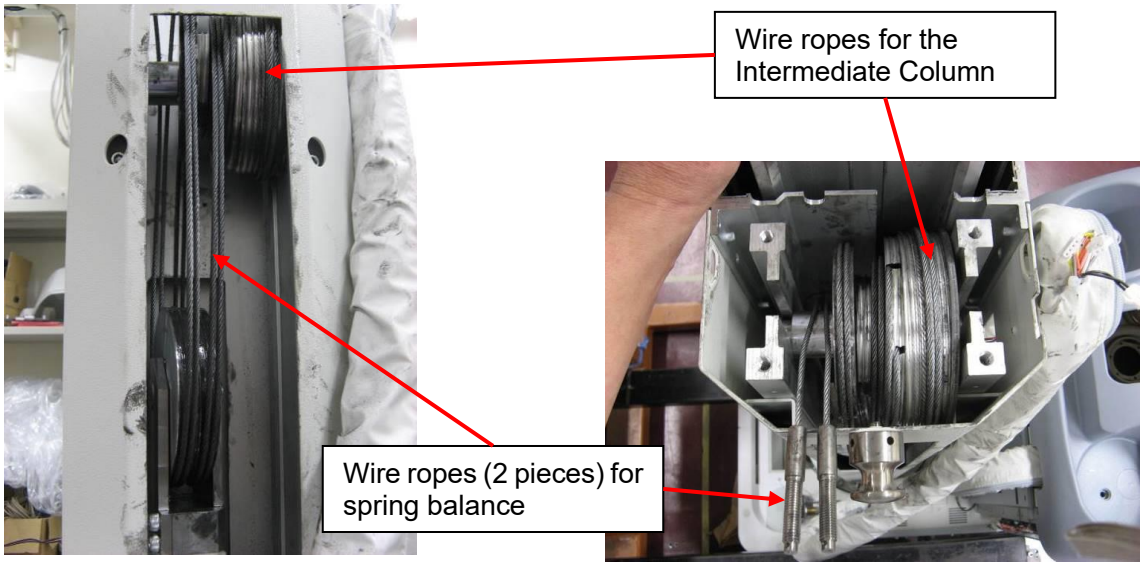


XVI. Wind the wire ropes (2 pieces) for spring balance onto the Pulley A ASSY and Helical Cam ASSY to the former states.

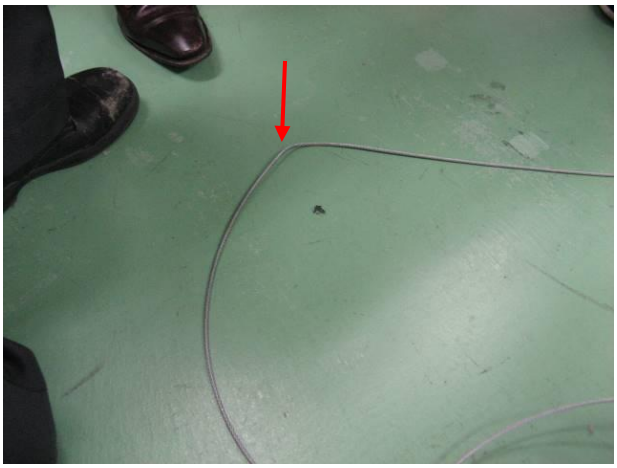




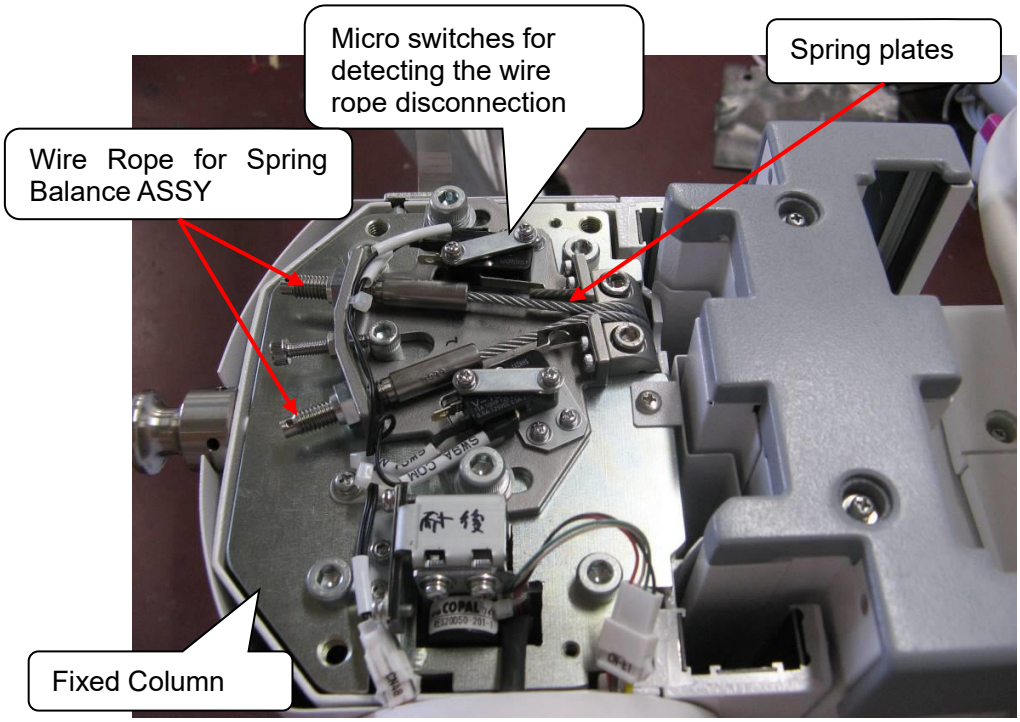
XVII. Check that the wire ropes do not run on the groove of pulley or each wire rope is not over lapped or they do not run off.



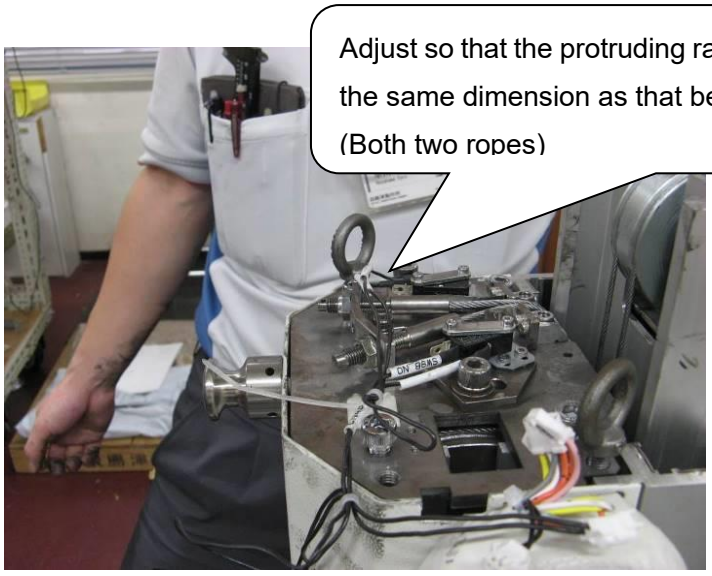
Note that the wire ropes can become deformed as shown on the below picture if the Arm up/down operation is done when the wire rope runs on the groove of pulley or runs off. In that case, the wire rope can be damaged earlier, and lifetime will become shorter.



XVIII. When the wire ropes for Spring Balance ASSY are fixed on the top surface of the Fixed Column, take care not to injure the L-shape spring plates with 0.3mm thickness, which are located between the wire ropes and the actuator of the micro switches for detecting the wire rope disconnection.



XIX. Perform the tension adjustment of the wire ropes for spring balance. Adjust so that the protruding rate of the wire rope end to the fixing bracket becomes the same dimension as that before the replacement.



XX. Recover the machine to the original condition in reverse order, then, confirm that Arm up/down operation can be done between the upper and lower limit position smoothly. Don't forget to remove the Spacer (Jig for wire rope replacement) which was inserted into Spring Balance ASSY. As well, apply the lubricant "Wirol R", which is included in 566-21399-01 "Wire rope set 1" or 566-21399-02 "Wire rope set 2", to the new wire ropes.

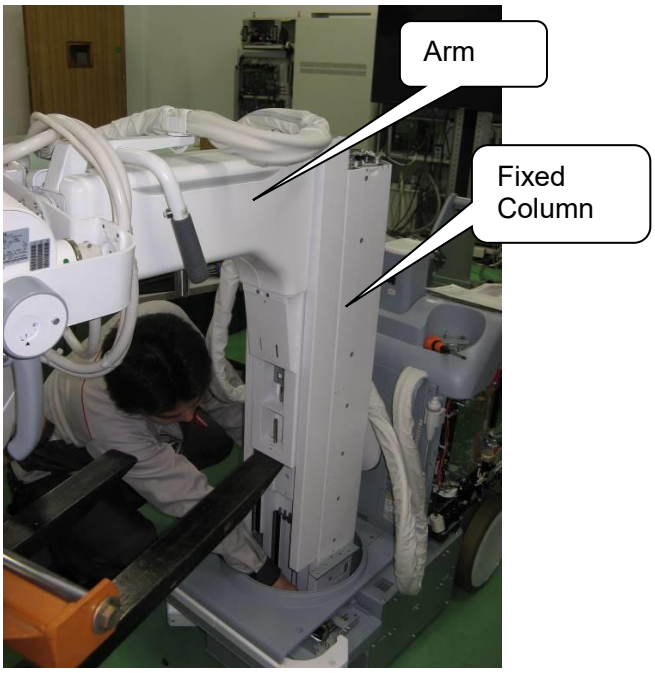
5.1.3 Replacement of Wire Rope for Intermediate Column ASSY

The procedure of replacing the Wire Rope Set for Intermediate Column ASSY (P/N:566-21399-02 "Wire Rope SET 2") is explained below. (This set includes 2 wire ropes for suspending the Arm.)

I. Support the Arm with a jig or lifter etc. so that no load is applied to the wire ropes.

＜How to support the Arm by using the Jig＞

- Support the Arm with lifter etc.

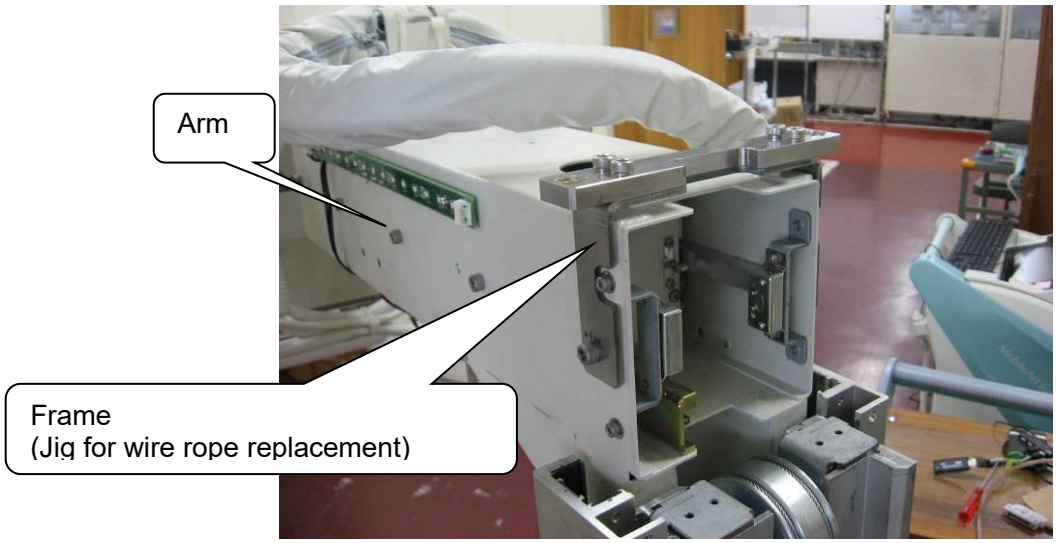


5. Parts replacement

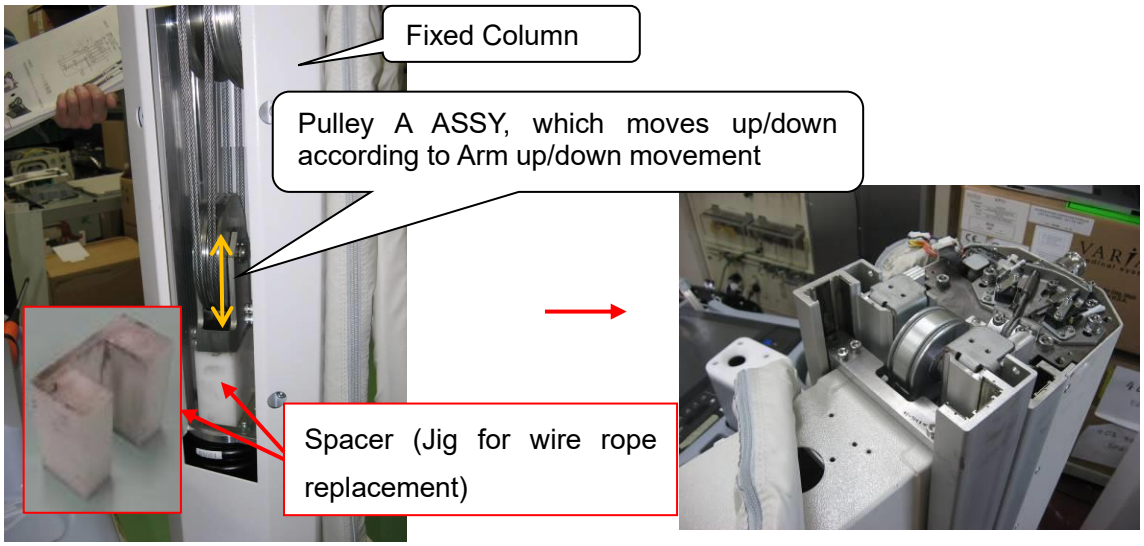
＜How to support the Arm by using the Jig＞

- Wire replacement jig for Intermediate Column ASSY shown below, "P/N:566-21946 Wire Replacement Jig, Arm", use to support.

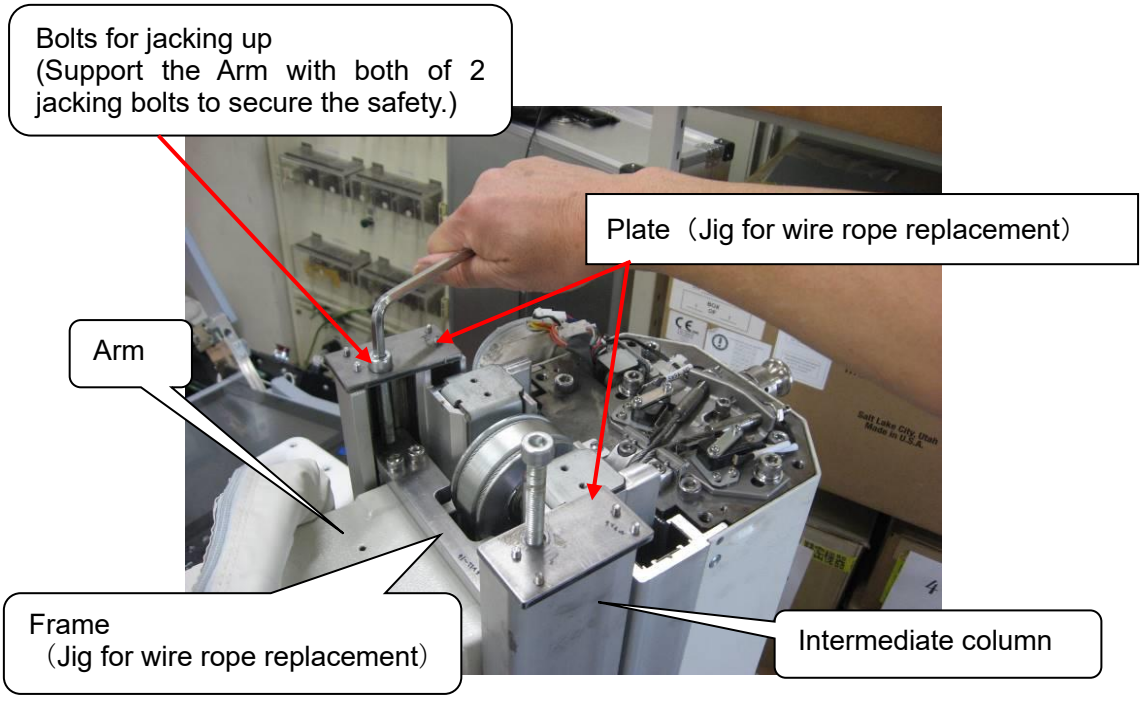
- Lift up the Arm, then, fix the Frame (Jig) on the top of the Arm as shown on the below picture.At this time, if the spacer (Jig for wire rope replacement) is inserted in the gap of Spring Balance ASSY in Fixed Column, remove the spacer once in order to lift up the Arm, then, fix the Frame (Jig).



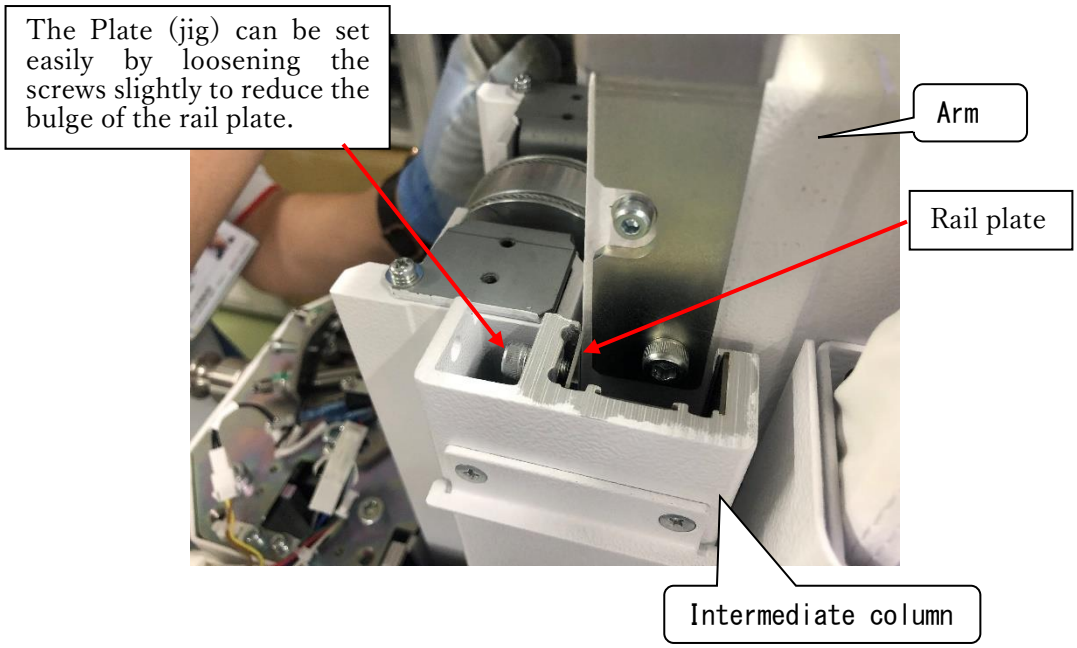
- Lift down the arm once and insert the spacer (Jig for wire rope replacement) into the gap of Spring Balance ASSY in Fixed Column in order to locate the Arm and Intermediate Column to the convenient position for wire rope replacement, then, lift up the Arm until the bottom of Pulley A ASSY touches the top of the spacer.



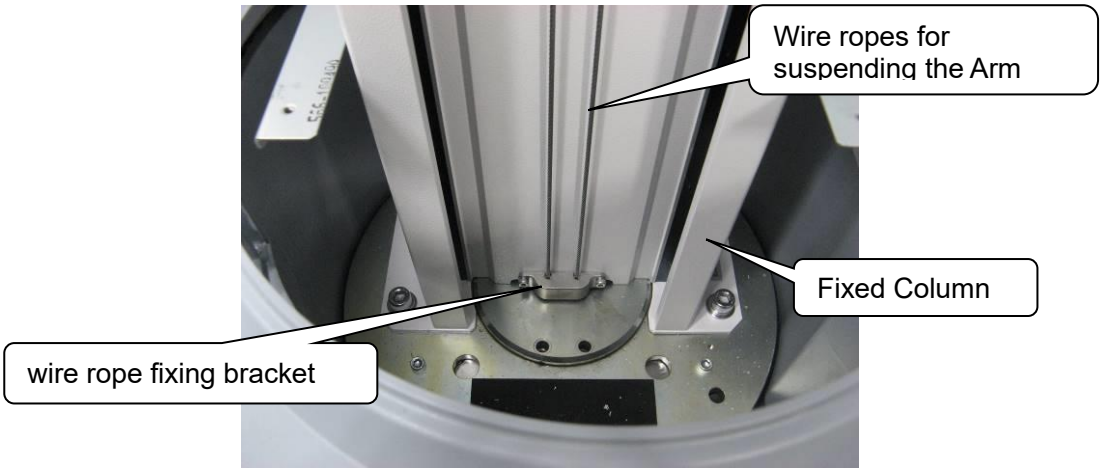
- Fix the Plate (Jig for wire rope replacement) on both left and right sides of the upper end of the Intermediate Column as shown on the below picture. Then, screw both left and right bolts for jacking up into the Frame (Jig) in order to lift up the Arm. (The Arm can be lifted up even with only one of jacking bolts, however, make sure to support the Arm with both of 2 jacking bolts to secure the safety.)



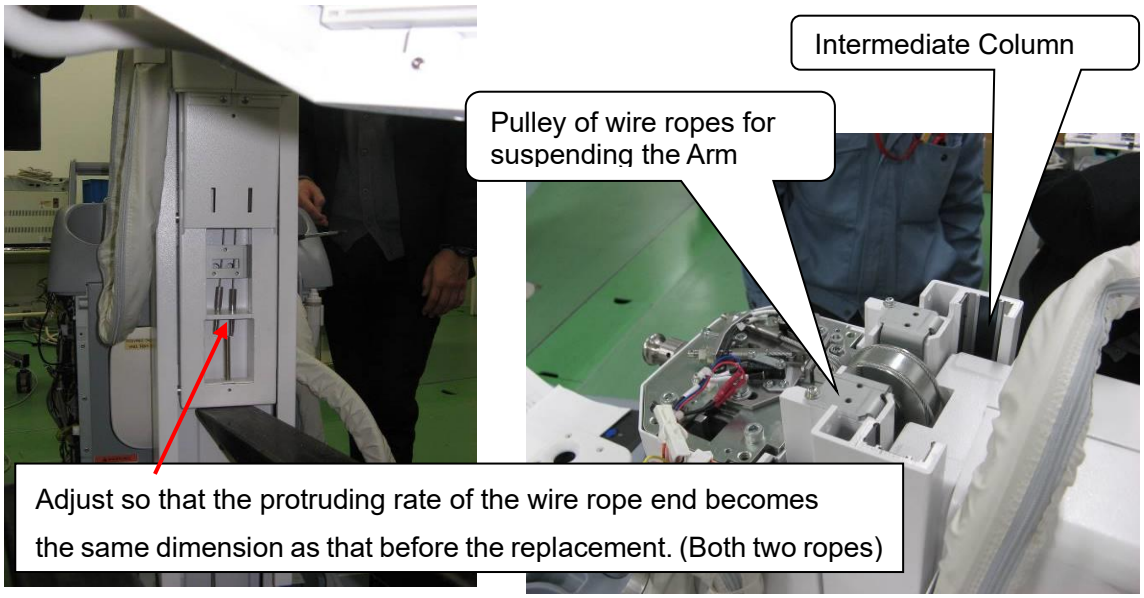
At this time, if the bulge of the Rail Plate at the top end of the Intermediate Column prevents you from setting the Plate (jig for wire rope replacement), loosen the screw slightly.



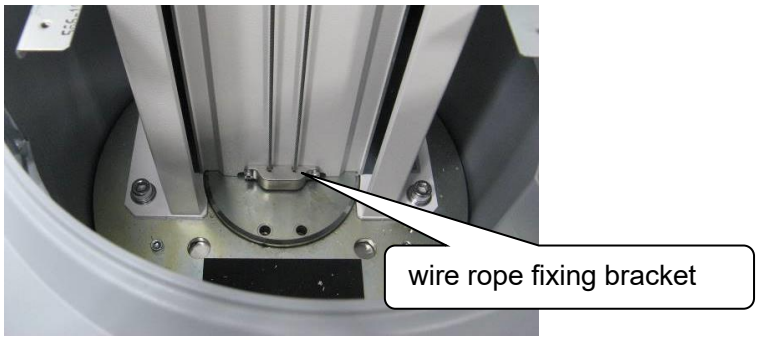
II. Remove the wire rope fixing bracket below the Fixed Column so that the wire ropes (2 pieces) can be free to the Fixed Column.



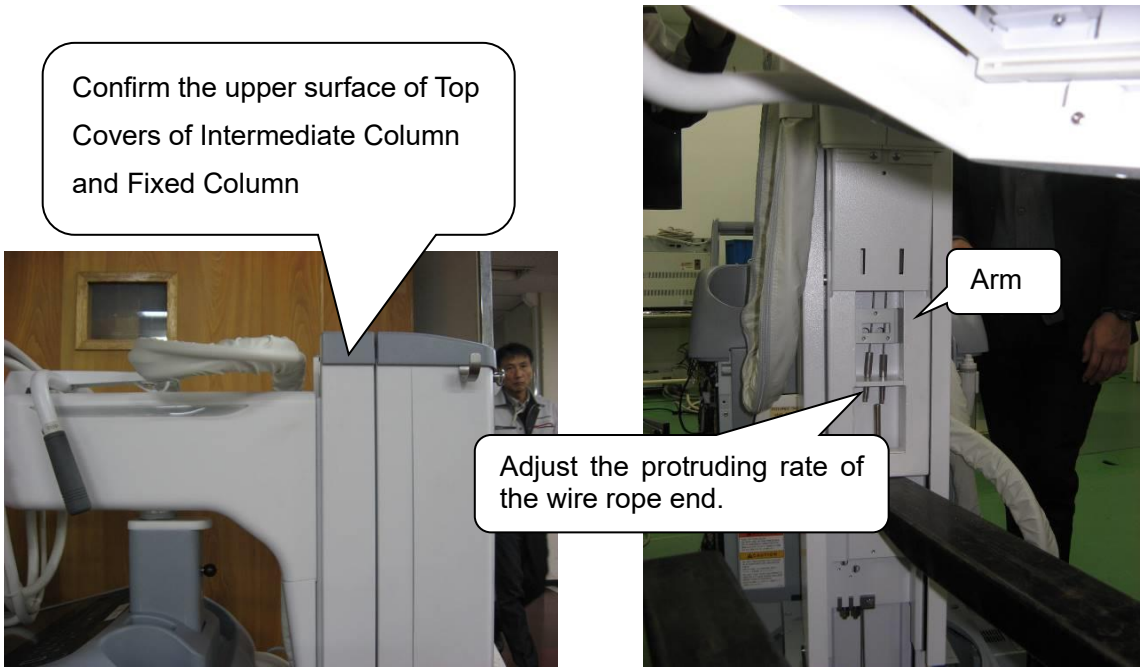
III. Remove the wire rope fixing bracket below the Arm so that the wire ropes (2 pieces) can be free to the Arm. **However, measure the protruded amount of the wire rope end against the fixing bracket to match the tension adjustment after the replacement of wire ropes.** Then, pull out the wire ropes from the pulley at the top of the Intermediate Column upward and replace to the new one. Adjust so that the protruding rate of the wire rope end becomes the same dimension as that before the replacement.



It is easier to fix the wire rope fixing bracket to the Fixed Column after removing the lifter from the Arm so that the tension is applied to the wire ropes (2 pieces).



IV. Confirm that the upper surfaces of Top Covers of Intermediate Column and Fixed Column are at the same height when the Arm is stored to the driving position. If they are not at the same height, adjust the protruding rate of the wire rope end. The height of the Intermediate Column decreases when the protruding rate of the wire rope end increases.



V. Recover the machine to the original condition in reverse order, then, confirm that Arm up/down operation can be done between the upper and lower limit position smoothly. Don't forget to remove the Spacer (Jig for wire rope replacement) which was inserted into Spring Balance ASSY.

VI. As well, apply the lubricant "Wirol R", which is included in 566-21399-01 "Wire rope set 1" or 566-21399-02 "Wire rope set 2", to the new wire ropes.

5.2 Replacement of FSR ASSY

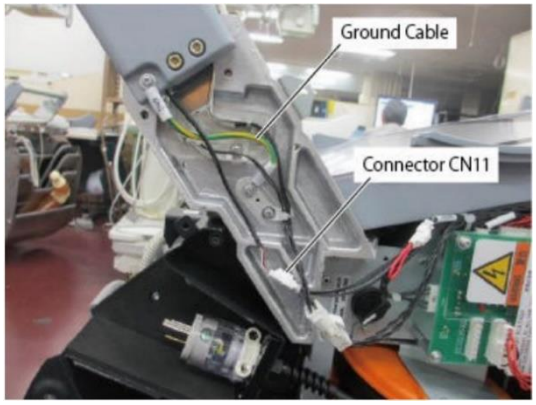
(Referendce work time 1 person × 1.5 hours in case of replacing FSR at both sides)

Make sure to set to OFF the main breaker and discharge the main capacitor before starting the work.

I. Turn off the power and turn off the main breaker.

II. Refer to "How to Open/Close Covers" in the installation manual to remove the left and right insidecovers and side covers.

III. Disconnect the connector (CN11) on the right side of the handle, the ground cable, and the tie wrap.

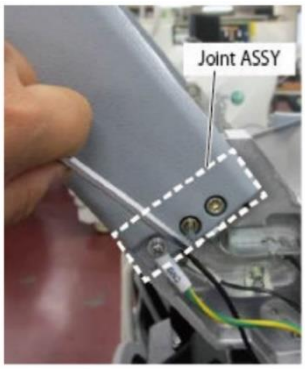


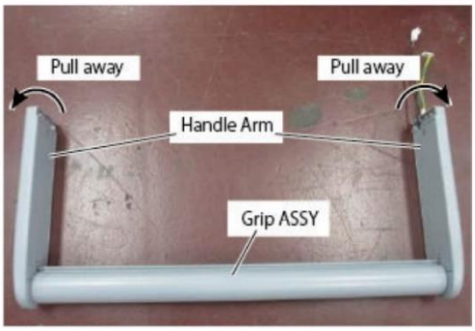
If [Changing the Grip Bar Height Kit] option is installed, remove the left and right connectors (CN38A/B).

IV. Remove two attachment bolts from each side of Joint ASSY fixing the handle arm, and remove the handle arm unit.

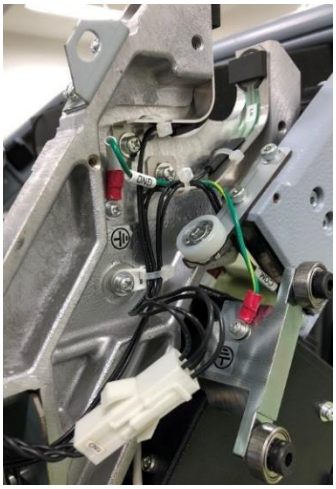
Note:

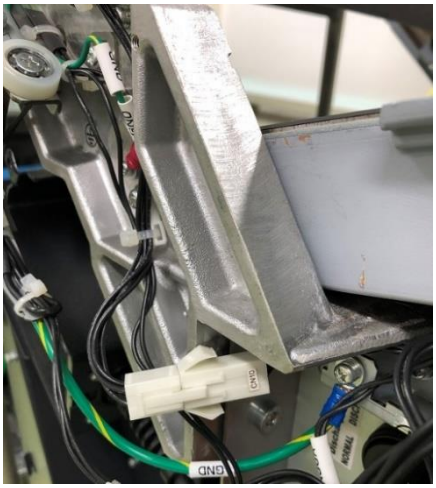
The handle arm is set in the projection of Joint ASSY. Pull both ends of the handle arm away from the projection to remove it.



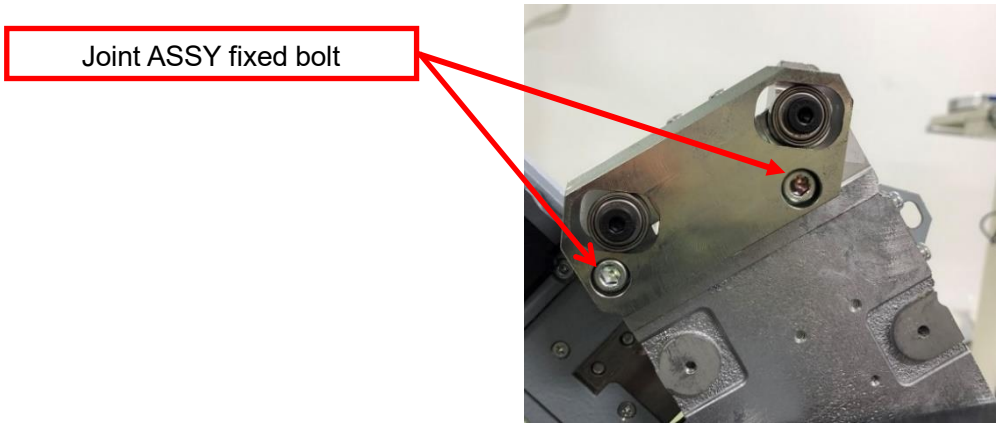


V. Disconnect the connectors for FSR ASSY Left (CN9) and Right (CN8). Make sure the cable is free by cutting off all the ties holding the associated cable.

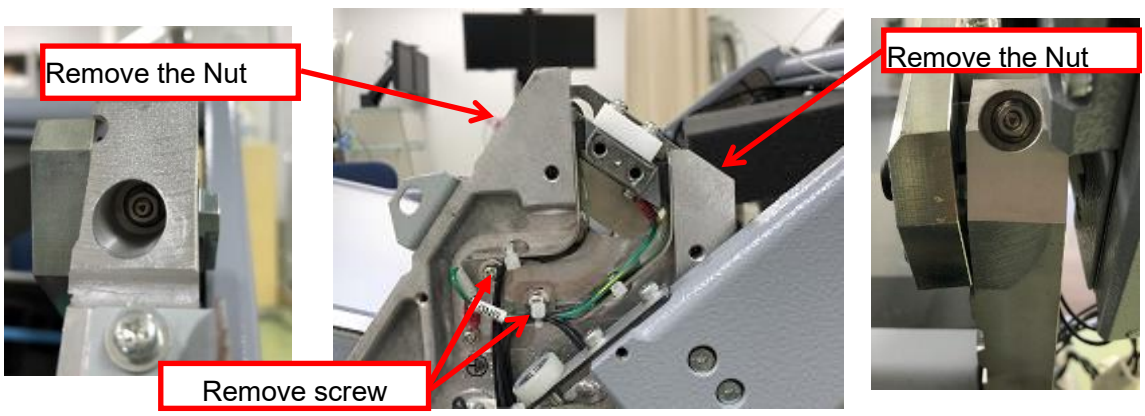




VI. Remove the two hex socket bolts on the bracket securing the joint ASSY and remove the joint ASSY. If [Changing the Grip Bar Height Kit] option is installed, remove the Ground cable which is attached to Joint ASSY.



VII. Remove the 2 plus screws and 2 Nuts which fix the FSR ASSY.



5. Parts replacement

VIII. Reverse the removal procedure to install the new FSR and return the unit to its original condition.

Note:
Apply Loctite 242 (Blue) to the hex socket bolts used to attach the Joint ASSY retaining bracket and the hex socket bolts used to attach the handle arm to the joint ASSY to prevent them from loosening.

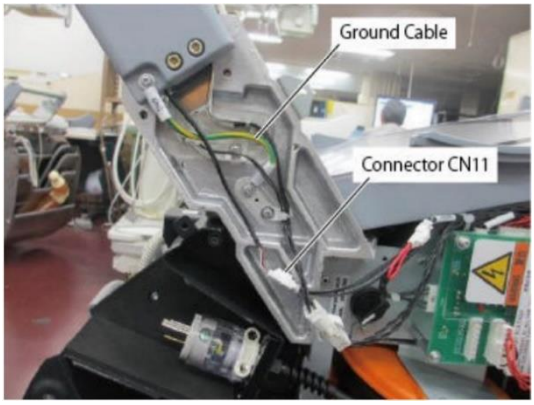
IX. After the assembly is completed, adjust the handle referring to the installation manual "Handle Adjustment".
X. Finally, make sure that you can drive forward, back and turn normally.

5.3 Replacement of the Photo Sensor

(Reference work time 1 person ×30min.)

Make sure to set to OFF the main breaker and discharge the main capacitor before starting the work.

I. Turn off the power and turn off the main breaker.
II. Refer to "How to Open/Close Covers" in the installation manual to remove the left and right inside covers and side covers.
III. Disconnect the connector (CN11) on the right side of the handle, the ground cable, and the tie wrap.

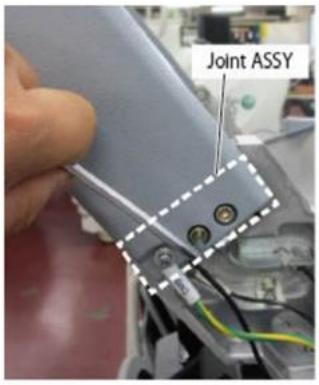


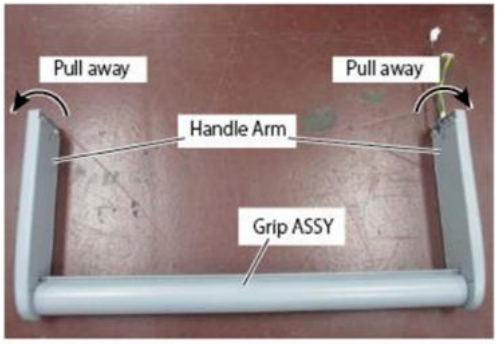
If [Changing the Grip Bar Height Kit] option is installed, remove the left and right connectors(CN38A/B).

IV. Remove two attachment bolts from each side of Joint ASSY fixing the handle arm and remove the handle arm unit.

Note:

The handle arm is set in the projection of Joint ASSY. Pull both ends of the handle arm away from the projection to remove it.

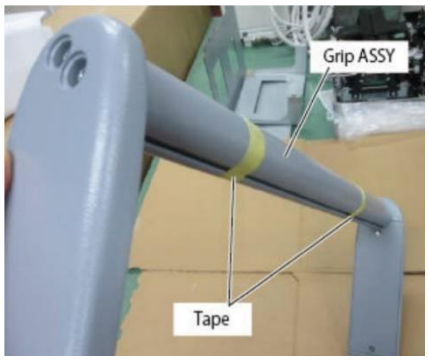




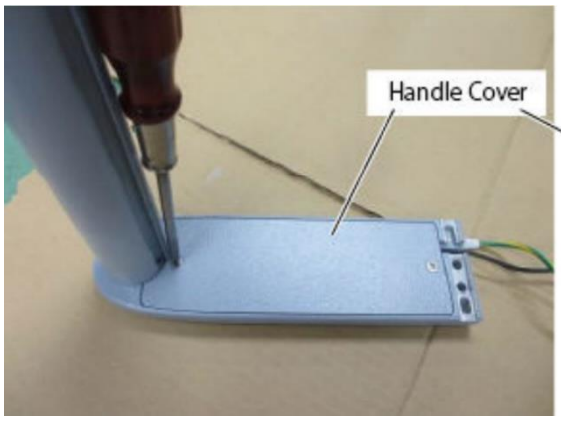
5. Parts replacement

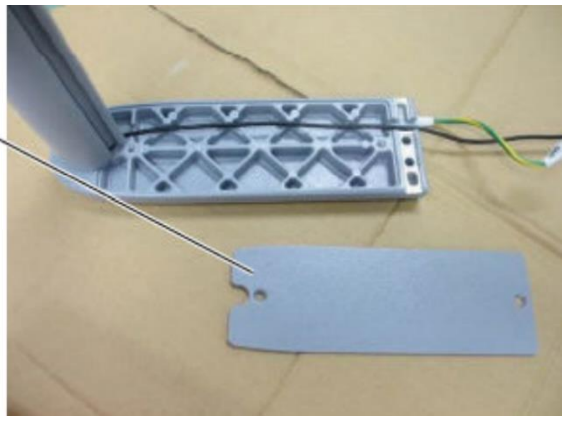
V. Secure Grip ASSY with masking tapes as shown below.

Note:
This step must be performed. If Grip ASSY turns inside out, the internal parts may spring out.

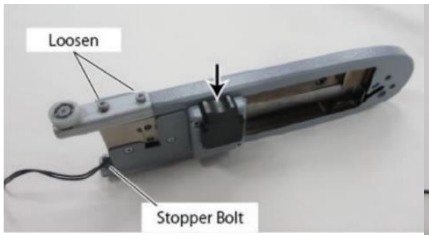


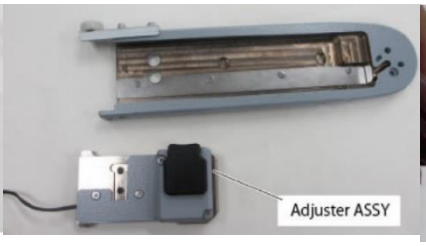
VI. Remove the handle cover from the cable-wired side (right side) of the handle

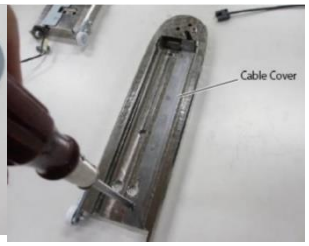




Note:
If [Changing the Grip Bar Height Kit] option is installed, remove the stopper bolt of the handle arm R and remove the adjuster ASSY while pressing the black button before removing the cable cover.

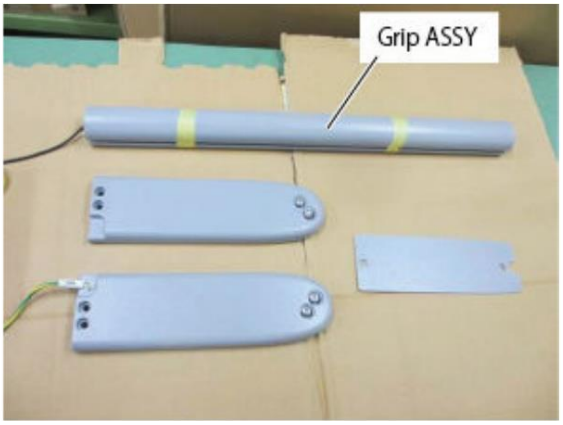




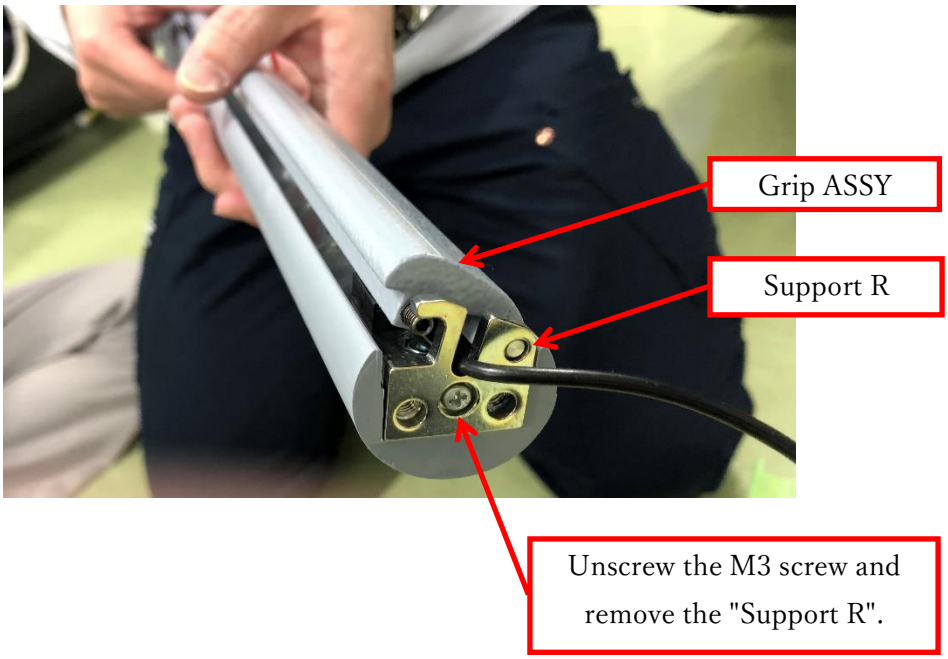


5. Parts replacement

VII. Remove two fixing bolts on each side of Grip ASSY, and disassemble the handle arm from Grip ASSY.



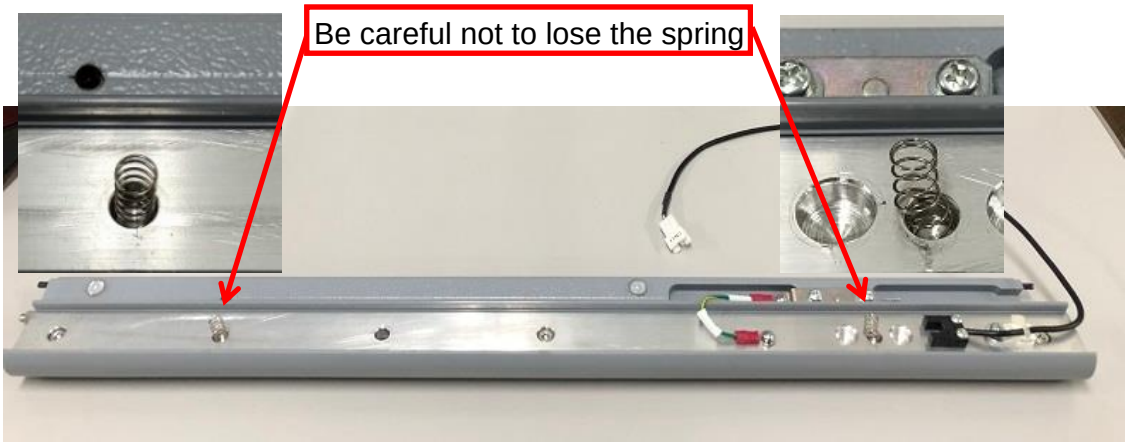
VIII. Unscrew the M3 screw and remove the "Support R" which is attached to the right end surface of the Grip ASSY.



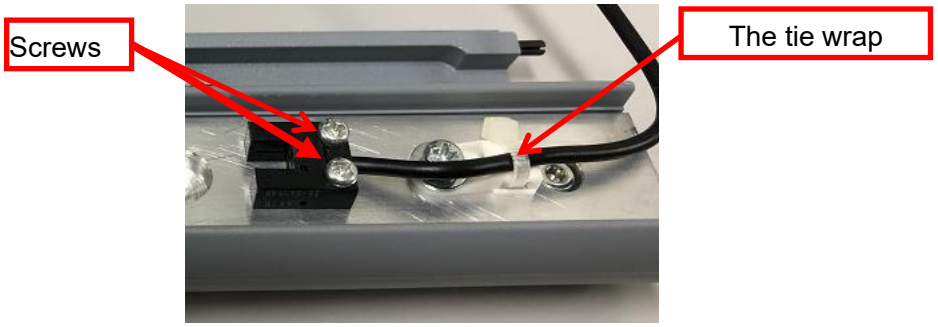
While grasping the Grip ASSY, remove the securing tape attached in step 5 and gently open the Grip ASSY.

Note:

There are 2 small springs inside, so please do not lose them.

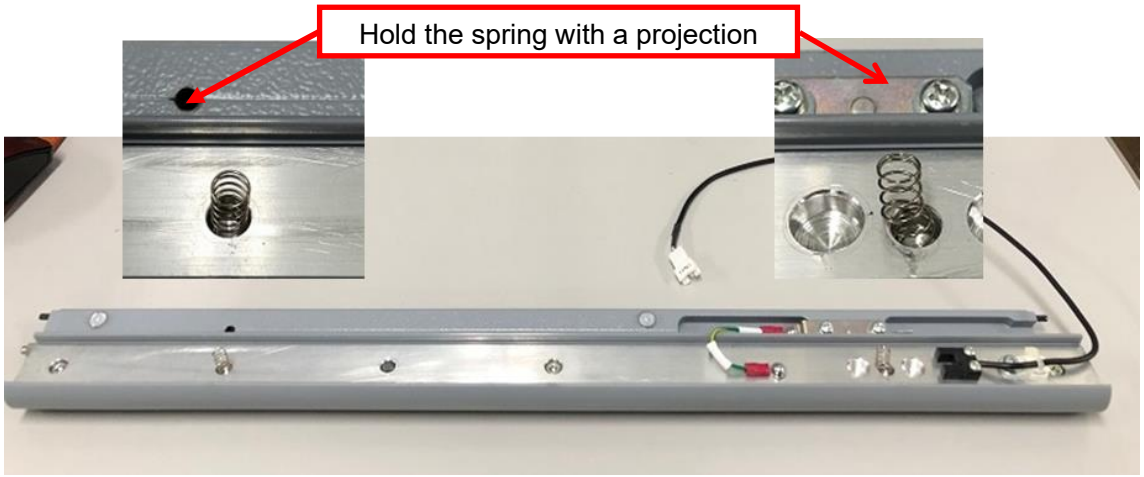


IX. Remove the tie wrap that secures the Photo Sensor cable and remove the two securing screws to remove the Photo Sensor.



X. Secure the new Photo Sensor with two screws and secure the cable with the tie wrap.

XI. Set the spring as shown in the figure, assemble the Grip ASSY, and fix it with masking tape to prevent the spring from coming loose.



XII. Assemble in the reverse order.

Note:

Be careful not to pinch the Photo Sensor cable when attaching the Handle arm R to the Grip ASSY.

To prevent the handle arm from loosening, apply LOCTITE242 (Blue) to the hex socket bolts that attach the Handle arm to the Grip ASSY and the hex socket bolts that attach the Handle arm to the Joint ASSY

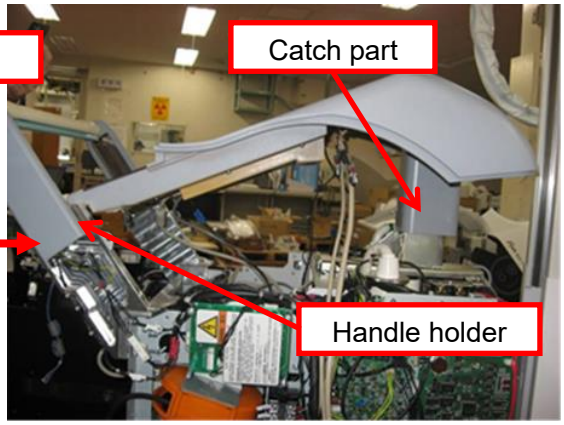
XIII. Turn on the system and check that the brake release action is normal by grasping the Grip ASSY. Also, confirm that the brake operates normally by releasing the Grip ASSY.

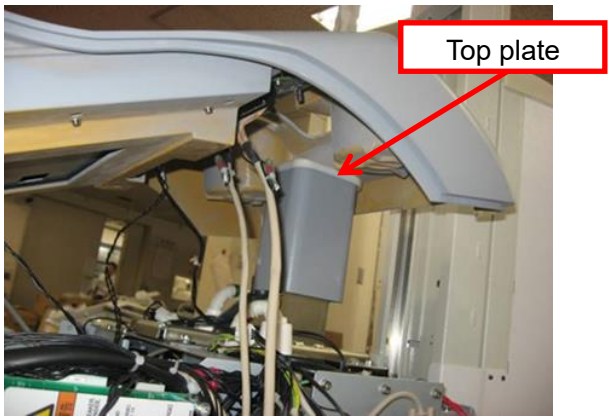
5.4 Replacement of LCD

(Reference work time 1 person × 1 hour.

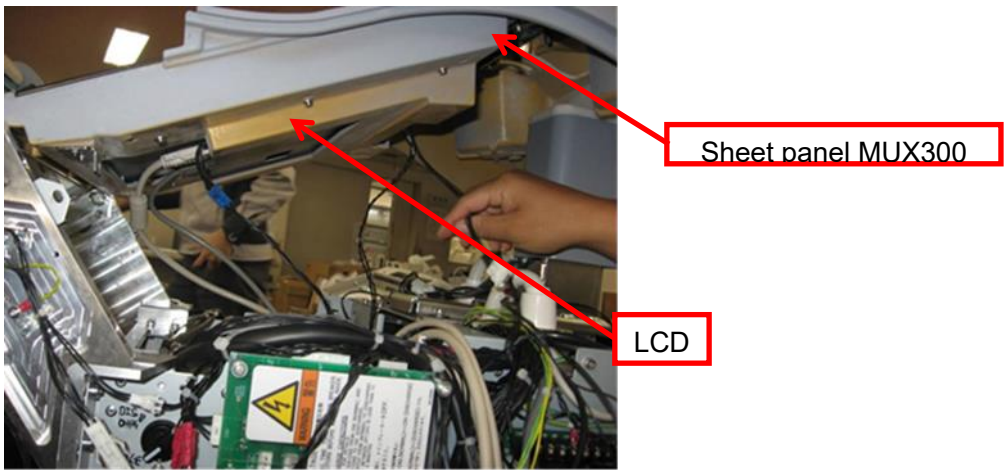
Make sure to set to OFF the main breaker and discharge the main capacitor before starting the work.

I. Refer to "How to Open/Close Covers" in the installation manual to remove the left and right inside covers and side covers, open the FPD box, remove the Top cover. Lift it up, place it on the Catch and handle holder. (If you lift the top cover straight up, it will interfere with the top plate of the catch. Lift it diagonally.)

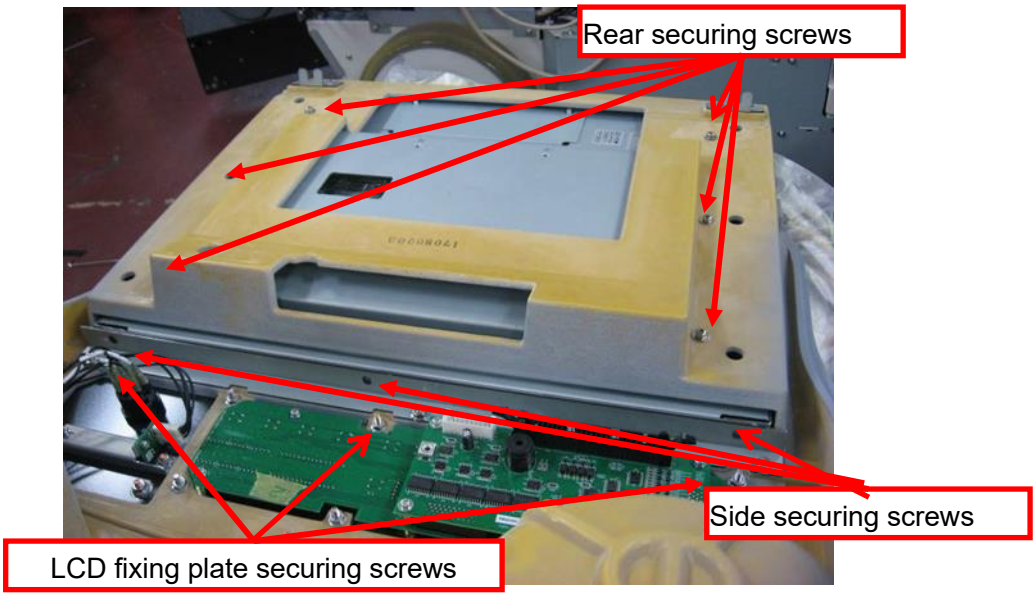




II. Disconnect all cables connected to the Sheet panel MUX 300 and LCD and remove the Top cover.



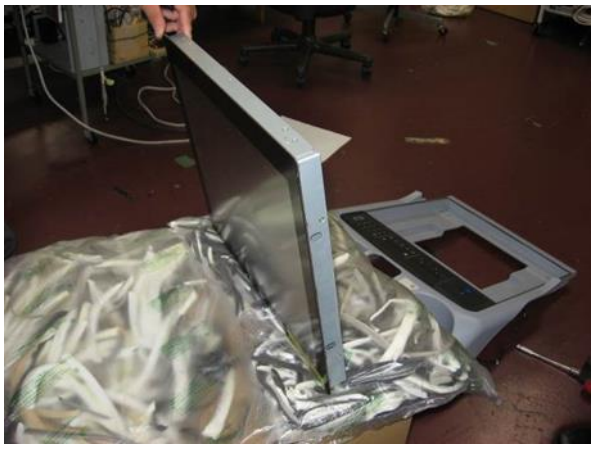
III. Turn over the top cover and place it on a cushion, etc., and loosen the three LCD bracket screws. Then remove the three LCD side securing screws and the six rear securing screws in that order.



IV. Remove the LCD from the Top cover and remove the two LCD mounts from the LCD. Install the removed LCD mount on a new LCD.

5. Parts replacement

V. Reverse the steps to install the new LCD on the Top cover.



VI. Fit the LCD end face to the step on the top cover surface.

VII. For fine adjustment of the LCD mounting, loosen the LCD mounting screws through the following six holes.

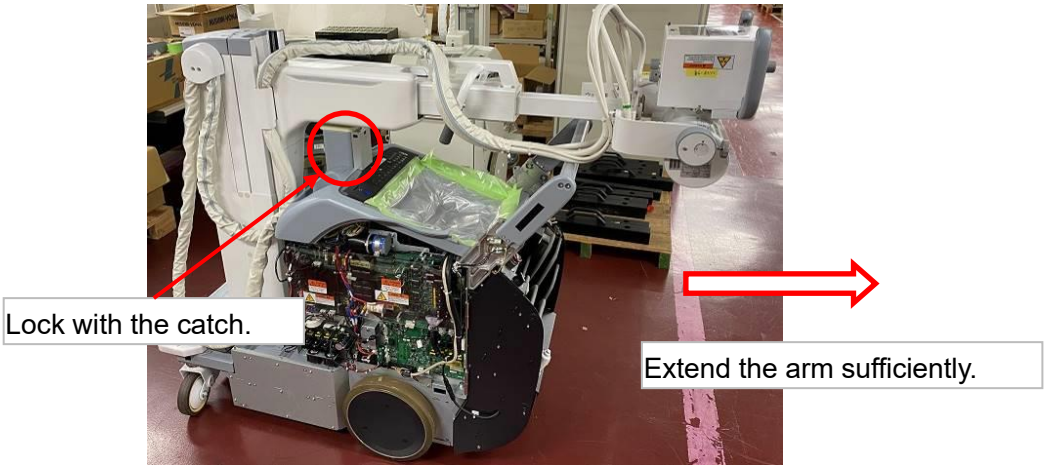
VIII. Refer to "Touch Screen Monitor Calibration" of the installation manual, perform calibration of the touch panel, and confirm that the touch panel operates normally.

5.5 Replacement of the X-ray tube

(Reference work time: 2 persons x 2.5 hours)

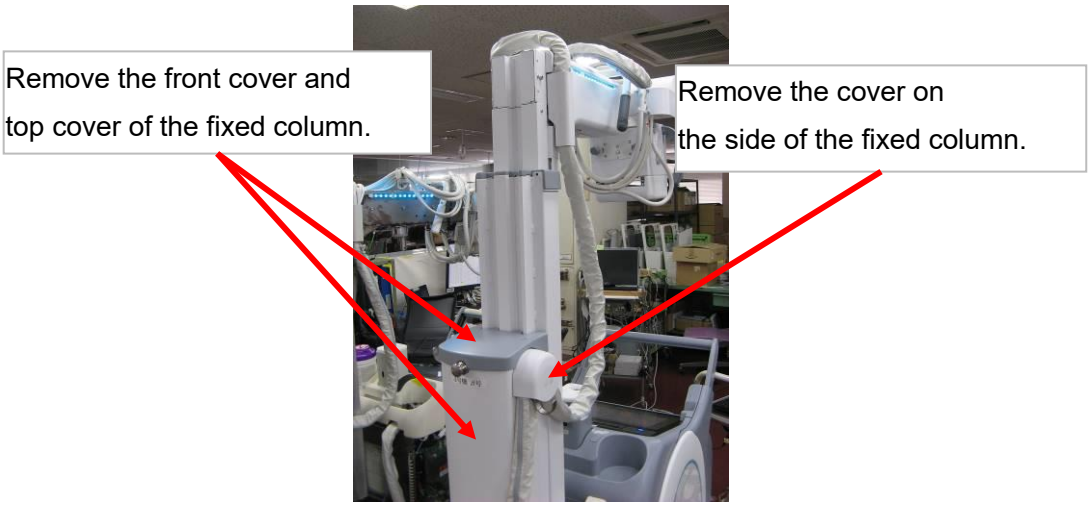
Make sure to set to OFF the main breaker and discharge the main capacitor before starting the work.

I. Lock to the catch with the arm fully extended. If the arm is not pulled out more than 1/3 of the whole, it will not be locked by the catch part, so extend it sufficiently.
Be sure to fix the vertical movement of the arm before carrying out this operation. There is a risk of losing balance and leading to an unexpected accident.

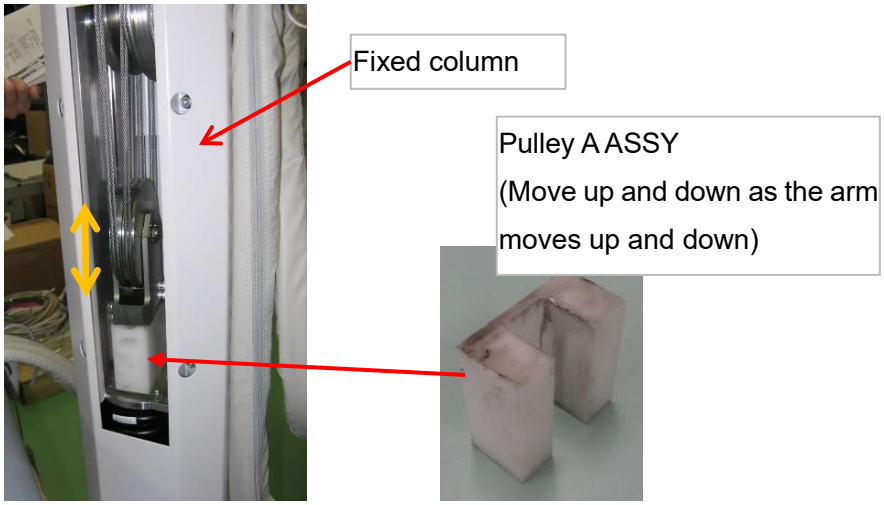


If the arm cannot be extended due to a malfunction, etc., fix the vertical movement of the arm according to the following steps (i) to (ii).

i. Remove the side covers on the fixed columns, then remove the front and top covers on the fixed columns.

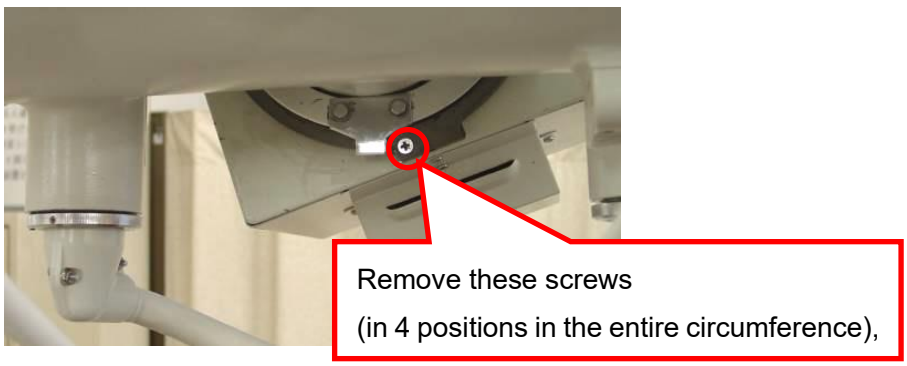


ii. Insert the spacer of the wire rope replacement jig into the gap of the spring balance assembly built in the fixed column to fix the arm so that it does not jump up when replacing the tube. Then lift the arm until the bottom of the Pulley A ASSY touches the top of the spacer.



This spacer is included in both of the following two types of wire rope replacement jigs.
P/N: 566-21942 WIRE REPLACEMENT JIG,COLUMN
→ For wire rope of spring balance assembly
P / N: 566-21946 WIRE REPLACEMENT JIG,ARM
→ For wire rope of Middle column
In addition, individual arrangements are also possible.
Single P / N: 566-20458-01 BLOCK L80

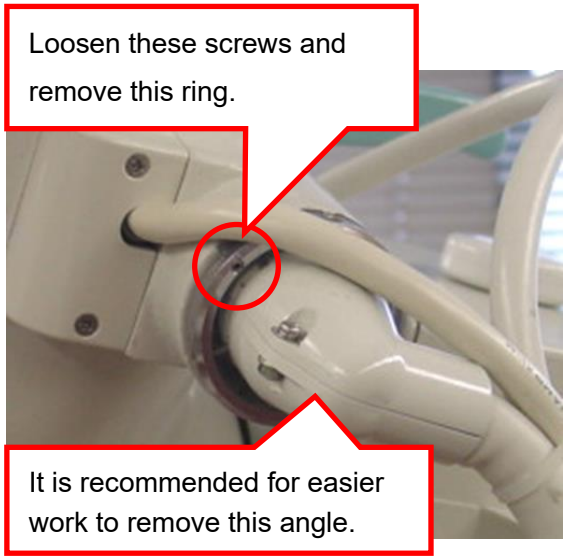
II. Remove the collimator from the rotation unit mounted in the tube. Put the removed collimator on cushion on the main unit or a base.



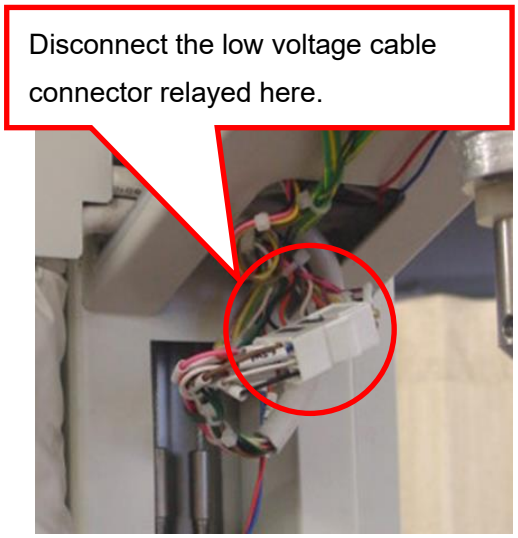
III. Disconnect the high voltage cable from the plug area. Disconnect the low voltage cable from the connector relay area under the arm. At this time, in disconnecting the high voltage cable, loosen the ring rotation stopper (set screws in 4 positions).

It is recommended for easier work to remove also the angle from the high voltage cable.(If this angle remains attached, it is difficult to connect the high voltage cable to the plug.)

High Voltage Cable Fixing Ring

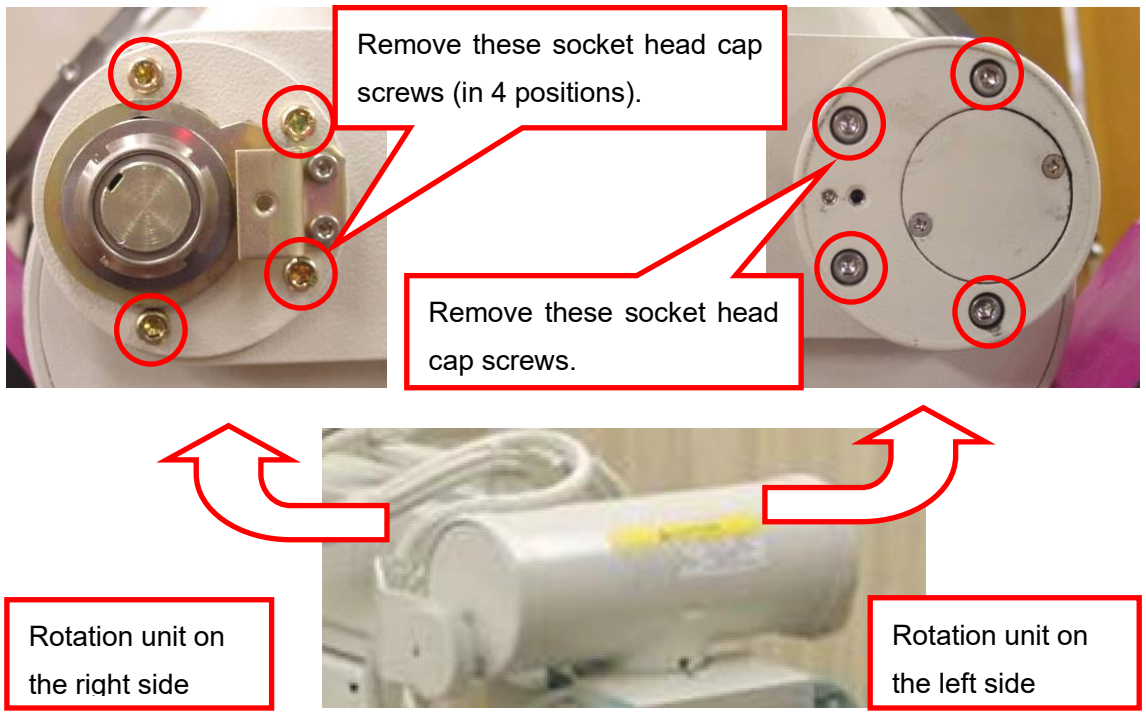


Disconnecting the High Voltage Cable



Connector Relay Area

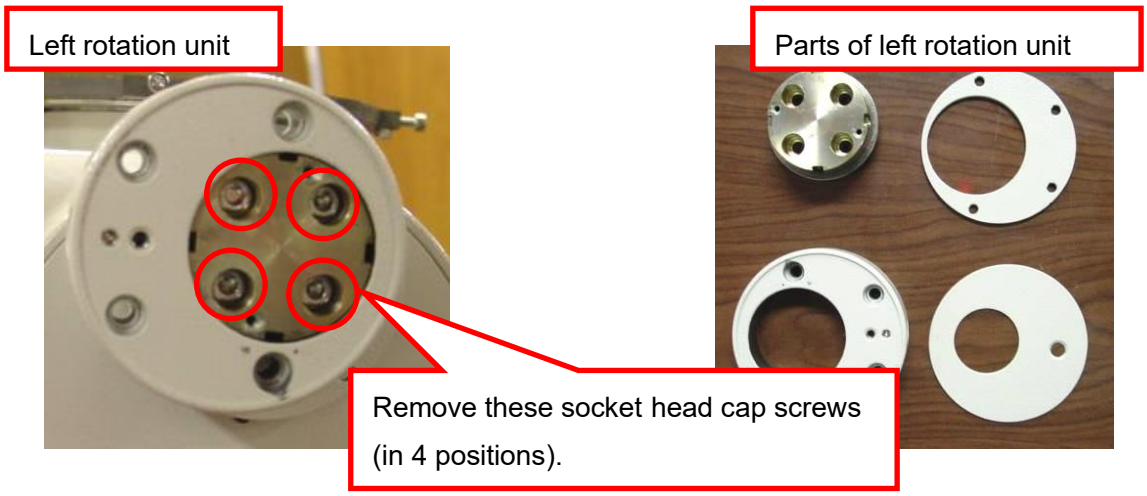
IV. Remove the cover from the tube rotation unit attaching the tube to the arm, remove socket head cap screws (in 4 positions on each of the left and right sides) inside, then remove and move down the tube from the holding frame.
The weight of the tube ASSY is approximately 13 kg. Pay rigid attention in moving it down from the arm.



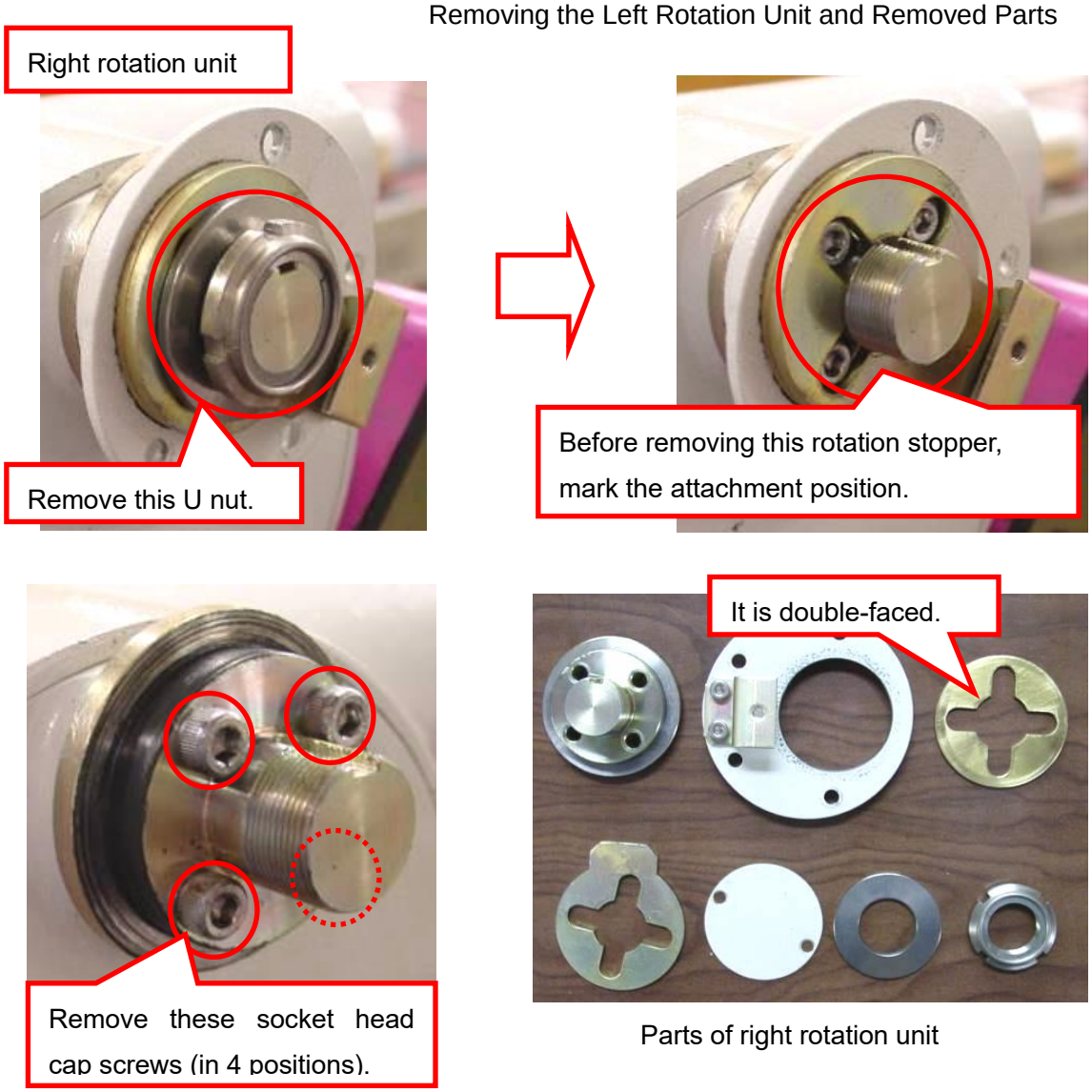
Tube Holder

V. Remove the left and right tube rotation units from the tube. As to the right side, loosen the U nut (new type lock nut not requiring a inner clip washer) in the center, then remove the rotation stopper, the spring washer, etc. After that, remove socket head cap screws from the shaft area, then remove the shaft area. As to the left side, remove the cover (countersunk head screws in 2 positions), then remove socket head cap screws from the inside.

Grease will be applied on sliding units at the end. Every time removing each part, clean the sliding units. Moreover, please be sure to decompose at the time of decomposition work, performing marking so that an attachment position and a direction may be known from back. It mistakes, and when assembling, it may not operate normally.



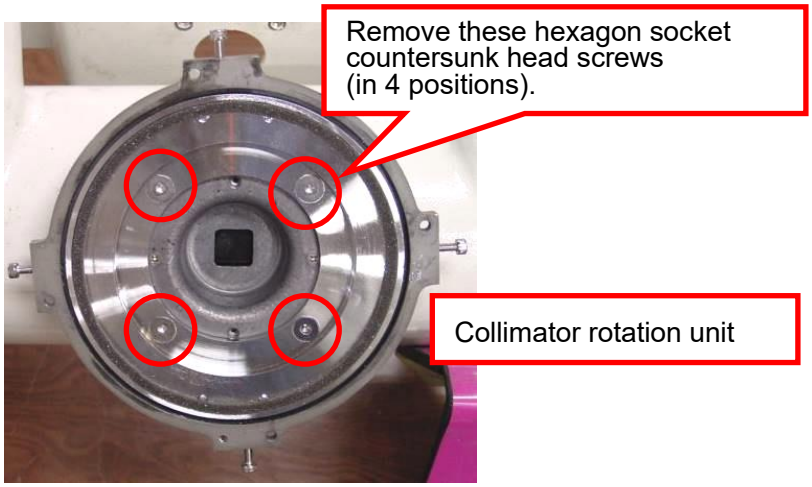
Removing the Left Rotation Unit and Removed Parts



Parts of right rotation unit

Removing the Right Rotation Unit and Removed Parts

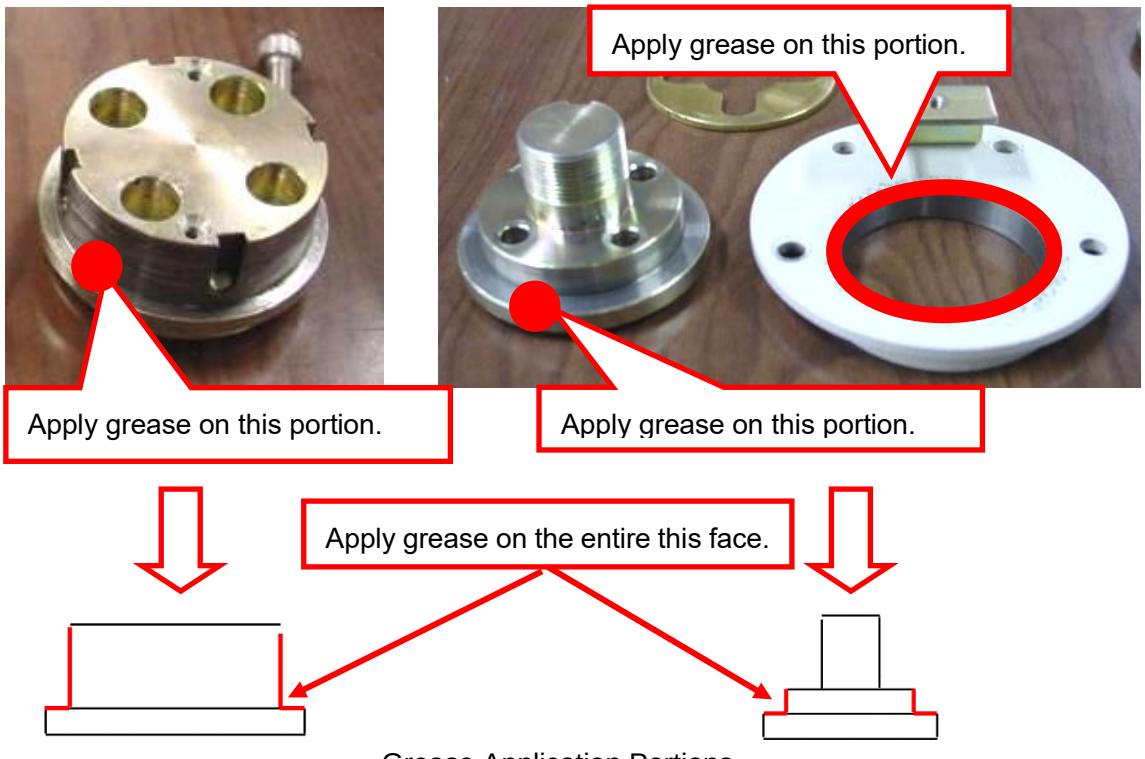
VI. Remove the collimator rotation unit and attach it to a new tube. Please be careful in the direction of stopper at this time.



Removing the Collimator Rotation Unit

VII. Attach the left and right rotation units to the new tube in the order of removal. At this time, clean the sliding units, and apply grease on the following portions before attachment.

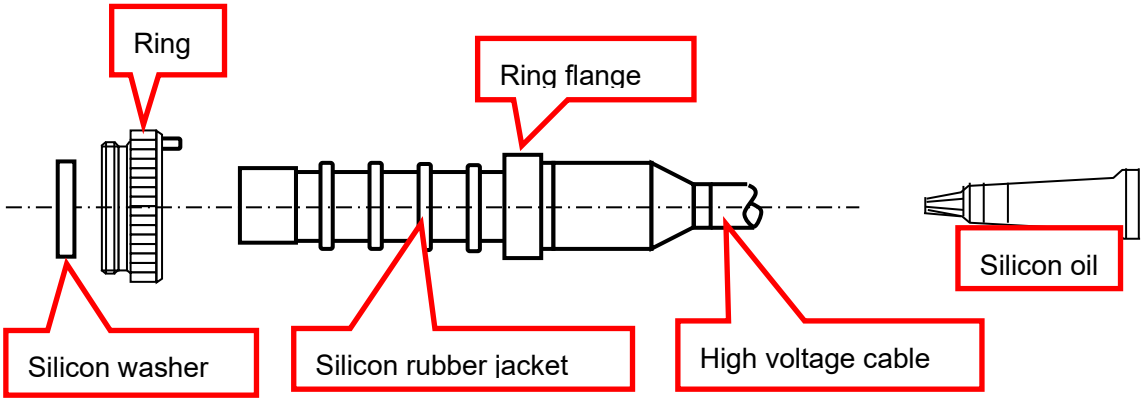
Make sure to use grease not containing abrasive. Recommended grease: Albania grease 2 (Showa Shell)



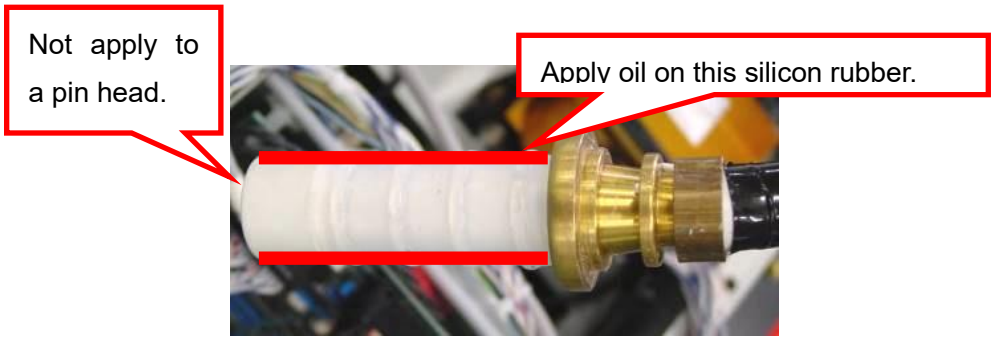
Grease Application Portions

VIII. At the end, attach the low voltage cable, the collimator and the high voltage plug, and treat the cables. At this time, attach the high voltage plug using the following procedure.

i. Make sure that the inside of the socket in the X-ray tube unit is not dirty.
ii. Make sure that the plug and the ring nut are not dirty.
(For the name of each part, refer to the below drawing.)
iii. Let the ring flange go through the ring nut.
iv. Apply silicon oil on the silicon rubber jacket and the silicon washer.
v. Fit the silicon washer into the tip of the plug.
vi. Put the plug into the socket. When the plug tip reaches the bottom of the socket, turn the plug to find the position in which the plug is fitted into the key way, then push the plug.
vii. Tighten the ring nut with hand.
viii. Tighten the rotation stopper screw so that the ring nut does not rotate.



High Voltage Plug



High Voltage Plug (Oil Application Position)

IX. After replacing the X-ray tube, perform warm-up operations (aging) and tube current adjustment in accordance with “System Warm-Up” and “Check tube current” in Installation Manual.

5.6 Replacement of the Obstruction Detector ASSY

(Reference work time 1 person ×45min.)

Make sure to set the main breaker to OFF and discharge the capacitor before starting work.

There are two types of Obstruction Detector ASSY (566-20821 and 566-25103).

Since they are not interchangeable, refer to the parts list carefully and order the correct parts.

The following picture shows the case of the new type 566-25103.

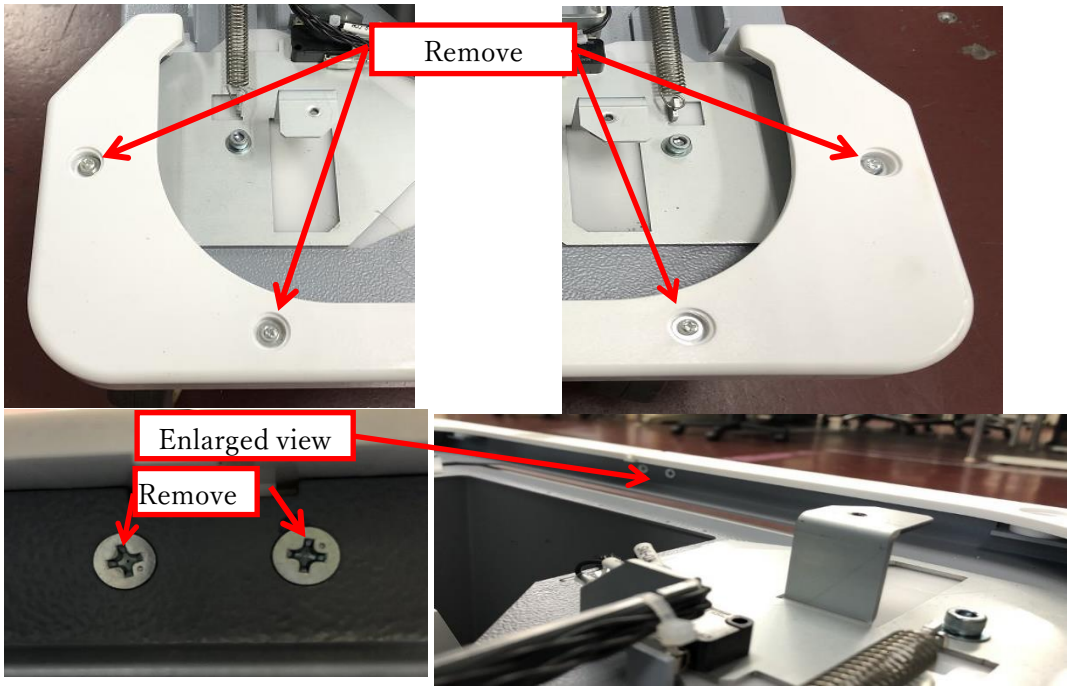
5.6.1 Remove the covers

Refer to "How to Open/Close Covers" in the installation manual to remove the left and right inside covers, side covers, Front rear-cover and Front cover.

5.6.2 Replacement of the Detector cover

Both [Detector cover L] and [Detector cover R] can be replaced by removing 3 screws.

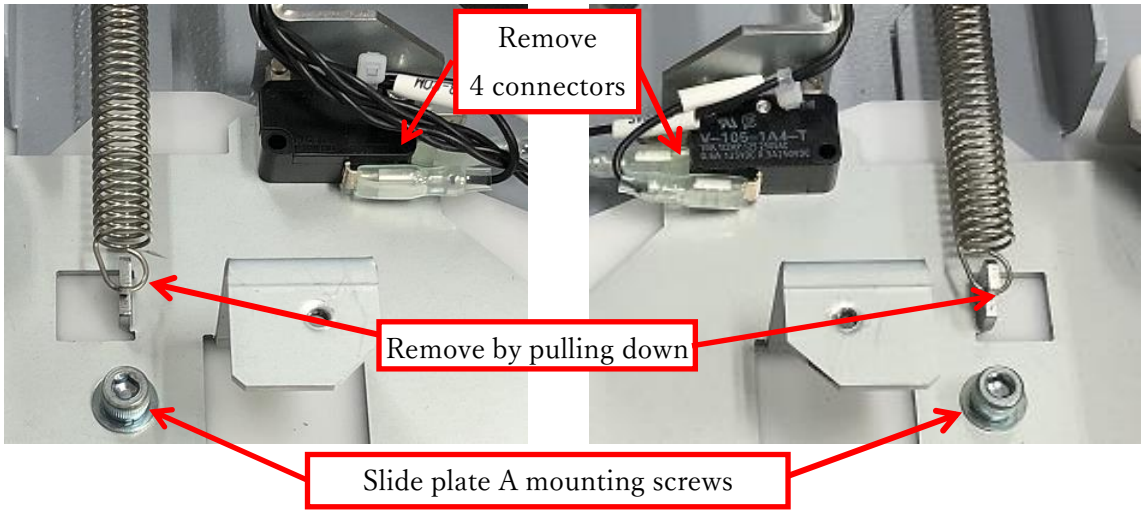
Replace the covers by removing the screws as shown below.



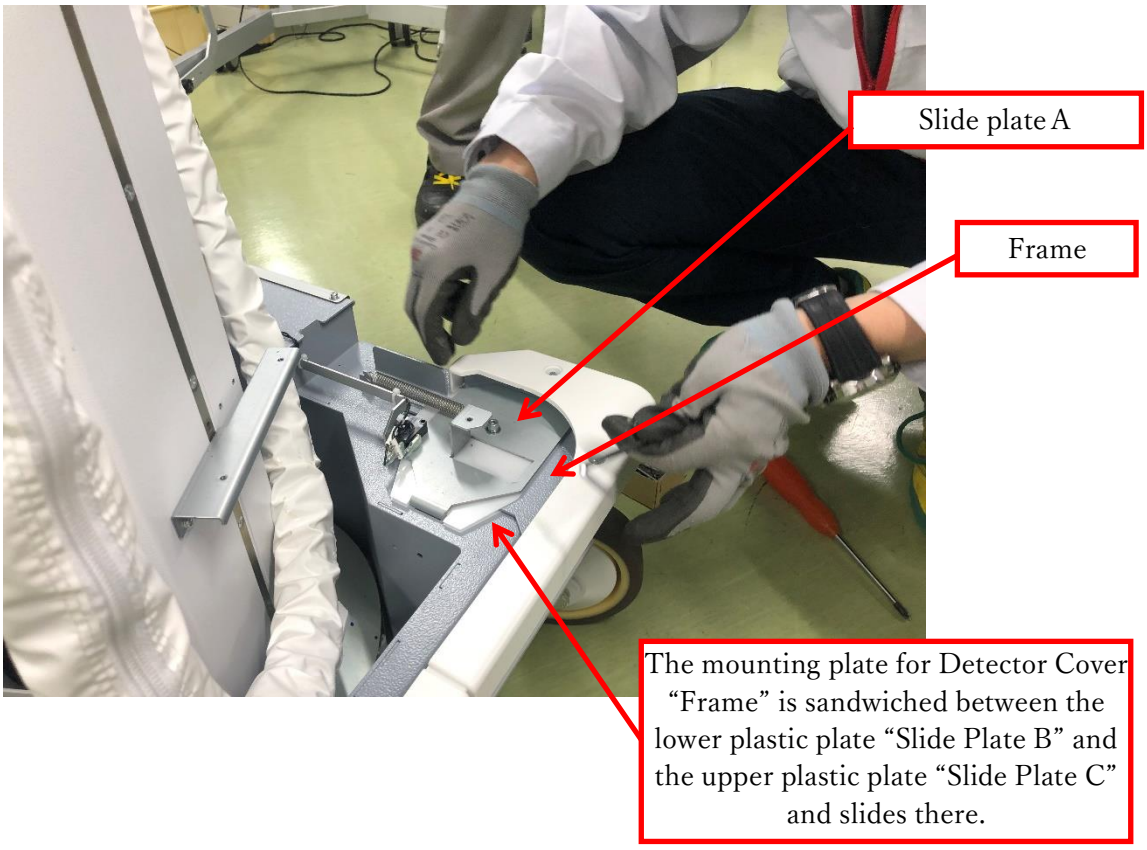
5.6.3 Replacement of the [Slide plate], [Spacer]

I. When replacing the "Slide Plate B", "Slide Plate C" or "Spacer", remove the left and right detector covers.

II. Remove the spring and microswitch harness from the left and right slide plates A. (Record harness marks and connection locations.)

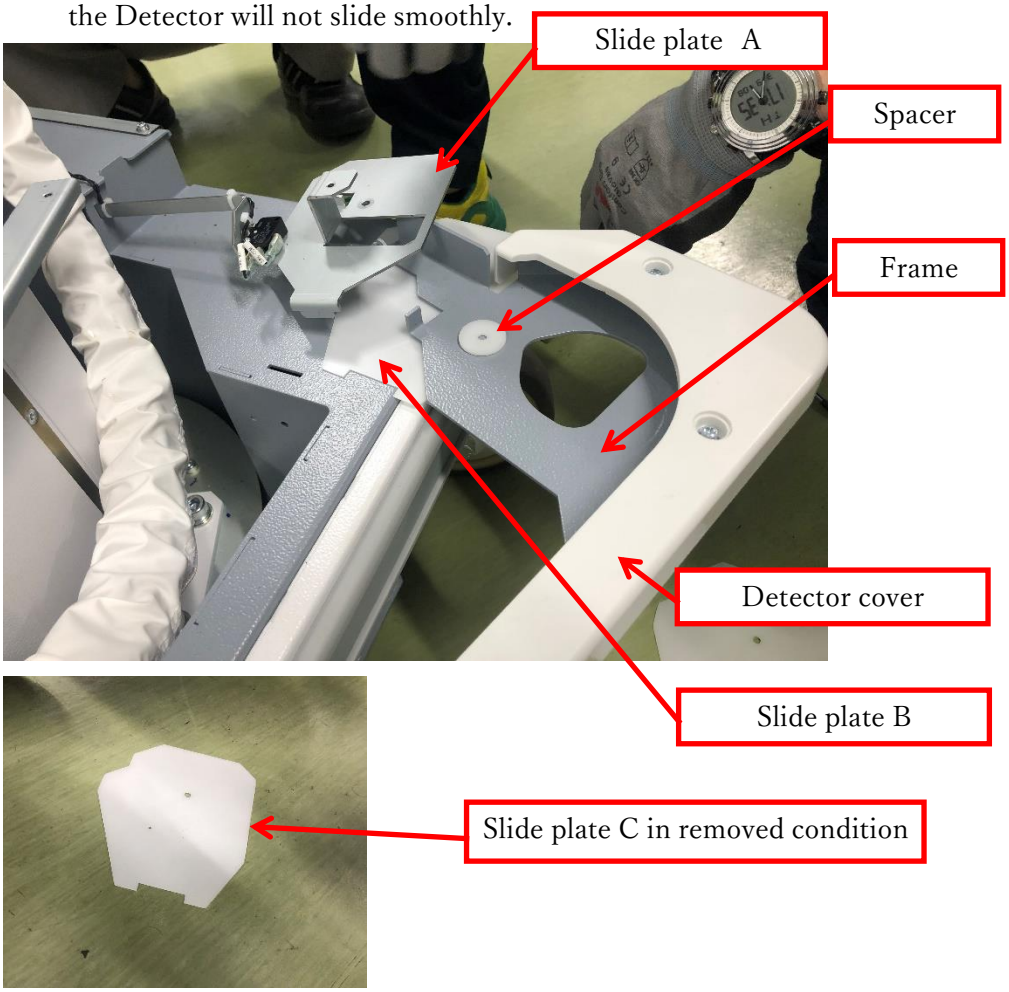


III. Remove two slide plate A mounting screws, remove slide plate A, replace "Slide Plate B" "Slide Plate C" or "Spacer", and reassemble in the original steps.



In case of the new-type Obstruction Detector ASSY 566-25103, the material of the spacer has been changed from resin to metal since around 2023. (The below picture shows the spacer made of resin.)

When installing the new type of the spacer 566-29082 (made of metal), apply Albanian grease to the entire circumference of the end face. If grease is not applied, the Detector will not slide smoothly.

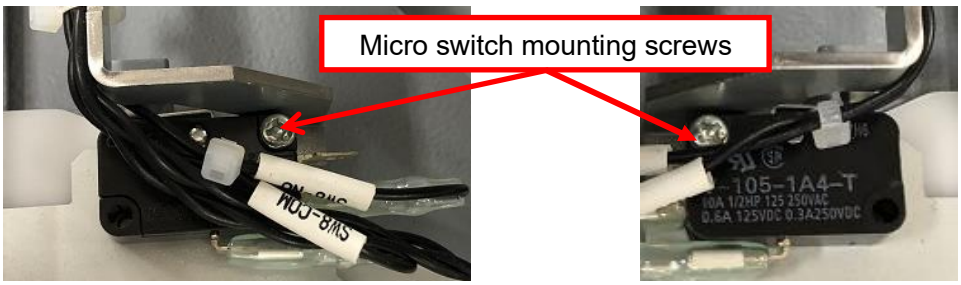


5.6.4 Replacement of the Micro switch

I. Remove the two harnesses from the microswitch to be replaced.

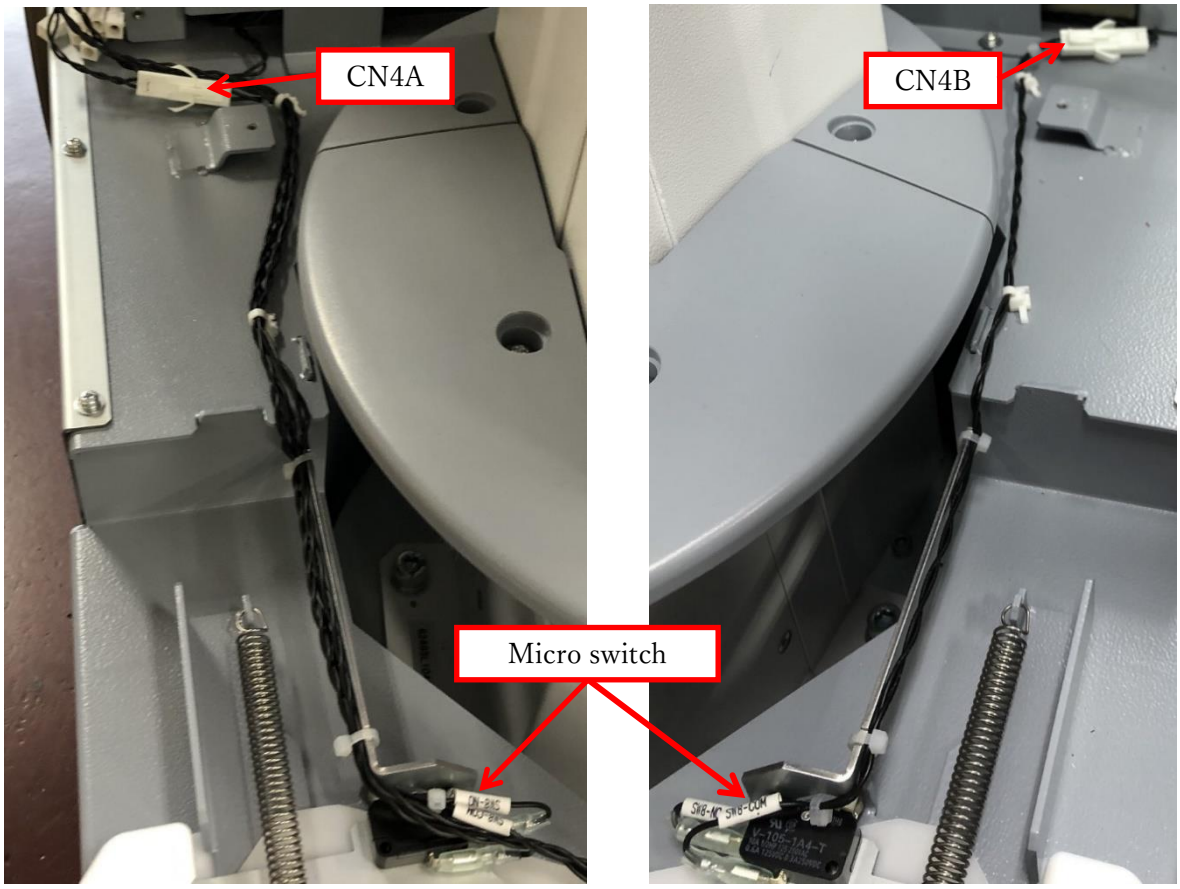
II. Remove the microswitch mounting screws and replace the microswitch with a new one.

III. Attach the harness to the microswitch.



5.6.5 Replacement of Micro switch harness

I. Cut the tie wrap that secures the harness of the microswitch being replaced.
II. Disconnect the two connectors on the microswitch side and the CN4A (or CN4B).
(Record harness marks and connection locations.)
III. Replace the herness to new one.
IV. Connect the two connectors on the microswitch side and the CN4A (or CN4B).
V. Fix the herness by using tie wrap.



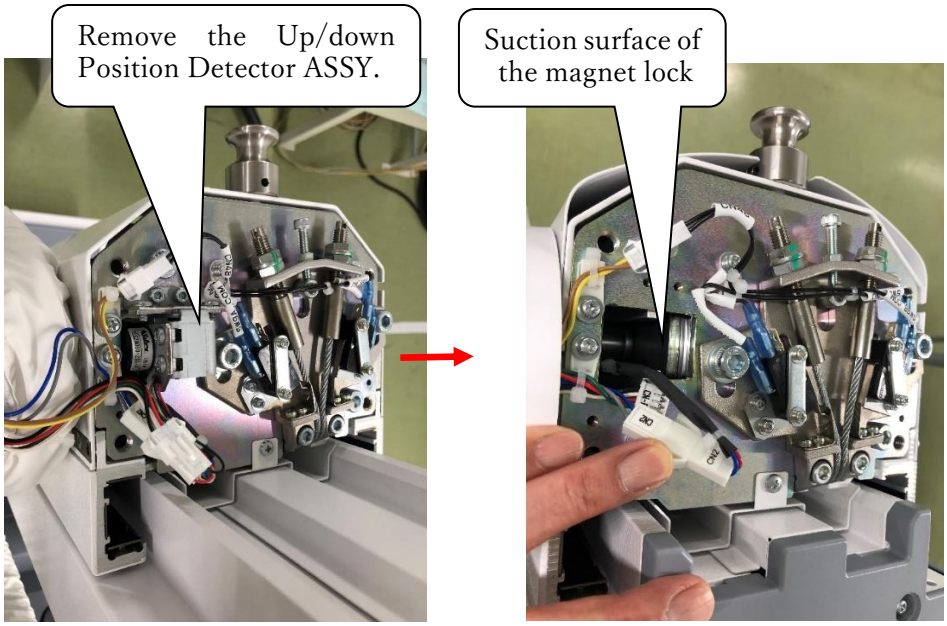
5.7 Replacement of the Magnet Lock

(Reference work time 1 person ×1.5 hours in case of replacing the Magnet Lock for up and down motion of the Arm only)

5.7.1 Replacement of the Magnet Lock for up and down motion of the Arm

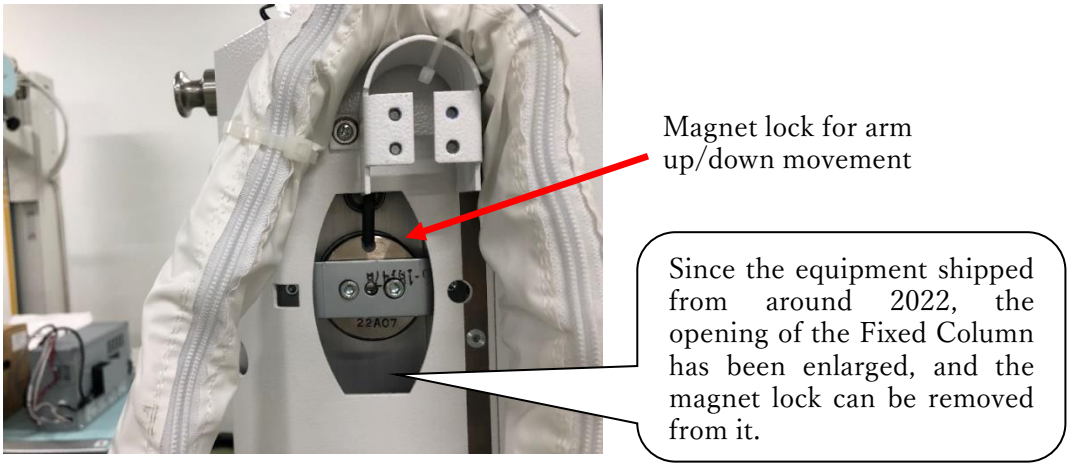
If the brake force of the up/down movement of the arm decreases, clean and degrease the surface of the brake plate (suction surface of magnet lock) before replacing the magnet lock, and wait for a while. (In many cases, the effect does not come out immediately after degreasing.)

You can access the suction surface of the magnet lock by removing the Up/down Position Detector ASSY attached to the top surface of the Fixed Column. In this state, move the arm up and down to degrease as wide area of the suction surface as possible.



Since the equipment shipped from around 2022, the opening of the Fixed Column has been enlarged, and it has become possible to replace the magnet lock for up/down movement of the arm without pulling up the Spring Balance ASSY from the Fixed Column.

The following explains replacement procedure of the magnet lock for up/down movement with pulling up and also without pulling up the Spring Balance ASSY.

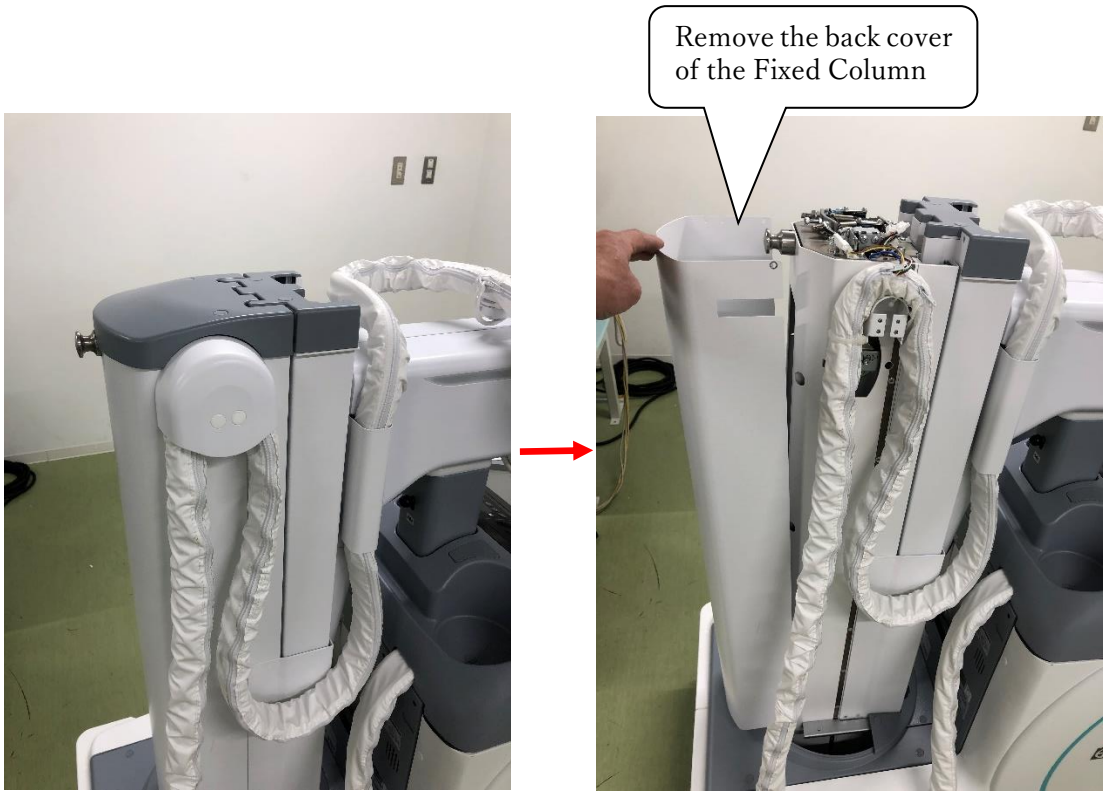


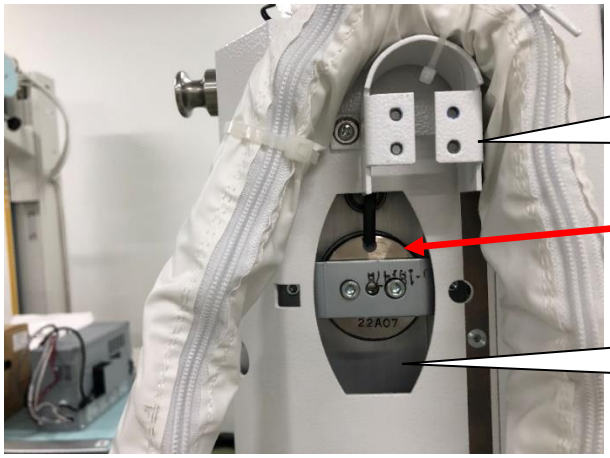
5.7.1.1 Work Procedure without lifting up Spring Balance ASSY (the equipment shipped from around 2022)

Since the equipment shipped from around 2022, the opening of the Fixed Column has been enlarged, and it has become possible to replace the magnet lock for up/down movement of the arm without pulling up the Spring Balance ASSY from the Fixed Column.

Remove the top cover, cable guide cover, etc.

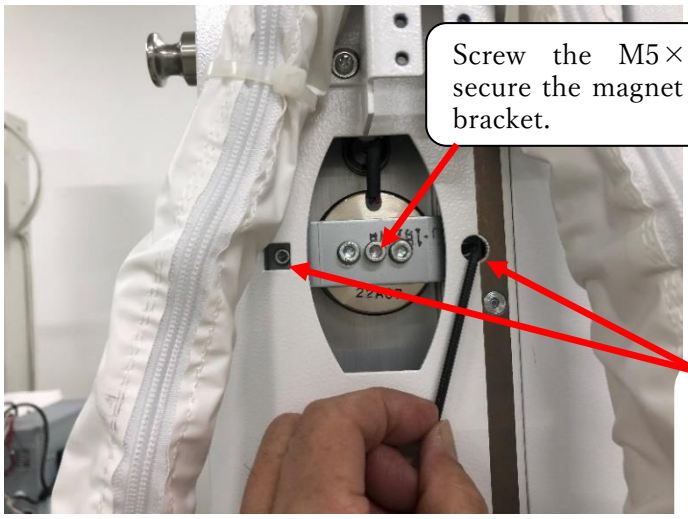
Then, remove the back cover of the Fixed Column.

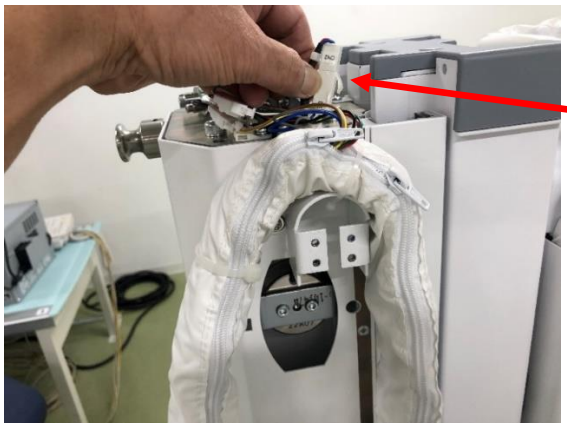




With the brake release operation in place, screw the M5X6 hexagonal socket head bolt into the screw hole in the center of the magnet lock to secure the magnet lock to the bracket. Remove the connector CN2 on the top of the Fixed Column.

Remove the two bracket fixing screws, remove the magnet lock from the opening of the Fixed Column and replace the magnet lock with a new one.



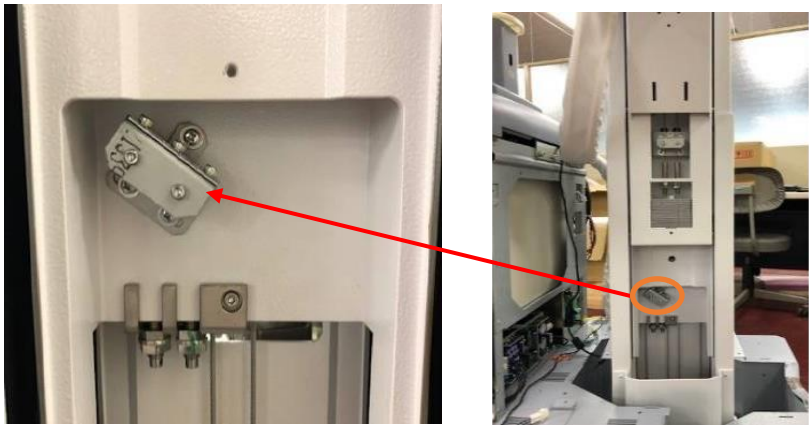


Disconnect connector CN2 before removing magnet lock + bracket.

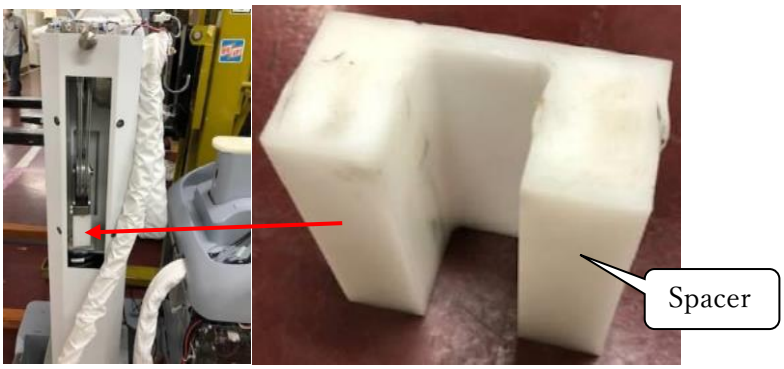
5.7.1.2 Work Procedure with lifting up Spring Balance ASSY (the equipment shipped before around 2022)

In the case of equipment shipped before about 2021, the Spring Balance ASSY must be pulled up from the Fixed Column in order to replace the magnet lock for up and down arm movement because the opening of the Fixed Column is small.

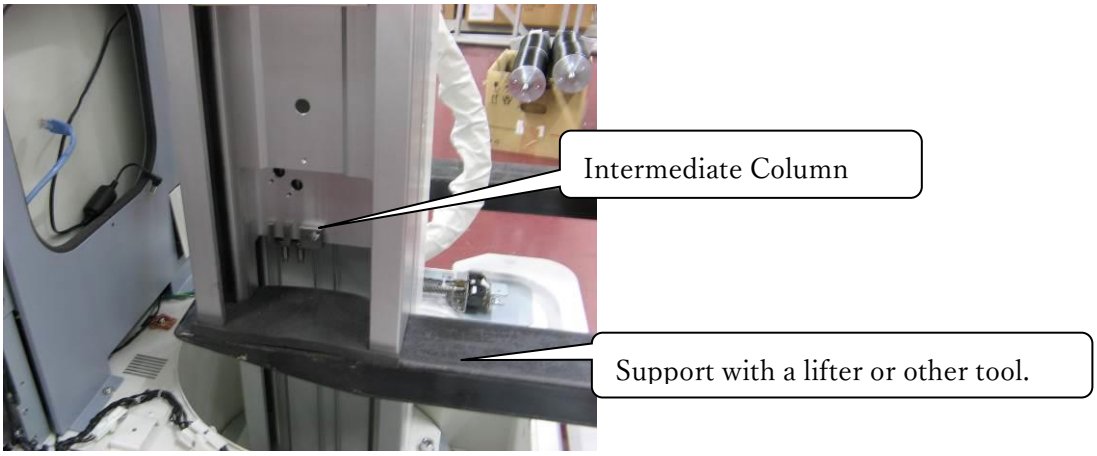
Ⅰ. Remove the break detection lock on the Intermediate Column.



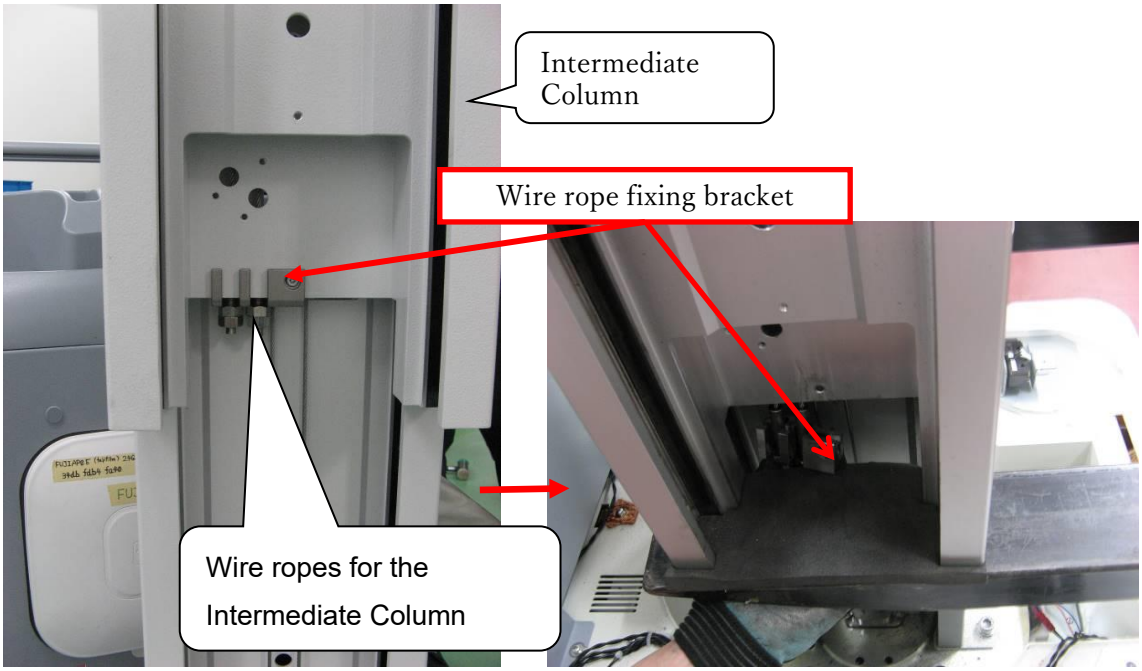
Ⅱ. Attach the Spacer to the Spring Balance ASSY. Lower the X-ray tube once to insert the Spacer, then raise X-ray tube again to sandwich the spacer.



III. Support the Intermediate Column with a lifter etc. so that no load is applied to the wire ropes. If you use Jig for wire rope replacement (P/N:566-21942 “Wire replacement Jig, Colum”) , you can support the Intermediate Column without using the lifter or other special equipment.
(Regarding Jig for wire rope replacement, refer to “5.1 Wire rope replacement”.)



IV. Remove the wire rope fixing bracket below the Intermediate Column so that the wire ropes (2 pieces) for the Intermediate Column can be free to the Intermediate Column.

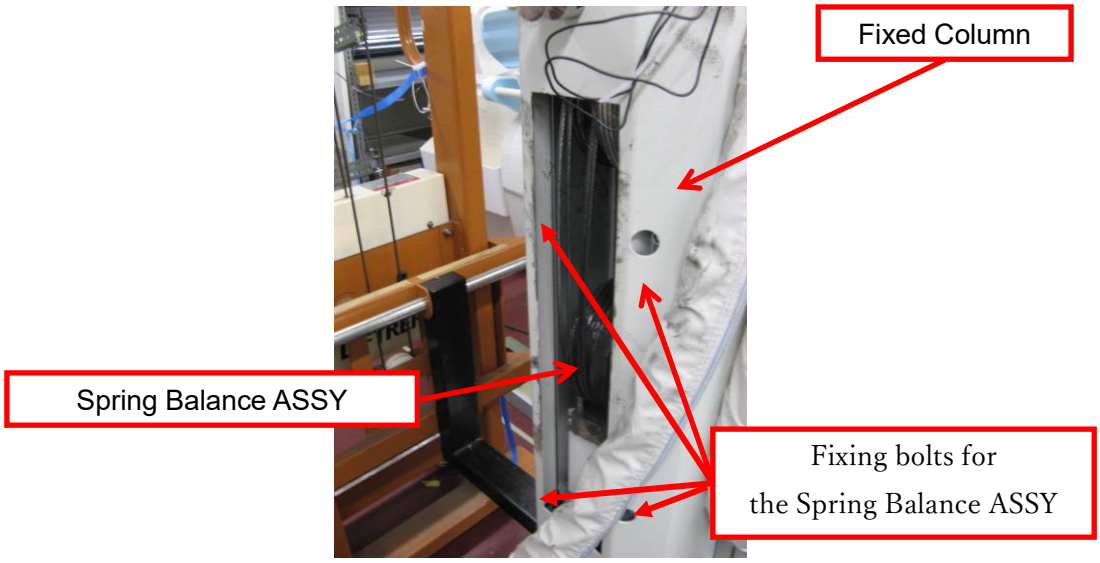


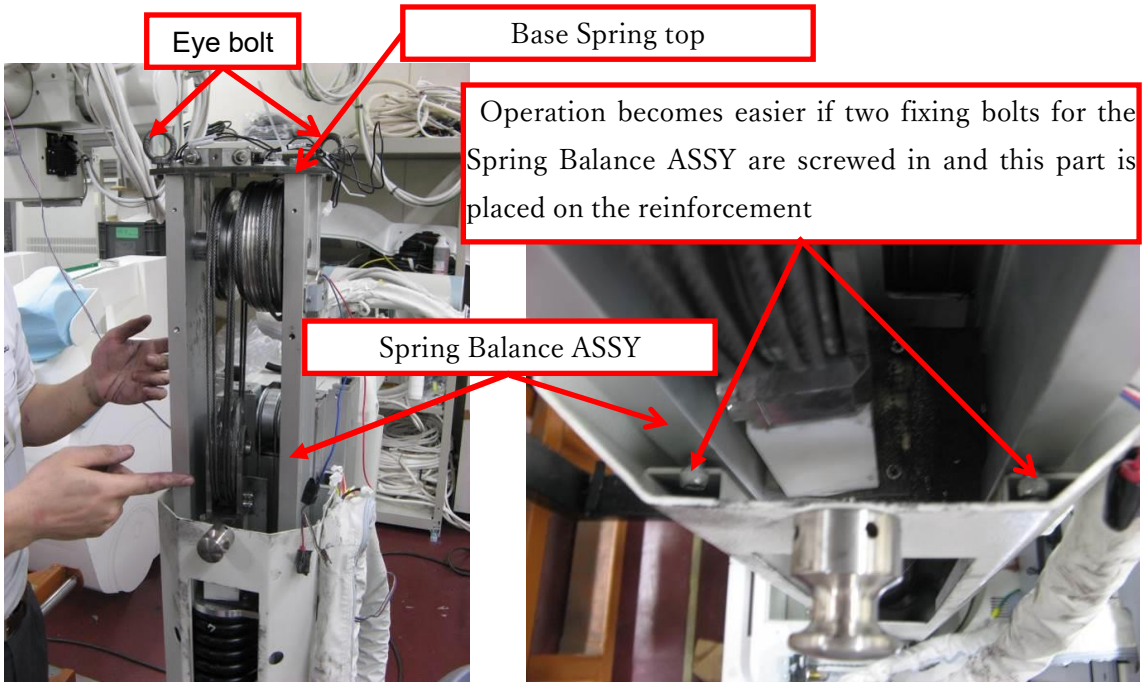
5. Parts replacement

V. Remove four socket head bolts for fixing the Spring Balance ASSY and pull the ASSY upward. The Spring Balance ASSY weighs approximately 25 kg. Work carefully because of heavy load.

(As shown in the picture below, it is easier to pull up if the eye bolts are screwed in thread holes for attaching the “Base, Spring top”.)

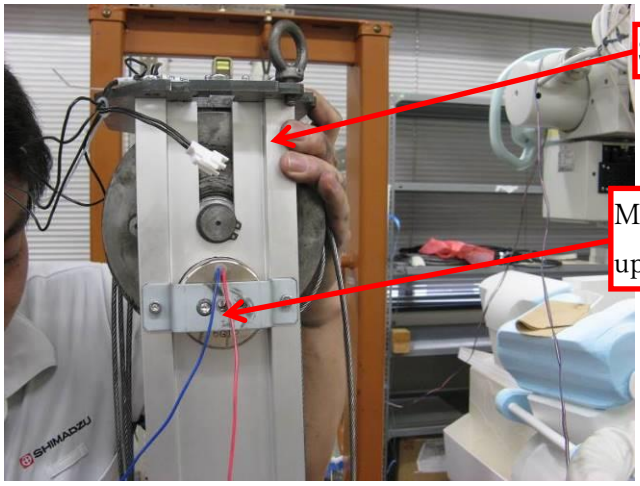
Additionally, the operation can be easier if two fixing bolts are screwed in at two places of attaching thread holes at the lower part of up and down balance section after pulling the Spring Balance ASSY up and this part is placed on the reinforcement plate part at the top end of Fixed Column.





5. Parts replacement

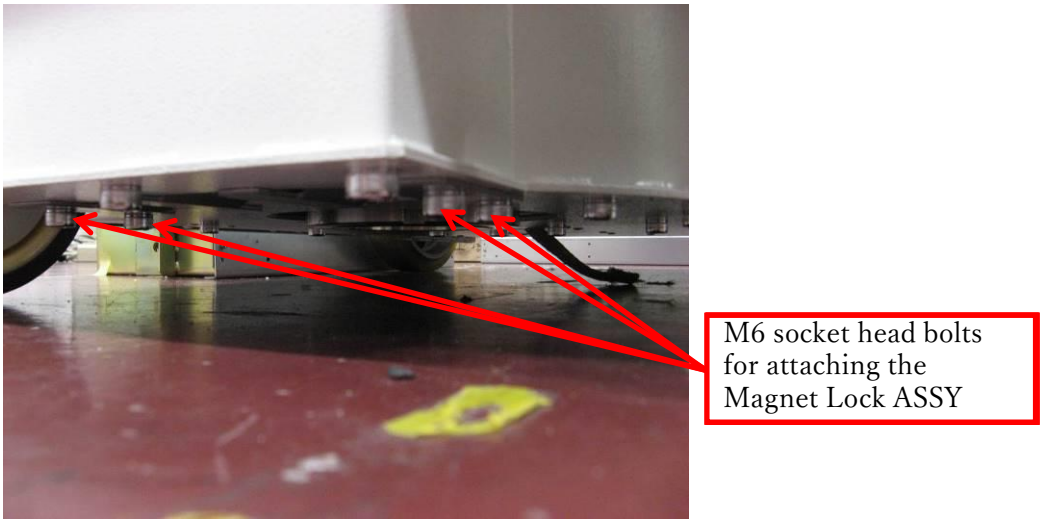
VI. First, perform the lock release operation and remove the Magnetic Lock and then remove the connector and replace with the new Magnetic Lock.



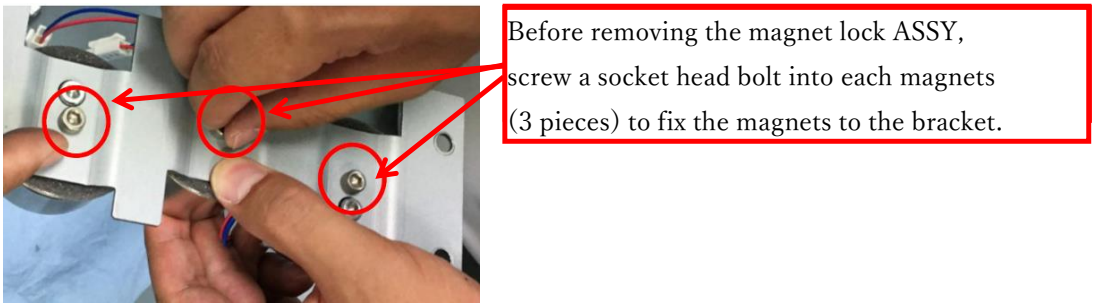
5.7.2 Replacement of the Magnetic lock for swing motion of the column

I. Remove four M6 socket head bolts for attaching the Magnet Lock ASSY which has been fixed at the bottom of the chassis.
(It is hard to see because the place between the bottom of the chassis and the floor is narrow, however you can recognize which screw should be removed by watching with the mirror or by feeling by hand touch because the M6 socket head bolts are only 4 pieces that attach the Magnet Lock ASSY.)

II. Carry out the lock release operation and then take out the Magnet Lock ASSY from the chassis and put it on the floor. You can remove the above 4 pieces of M6 socket head bolts by using normal hexagonal wrench without jacking up the chassis.

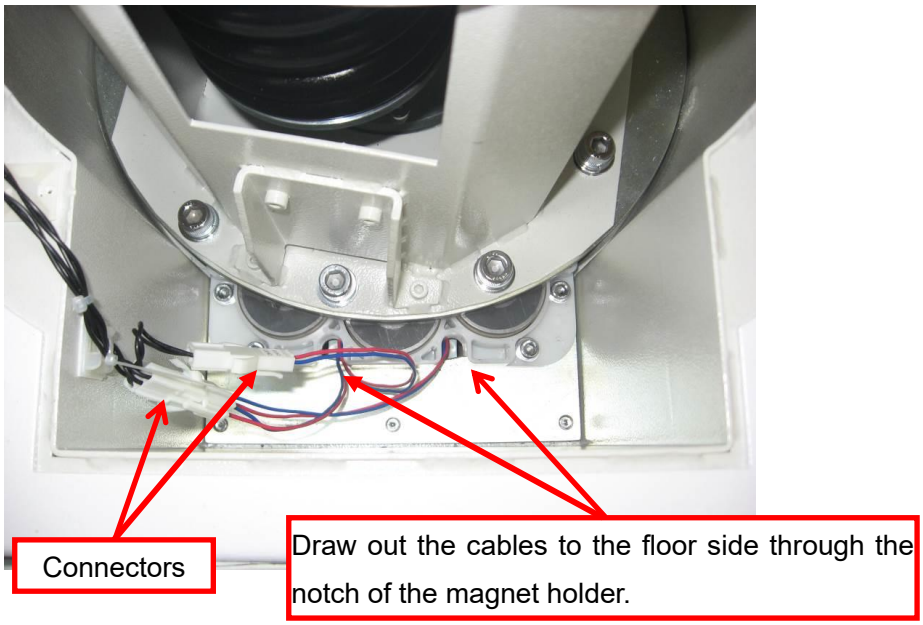


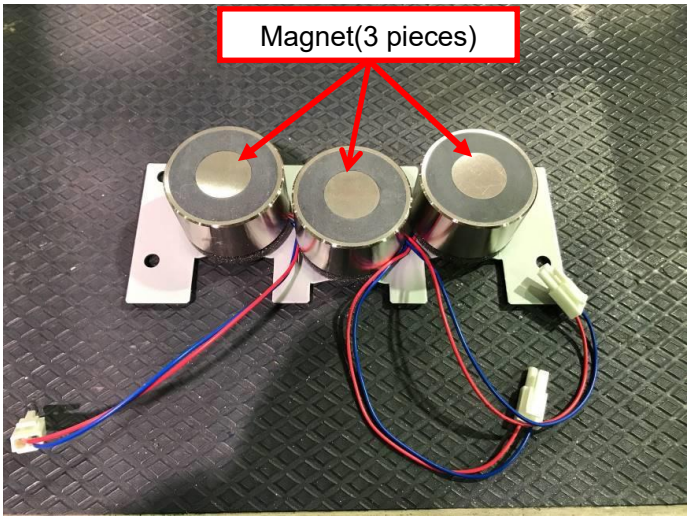
Before removing the magnet lock ASSY, screw a socket head bolt M5 x 6 mm into the center screw hole of each magnets to fix the magnet to the bracket. Do not forget to remove the socket head bolts after replacing the magnet.



5. Parts replacement

III. Referring to “How to Open/Close the Covers” in the installation manual, remove all the covers under the column (i.e. the Post base cover, Front back cover and Front cover) and then remove the connectors to pull out the cable of the Magnet Lock toward the floor side, and replace the Magnetic Lock with new one.





IV. Magnet Lock ASSY which is removed from the chassis and placed on the floor.

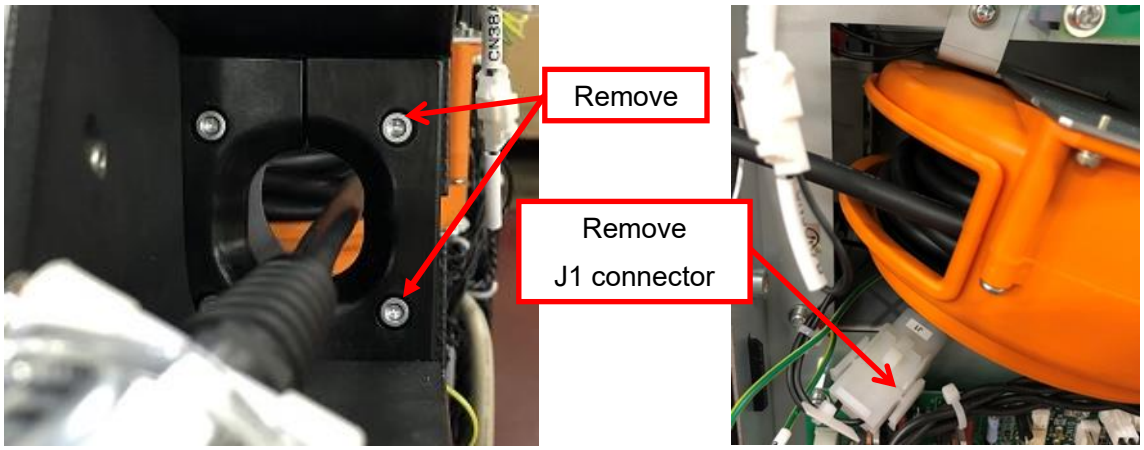
5.8 Replacement of the cord reel

(Reference work time 1 person × 60min.)

Make sure to set to OFF the main breaker and discharge the main capacitor before starting the work.

I. Referring to "How to Open/Close the Covers" in the installation manual, remove the right and left inside covers and side covers.

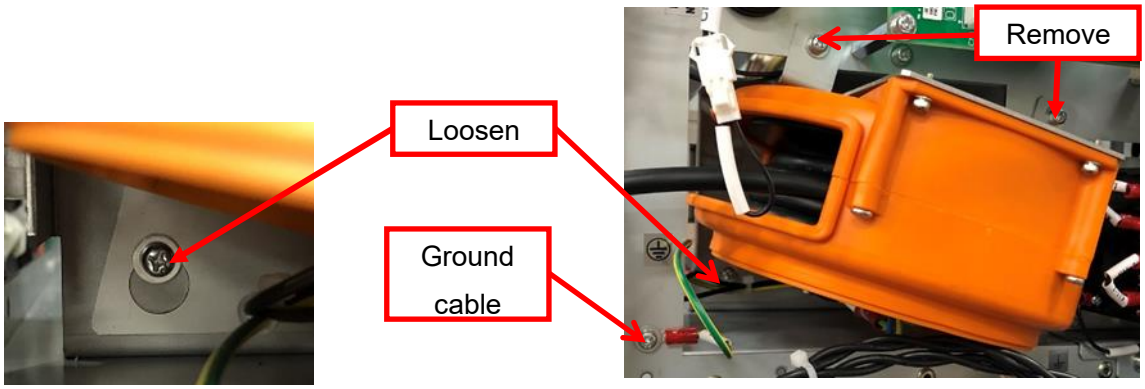
II. Remove the two hexagon socket head cap screws on the left side of the following illustration. Disconnect the J1 connector on the right in the illustration below.



III. Remove the ground cable of Cord reel.

IV. Loosen one of the screws shown below.

V. Cord reel can be removed by removing the two screws shown below. Lift the cord reel up slightly and pull it toward you.



VI. Remove the bracket from the cord reel by removing the four screws as shown in the figure.

5. Parts replacement

VII. Attach the bracket to the new cord reel.

VIII. Install the cord reel in the reverse order and replace the unit.

IX. Refer to the installation manual "Cord Reel Usage Information Setting Mode" and reset the usage information.

5.9 Replacement of PC

5.9.1 Use recovery media

(Referendce work time 1 person ・ 60min.)

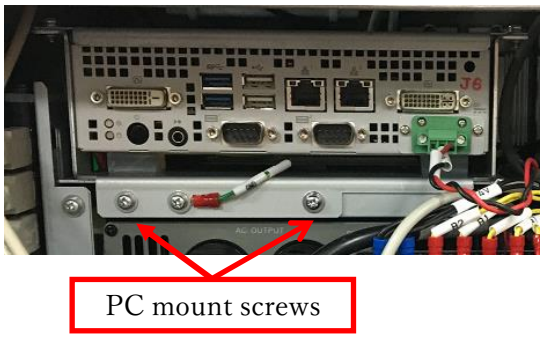
Make sure to set to OFF the main breaker and discharge the main capacitor before starting the work.

I. Referring to “How to Open/Close the Covers” in the installation manual, remove the right and left inside covers and side covers.Open the FPD box.

II. Record the cables connected to the PC so that you can see them, and then disconnect them all (Photos, etc.).

III. Remove the PC mount screws.

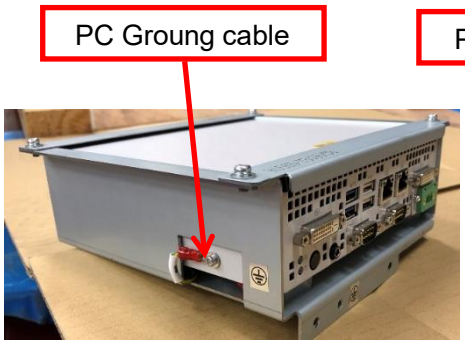
IV. Pull out the PC mount slightly and lift the back to remove the PC.

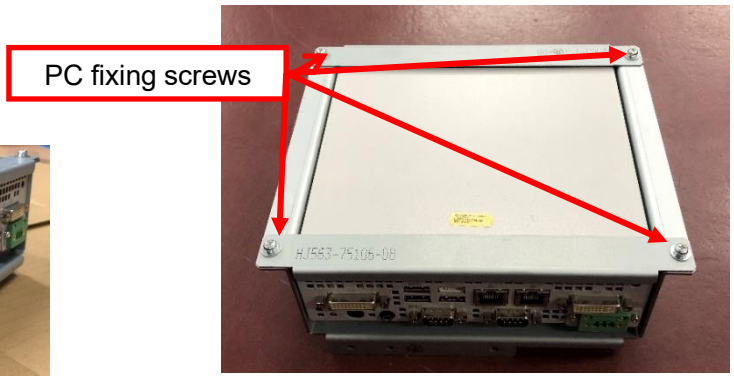


V. Pull out the PC mount.

VI. Remove the grounding cable connected to the PC as shown on the left below.

VII. Refer to the right side of the figure below and remove the PC by removing the four screws on the top of the PC.





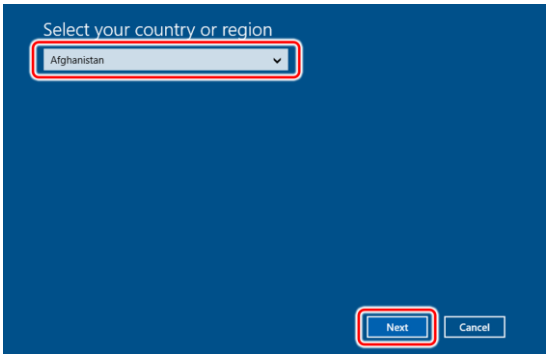
VIII. Refer to "10.9.4 Recovery method" in the installation instructions and recover the PC data from the recovery media you created beforehand.

IX. Make sure Windows boots properly and log in with a Windows account "shimadzu" (PW: service).

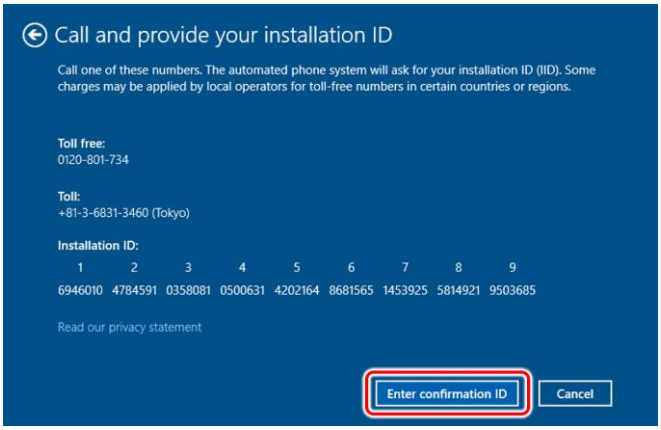
5. Parts replacement

X. Activate Windows

i. Double-click "Open the license activation screen.bat" on the desktop. Under Select your country or region, select your country and click Next.

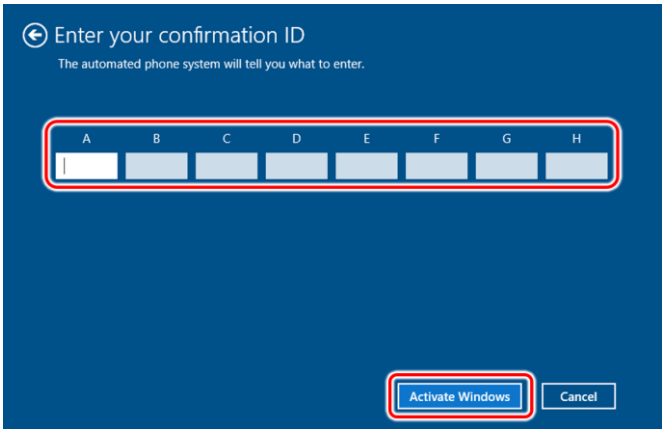


ii. Call one of the numbers provided and follow the directions to obtain an ID.

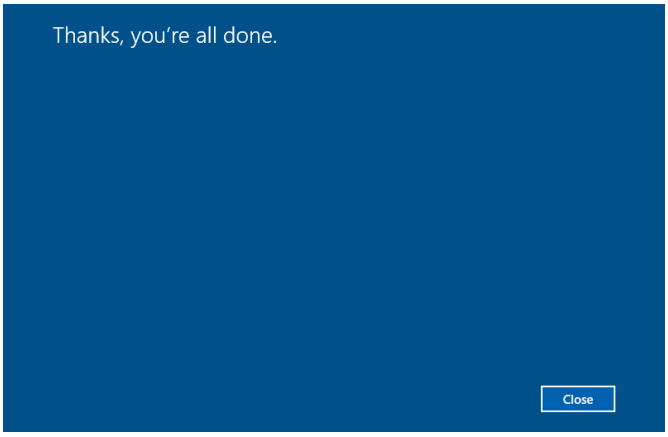


iii. Click "Enter confirmation ID" to go to the ID entry screen.

iv. Enter your ID when prompted on the phone. When you're done, click Activate Windows.



v. When the next screen appears, activation is complete. Click Close.



XI. If the PC is changed, the DR software license will need to be reissued.

XII. Finally, activate the unit to make sure it works properly.

XIII. Set up DR as needed.

5.9.2 Not use recovery media

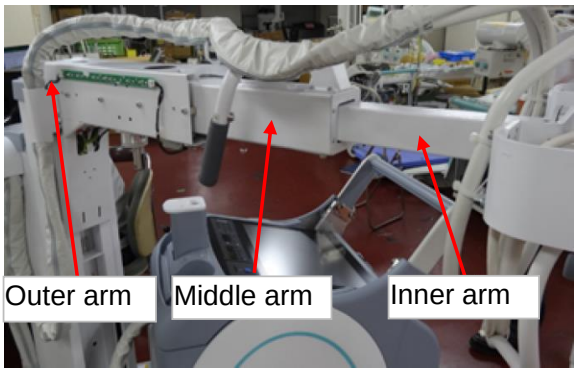
You'll need media and a PC with the Windows 10 OS. You also need to install DR software locally.
For each of these, please refer to the table below for necessary technical documentation.

System	Document
MX8c	M517-2750/M517-E845
MX8s	M517-2751/M517-E846
MX8k	M517-2778/
MX8i	M517-2779/

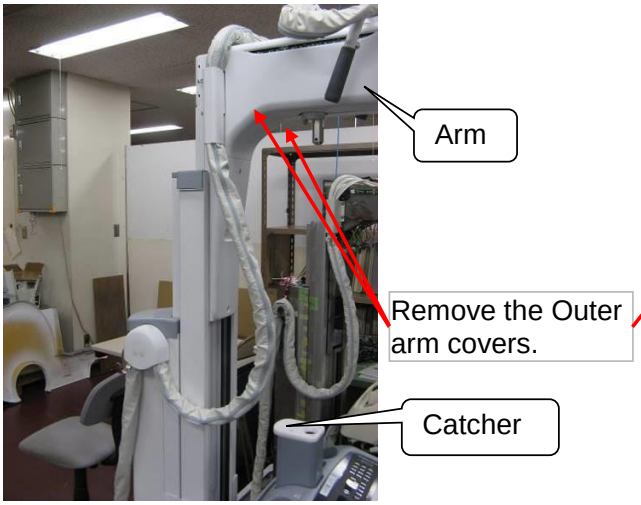
5.10 Replacement of Inner arm

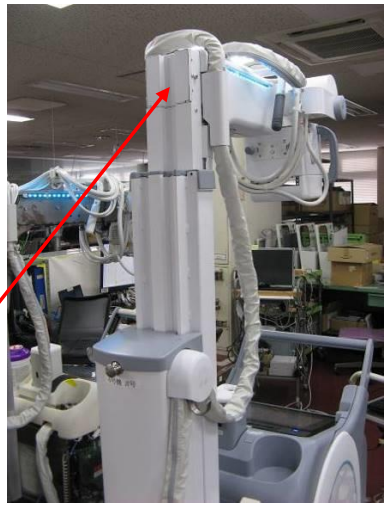
(Referece work time: 2 persons × 2.5 hours)

I. The names of the arms are as follows.



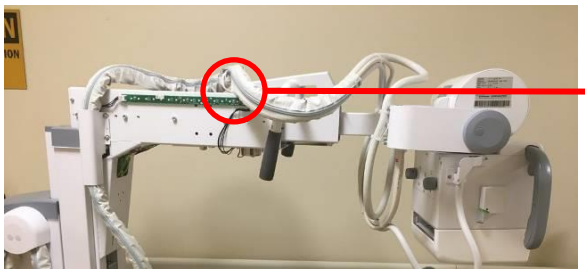
II. Remove the three Outer arm covers.

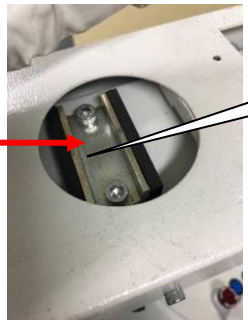




III. Remove the collimator and X-ray tube after locking the arm to the catcher referring to Chapter 5.6. Before locking the arm assembly to the catcher, make sure to remove the fixing bolt for the connecting bracket from the rear side of the arm which is shown on the top middle picture on the next page. Because it will not be able to be removed after X-ray tube is locked to the catcher.

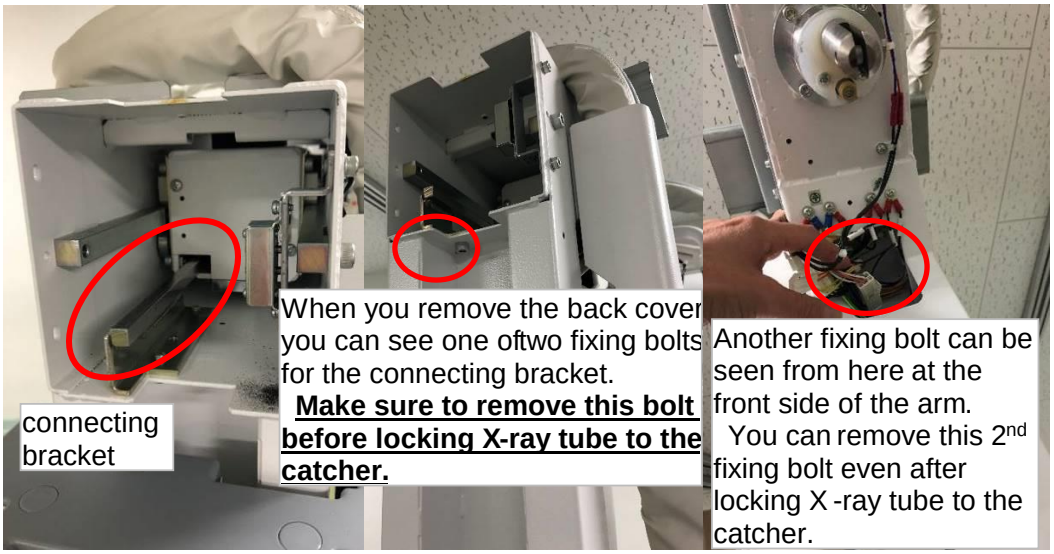
IV. Remove the two bolts from the large round hole on the upper side of the outer arm to remove the stopper.



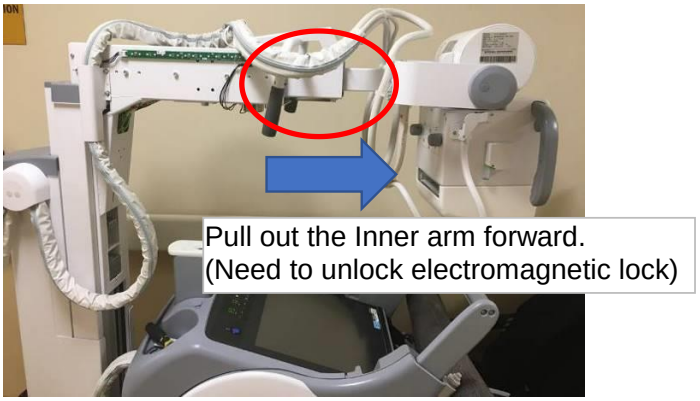


V. Remove the two bolts that secure the Outer arm and the connecting bracket.

5. Parts replacement



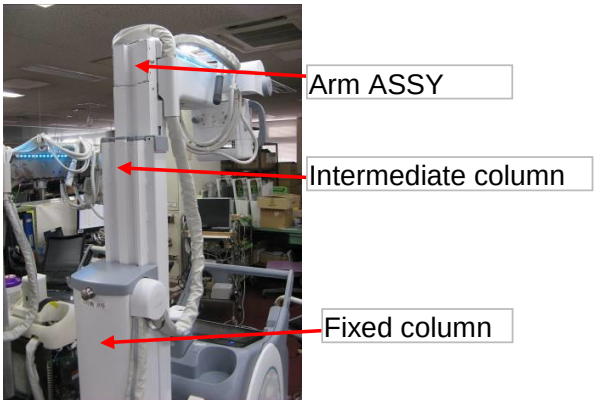
VI. Pull out the Inner arm together with the connecting bracket. (Middle arm will come out together with Inner arm.)



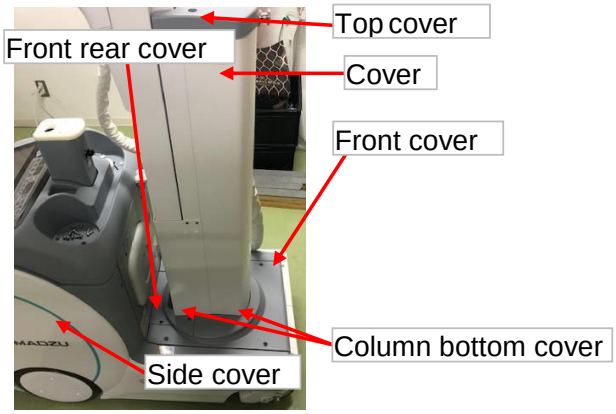
5. Parts replacement

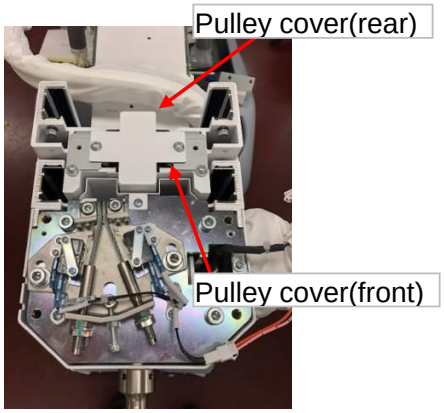
5.11 Replacement of Middle column

I. The names of the telescopic columns are as shown in the figure below.

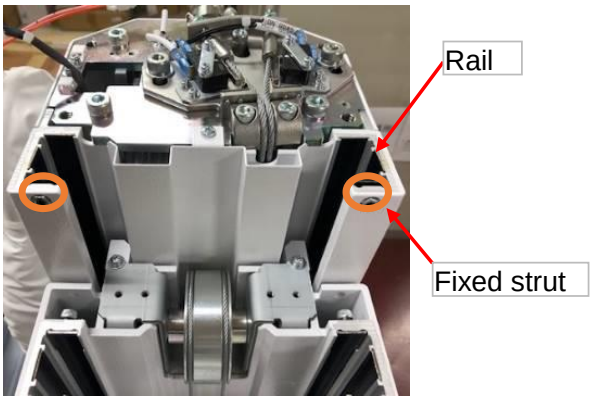


II. Remove the Top cover and Cover of the fixed support, the front and rear Pulley cover of the intermediate column, the Front cover, the Front rear cover, and the Column bottom cover. There is no need to remove the Side cover of main body.



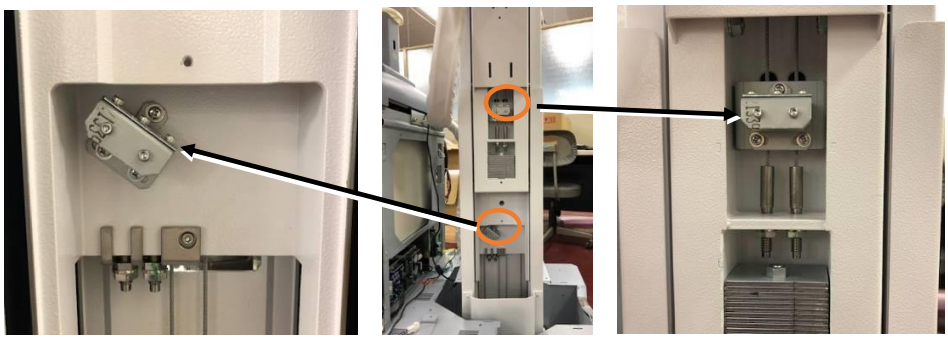


III. Turn the arm about 90 degrees and lower the X-ray tube position. Then remove the set screw that prevents the intermediate column from coming off, which is pushing the Rail of the Fixed strut.



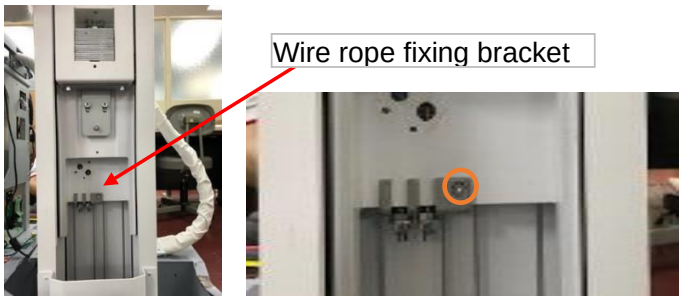
5. Parts replacement

IV. Remove the disconnection detection lock on the intermediate column and arm.



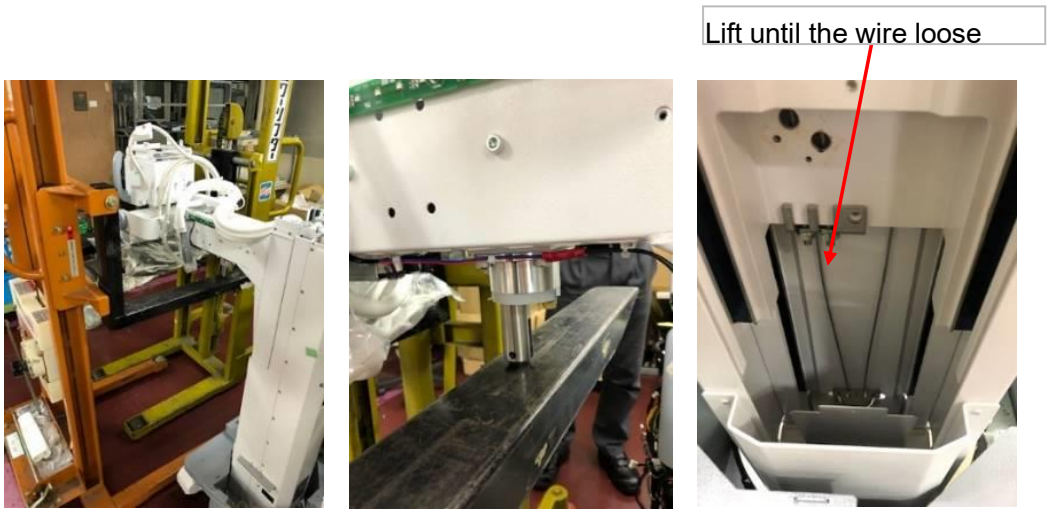
V. Attach the Block to the spring balance assembly. Lower the X-ray tube once to insert the Block, then raise the X-ray tube again to completely hold the Block.

VI. Remove the bolt of the Wire rope fixing bracket of the intermediate column.



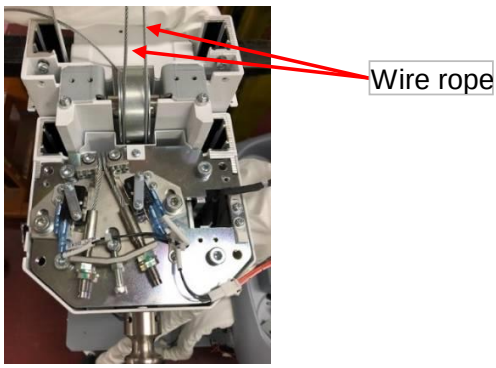
5. Parts replacement

VII. Rotate the X-ray tube 180 degrees around the horizontal axis and point the collimator up.On a horizontal and stable floor, lift the tip of the catch pin on the arm with the first lifter. Lift the arm wire rope until it loosens.



VIII. Remove the wire rope fixing bracket at the bottom of the fixing column.

IX. Pull out the Wire rope removed in step 8.

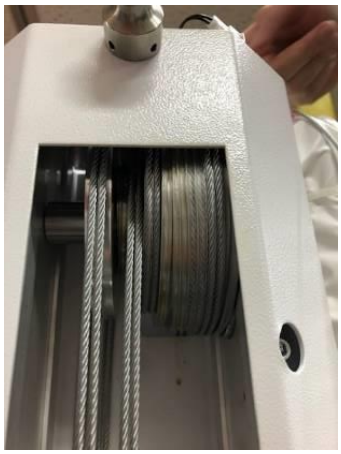
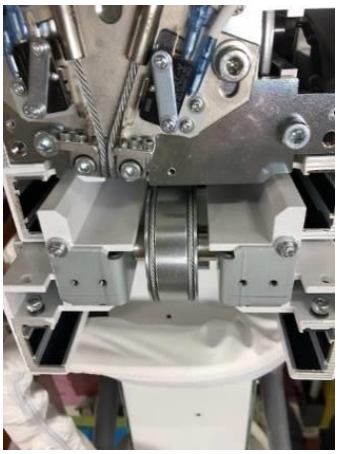


5. Parts replacement

X. Support the bottom of the X-ray tube with a second lifter. If the tube holder and lifter are wrapped with a band or the like and fixed, the balance will not be easily lost.

XI. Lift the intermediate column and pull it out. Please note that if the intermediate column is completely pulled out, the arm part will be completely free and the balance will be lost.

XII. Refer to steps 2 to 11 and reverse the procedure to replace the device. Check that the wire rope is properly hooked on the fixing bracket and pulley, and that there is no slack or pinching, and work carefully.

5. Parts replacement

XIII. Before installing the covers, check that there are no problems with the vertical axis rotation operation or vertical expansion / contraction operation of the column. Then install the cover removed in step 1.

5.12 Replacement of the motor

Replace the motor while referring to “Replacing the Motor” in Installation Manual.

<Motor replacement standard>

If abnormal noise or vibration from the motor is found during driving, it is highly possible that the motor is defective.

Lift the wheel from the floor and turn the wheel in the rotation direction by hand. If a backlash of 10 degrees (about 26 mm on the wheel perimeter) or more is found, because the motor output shaft may be worn significantly, it is necessary to replace the motor promptly.

5.13 Replacement of the battery

Replace the battery while referring to “Replacing the Battery” in Installation Manual.

<How to judge whether the battery is normal or defective>

There are cases where tube voltage dropping trouble (errors F12, F15 and F16) occurs especially under a high X-ray condition when the battery level is low with the battery degraded.

In such a case, if recharging the battery for a short time improves the trouble condition, it is possible to judge that the trouble is attributable to a defect of the battery without a doubt.

Even if all of the five battery level indicator LEDs light, the battery is not necessarily in the fully charged condition and there are cases where F12, F15 or F16 occurs due to insufficient capacity. In such a case, if additional charging brings about obvious improvement, it is highly possible that the battery is defective.

NOTE

Because there were cases where, in spite of the key switch being turned OFF, a halfway voltage was supplied from the battery to the control circuit due to a contact fault of the power relay K7 or K8 on the MUX POWER-32K PCB and abnormal noise was emitted from the inside of the system or the battery voltage was unbalanced, it is necessary to check the MUX POWER-32K PCB for a malfunction.

If the battery voltage extremely drops (over discharge) repeatedly or shallow charging/discharging is repeated frequently, the life of the battery will be decreased. While referring to “4.2.48 bbb”, pay sufficient attention to how to use the system so as to prevent the recurrence of a defect of the battery.

If the battery voltage during radiography is unbalanced by 10V or more in spite of ± balancing in a stand-by state, it is possible to judge that the battery is defective without a doubt. Usually the battery voltage during radiography is hardly unbalanced: it can only be unbalanced by about ±1V at most.

In the following case, for example, some of the 10 batteries on the - side may be defective.

+ side: 126V (stand-by) → 100V (during radiography)

- side: 126V (stand-by) → 60V (during radiography)

In such a case, in addition to errors F12, F15, F16, etc. that mean tube voltage dropping, errors F55, F56, etc. that mean unbalancing may be apt to occur.

In the following halfway case experienced rarely, for example, the batteries on the - side may be on the point of getting defective.

+ side: 126V (stand-by) → 100V (during radiography)

- side: 126V (stand-by) → 92V (during radiography)

5.14 Replacement of LED lamp

(Reference work time: 1 persons × 1.5 hours)

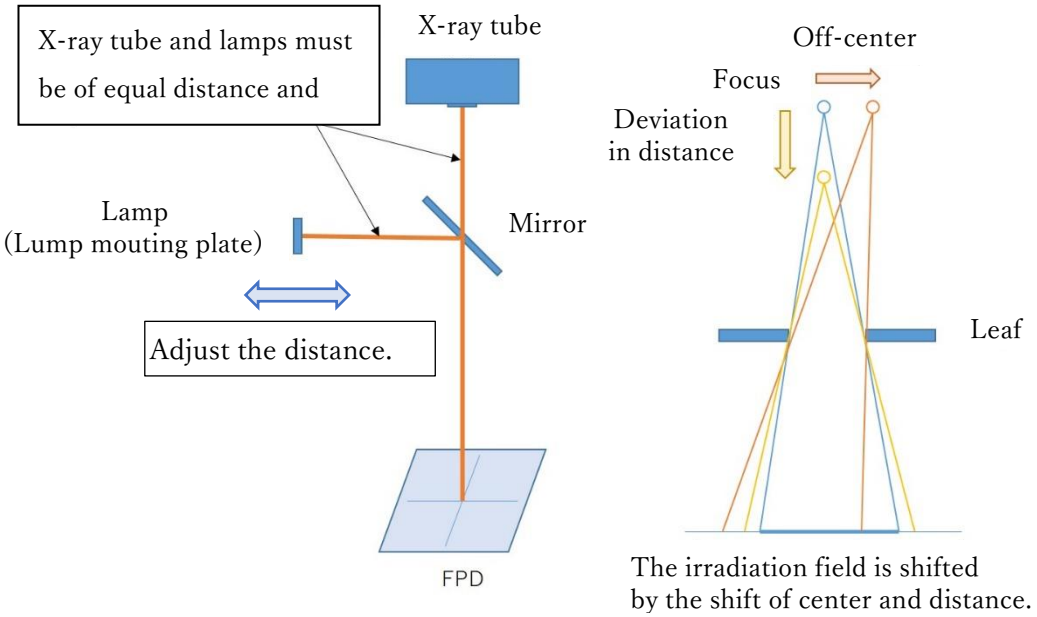
I. First, make sure that the center of the collimator illumination field matches the center of the X-ray irradiation field. Also, please carefully check the following precautions.

NOTE

There are following two kinds of discrepancy between the collimator illumination field and X-ray irradiation field. When installing the apparatus, make sure which discrepancy is occurring and adjust the discrepancy:

① Discrepancy between the collimator illumination field center and X-ray irradiation field center.
This is adjustable.

② Discrepancy between the collimator illumination field size and X-ray irradiation field size.
As a system specification, this does not have an adjustment mechanism.



NOTE

In the case of a new installation, adjust the center position of the illumination field and X-ray irradiation field at the installation position of the rotating ring of the collimator.

On the other hand, when replacing the LED ASSY, adjust the mounting position of the LED ASSY based on the assumption that the collimator is installed in the correct position (the center of the light field is aligned with the center of the X-ray field).

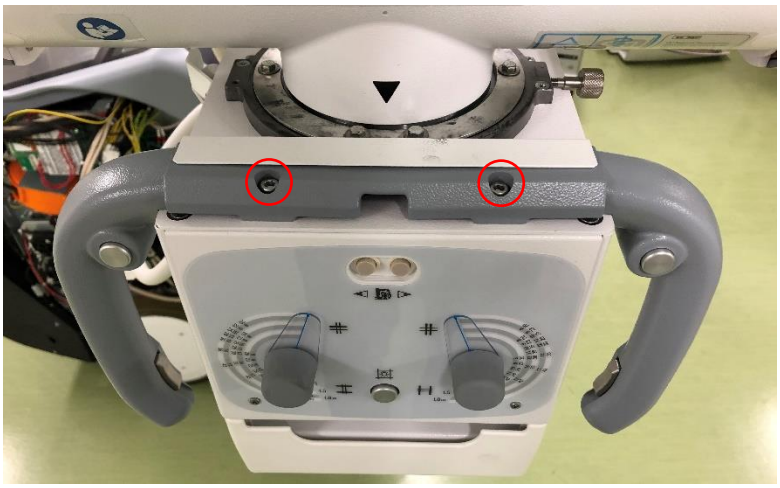
Before replacing the LED ASSY, be sure that the center of the light field and X-ray fields are aligned.

When replacing the lamp, never change the collimator position.

5. Parts replacement

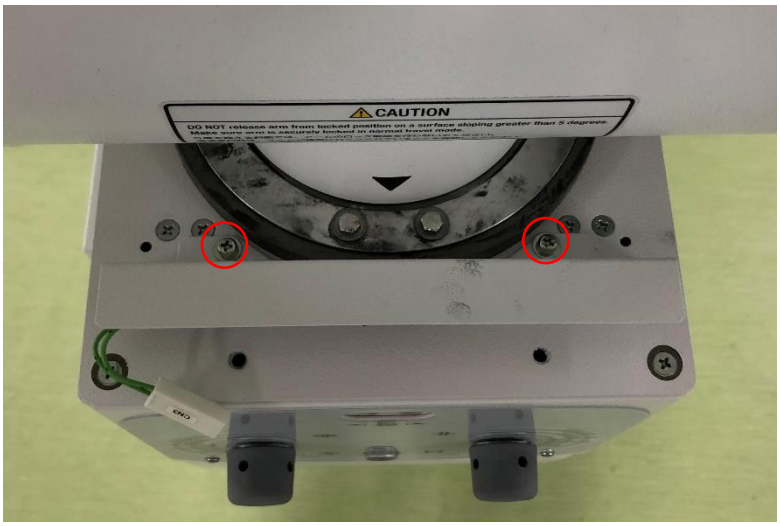
II. Remove the collimator handle cover.

III. Remove the handle.



5. Parts replacement

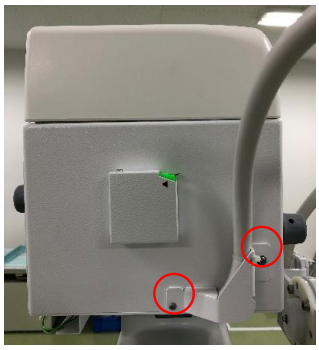
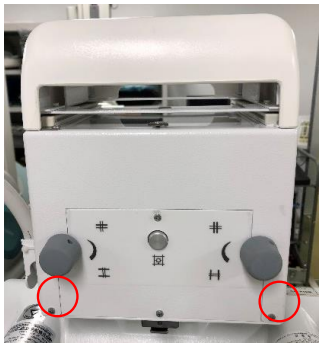
IV. Remove the cable cover.



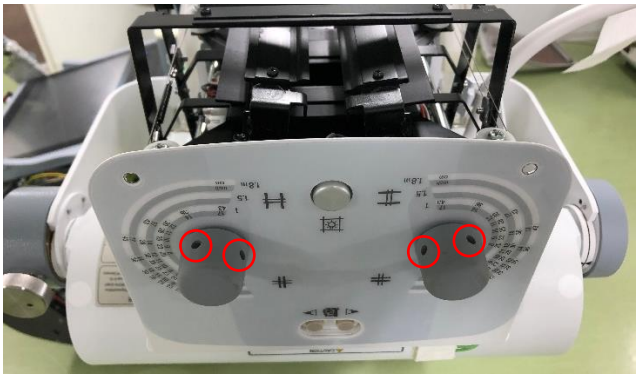
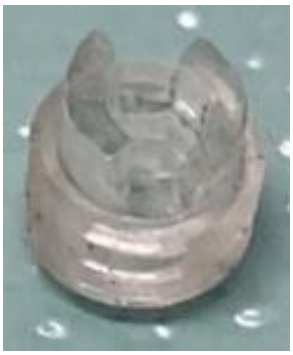
V. Rotate the collimator 180 degrees and flip it upside down.

5. Parts replacement

VI. Remove the seven screws and remove the collimator cover.

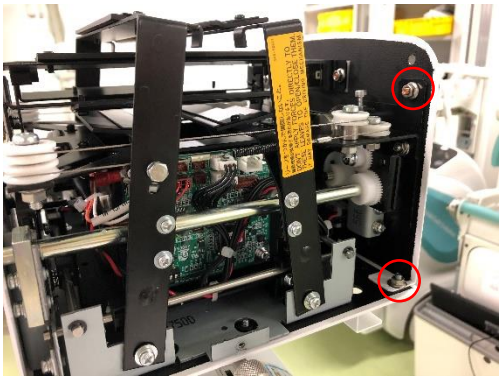



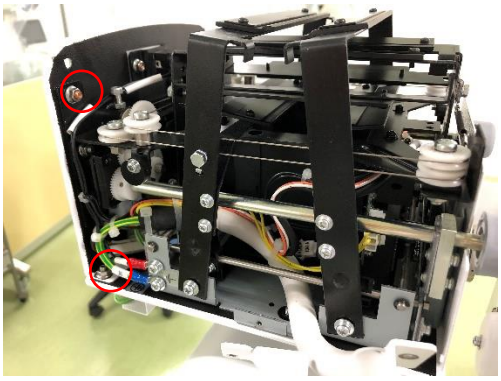
VII. Remove the knob. The knob is secured in two places with set screws. It is attached with a metal set screw and a plastic set screw over it. Remove the plastic set screw first, then remove the metal set screw in.

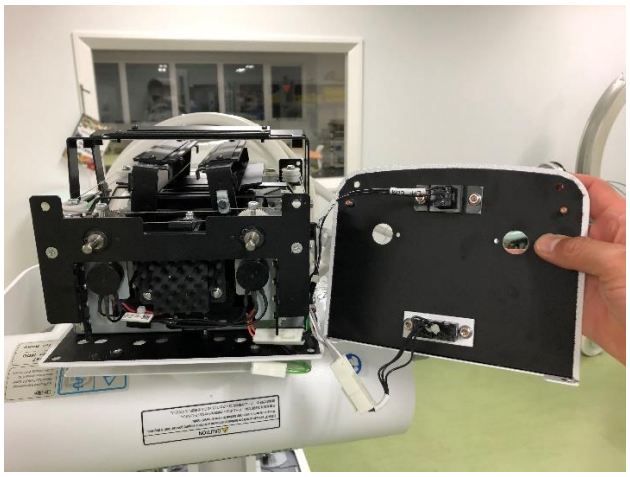



5. Parts replacement

VIII. Remove the front cover. Flange nuts on the two lower sides are secured with countersunk screws.

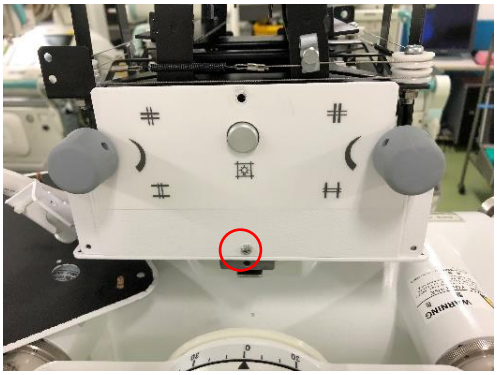
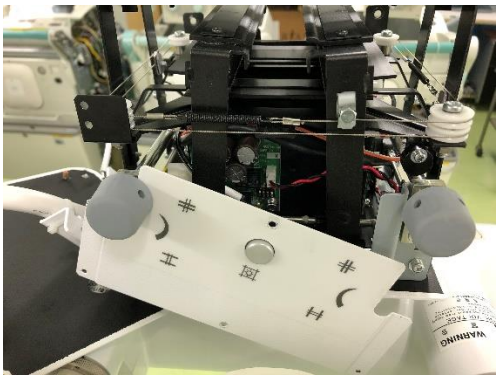




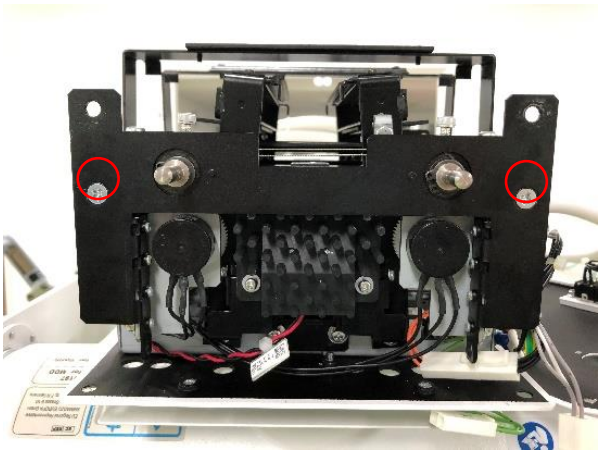
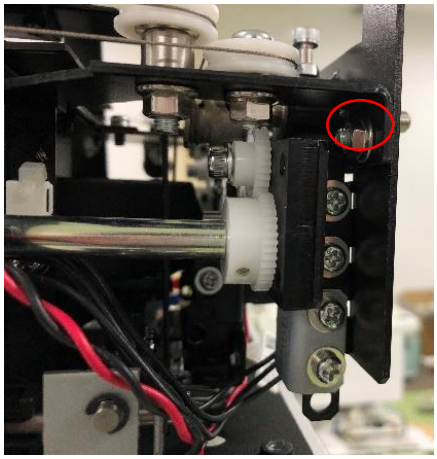


5. Parts replacement

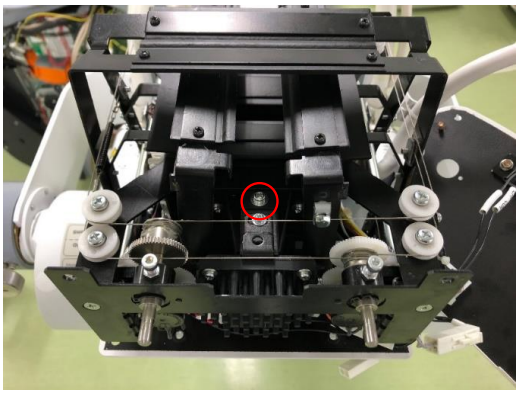
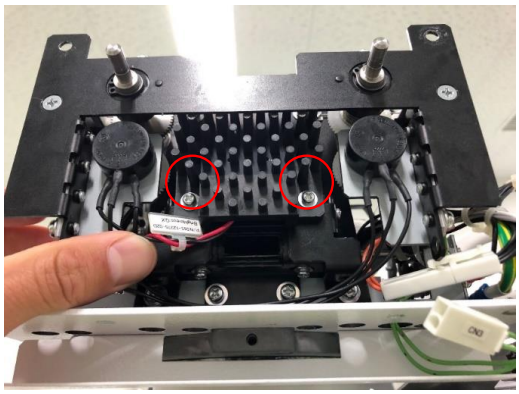
IX. Remove the LED lamp connector from the board. Remove the screw that secures the collimator rear cover and slide the cover away from the cable. When moving the cover, be careful not to break the cable because it is connected.

X. Remove the screws that secures the plate with the knob shaft.

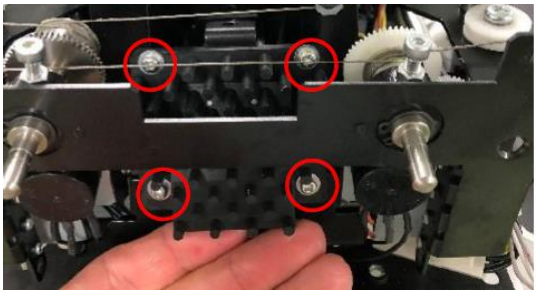
XI. Remove the three screws securing the LED lamp ASSY.

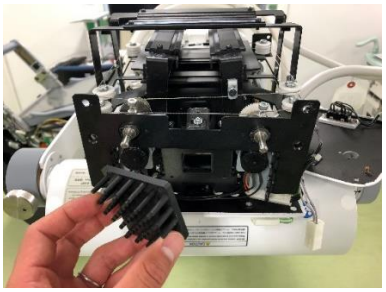



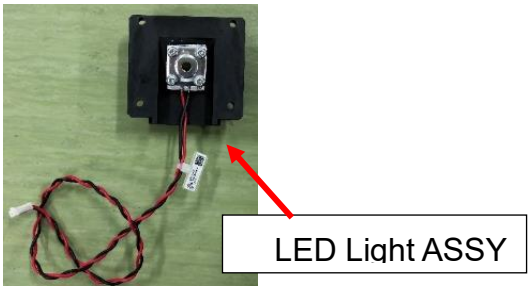
5. Parts replacement

XII. Remove the heat sink.

(The heat sink should be replaced together with LED LIGHT ASSY to the new one. LED LIGHT ASSY is fixed at the rear side of the heat sink. Though 2 pieces of the mounting plates are also included in LED lamp ASSY, it is not necessary to replace those plates to the new one.)

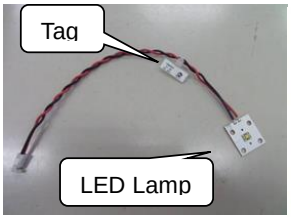




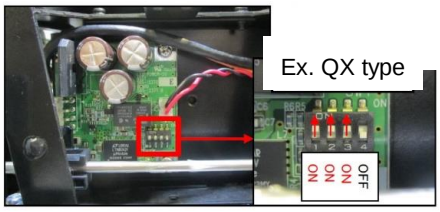


XIII. Install the new LED LIGHT ASSY with the heat sink and connect the cables. Secure the cable with a tie band so that it does not interfere with leaf movement or opening and closing of the cover.

XIV. Check the symbol on the tag attached to the LED LIGHT ASSY and set the dip switch SW1 of the LED POWER -20 board referring to the table below.



LED LIGHT ASSY



LED POWER-20 PCB

LED rank

	Dip SW1 -No			
	1	2	3	4
PX	OFF	ON	OFF	OFF
PY	OFF	ON	ON	OFF
PZ	ON	OFF	ON	OFF
QX	ON	ON	OFF (※ON)	OFF

DIP S/W setting

On the other hand, for LED lamps shipped from around April 2022, the LED rank symbol is not available on the tag attached to the LED LIGHT ASSY. In this case, set the DIP switch SW1 on the LED POWER-20 board as shown below.

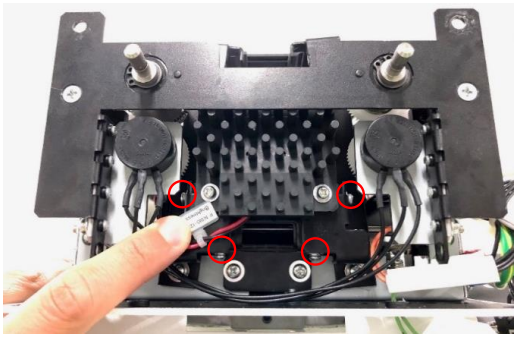
SW1-1 ON、SW1-2 ON、SW1-3 OFF、SW1-4 OFF

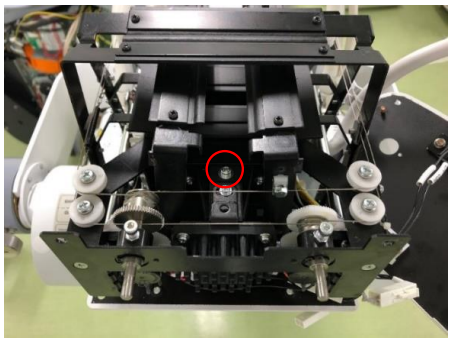
XV. Install the collimator rear and front covers.

XVI. Install and temporarily fix the knob.

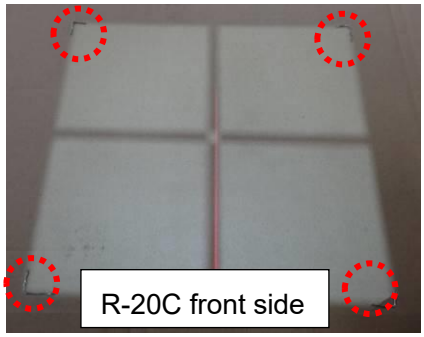
XVII. Rotate the collimator 180 degrees so that the field faces the floor.

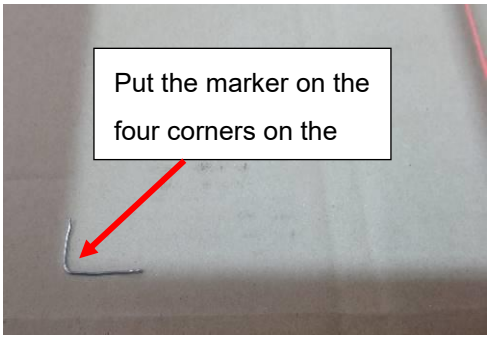
XVIII. Aligns the illumination field and x-ray irradiation field. Loosen the five screws holding the lamp assembly in place.



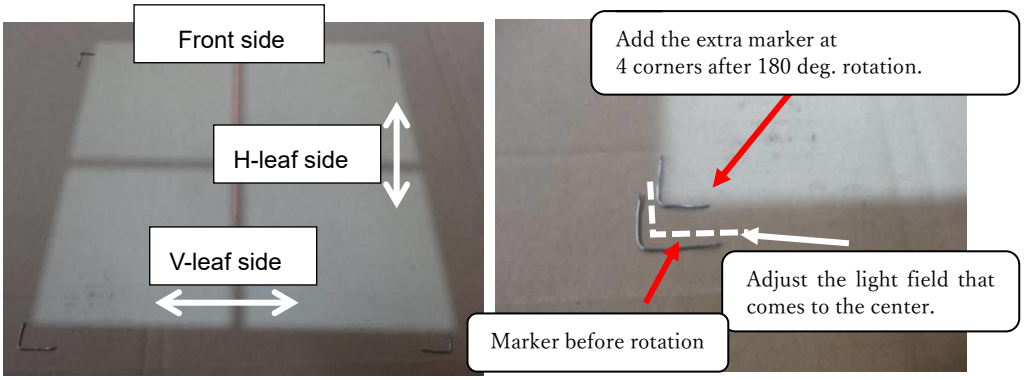


XIX. Move the X-ray tube SID 1 meter above the table or floor, open the collimator 35 cm square, and put the marker on the four corners.

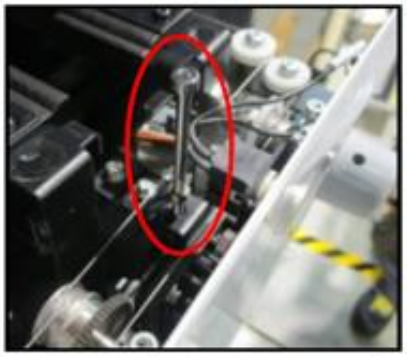


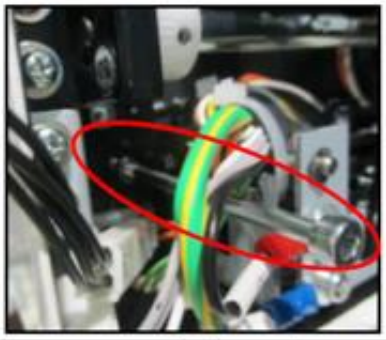


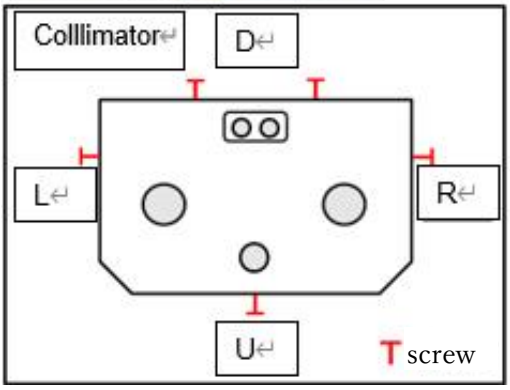
XX. Rotate the collimator body 180 degrees and make sure it matches the 4 corner marks. If they do not match, place marks at the 4 corners after rotation, and adjust the light field so that it is positioned in the middle between those marks and the marks placed in step 19.



XXI. Install the adjustment screws (M4x12 screws included with the device, or equivalent screws or bolts) and align the illumination field and x-ray irradiation field. Refer to the figure below for the movement of the illumination field when the bolt is tightened.







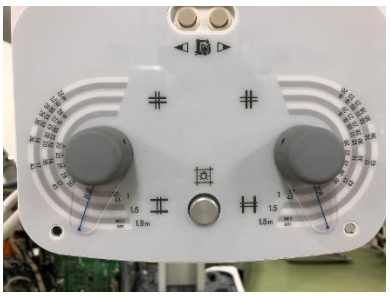
1: Tighten the adjustment screws so that they all touch.
2: Adjust using the example below

Ex.1: ↓move illumination field to down
Loosen U adjust bolt then tightn D adjust bolt

EX.2: →move illumination field to right
Loosen L adjust bolt then tightn R adjust bolt

XXII. After aligning the center of illumination field and x-ray irradiation field, tighten the five screws that secure the lamp ASSY and remove the adjustment bolts. Be sure to tighten the lamp assembly screws.

XXIII. Adjusts the opening display. For a SID 1 m, adjust the illumination area to 14 inches square. Adjust and secure the knob needle to point 14 inches at 1 meter SID.
Be sure to install the resin set screw.



XXIV. Replace the cover in the reverse of step 2~5.

XXV. Make sure that the illumination field match the x-ray irradiation field.
If adjustment is necessary, refer to the "Centering the Collimator and X-Ray Focus " section of the Installation manual.
See also "5.16 Supplement" for information on " Centering the Collimator and X-Ray Focus".

XXVI. When MX9 version does not have the 3D camera option, the angle of the collimator detected by the angle sensor of this CL MEAS-17 board is displayed.
Since the attitude of the CL MEAS board may be shifted when the collimator cover is removed, it is necessary to check whether the angle display is not shifted after replacing LED POWER-20 board. Follow the procedure "7.26.3 Checking the collimator angle" in the Installation Manual to check the angle, and if necessary, adjust the collimator angle detection again.

5.15 Supplement of "Centering the Collimator and X-ray Focus"

"Centering the Collimator and X-ray Focus" in the Installation Manual is supplemented with photographs here.

If the center of the X-ray field and the center of the light field are out of alignment, adjust the mounting position of the collimator body to the X-ray tube according to Installation Manual instead of adjusting the lamp position for LED mounting device. (Adjustment of the lamp position is necessary only when replacing the lamp.)
When adjusting the mounting position of the collimator body with the "screw" for adjustment (4 places at 90 deg. pitch) described below, be sure to loosen all the "bolt" (6 pieces) before adjustment. After adjustment, be sure to retighten the “bolt”.

Excerpt from Installation Manual

6 **Attach the screw and loosen the bolt a little, and then correct the discrepancy measured in the step 5 by adjusting the position where the screw is tightened. After adjustment, tighten the bolt firmly, and then remove the screw.**

⚠ CAUTION

Because the center of the collimator blades and the illumination field center are aligned each other using a dedicated device in factory assembling, never move the lamp mounting board.

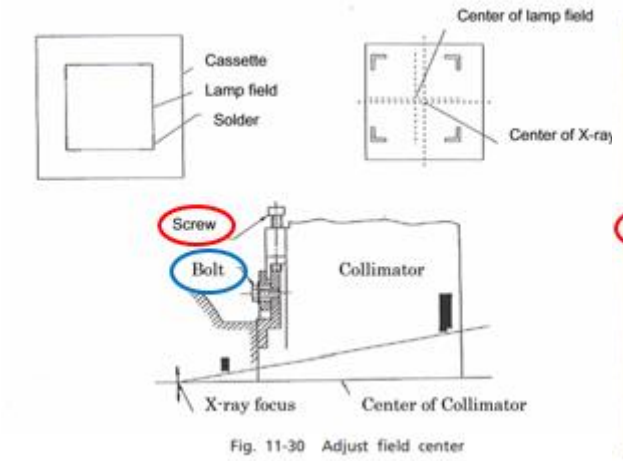
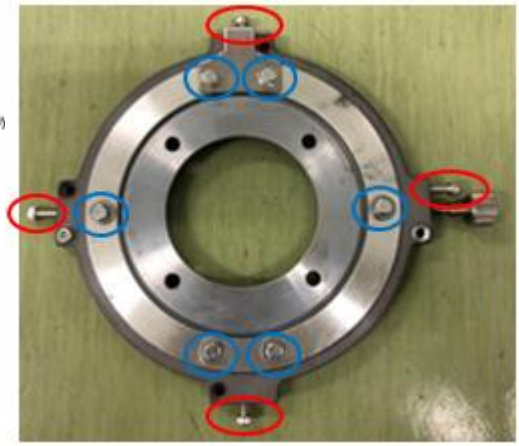


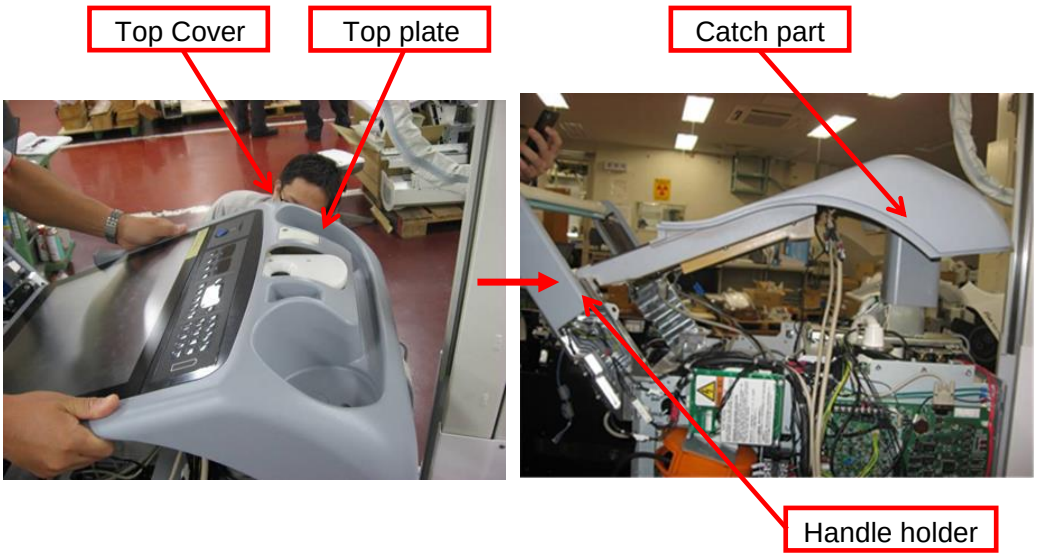
Fig. 11-30 Adjust field center

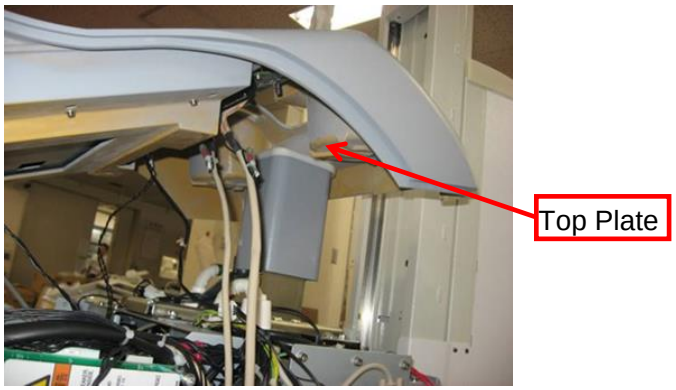


5.16 Replacement of Sheet panel

(Referece work time: 1 person × 0.5 hours)

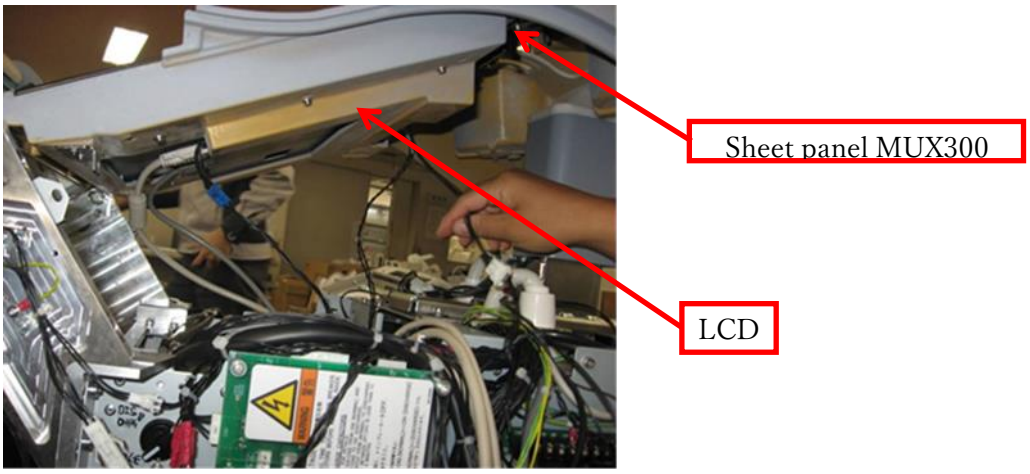
I. Refer to "How to Open/Close Covers" in the installation manual to remove the left and right insidecovers and side covers, open the FPD box, remove the Top cover.
Lift it up, place it on the Catch and handle holder. (If you lift the top cover straight up, it will interfere with the top plate of the catch. Lift it diagonally.)





5. Parts replacement

II. Disconnect all cables connected to the Sheet panel MUX 300 and LCD and remove the Top cover.



III. For DaRt type, replace it according to the following procedure.

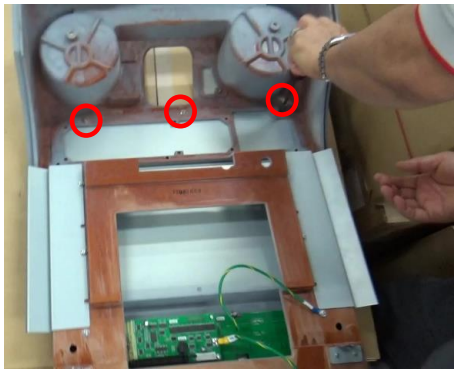
i. Remove the 6 flange nuts and remove the seat panel.

ii. Install a new seat panel and reassemble in the reverse order of the above.

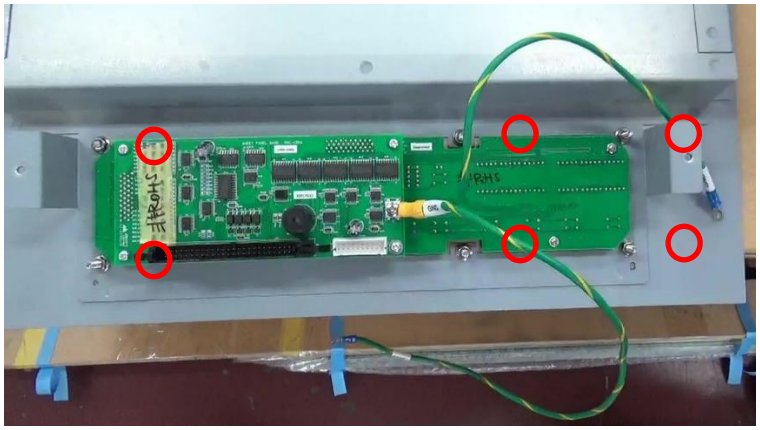
5. Parts replacement

IV. For Art type, replace it according to the following procedure.

i. Remove 3 flange nuts and 2 screws. Then remove the resin cover from the base.



ii. Remove the 6 flange nuts and remove the seat panel from the base.



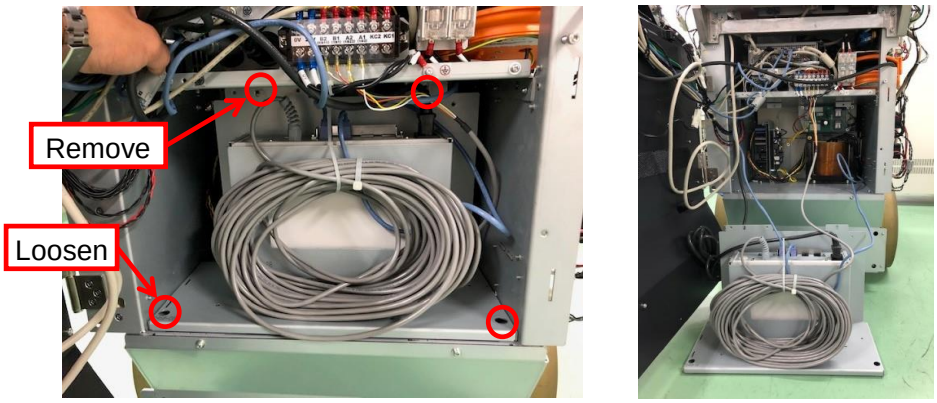
iii. Install a new seat panel and reassemble in the reverse order of the above.

5.17 Replacement of DC power supply for DR, POWER-200D and DR POWER CONT

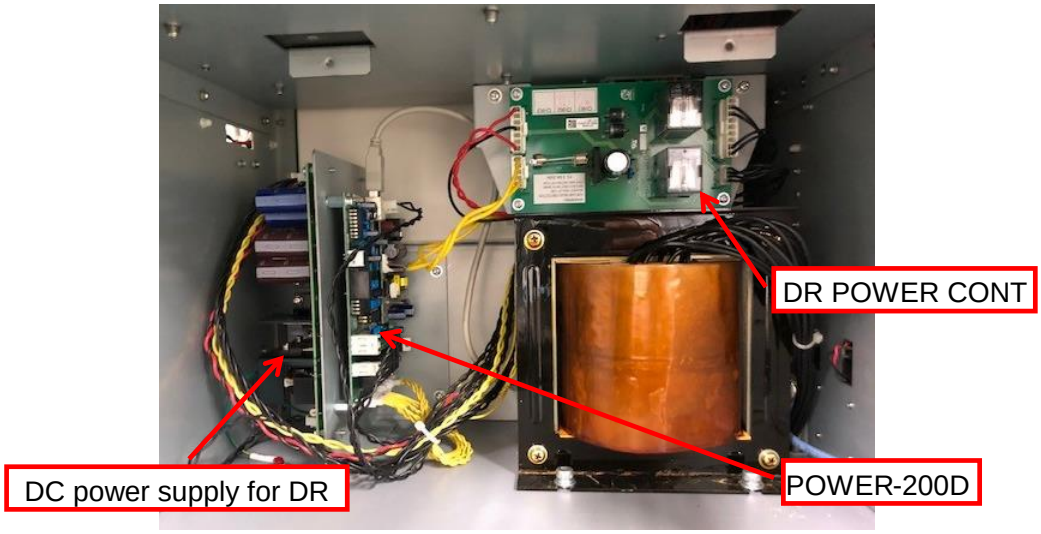
(Referece work time: 1 person × 0.75 hours)

I. Referring to “How to Open/Close the Covers” in the installation manual to remove the right and left inside covers and side covers. Then open the FPD box.

II. Remove the 2 upper screws and loosen the 2 lower screws that attach the I / F BOX. Then remove the I / F BOX.



III. The positions of the DC power supply for DR, POWER-200D, and DR POWER CONT are as shown in the figure below. Remove the replacement target.



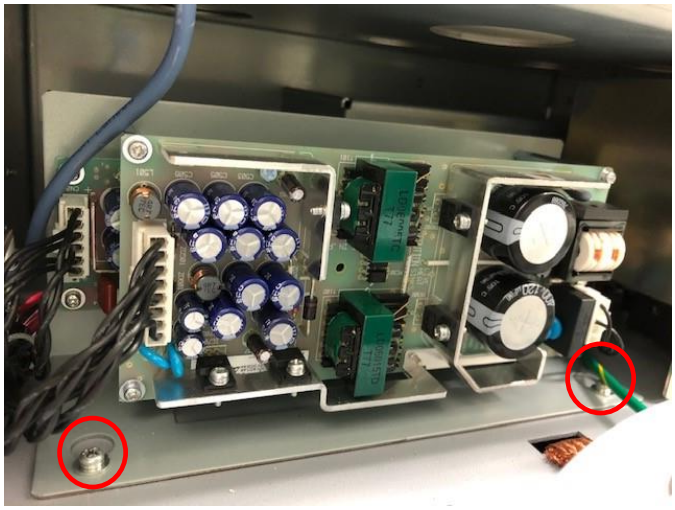
IV. Install new parts and reassemble in the reverse order of the above.

5.18 Replacement of Power supply

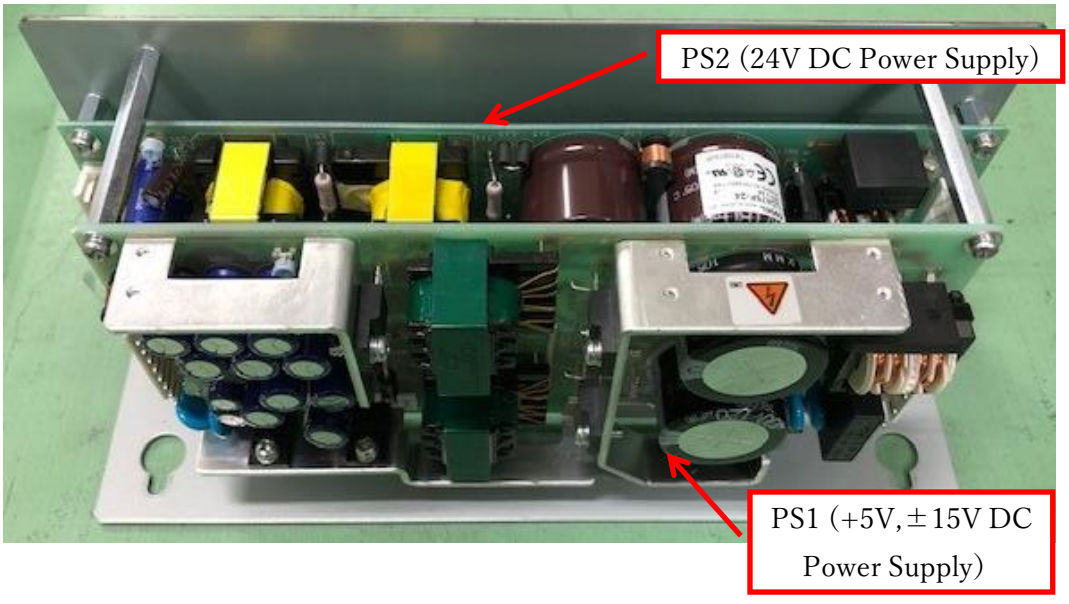
(Reference work time: 1 person × 0.5 hours)

I. Referring to “How to Open/Close Front Cover” in the installation manual to remove front cover.

II. Loosen the 2 screws that secure the power supply unit.



III. Remove the 4 connectors fixed to the power supply and take out the power supply unit.



IV. Replace the replacement target power supply and reassemble in the reverse procedure above.

5.19 Replacement of the XCONT PCB, the INVERTER UNIT PCB and the H.V.Transformer

5.19.1 Combination of parts

In each system of MX8 DaRt / Art (32kW / 12.5kW), the combination of the XCONT PCB, the INVERTER UNIT PCB, and the H.V.Transformer is one of the following.

<table>
<tr><th colspan="2">System</th><th>#</th><th>XCONT PCB</th><th>INVERTER UNIT PCB</th><th>H.V.Transformer</th></tr>
<tr><td colspan="2" rowspan="2">MX8
DaRt</td><td>1-1</td><td>XCONT-2008X:
P/N: 565-12406-01,
XCONT-2018:
Except for 565-15000-03</td><td>INVERTER UNIT-32K
P/N : 562-26303</td><td>D150MH-40
P/N : 562-25900</td></tr>
<tr><td>1-2</td><td rowspan="3">XCONT-2018
P/N : 565-15000-01/02
or Rev.K or later of
565-15000</td><td>INVERTER UNIT-32K
P/N : 562-26303-01</td><td>D150MH-40(S)
P/N : 565-16863/-01</td></tr>
<tr><td rowspan="2">MX8
Art</td><td>32kW</td><td>2-1</td><td rowspan="2">INVERTER UNIT-C1
P/N : 562-22596</td><td rowspan="2">D125PH-C1
P/N : 562-20700-02</td></tr>
<tr><td>12.5kW</td><td>3-1</td></tr>
<tr><td colspan="2">MX9
DaRt</td><td>4-1</td><td>XCONT-2018
P/N : 565-15000-03</td><td>INVERTER UNIT-32K
P/N : 562-26303-01</td><td>D150MH-40(S)
P/N：565-16863/-01</td></tr>
</table>

< Regarding MX8 DaRt >

In the combination of 1-1, if the same H.V.Transformer or INVERTER UNIT PCB as the original ones are not available, it is necessary to replace all the parts so that the combination is 1-2. On the other hand, any XCONT-2018 PCB except for 565-15000-03 is compatible with the XCONT-2008X PCB and can be used for 1-1 combinations.

< Regarding MX8 Art >

The combination of each part is only 2-1 for the 32kW specification and 3-1 for the 12.5kW specification. It is not possible to combine the XCONT2008X PCB or the combination of 1-1 in the 32kW specification.

< Regarding MX9 DaRt >

The combination of each part is only 4-1.

Regarding MX7 or earlier versions of DaRt or Art 32kW, the combination of each part is only 1-1, and 1-2 is not allowed.

5.19.2 When there are the same parts as the original ones.

(In the case of 1-1, 1-2, 2-1, 3-1 and 4-1)

5.19.2.1 Introduction

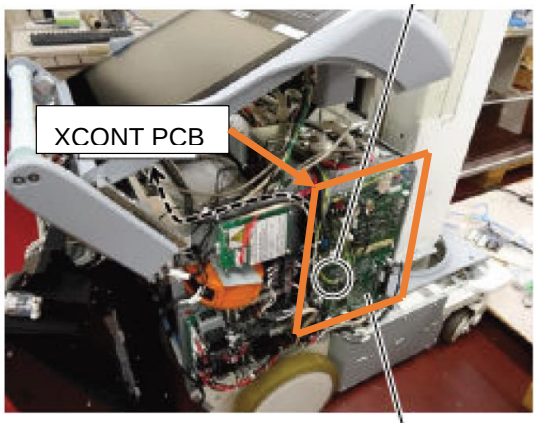
If there are the same maintenance parts as the original ones, it can be replaced the parts individually. Refer to the replacement procedure for each parts. The X-ray tube voltmeter (kV meter) is required to replace the H.V.Transformer.

5.19.2.2 Replacement of the XCONT PCB

(Reference work time: 1 person × 0.5 hours)

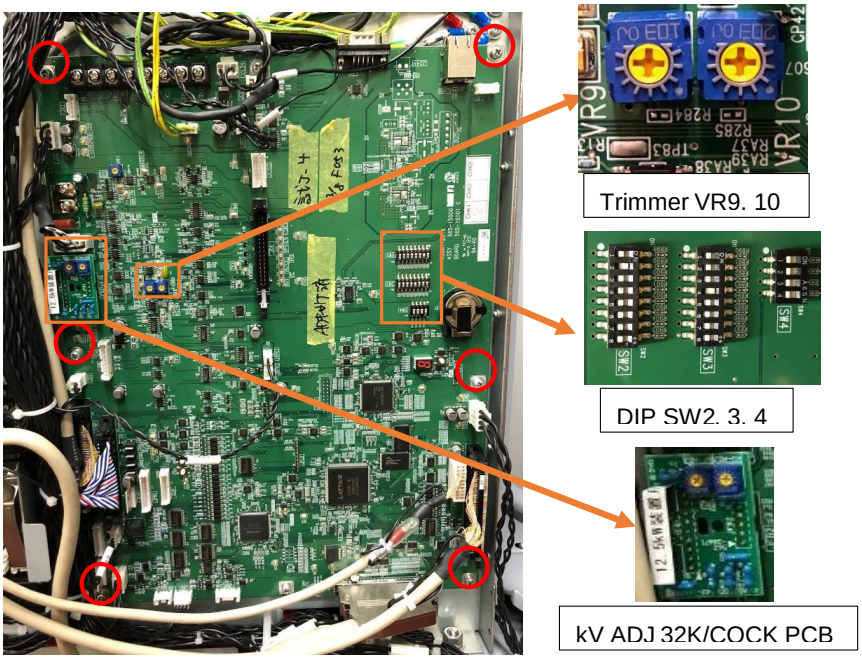
I. Turn off the power and turn off the main breaker.

II. Referring to “How to Open/Close the Covers” in the installation manual, remove the right inside covers and side covers.



5. Parts replacement

III. Remove all cables connected to the XCONT PCB, remove the 6 screws that secure that PCB, and replace with a new one.



IV. Remove the kV ADJ 32K PCB for 32kW type or the kV ADJ COCK PCB for 12.5kW type from the original XCONT PCB and attach it to the new XCONT PCB after replacement.

V. Make sure that DIP SW2,3 and 4 have the same settings as the original board.

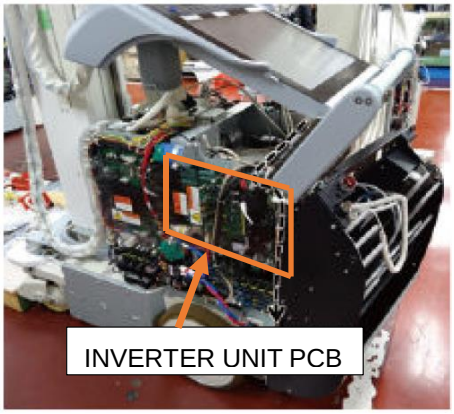
VI. Next, refer to Chapter 2.5.3 and the Installation Manual to initialize the EEPROM by setting DIP SW2-8 ON and SW3-1 ON, adjust the trimmers VR9 and VR10, upload the backed-up data, adjust the FVR for confirmation, adjust the mAs circuit, adjust the inverter control circuit, and adjust the battery detection voltage. Note that the adjustment value differs depending on the type of H.V.Transformer.

5.19.2.3Replacement of the INVERTER UNIT PCB

(Reference work time: 1 person × 0.5 hours)

I. Turn off the power and turn off the main breaker.

II. Referring to "How to Open/Close the Covers" in the installation manual, remove the left inside covers and side covers.



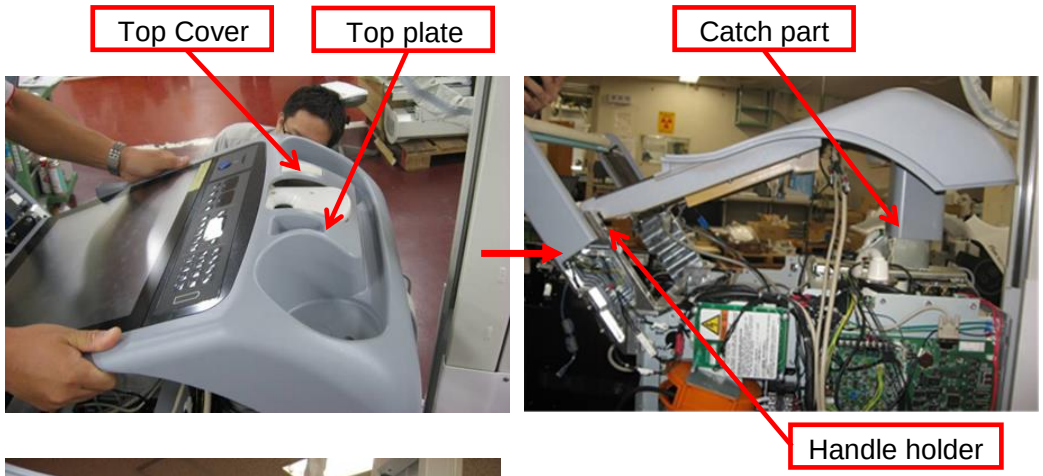
III. Remove all cables connected to the INVERTERT UNIT PCB, remove the 6 screws that secure the PCB, and replace with a new one.

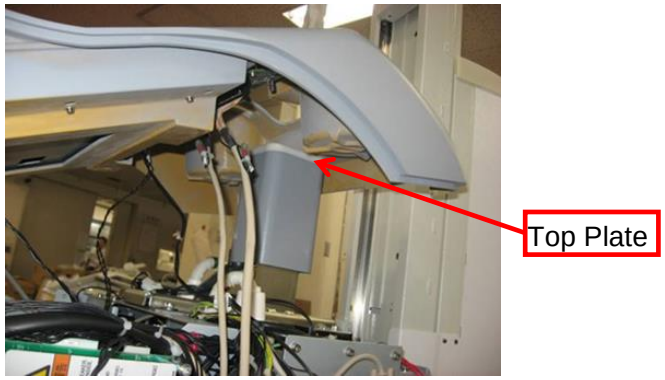
5.19.2.4 Replacement of the H.V.Transformer

(Reference work time: 2 person ・ 2.5 hours, X-ray tube voltmeter required)

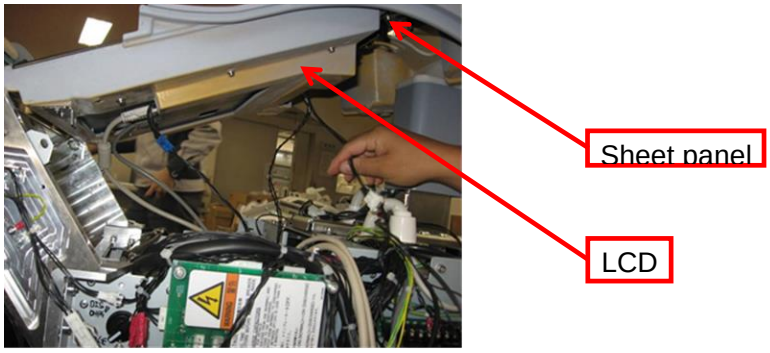
I. Turn off the main breaker, and in the case of DaRt and Art 32kW type, discharge the capacitor for safety.

II. Refer to "How to Open/Close Covers" in the installation manual to remove the left and right inside covers and side covers, open the FPD box, remove the Top cover. Lift it up, place it on the Catch and handle holder. (If you lift the top cover straight up, it will interfere with the top plate of the catch. Lift it diagonally.)

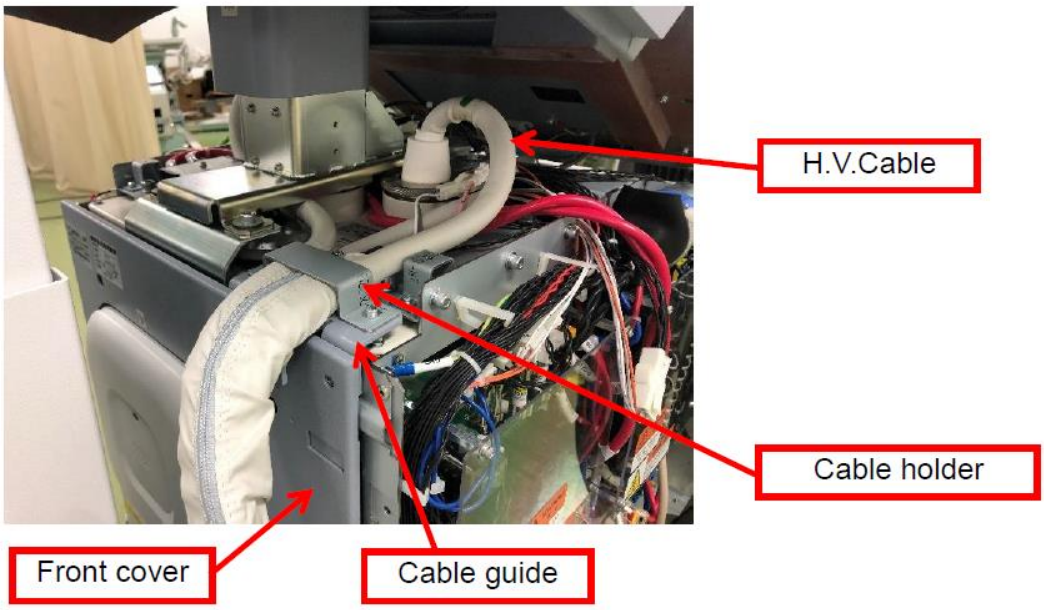




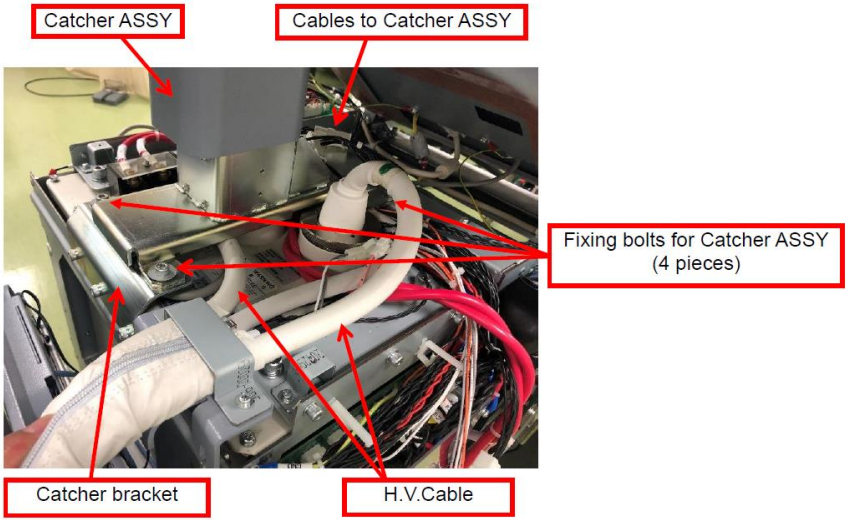
III. Disconnect all cables connected to the Sheet panel and LCD (only DaRt) and remove the Top cover.



IV. Remove the front cover, cable guide and cable holder referring to “How to Open/Close the Covers” in Installation Manual.
Before removing the cable holder and cable guide, mark their current location onto the surface of H.V.Cable for easy cable treatment after replacing H.V.Transformer.

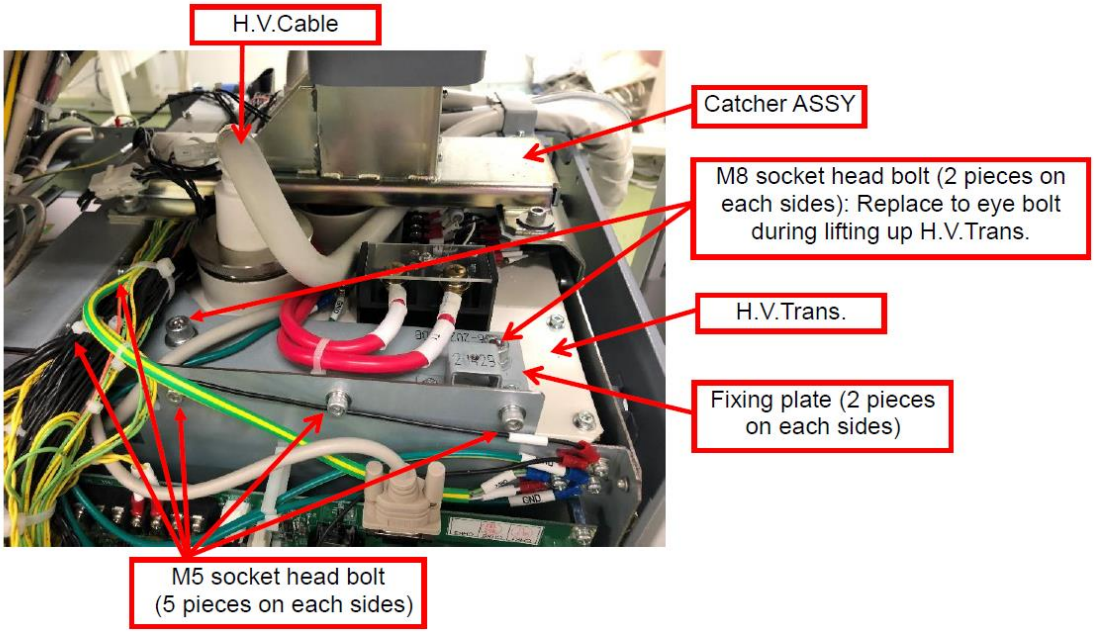


V. Disconnect the cables at the connector from Catcher ASSY, then, remove Catcher ASSY by removing 4 fixing bolts.
Before removing Catcher ASSY, mark the current fixing position of it to return it to the original position after replacing H.V.Transformer. (In case of forward/backward direction, Arm Joint Pin should come into the hole on Catcher ASSY smoothly. As well, in case of left/right direction, X-ray tube should locate at the center of the machine.)
After that, remove Catcher Bracket by removing 3 fixing bolts.



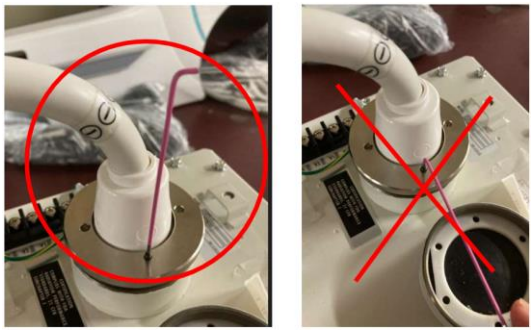
5. Parts replacement

VI. Disconnect all cables from H.V.Transformer. (Before disconnection, take pictures of the original cable wiring condition for easy work after replacing H.V.Transformer.)
Remove the left and right fixing plates by removing 2 pieces of M8 socket head bolts and 5 pieces of M5 socket head bolts for each.

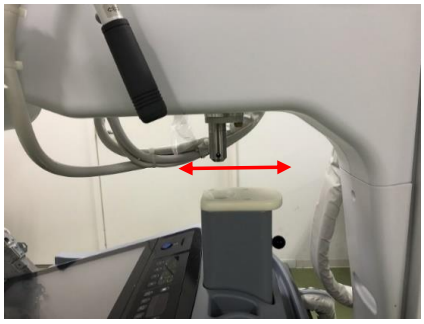


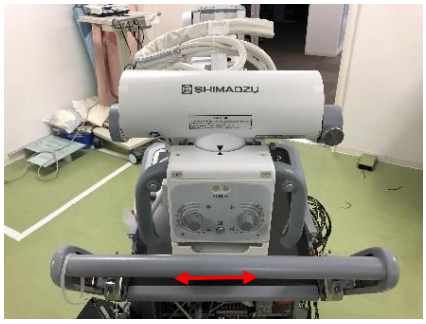
VII. Attach 4 pieces of M8 eye bolts to the screw holes for the left and right fixing plates which were removed as mentioned above.
Grip the eye bolts by 2 people, then, lift up and replace H.V.Trans. to the new one. As H.V.Transformer has about 39kg weight, be careful during replacement.

VIII. After replacing H.V.Transformer, connect H.V.cables to H.V.Transformer.
Apply the tightening torque of 0.5N･m or less, and be careful not to over-tighten the H.V.cables. Tightening with a strong force of about 1.0Nm, such as using a horizontal L-shaped wrench, may cause the socket to crack.



IX. After replacing H.V.Transformer, fix Catcher ASSY to its original position and confirm the following.
<Forward/backward direction>
Arm Joint Pin should come into the hole on Catcher ASSY smoothly.
<Left/right direction>
X-ray tube should locate at the center of the machine





X. At last, confirm the normal radiography operation and adjust tube voltage waveform by using the potentiometer VR3 and VR4 on KV ADJ-32K PCB referring to “Checking the X-ray exposure” in “Adjusting Methods” of Installation Manual.
The adjustment point (tube current value) at this time differs depending on the type of H.V.Transformer. (Refer to Chapter 2.5.3.)

5.19.3 If there are no parts that are the same as the original ones (In the case of 1-1)

5.19.3.1 Introduction

If the same type of H.V.Transformer or INVERTER UNIT PCB as the original ones are not available in the combination of MX8 DaRt 1-1, order all the parts so that the combination is 1-2 and replace following the procedure below. The X-ray tube voltmeter (kV meter) is required to replace the H.V.Transformer.

5.19.3.2 Replacement of the XCONT PCB

(Reference work time: 1 person × 0.5 hours)

I. Refer to Chapter 2.5.3 and 2.5.4 of this Service Manual for the PCB replacement procedure. Chapter 2.5.4 provides a flowchart of the XCONT board replacement procedure when changing the type of H.V.Transformer.

II. Before replacing the PCB, use the "all" command to get the backup data of "all EPROM storage data" of the XCONT PCB. Refer to "PC maintenance function" in the installation manual.

> **NOTE**
>
> The output ratio data corresponding to the combined H.V.Transformer is recorded on the XCONT PCB. In this work, the H.V.Transformer will be replaced from D150MH-40 (P/N: 562-25900) to D150MH-40(S) (P/N: 565-16863 or 565-16863-01). So, it is needed to update the output ratio data for D150MH-40(S) to the backup data.

III. Replace the XCONT PCB by referring to the procedure in "5.19.2.2 Replacement of the XCONT PCB". For the DIP SW settings, refer to "XCONT" in Chapter 2 of this service manual or the installation manual. (Turn on DIP SW2-6 when D150MH-40(S) is combined.)

IV. Initialize the XCONT PCB after replacement. Set DIP SW2-2 to OFF and DIP SW2-6 to ON to perform initialization. For the detailed procedure, refer to "Initialization procedure" in the installation manual.

V. Use the "heratio" command to acquire the output ratio data of the XCONT board. Refer to "PC maintenance function" in the installation manual. In this work, the output ratio data corresponding to D150MH-40(S) (P/N: 565-16863/-01) is acquired.

VI. Write the backup data to the XCONT PCB with the "all" command.

VII. Write the output ratio data to the XCONT PCB with the "heratio" command. In this procedure, the output ratio data of the backup data is updated to the corresponding data to D150MH-40(S) (P/N: 565-16863/-01).
(Since the output ratio data is rewritten by initializing XCONT PCB with the DIP SW setting corresponding to each H.V.Transformer type, the above procedure is necessary.)

VIII. Next, adjust the trimmer VR9 and VR10, perform FVR adjustment for confirmation, perform “Adjusting mAs circuit”, “Adjustment of the inverter control circuit” for confirmation, “Adjustment of battery voltage detection circuit” according to Chapter 2.5.3 of this service manual or Installation Manual.
Each adjustment methods are different depending on the type of H.V.Transformer.

5.19.3.3Replacement of the INVERTER UNIT PCB

(Reference work time: 1 person × 0.5 hours)

Replace the INVERTER UNIT PCB by referring to the procedure in "5.19.2.3 Replacement of the INVERTER UNIT PCB".

5.19.3.4Replacement of the H.V.Transformer

(Reference work time: 2 person × 2.5 hours, X-ray tube voltmeter required)

Replace the H.V.Transformer by referring to the procedure in "5.19.2.4 Replacement of H.V.Transformer".

When you replace the H.V.Transformer to the different model, INVERTER UNIT and XCONT should be replaced according to the combination list on “5.19.1 Combination of parts”. In addition, the DIP SW setting on the XCONT -2018 board and some adjustment procedure has been changed as shown below.

(1) About the Dip SW setting

Take care of the Dip SW setting on the XCONT-2018 board depending on the model of High Voltage Transformer according to the below table.

High Voltage Transformer	DIP SW setting on XCONT-2018
D150MH-40（P/N:562-25900）	S2-6 OFF
D150MH-40（S）（P/N:565-16863/-01）	S2-6 ON

(2)Adjusting the Inverter

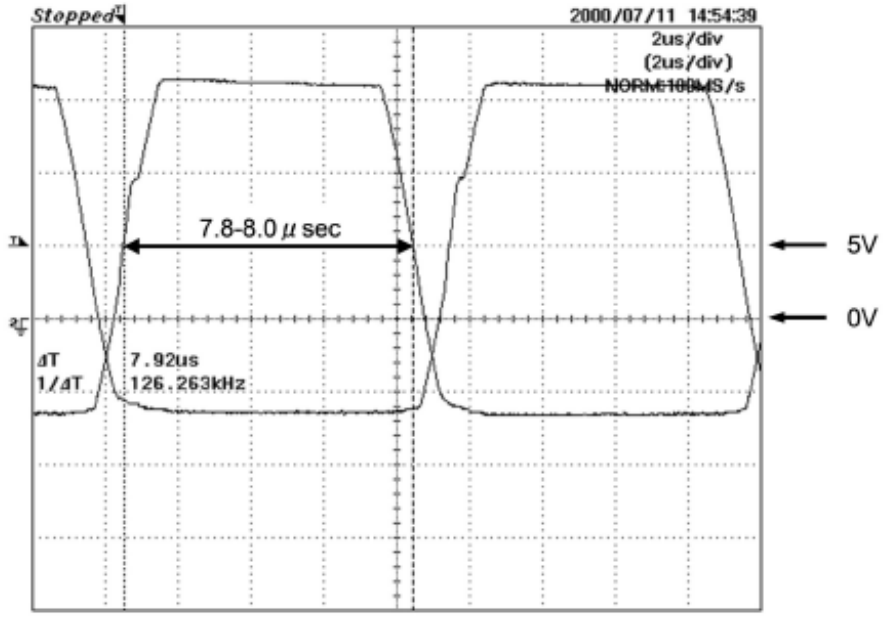
Note that the method of adjusting the IGBT gate waveform differs between the old and new High Voltage Transformers as shown below.

High Voltage Transformer	Pulse width
D150MH-40（P/N:562-25900）	7.8~8.0μs
D150MH-40（S） （P/N:565-16863/-01）	10.9~11.1μs

Check the inverter pulse width.

1. Connect oscilloscope probes between CP5 [Ga] - CP6 [Ea] and CP10 [Gb] - CP12 [Eb] on INVERTER UNIT board.
2. Perform READY and XRAY operations, and check the gate waveform of IGBT.
3. Confirm that the pulse width at +5 V is as shown in the following table.

If necessary, readjust the pulse width using the procedure in “Adjusting inverter pulse width of XCONT-2018 board” in installation manual.



Waveform between CP[Ga] – CP[Ea], CP[Gb] – CP[Eb] on INVERTER UNIT board

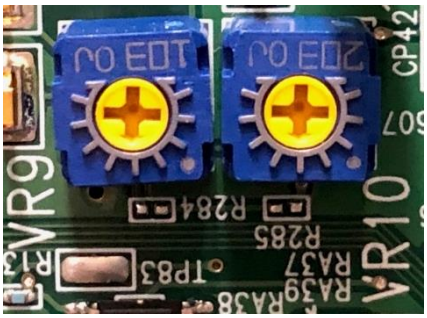
(Waveform for D150MH-40 is shown above.)

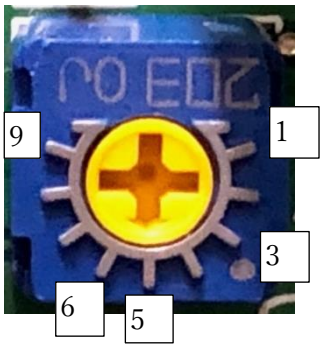
(3)Adjusting VR9, VR10 on XCONT-2018

The way to adjust the trimmer VR9 and VR10 on the XCONT-2018 board is different for the old and new High Voltage Transformers.

Only one of the trimmer's crosses is an arrow. Adjust the arrow to point to the memory location in the table below. Regarding VR10, it should be adjusted by measuring the resistance value between CP61［FB RES］and CP42［VINCHK3］on XCONT-2018 board. Note that the adjustment value varies depending on the type of transformer.

High Voltage Transformer	VR9	VR10
D150MH-40（P/N:562-25900）	Turn 5 scales in CW direction	5.8kΩ ±5%
D150MH-40（S）（P/N:565-16863/-01）	Turn 6 scales in CW direction	10 kΩ（Turn 5 scales in CW direction）





(4)Adjusting the tube voltage waveform

In the tube current 2 point adjustment mode, adjust VR4 (kV-ADJ) of the KVADJ board so that the tube voltmeter indicates 80 kV +/- 1 kV, and adjust VR3 (WF-ADJ) so that the tube voltmeter waveform is horizontal.

The adjustment point (tube current value: 50mA or 200mA) at this time differs between the old and new model of High Voltage Transformers.

Set X-ray condition as below, then, perform exposure.

High Voltage Transformer	Adjustment point
D150MH-40（P/N:562-25900）	80kV 50mA
D150MH-40（S）（P/N:565-16863/-01）	80kV 200mA

5.20 Replacement FPD-guide right frame ASSY

(Reference work time 1 person ・ 60min.)

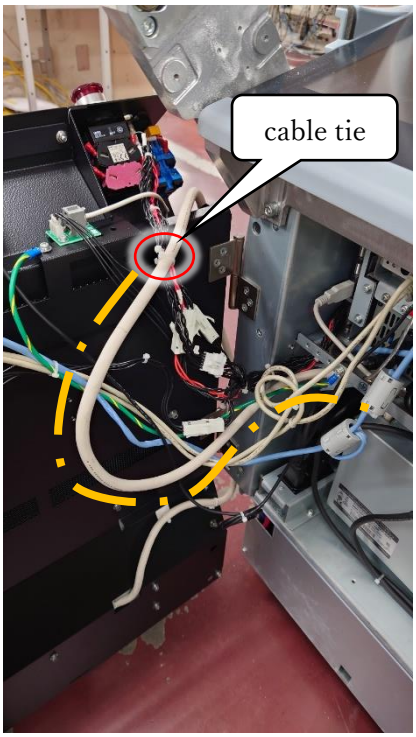
I. The configuration of the FPD-guided right frame ASSY is shown below.

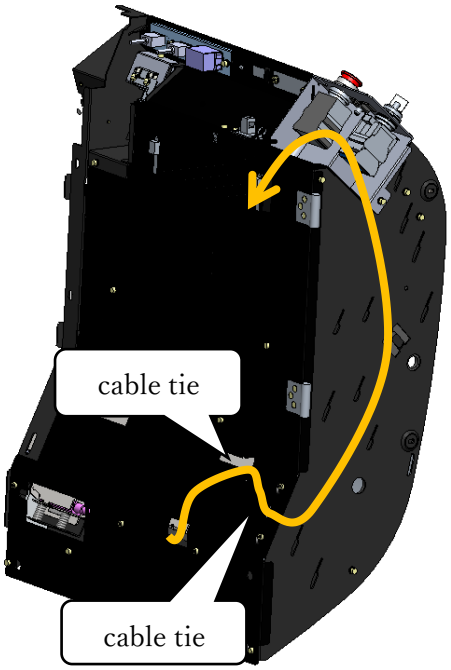
NOTE

Do not disassemble the FPD-guided right frame ASSY because the dimensions are adjusted when the parts are assembled at the factory.

II. Referring to “How to Open/Close the Covers” in the installation manual, remove the right and left inside covers and side covers. And then open the FPD box.

III. Cut the cable tie that secures the IF cable, and let the IF cable slack.

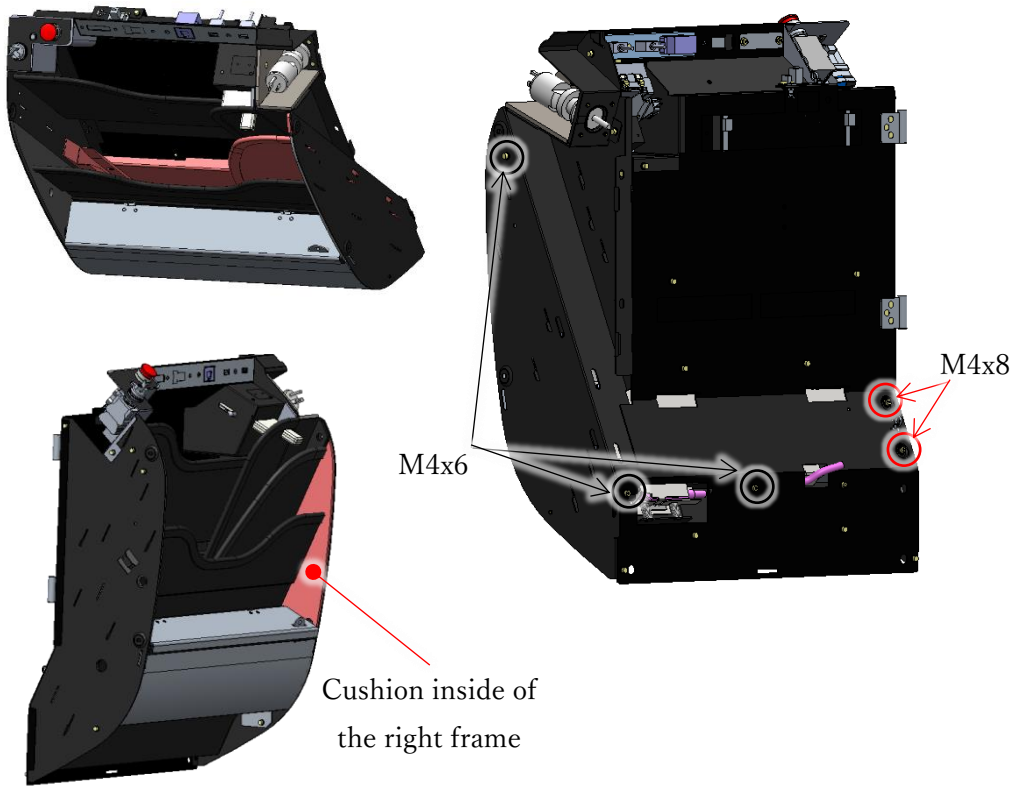




IV. Remove the frame 7 to the FPD box. Frame 7 is fixed with one screw on the back and one screw on the side by inserting the claws into the square holes of the box frame.

V. Remove the left frame ASSY to the FPD box. The left frame ASSY is fixed with one screw on the back and one screw on the side by inserting the claws into the square holes of the box frame.

VI. Remove the FPD-guided right frame ASSY to the FPD box. The FPD-guided right frame ASSY is fixed with four screws on the back and one screw on the side by inserting the claws into the square holes of the box frame.

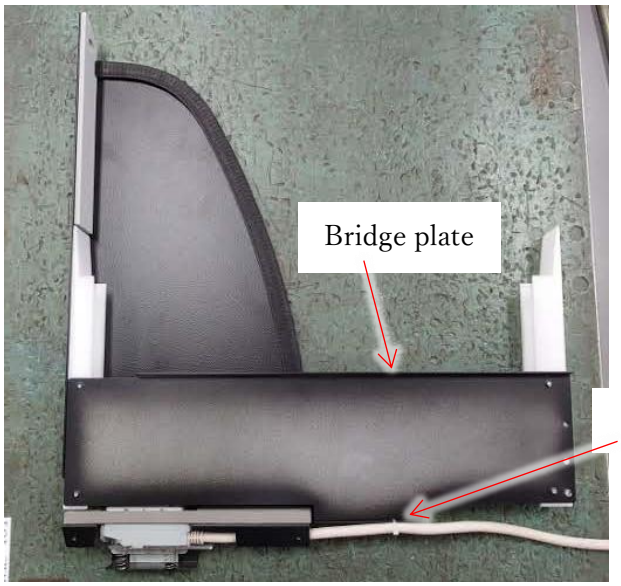


NOTE

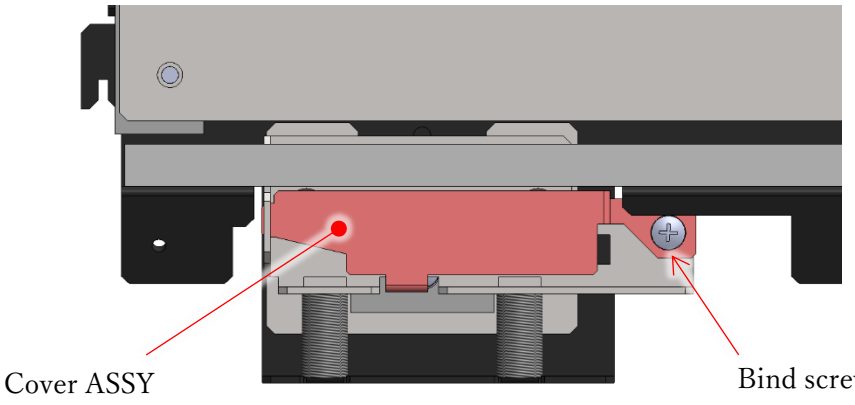
The IF cable is fixed to the FPD-guided right frame ASSY. When removing the FPD-guided right frame ASSY from the FPD box, work while pulling out the IF cable from the FPD box.

When removing the FPD-guided right frame ASSY from the FPD box, be careful that the claw of the FPD-guided right frame ASSY does not damage the cushion inside the right frame of the FPD box.

VII. Cut the cable tie the IF cable fixed to the square hole and slit of the bridge plate.



VIII. Remove the IF cable connector from the FPD-guided right frame ASSY. The connector is pasted on the cover ASSY made by sheet metal with double-sided tape and fixed with one bind screw.



IX. Reverse order of step 6 and 7 to attach the connector of the IF cable to the new FPD-guided right frame ASSY and secure it with a cable tie.

X. Refer to step 5, insert the claw of the FPD-guided right frame ASSY into the square hole of the box frame, and fasten four screws on the rear side and one screw on the right side temporarily.

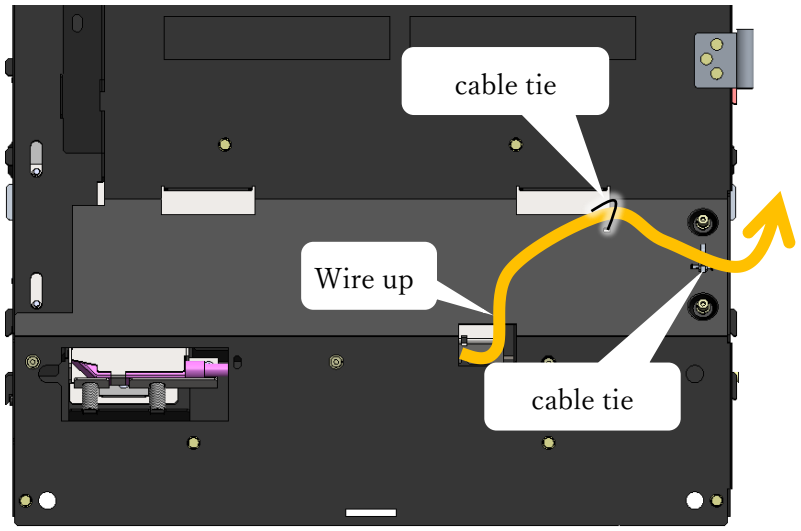
NOTE

Be careful not to crush the IF cable by inserting the FPD-guided right frame ASSY into the FPD box while simultaneously pulling the IF cable out of the FPD box.
When inserting the FPD-guided right frame ASSY into the FPD box, be careful that the claw of the FPD-guided right frame ASSY does not damage the cushion inside the right frame of the FPD box.

XI. Refer to step 4, Insert the claw of the left frame ASSY into the square hole of the box frame, and fasten one screw on the rear side and one screw on the left side temporarily.

XII. Refer to step 3, Insert the claw of the frame 7 into the square hole of the box frame, and fasten one screw on the rear side and one screw on the left side temporarily.

XIII. Reverse order of step 2, wiring the cable and fix it with cable ties.



XIV. Attach the FPD box and the covers in reverse order of step 1.

5.21 Replacement of Capacitor

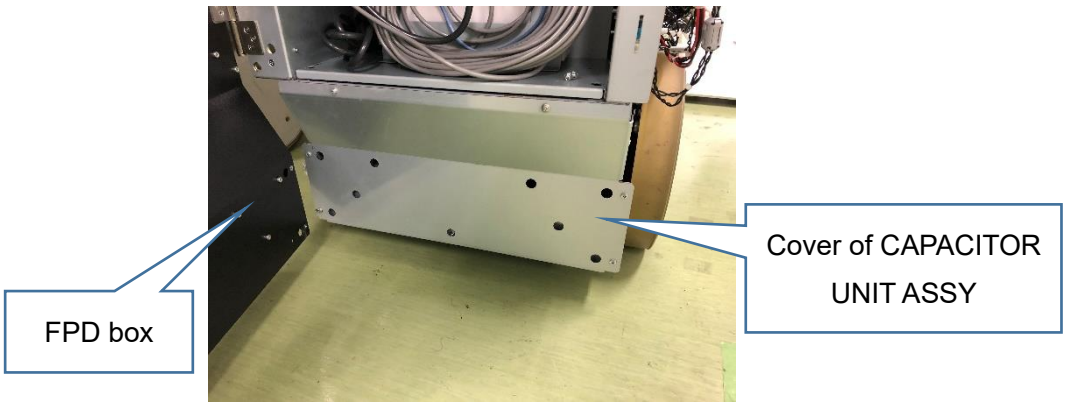
(Reference work time 1 person ・ 60min.)

Here is explanation how to replace the high-capacity capacitors for MobileDaRt Evolution or 32 kW type of MobileArt Evolution.

There are two types according to the manufacturing date, and you can visually check the shape of the cover by opening the FPD box as shown in the figure below.

1) Initial type shipped before about 2017

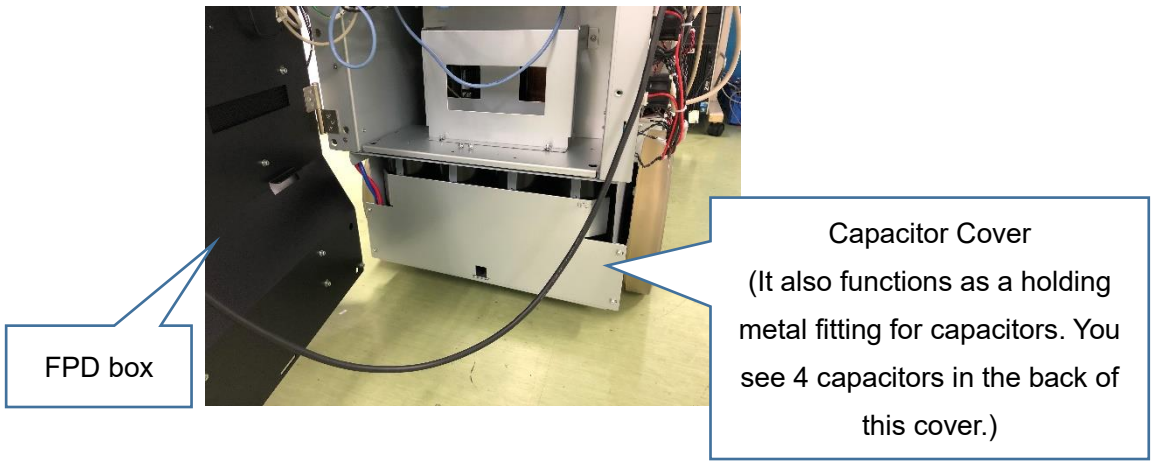
Replace CAPACITOR UNIT ASSY (566-20818) as the assembly which includes 4 capacitors and their brackets.



2) Current Type

Replace the capacitors “C, LNR2V223ASEACQ” (056-44371-07) separately.

The bracket for the capacitors does not exist.



5. Parts replacement

(1) Preparation

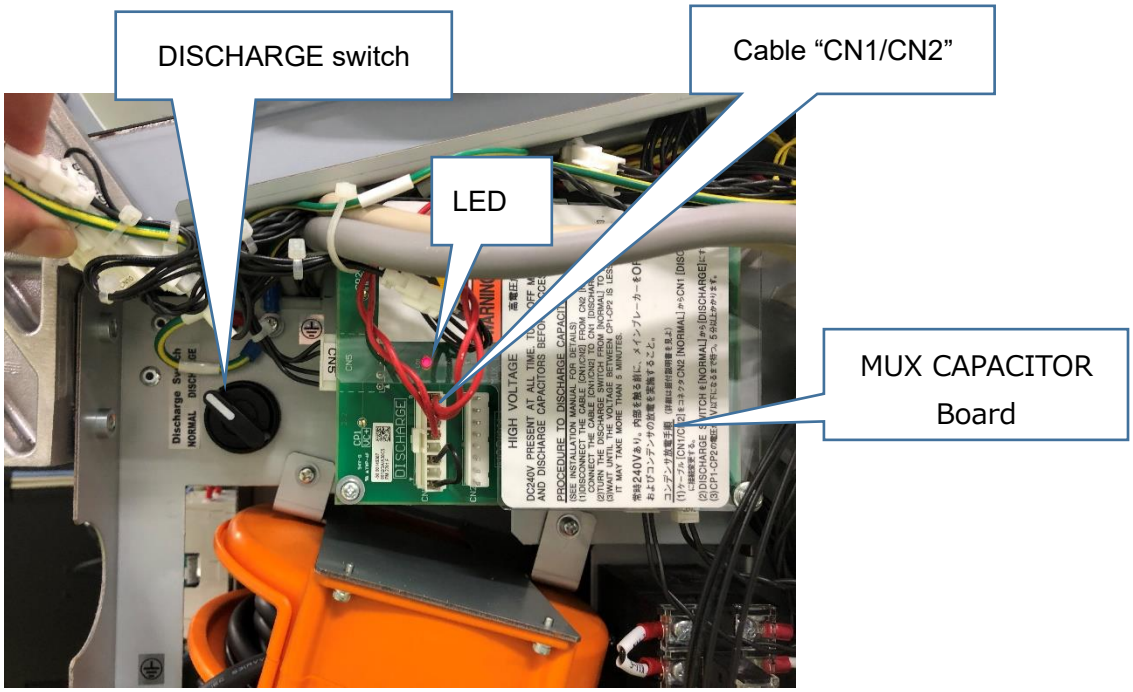
This section is common to 1) Initial Type and 2) Current Type.

Turn off the main breaker.

There is still some charge in the high-capacity capacitors, so follow the steps below to discharge it.

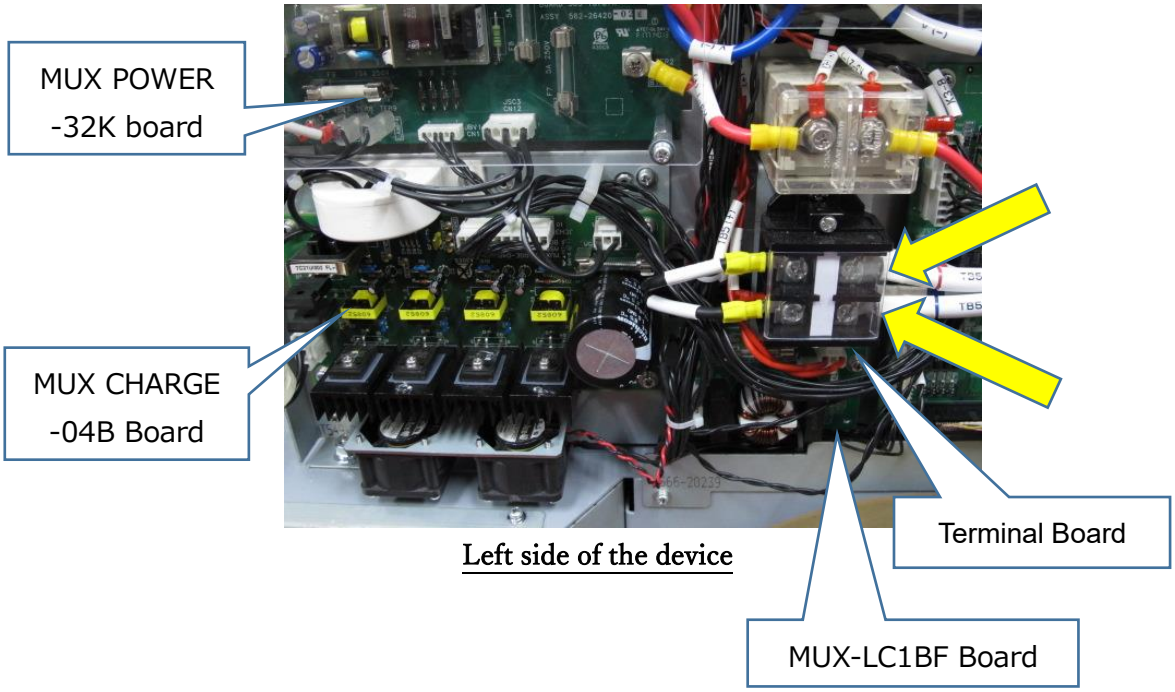
First, disconnect the cable “CN1/CN2” connected to connector “CN2 (NORMAL)” on the MUX CAPACITOR board on the right side of the device from CN2 and connect it to connector “CN1(DISCHARGE)” side. - > The LED on the board will light while the charge remains in the capacitors.

Then switch the DISCHARGE switch from “NORMAL” to “DISCHARGE” to start discharging.



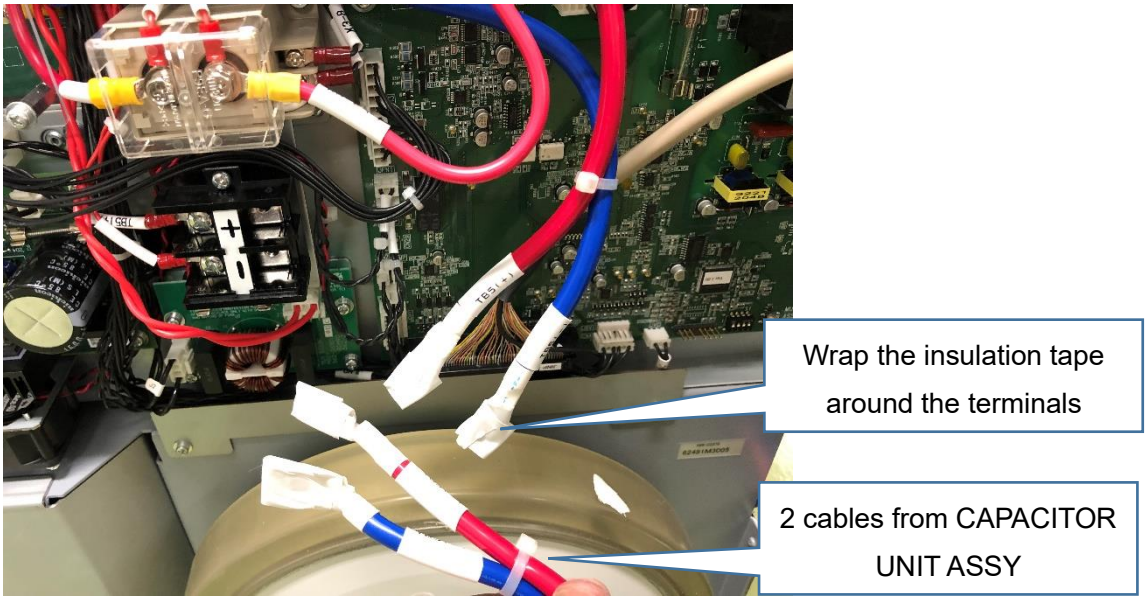
Right side of the device

Measure the capacitor voltage at the terminal board (yellow arrow in the figure below) above the MUX-LC1BF board on the left side of the device, and wait until the voltage drops below 0.5 VDC (it usually drops below 0.5 VDC about 10 minutes after the start of discharge treatment).



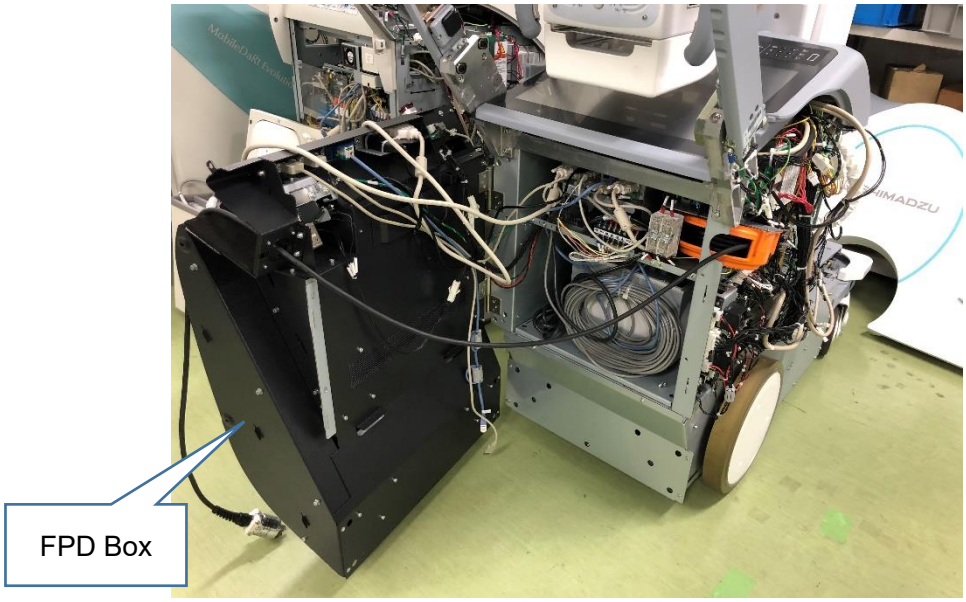
Left side of the device

Remove all of 4 cables connected from CAPACITOR UNIT ASSY etc., from the two right terminals of the terminal board above, and wrap the insulation tape around the amplifier terminals for insulation just in case. 2 red cables "TB5 (+)" are connected to the upper terminal and 2 blue cables "TB5 (-)" are connected to the lower terminal.
In case of 2) Current Type, the cable routing is slightly different, so it is possible to replace the capacitors without removing these cable.

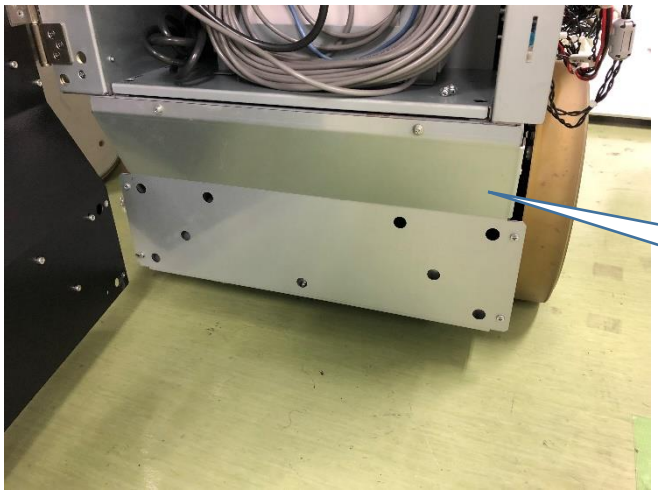


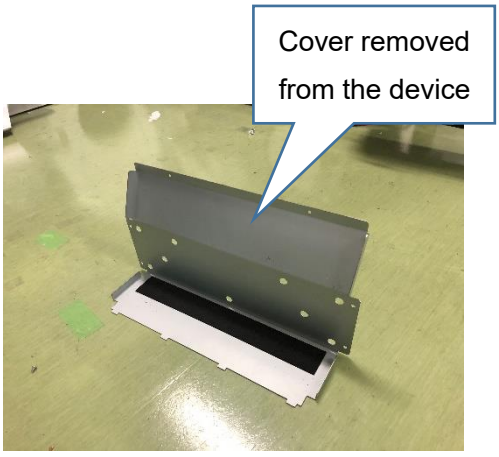
(2) Replacement of CAPACITY UNIT ASSY・・・In case of 1) Initial Type shipped before about 2017

Remove the left and right side covers and open the FPD box.



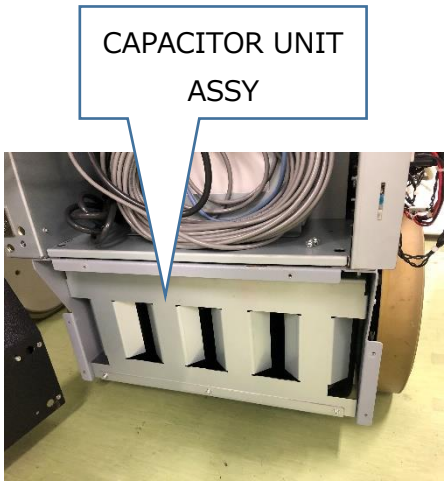
Remove the cover for CAPACITOR UNIT ASSY.

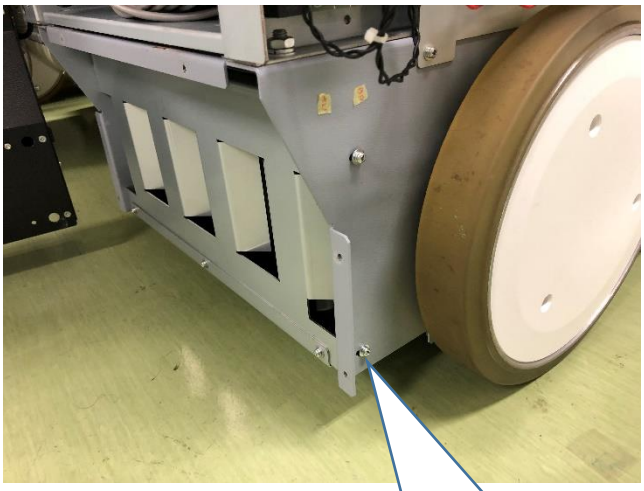




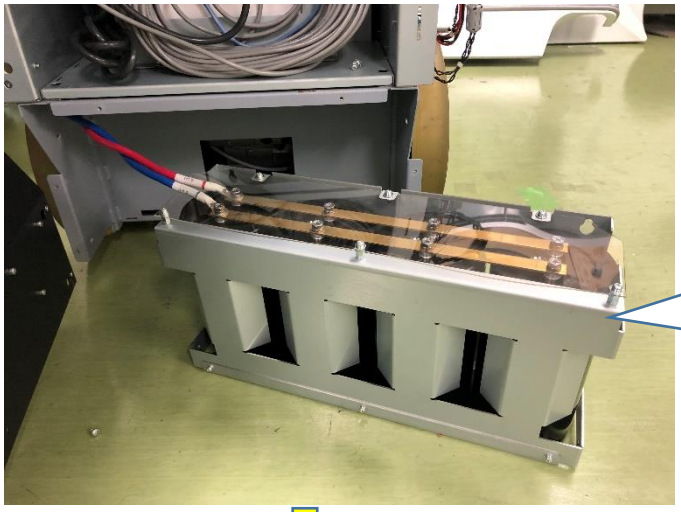
5. Parts replacement

Remove the CAPACITOR UNIT ASSY from the device chassis by removing the four M4 screws.

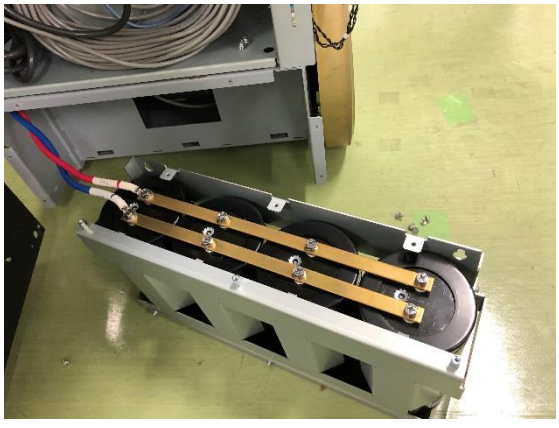




Fixing Screws for CAPACITOR UNIT ASSY (4 pieces)



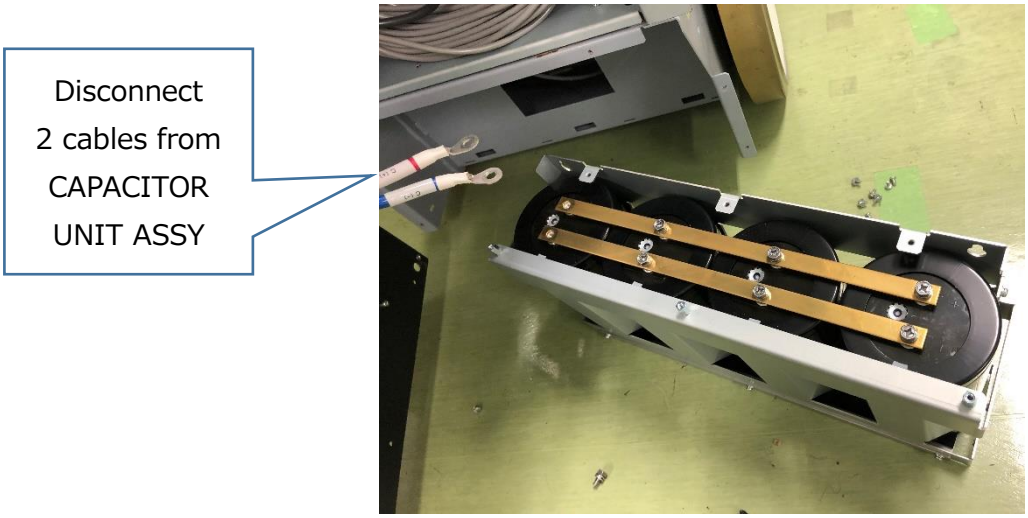
CAPACITOR UNIT ASSY removed from the device chassis
(Bracket shapes vary slightly depending on product date. This picture shows the early type before 2017.)



Remove the transparent acrylic plate cover on the top of CAPACITOR UNIT ASSY by removing six M4 screws.

5. Parts replacement

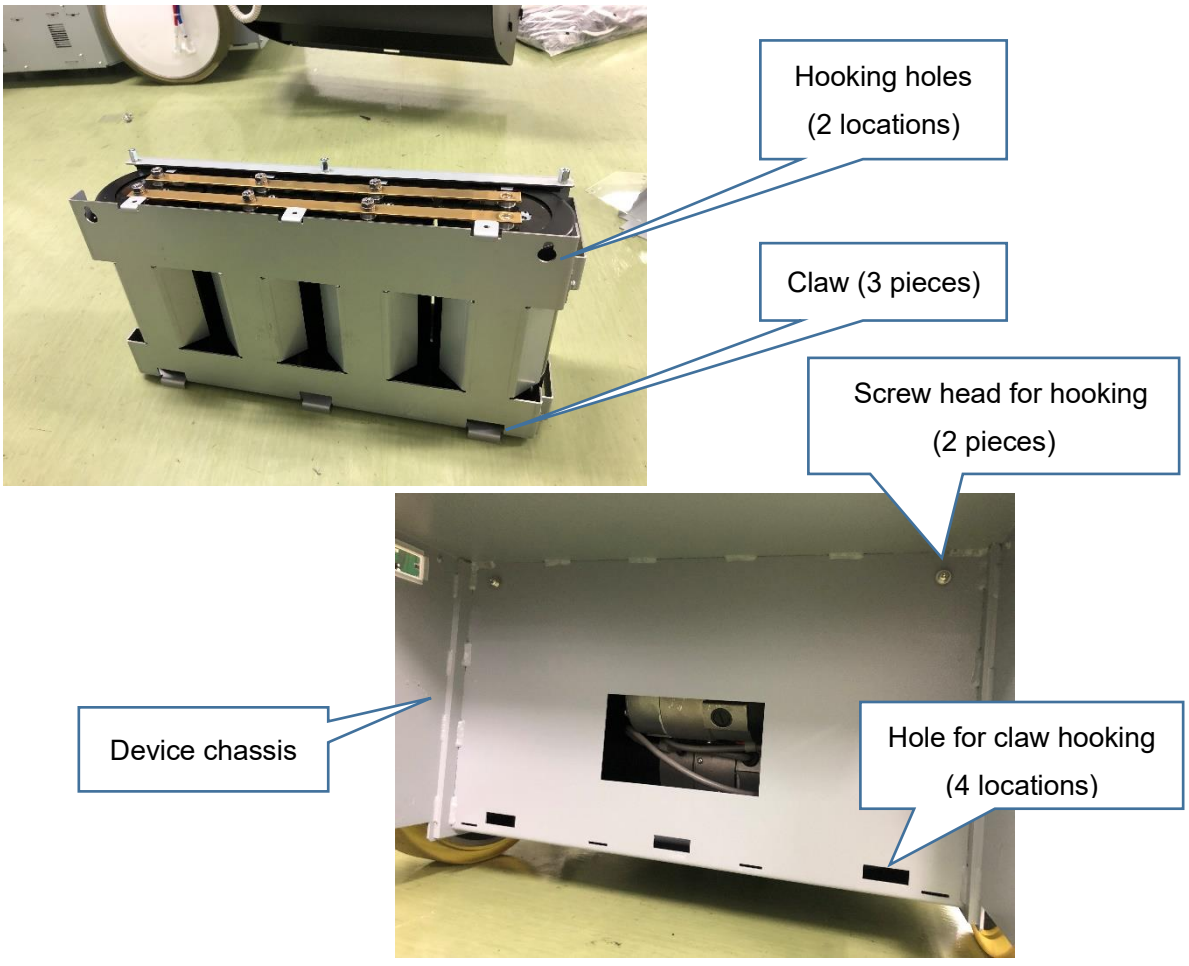
Disconnect the 2 cables connected to the CAPACITOR UNIT ASSY.



Use the reverse procedure to install a new CAPACITOR UNIT ASSY.

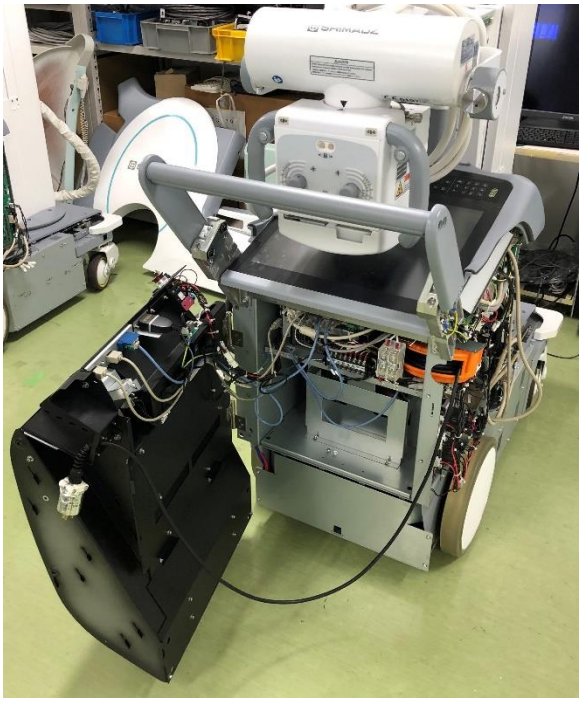
Connect the 2 cables and insert the CAPACITOR UNIT ASSY so that the screws on the device chassis fit into the hooking holes shown below.

Lift it all the way up, push it in, and secure it to the device chassis with 4 screws.



(3) Replacement of capacitors・・・In case of 2) Current Type

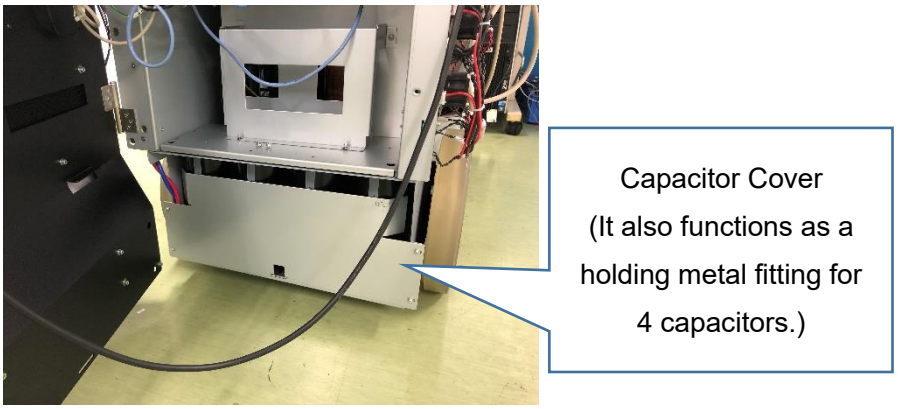
Remove the left and right side covers and open the FPD box.



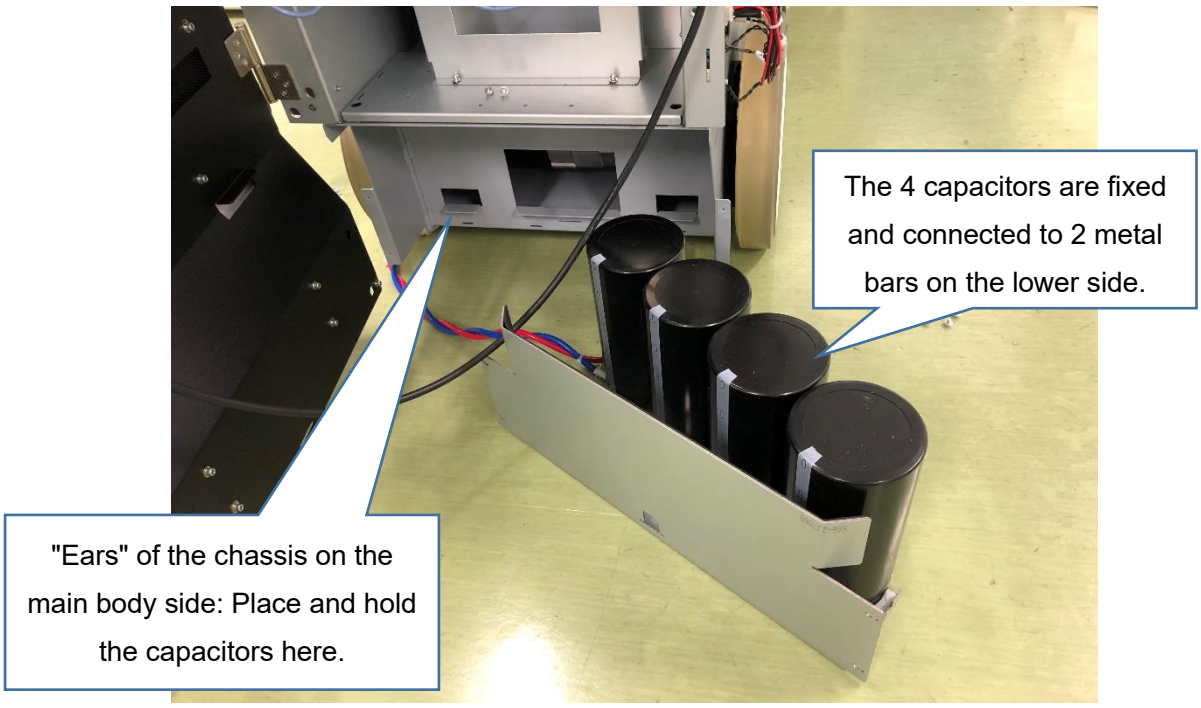
FPD box

5. Parts replacement

Remove the Capacitor Cover.
However, this cover also functions to hold 4 capacitors by pressing them against the chassis. If you inadvertently remove this cover, the capacitors will be free and roll down to the floor, so remove them carefully.



The 4 capacitors are connected via two metal bars in the upside-down state of 1) Initial Type. Remove each pieces of the capacitors from the cable and metal bar and replace with the new capacitor. (Refer also to the picture of CAPACITOR UNIT ASSY of 1) Initial Type.)

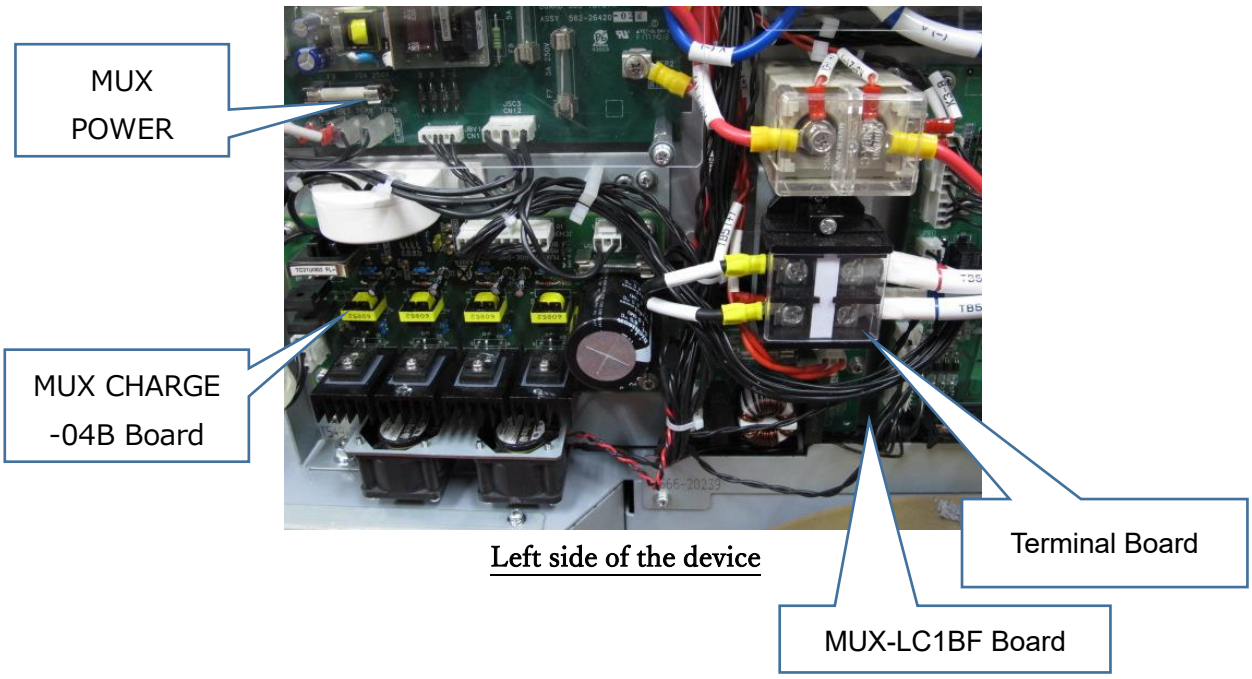


When reattaching the 4 capacitors and Capacitor Cover to the chassis, place and hold the capacitors on the “Ears” of the chassis on the main body of the device before reattaching Capacitor Cover.

(4) **Check operation after replacement**

Connect all of 4 cables from CAPACITOR UNIT ASSY etc. to the terminal board above the MUX-LC1BF board on the left side of the device as before.

Connect 2 red cables "TB5 (+)" to the upper terminal and connect 2 blue cables "TB5 (-)" to the lower terminal.

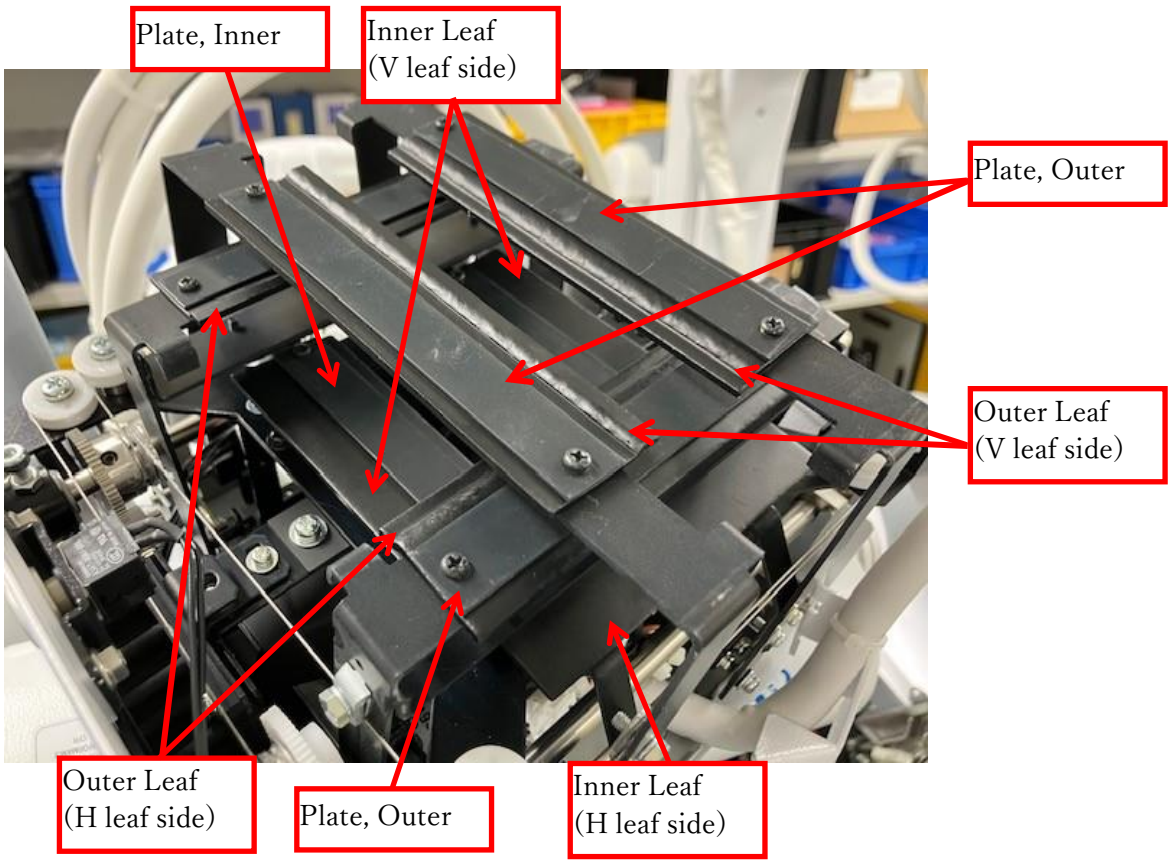


Left side of the device

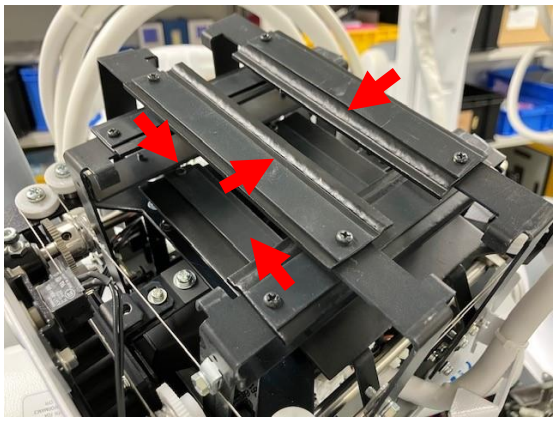
Release the discharge treatment, turn on the main breaker, and turn on the key switch to make sure that you can perform radiography properly.

5.22 Replacement of Collimator Outer/Inner leaf

This section describes the procedure for replacing “Outer Leaf”/ “Inner Leaf” and the “Plate, Outer”/ “Plate, Inner” of the collimator.
The above four parts are used at totally 4 places, 2 each on the H/V leaf side.



The mounting holes for each leaves are larger compared to the diameter of the screw. However, all the mounting holes should be fixed in the direction indicated by the arrow in the figure below, as close as possible to the inside (the direction in which the leaf closes).
Loctite (242 or 243 blue) is applied to the mounting screws. When replacing the leaf, degrease the screws with Loctite SF 7063 degreaser (Spray with liquid, and wipe with a cloth or air), apply Loctite again, and tighten the screws.



When MX9 version does not have the 3D camera option, the angle of the collimator detected by the angle sensor of this CL MEAS-17 board is displayed.
Since the attitude of the CL MEAS board may be shifted when the collimator cover is removed, it is necessary to check whether the angle display is not shifted after replacing Collimator Outer/Inner Leaf. Follow the procedure "7.26.3 Checking the collimator angle" in the Installation Manual to check the angle, and if necessary, adjust the collimator angle detection again.

5.23 Replacement of Charging Unit SF-DM9

●Service Parts List

DS Cable and DS Connector Guide are included in DR-ID 800.
Other parts below are included in Charging Unit SF-DM9.

Part No.（Shimadzu）	Part No.（FUJI）	Number	Part Name
572-55015-50	136Y204651B	1	DS Unit Cable
572-55015-51	898Y202554	1	DS Connector Replacement Kit
572-55015-52	363N131302C	1	Pocket_Guide_1
572-55015-53	363N131303D	1	Pocket_Guide_2
572-55015-57	382N130469	1	Front_Guide_Sheet
572-55015-54	898Y202552	1	Bottom Damper L Kit
572-55015-55	898Y202553	1	Bottom Damper S Kit
572-55015-56	898Y202600	1	SF-DM9_DS_Shutter_Unit
572-55015-58	308S0517	1	TP Shoulder Bolt

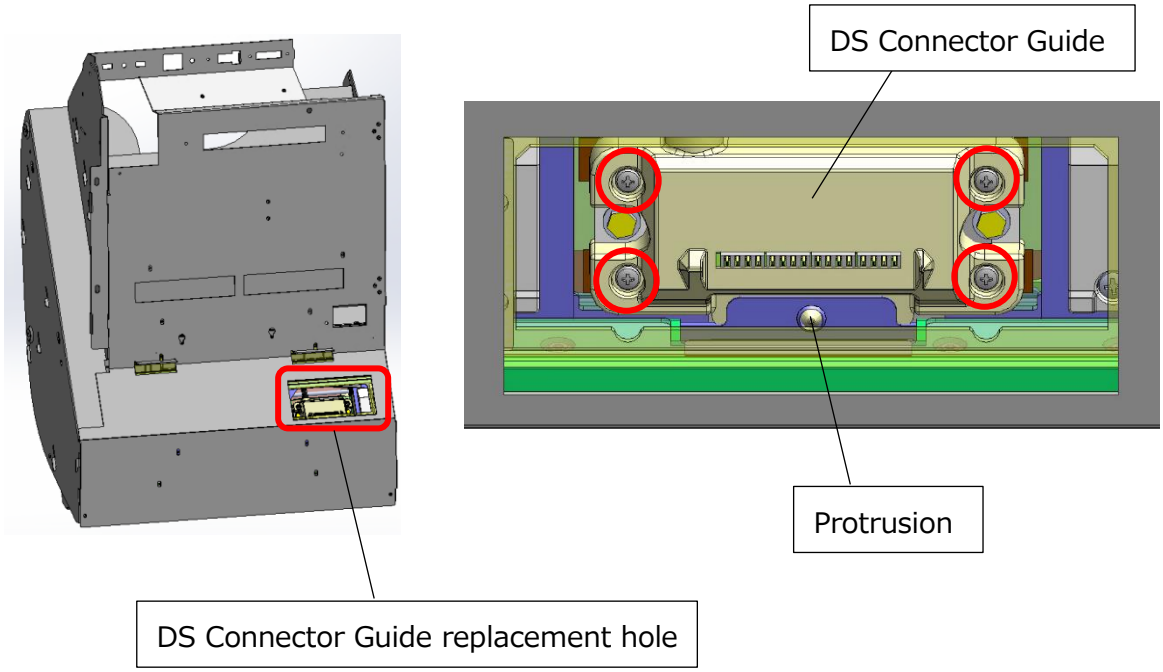
5. Parts replacement

● Replacement Procedure

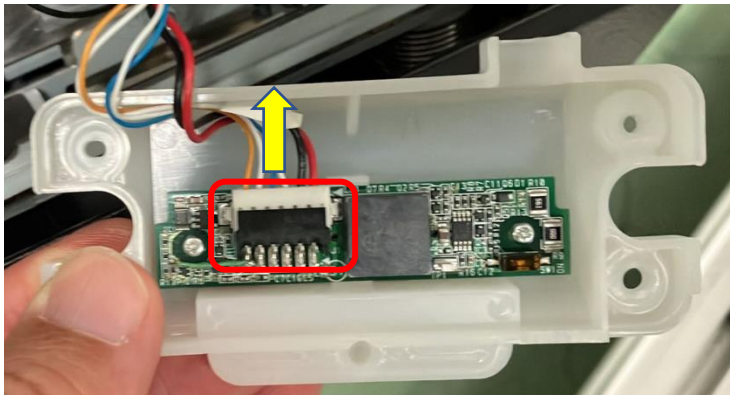
＜DS Connector Replacement Kit (DS Connector Guide)＞

The procedure for replacing the DS Connector Guide through the DS Connector Guide replacement hole is described below.

1. Remove the 4 fixing screws (M2.5 L6) and remove the DS Connector Guide.
   ※When removing the DS Connector Guide, the protrusion will get caught in the sheet metal hole, so remove the DS Connector Guide while pushing the lower part where the protrusion is located.

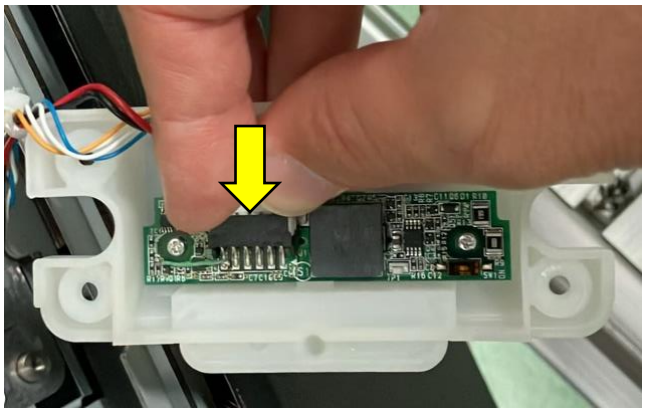


2. Remove the connector of DS Cable.



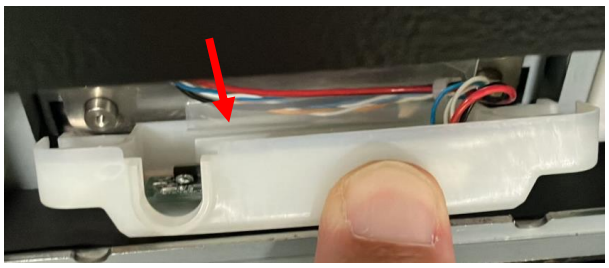
5. Parts replacement

3. Insert the DS cable connector into the board on the new DS Connector Guide board.
※Insert the connector all the way into the board.

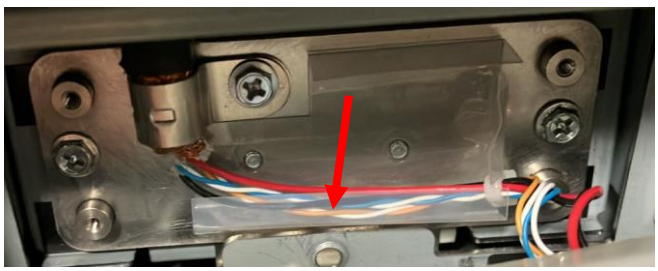


4. Return the DS Connector Guide to its original position and cover the base sheet metal.
*Fit it from the lower part where the protrusion is located.

※Look inside the DS Connector Guide to verify that the lower part of the seat inside is inside the DS Connector Guide.

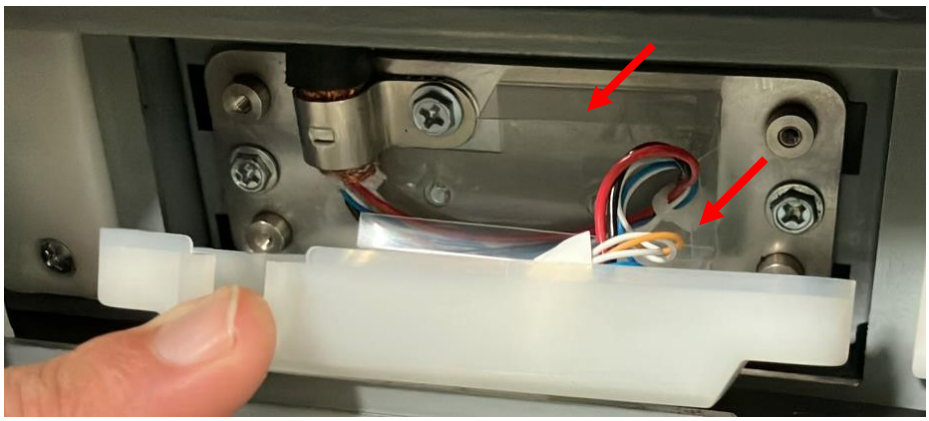


※The DS cable should be stretched along the inside of the sheet.
(Photo: DS Connector Guide omitted for easy viewing)



5. Parts replacement

※Check that the DS cable has a loose wire inside the right fold of the inner sheet and that the right and upper folds of the inner sheet are folded inward.

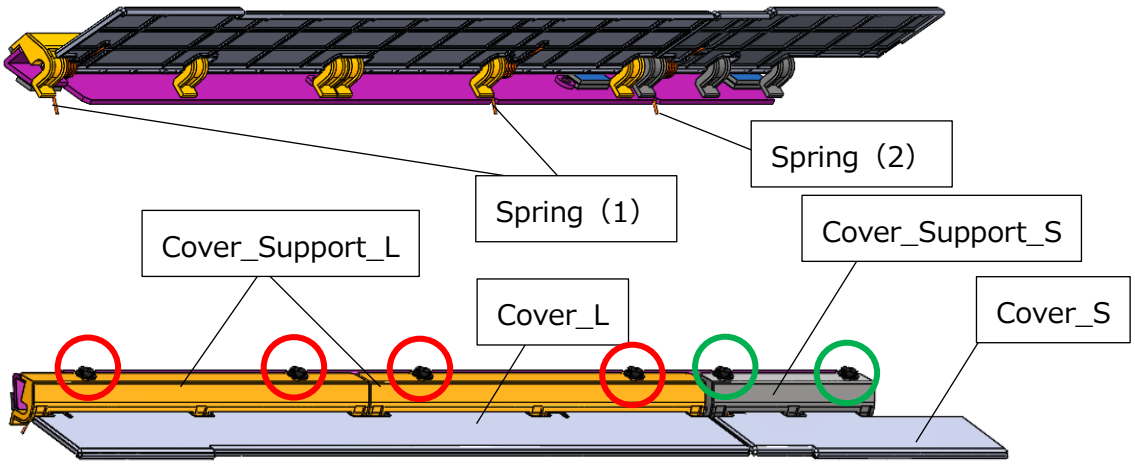


※When the DS Connector Guide is placed on the base sheet metal, visually check that there is no pinching of the internal sheet and DS cable on the right and upper sides.

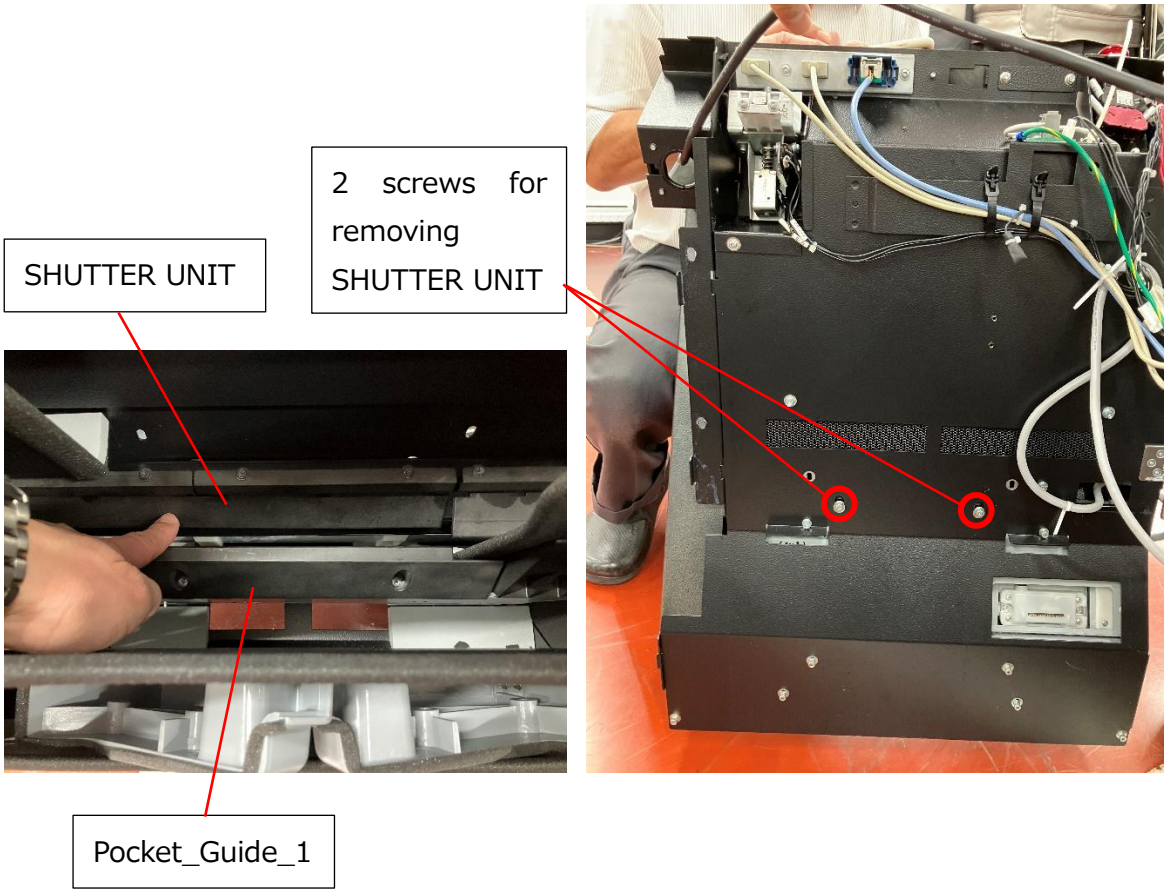
5. Tighten the DS Connector Guide with 4 screws (M2.5 L6) with Torque: 45 N・cm.
※In the above picture, tighten the fixing screws for DS Connector Guide in the order of upper left → upper right → lower 2 screws.

5. Parts replacement

＜SHUTTER UNIT（898Y202600）＞

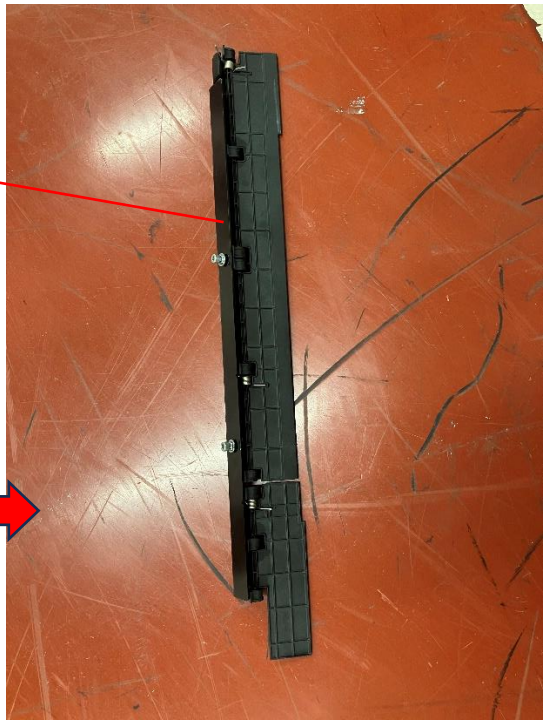


Remove the two red ○ screws shown below from the FPD box and remove the Shutter Unit.



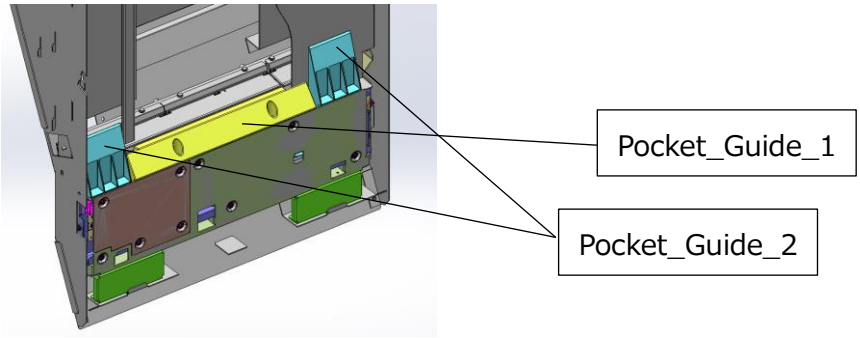
5. Parts replacement

SHUTTER UNIT removed from FPD BOX

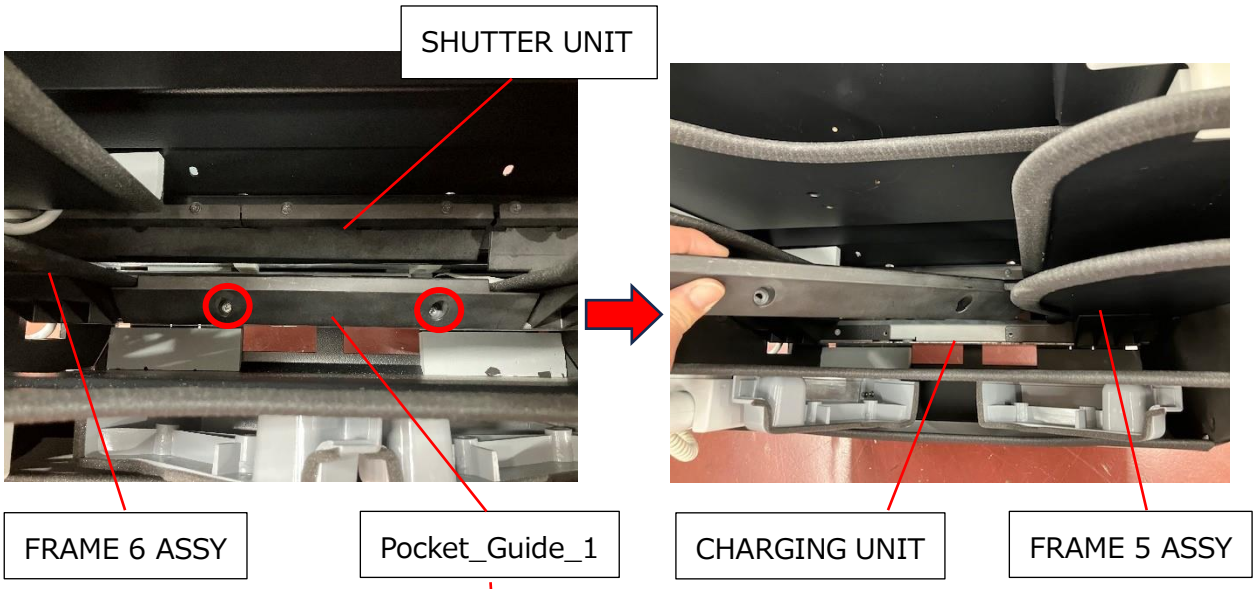


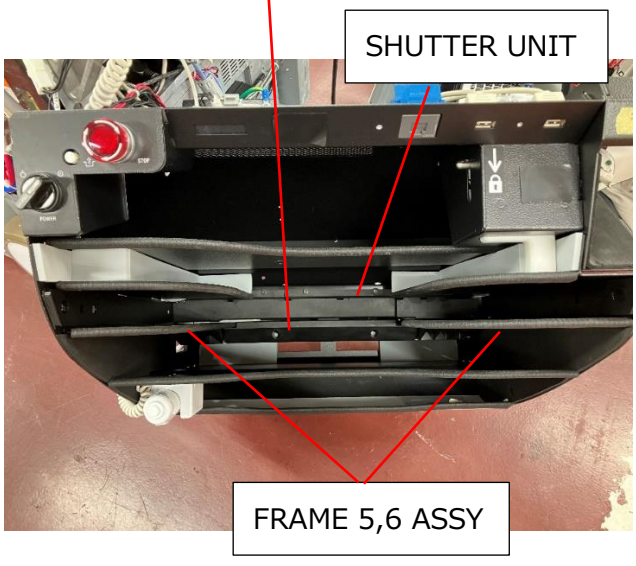
5. Parts replacement

＜Pocket_Guide_1（363N131302C）、Pocket_Guide_2（363N131303D）＞



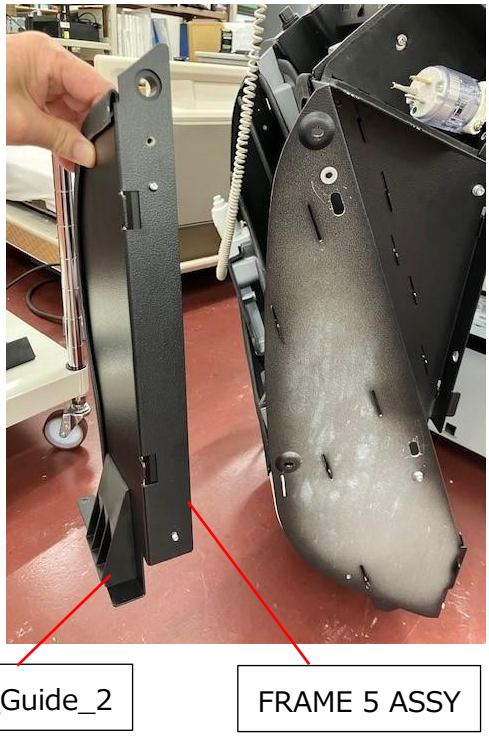
Remove the two red ○ screws shown below from the FPD box and remove Pocket_Guide_1.
Pocket_Guide_1 is fastened to the charging unit together with the frames 5 and 6 ASSY.



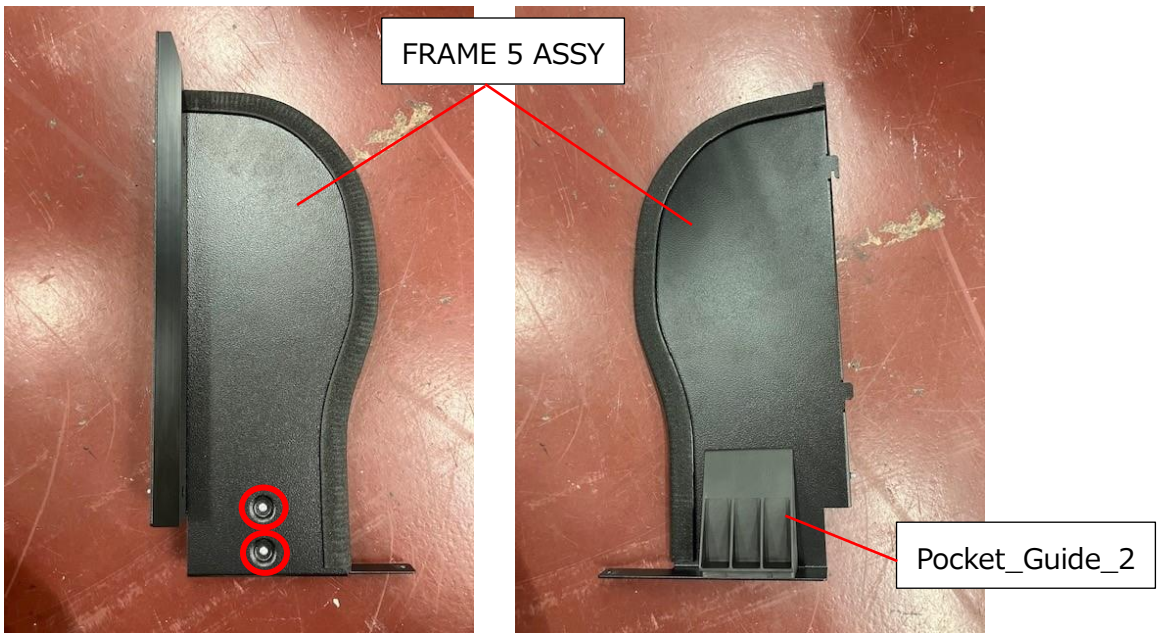


5. Parts replacement

Remove each of the mounting bolts (shown in red ○) on the side of the FPD box, and remove frames 5 and 6 ASSY.



Remove the two red ○ screws shown below from FRAME 5 or 6 ASSY and remove Pocket_Guide_2.

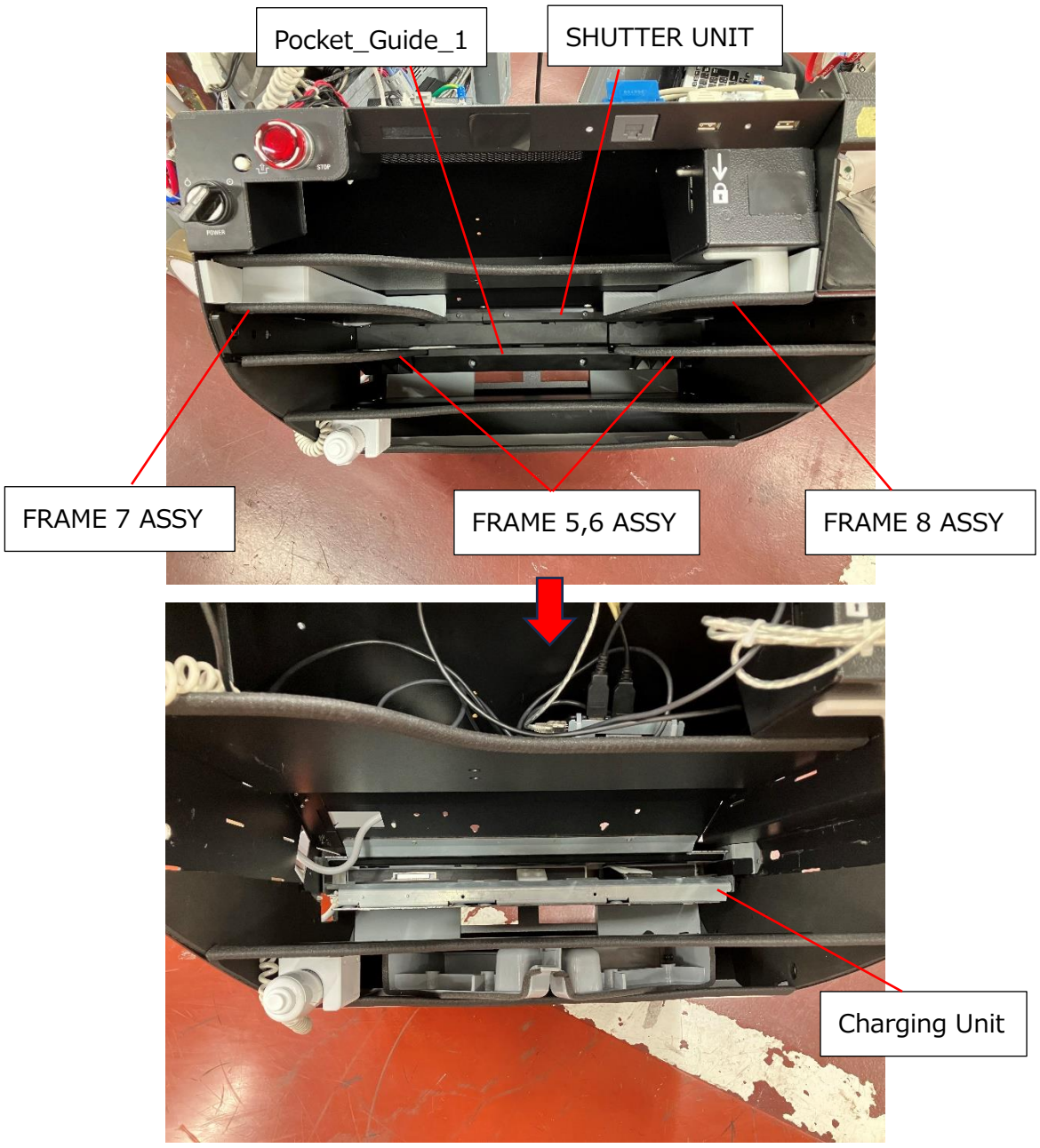


5. Parts replacement

＜DS Unit（Charging Unit）＞

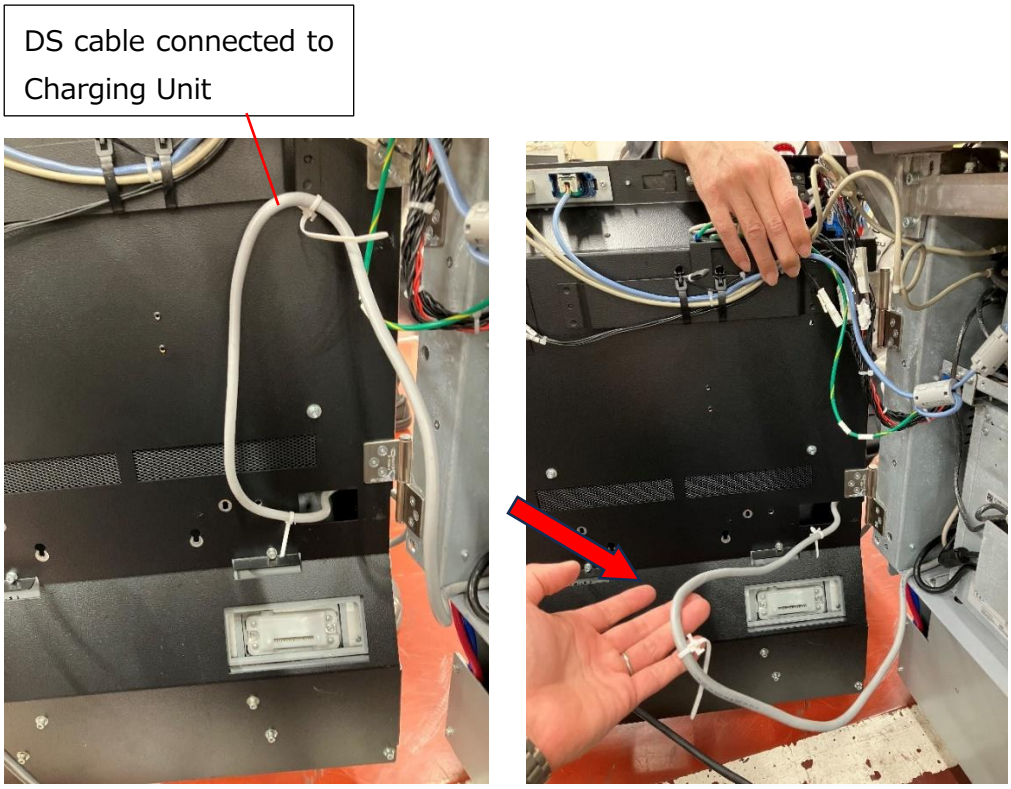
In order to replace the parts of the Charging Unit, it is necessary to remove the Charging Unit from the FPD BOX.

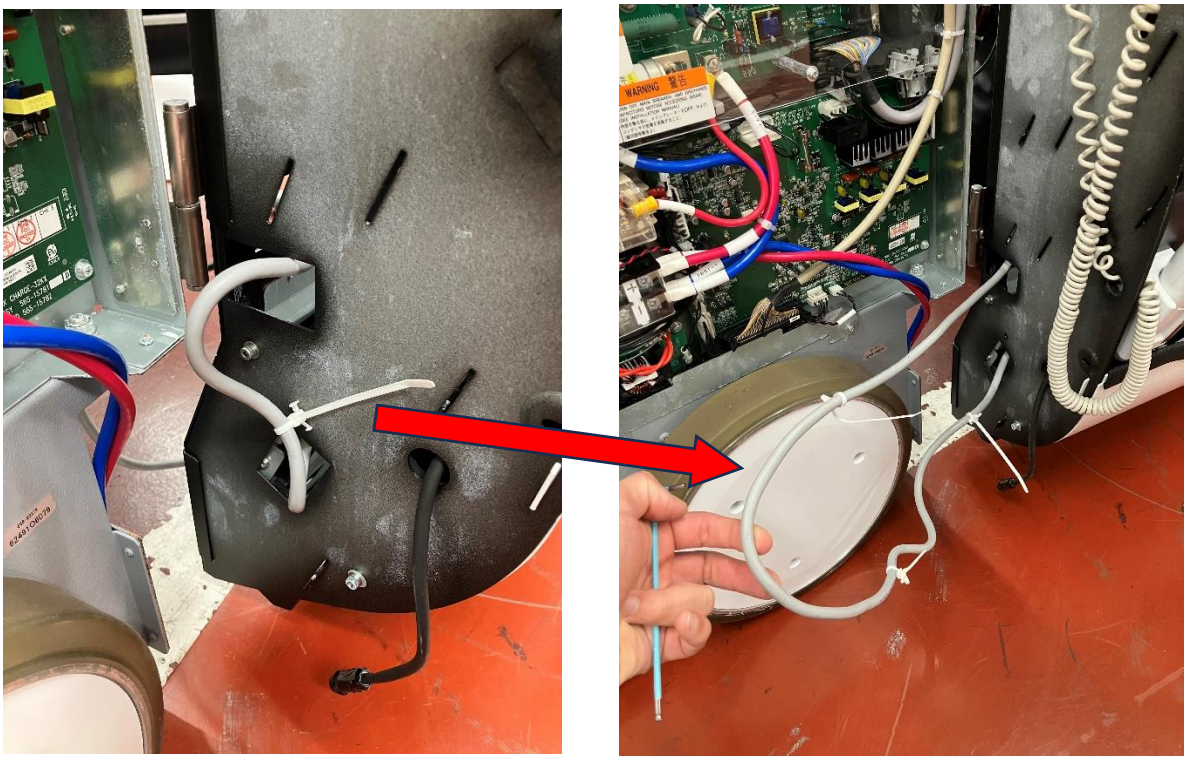
The Charging Unit appears when the SHUTTER UNIT, Pocket_Guide_1, and FRAME 5, 6, 7, and 8 ASSY are removed by the procedure described in the previous pages.



5. Parts replacement

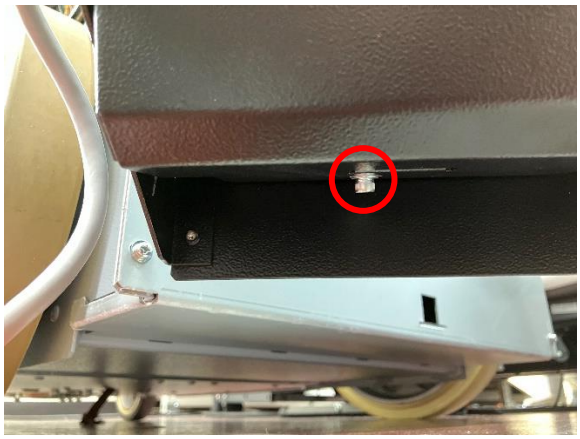
Change the wiring of the DS cable and pull out it, so that the Charging Unit can be removed from FPD BOX and placed on the floor.

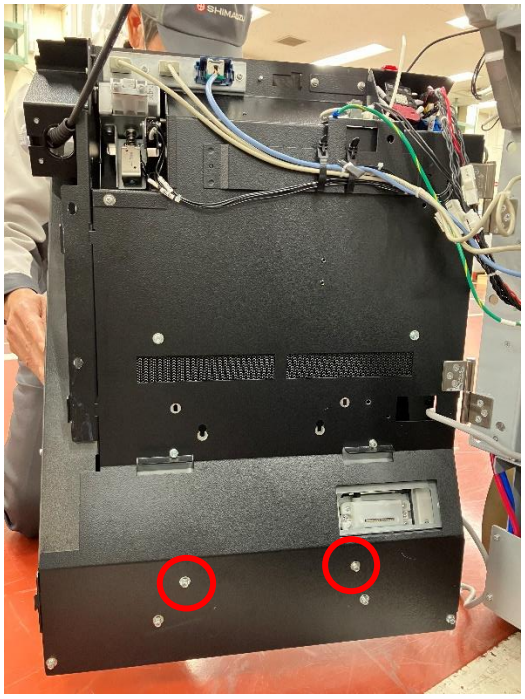




5. Parts replacement

Remove the 4 bolts (two bolts on the bottom of the FPD box and two bolts on the back) and remove the Charging Unit.

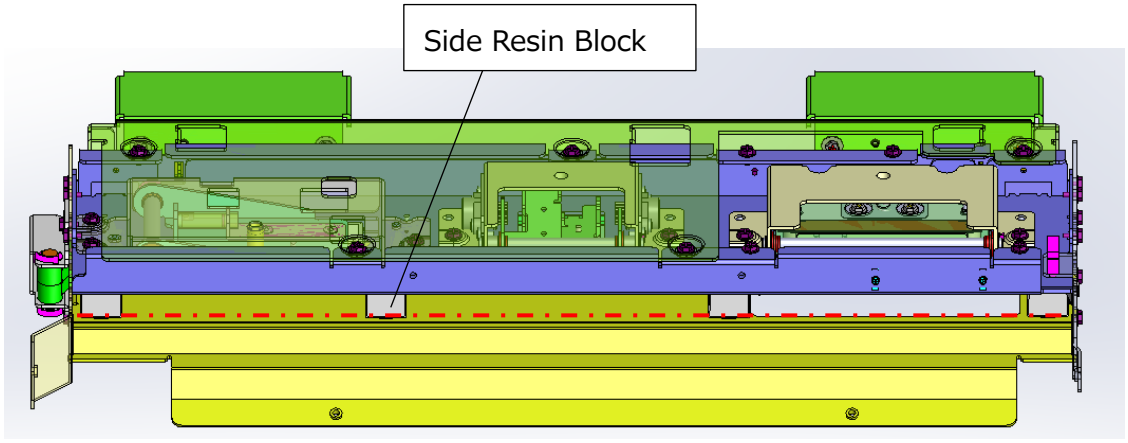




Charging Unit removed from FPD BOX

5. Parts replacement

・Replacement of 382N130469：Front_Guide_Sheet



1.Place the Charging Unit flat in the above position, and peel off the Front_Guide_Sheet.

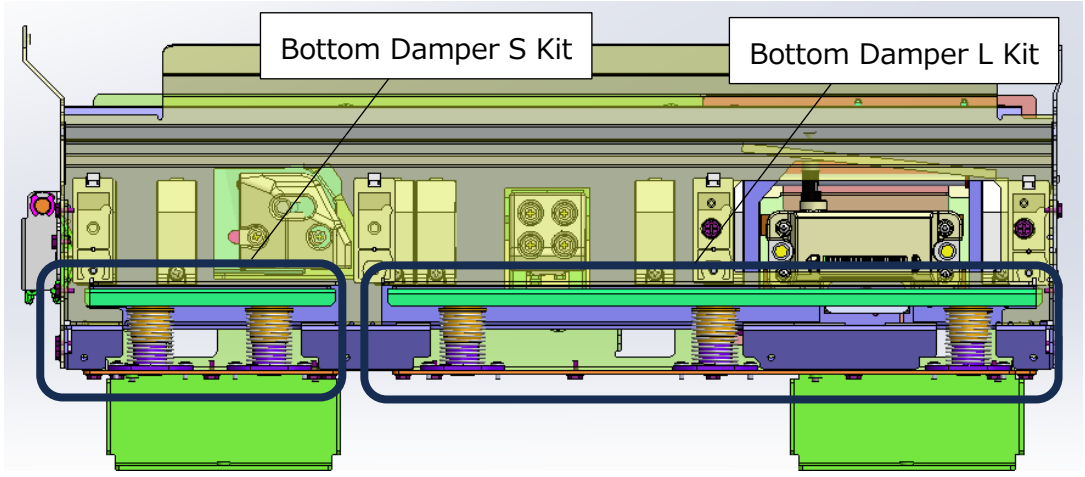
2.Clean the adhesive surface.

3.Attach the new Front_Guide_Sheet with the front end (red dashed line) of the Side Resin Block as a reference.

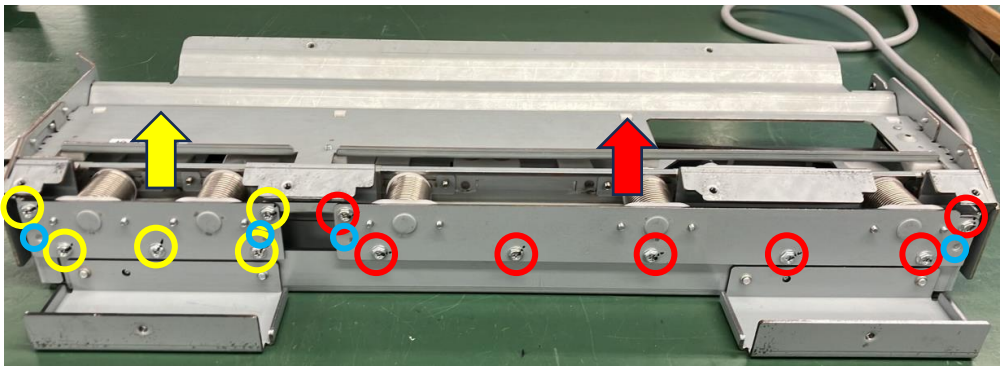
*Do not place both ends of the Front_Guide_Sheet on the side sheet metal. (red ○)

・898Y202552：Bottom Damper L Kit

・898Y202553：Bottom Damper S Kit



5. Parts replacement



・Replacement of 898Y202552：Bottom Damper L Kit

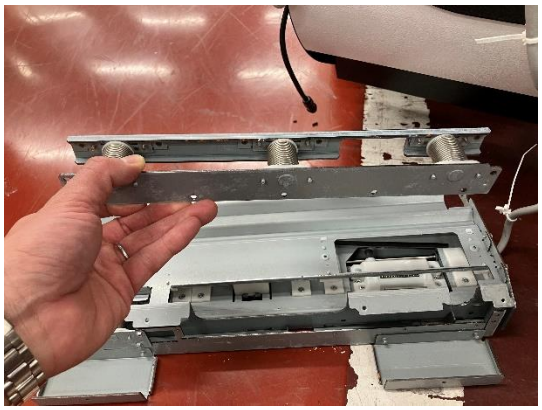
1.Place the Charging Unit flat in the above position, and remove the M3 x 7 (TP) screws with red ○.

2.Remove the Bottom_Damper_L_Kit from the opening in the direction of the red arrow.

3.Assemble the new Bottom_Damper_L_Kit using the reverse procedure. Attach the M3x7 (TP) screws with red ○. (Torque: 90 N・cm)

*Embossed mounting position on blue ○.

*Make sure that the Bottom_Damper_L_Kit does not float away from the screw mounting surface.



・Replacement of 898Y202553：Bottom Damper S Kit

1.Place the Charging Unit flat in the above position, and remove the M3 x 5 (TP) screws with yellow.

2.Remove the Bottom_Damper_S_Kit from the opening in the direction of the yellow arrow.

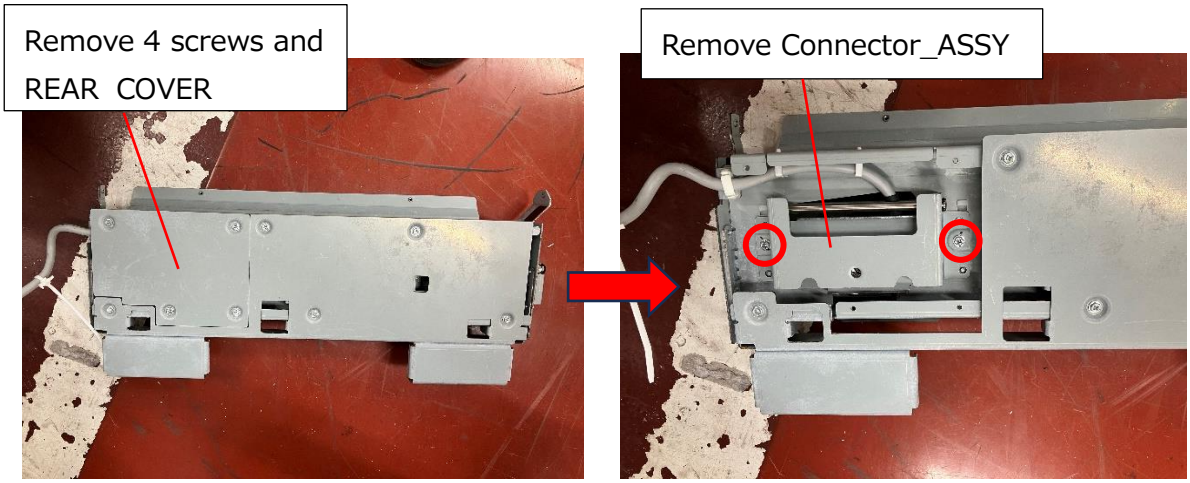
3.Assemble the new Bottom_Damper_S_Kit using the reverse procedure. Attach the M3x5 (TP) screws with yellow screws. (Torque: 90 N・cm)

*Embossed mounting position on blue ○.

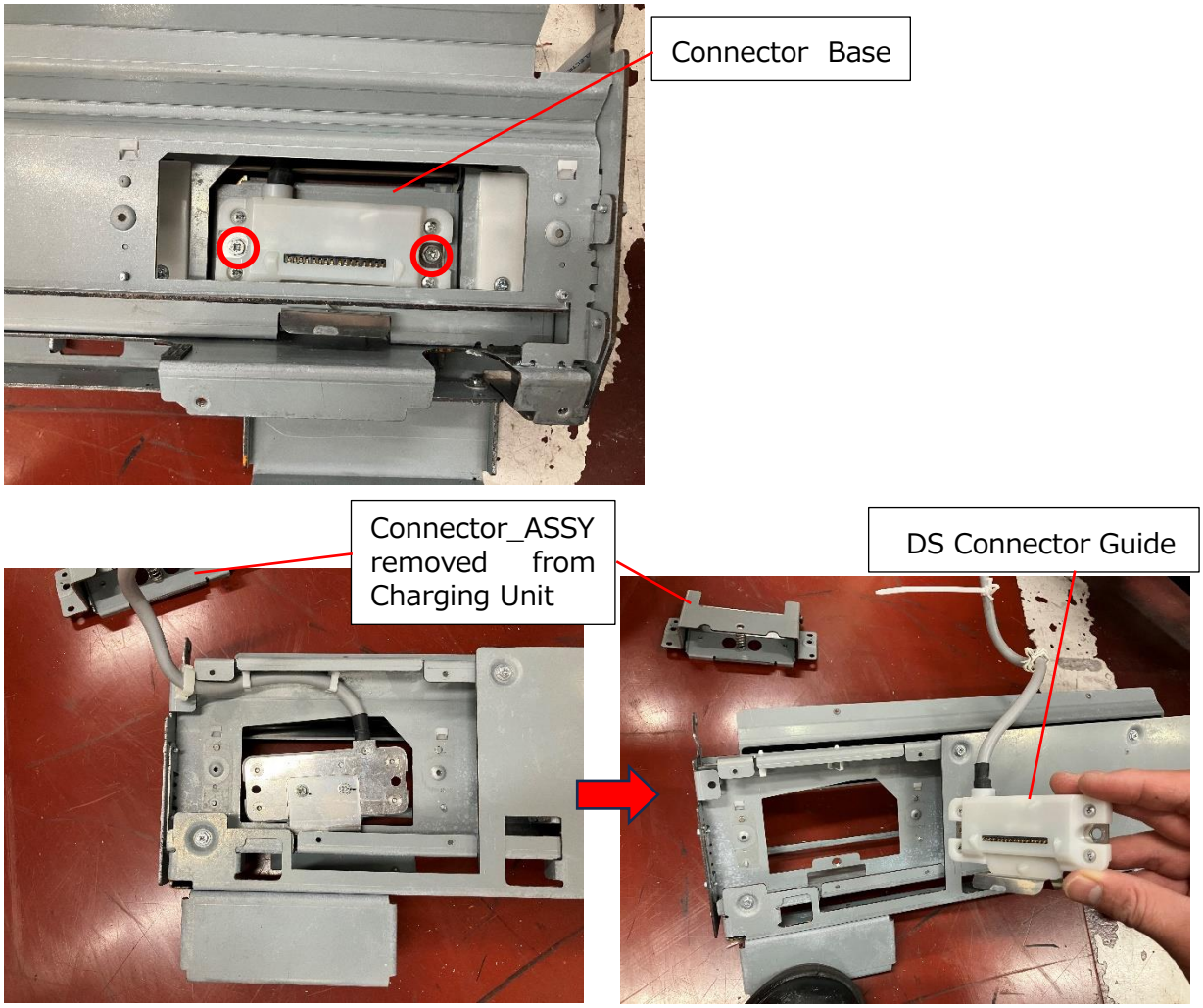
*Make sure that the Bottom_Damper_S_Kit does not float away from the screw mounting surface.

5. Parts replacement

When replacing the whole Charging Unit (DS unit) or the charging cable (DS unit cable), it is necessary to remove the charging connector from the Charging Unit according to the following. Remove the REAR_COVER, then remove the 2 screws red ○ (TP) and remove the Connector_ASSY.

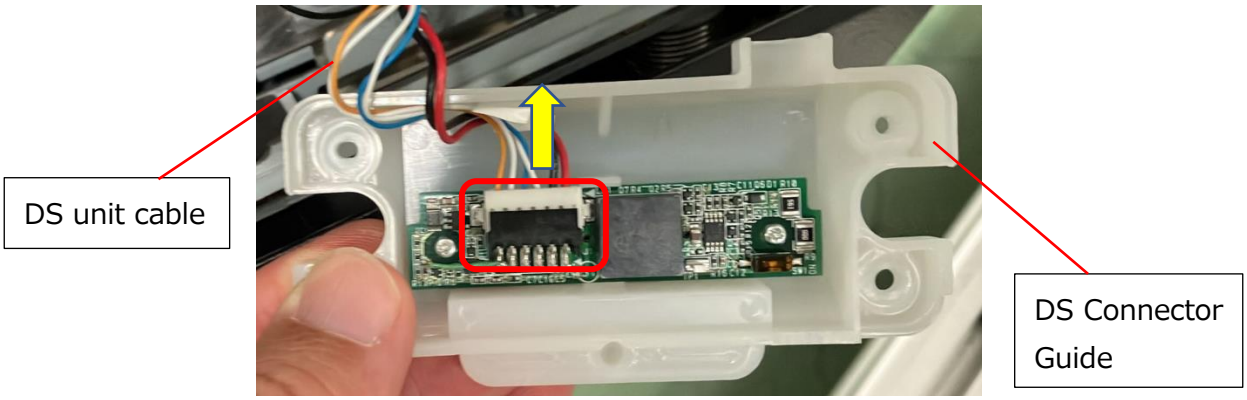


Remove the 2 screws red ○ (TP shoulder bolts) below and remove the DS Connector Guide and DS cable from the Connector Base (a part of Connector ASSY).

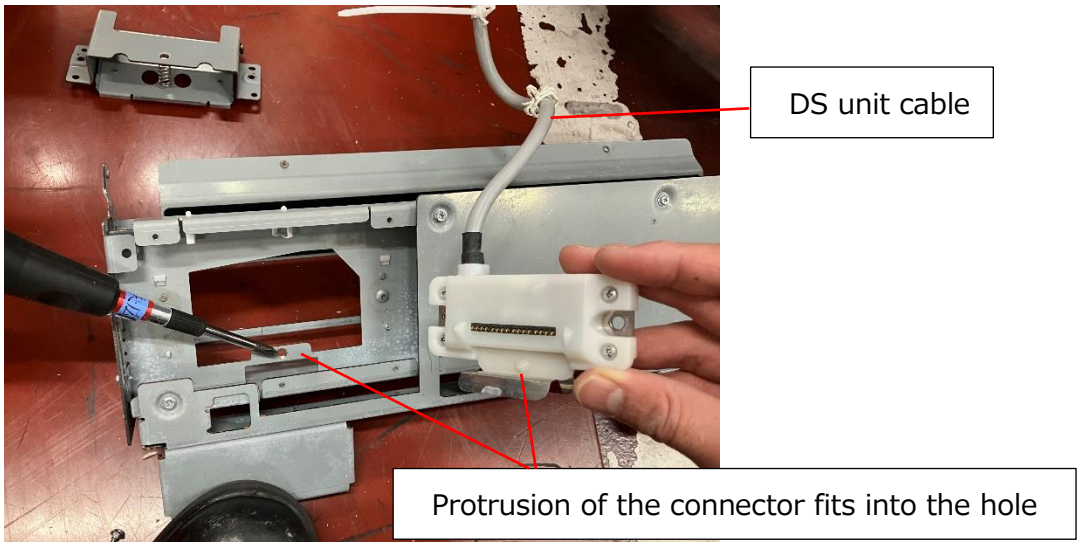


5. Parts replacement

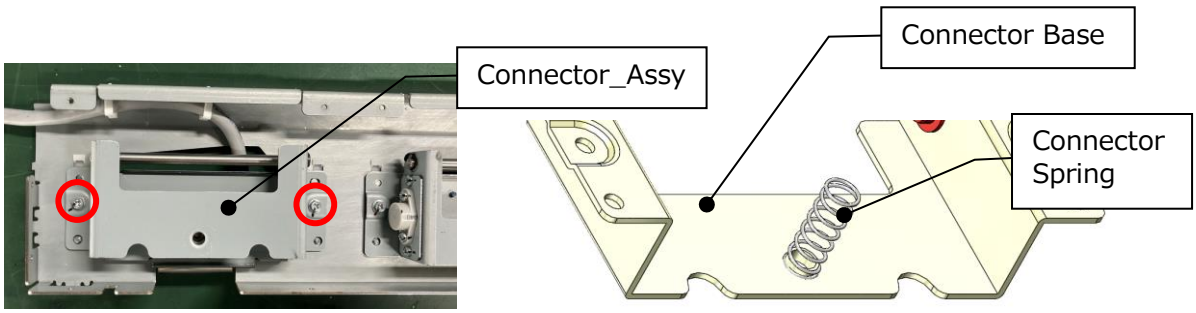
Refer to "Replacement Procedure for DS Connector Replacement Kit (DS Connector Guide)" above for instructions on removing and reconnecting DS Connector Guide from DS unit cable.



When DS Connector Guide is attached to the Charging Unit, DS Connector Guide is attached so that the protrusion of the connector fits into the hole of the base sheet metal.

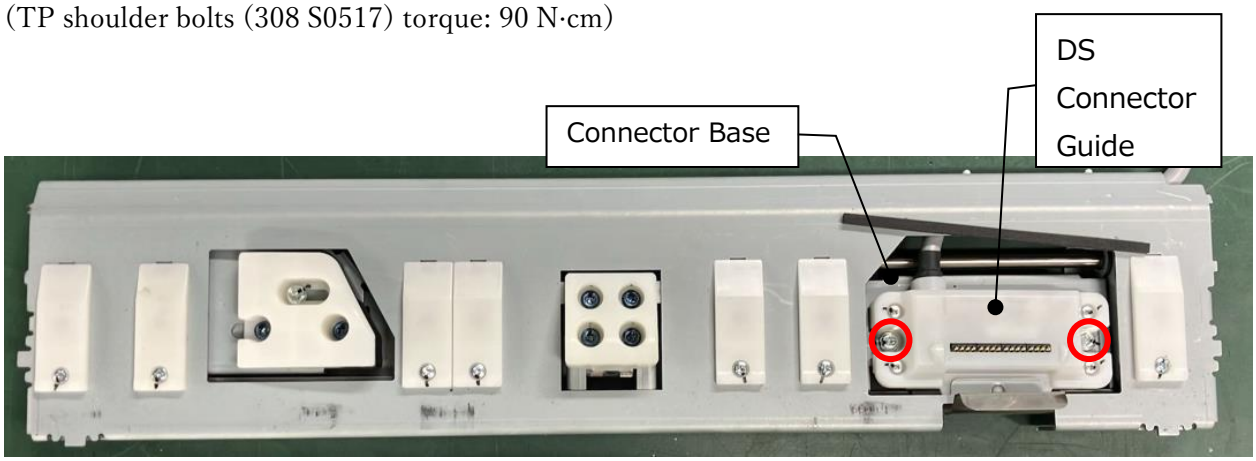


The Connector Spring is installed in the Connector_ASSY, so make sure that it is installed correctly. The connector spring is fitted to the protrusion of the connector base.

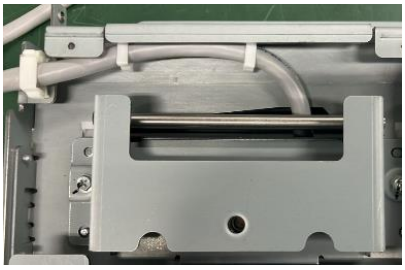


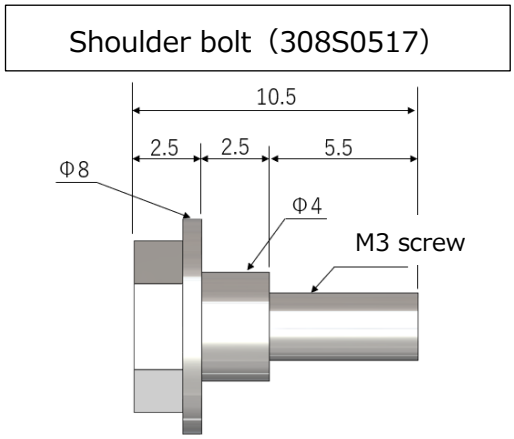
5. Parts replacement

To connect DS Connector Guide and DS unit cable of DR-ID 800 to the Charging Unit, fasten the Connector Guide and the Connector Base with M3 x 2 TP shoulder bolts (308 S0517) in red ○.
(TP shoulder bolts (308 S0517) torque: 90 N·cm)



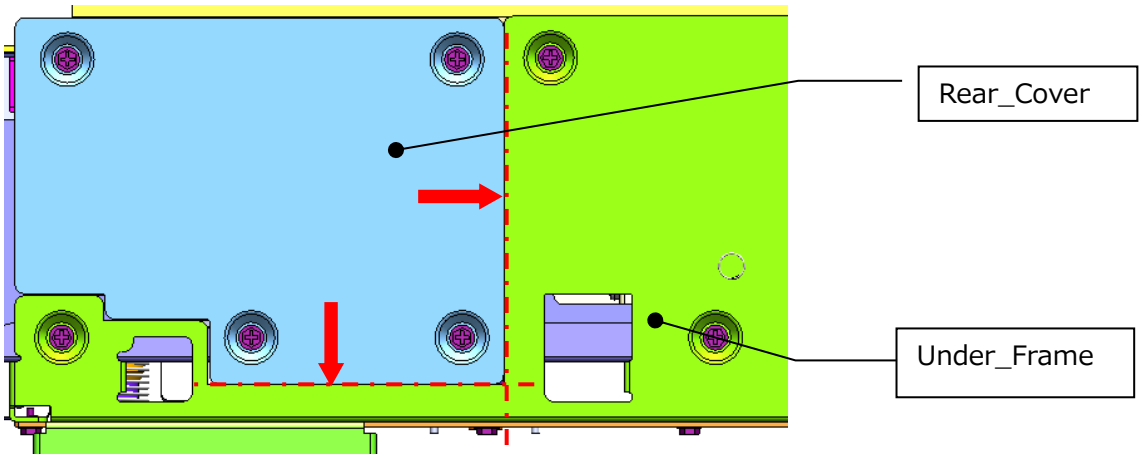
※The DS unit cable should be wired to the clamp as shown in the picture on the right.

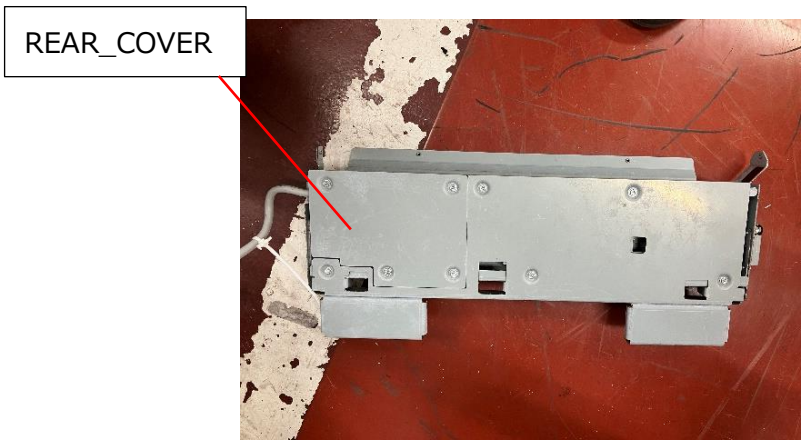




5. Parts replacement

Rear_Cover must be screwed while touching the Under_Frame.
(TP Screws M3 x 4, Torque: 90 N・cm)





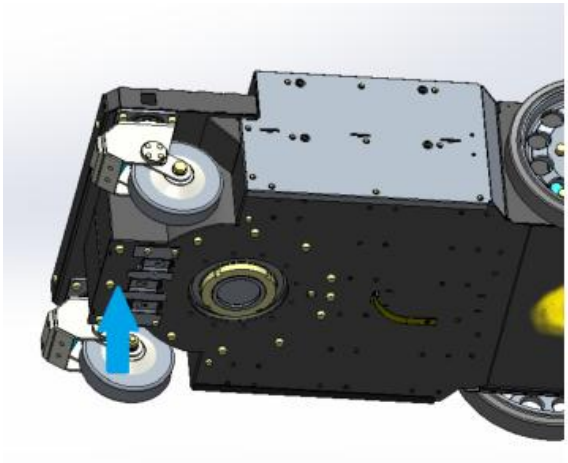
5. Parts replacement

5.24 Replacement of front wheels

(Referece work time: 1 person ´ 0.75 hours)

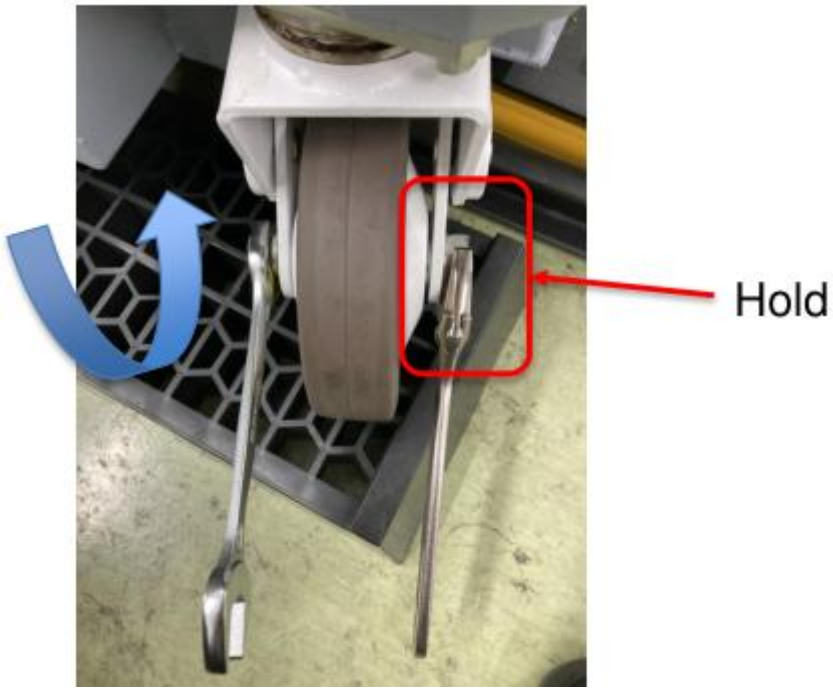
1. Place the front wheels on a ramp to create a gap for inserting the jack.

2. Jack up Unit at the location indicated by the blue arrow in the image below.(Raise it until the front wheels can be turned by hand.)



5. Parts replacement

3. Hold bolt with a wrench or similar tool , and loosen the nut on the opposite side.



4. Remove the wheel from the caster ASSY and replace it with a new one.
When reassembling the parts, please attach them to the caster ASSY in the order shown in the image below.
*Apply LOCTITE (Blue) on bolts.

6 PC maintenance

6.1 List of the usable commands

Here are the commands you can use when MAIN > appears on your PC's display.
Do not use any commands other than those listed below.
Instructions for using commands other than dtout are provided in the installation instructions.

Command	Contents
ver	The software version of XCONT-2018 is displayed
cls	Clear screen
date	Year/Month/Date is displayed.
time	Time is displayed.
maintenance	Download/ Upload of data of XCONT-2018 (Refer to “11.12 PC Maintenance Function” in Installation Manual.)
dtout	Output of text data

6.2 Explanation of command “dtout”

The dtout command outputs data stored in the CPU as comma-separated text data. There are 14 commands listed below.

The logs that can be obtained by the factory command are important for troubleshooting.

	Command	Content of output data
1.	fvr	Data of FVR adjustment
2.	handle	Data of Handle adjustment
3.	batt	Data of the battery
4.	userset	Data of User Setting
5.	replace	Data of Notification Function
6.	error	Error Log
7.	apr	Data of APR registration
8.	dap	Data of DAP adjustment (option)
9.	calcdap	Data of DAP calculation adjustment
10.	sesd	No use
11.	sod	No use
12.	drset	Data of DR setting mode
13.	factry	ALL data
14.	exit	Exit from the command “dtout”

6.3 How to use “dtout” command

6.3.1 Preparation to start command “dtout”

I. Connect the serial cross cable between maintenance PC and the connector J4 of XCONT.

II. Change the DIPSW settings. (Refer to “11.12 PC Maintenance Function” in Installation Manual.)

- MX9: DIPSW2-4 ON, 2-7 OFF and 3-8 ON
- MX8: DIPSW2-4 ON and 2-7 OFF
  (Since XCONT-2018 Rev.1.85: DIPSW2-4 ON, 2-7 OFF and 3-8 ON)
- MX7: DIPSW2-4 OFF and 2-7 OFF

III. Use [Tera Term] as the serial communication tool between maintenance PC and XCONT.

6.3.2 Serial port setting

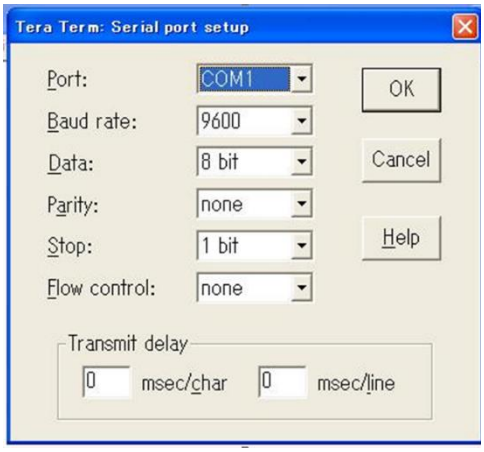
(See "PC Maintenance Function" in the installation instructions)

I. Select the TeraTerm Setup tab.

II. Select the Serial port tab.

III. Set the serial communication settings between PC and CPU as follows for the ports to be used and click OK.
(Ex. Communicate COM port “1”)

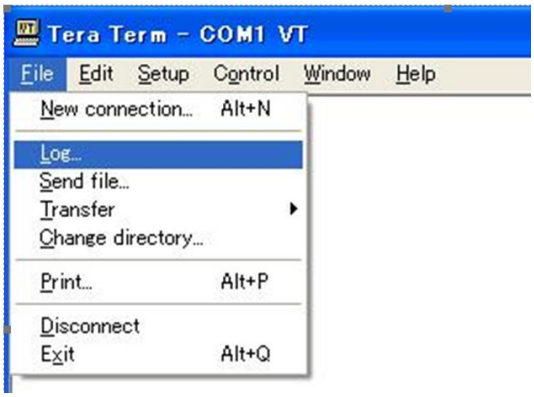
6. PC maintenance



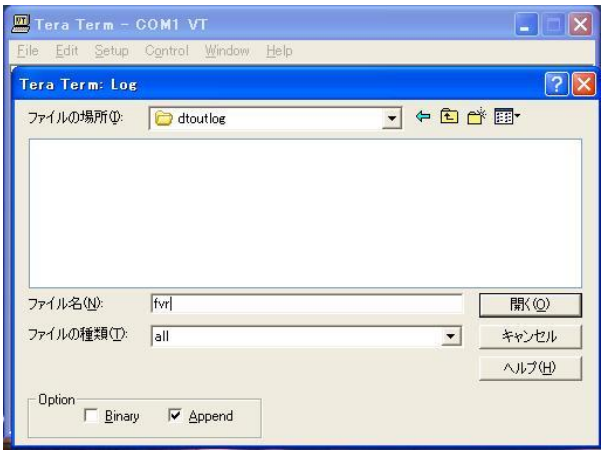
6.3.3 How to create a log file

If a log file is created during data output, the output data can be saved.

I. Select the File tab in TeraTerm. Select the Log tab.



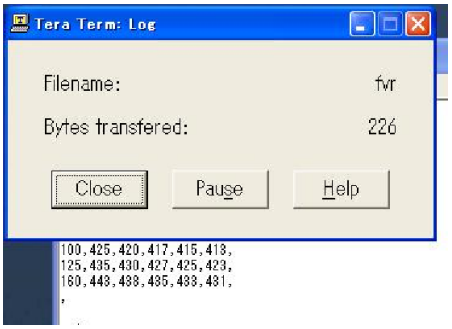
II. Specify where to save the file.
(Ex: Save as fvr in dtoutlog folder)



III. Enter a command.

(Ex: fvr command input)

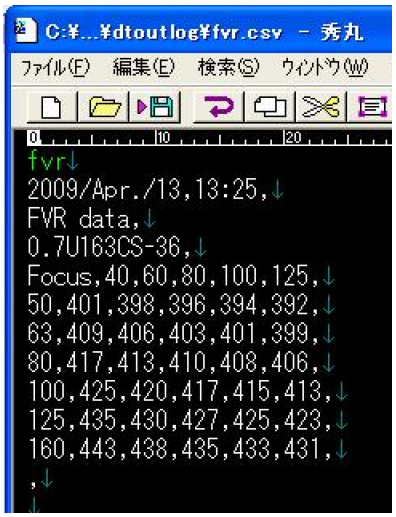
IV. When finished outputting data, select close to close the log file.



6.3.4 How to open a log file

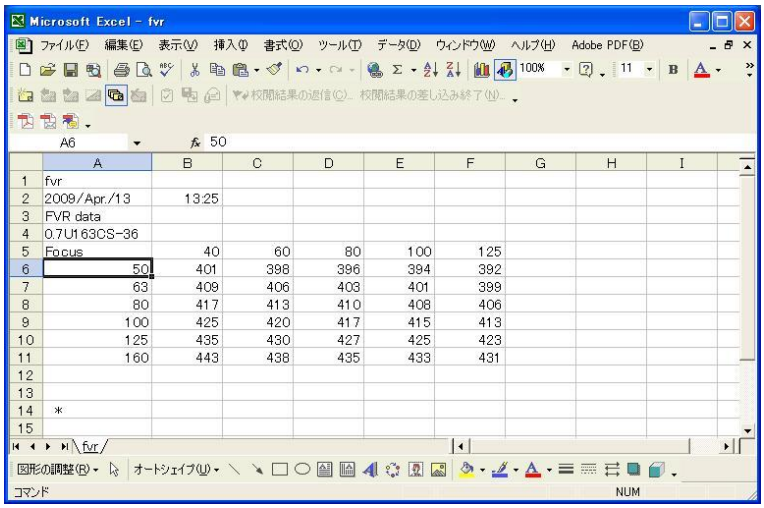
If you create a log file when outputting data and save the data, you can open the saved data in a text editor such as notepad.

(Example: FVR data log file opened by Hidemaru)



You can also open the saved data in Excel CSV format by changing the extension of the saved data to CSV.

6. PC maintenance



6.3.5 Data output method

Here's how to use the dtout command to output data:

I. Turn on the device

II. Press enter on PC keyboard and MAIN > appears.

Note:

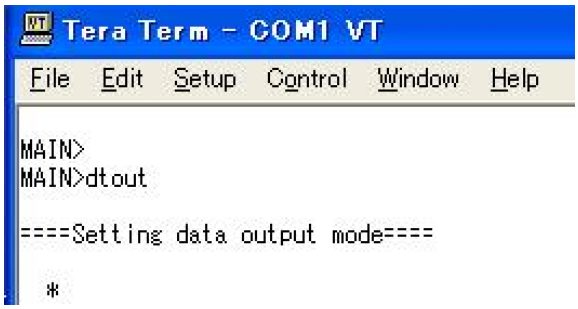
If MAIN > does not appear when you press the enter key, turn off the device and press

Check the following items:

1. Is DIPSW on the CPU board set correctly?
2. Does the serial communication cable between the PC and the CPU board use a crossover cable?
3. Are the serial communication cables between the PC and the CPU board properly connected?
4. Are the ports used for communication tools and communication settings correct?

III. Type dtout from the keyboard and press enter to activate the dtout command.

The PC display says "= = = = Setting data output mode = = =."



IV. Output data by entering a command from the keyboard that outputs the required data.

6.4 Examples of various data output

6.4.1 fvr：FVR Adjustment Data

Data is output in year/month/day and time in 24 H notation from the top.
Subsequently, the tube name, focus, and set FVR data (Setting kV, setting mA, setting data of each point) are output.

From the keyboard, type fvr and press enter.

```
2021/Sep./06,17:15,
FVR data,
0.7/1.3U163C-36,
Large,40,60,80,100,125,133,
50,382,380,379,377,375,374,
63,389,387,385,383,381,380,
80,395,393,391,389,387,386,
100,404,401,399,397,395,394,
125,412,408,406,404,402,401,
160,421,417,413,411,409,408,
200,430,425,421,419,417,416,
250,440,433,429,427,425,424,
320,454,444,439,437,435,434,
400,467,456,450,447,445,444,
Small,40,60,80,100,125,133,
50,387,385,383,381,379,378,
63,393,391,389,387,385,384,
80,402,399,397,395,393,392,
100,410,406,404,402,400,399,
125,419,414,412,410,408,407,
160,431,424,421,419,417,416,
200,442,433,429,427,425,424,
```

6.4.2 handle：Handle Adjustment data.

Data is output in year/month/day and time in 24 H notation from the top.
Subsequently, handle adjustment data for each point is output.
Data is displayed as 00 to 11, which corresponds to 1 ~ 12 when adjusted.

From the keyboard, type handle and press enter.

```
2021/Sep./06,17:15,
Handle data,
00,609,
01,714,
02,758,
03,345,
04,217,
05,179,
06,596,
07,704,
08,742,
09,353,
10,262,
11,194,
,
```

6.4.3 batt：Battery data

Data is output in year/month/day and time in 24 H notation from the top.
Then, the set battery code, remaining step reference voltage, charging step charging current, charging step reference voltage, VSVN correction factor, additional shooting permission voltage, PC-ON correction, AC-ON correction and charging suppression ON/OFF state data are output.

From the keyboard, type batt and press enter.

```
2021/Sep./06,17:15,
Battery data,
Code,6,
Code-E,
Disp,500,2413,2473,2487,2503,2518,2533,
Charge,50,200,100,40,18,10,
Charge Disp,500,1390,1420,1440,1450,1490,
Adjust,265,200,
Add EXP,2393,
PC-ON,30,
AC-ON,5,
Restriction,on,
```

6.4.4 userset：User settings data

Data is output in year/month/day and time in 24 H notation from the top.

Next, turn the status of DIPSW 3 -2 ON/OFF, and output data for 100 PIDs including passwords.

Data is output in the following order: Personal ID, Password, Power Assist, Acceleration, Turning, Backward Assist, Auto Power Off Time, Buzzer, Collimator Lamp Control, APR at startup, setting data for operating sound, Auto charge start time, FPD unlock password, and Silence time.

Out of the operation sounds, the running attention sound - ON when starting and - OFF when starting is output.

```
2021/Sep./06,17:15,
User set data,
DIPSW3-2,OFF,
Pid,0,PWD,      ,User
PAS,070,
Acc,060,
Stc,100,
bAc,080,
APO,05,
BON,on,
LAP,O-C,
Snd,1-1-2,2-2-3,3-7-2-ON,
AbC,60,
FPd,0,
ndr,18,08,
Pid,1,PWD,      ,User.
    ・
    ・
    ・(Pid, 1 to 98)
    ・
    ・
ndr,18,08,
Pid,99,PWD,     ,User
PAS,070,
Acc,060,
Stc,100,
bAc,080,
APO,05,
BON,on,
LAP,O-C,
Snd,1-1-3,2-2-3,3-3-3-ON,
AbC,60,
FPd,0,
ndr,18,08,
```

6.4.5 replace：Exchange notification function data

Output in 24 H notation. It then outputs the following in the order of output:

- Battery Start Date
- Charging times
- Number of full charges
- Number of charge limits,
- Low battery frequency
- bbb occurrences
- bb0 number of occurrences
- bb1 number of occurrences
- Start date of CPU backup battery
- Start date of emergency brake release battery
- Start date of tube use
- Number of shots
- Cumulative mAs (unit: 1000 mAs)
- Number of tube bulb discharges
- Integrated filament heating time (in 100 sec units)
- Cumulative starter rotation time (in 100 sec units)
- Start date of motor use
- Accumulated mileage (in 1-km increments)
- Start wire date
- Arm Upward Movement Distance (in 1 km)
- Arm downward travel distance (in 1 km)
- Code reel start date,
- Number of outlet connections,
- FSR start date,
- Number of times the left FSR was pushed hard
- Number of times the right FSR was pushed hard
- Number of times of obstacle detection part Kenschutz
- Number of arm locks

From the keyboard, type replace and press enter.

6. PC maintenance

2021/Sep./06,17:15,
Replace parts,
BATT,2021/Jul./26,0,
Charge,47,
Full,3,
Restriction,27,
Low,0,
bbb,0,
bb0,0,
bb1,0,
CPUBATT,2021/Jul./26,0,
MOTBAT,2021/Jul./26,0,
TUBE,2021/Jul./26,0,
Exp,56,
mAs,1,
Discharge,0,
Filament,2,
Stator,1,
MOTOR,2021/Jul./26,0,
Trip Meter,3,
Wire,2021/Jul./02,0,
Arm up,0,
Arm down,0,
Cordreel,2021/Jul./02,0,
ConnectAC,27,
FSR,2021/Jul./02,0,
Left handle,2846,
Right handle,1939,
Extra Log,
Obstruction Detect,30,
Arm lock,118,

6.4.6 error：Error log

Data is output in year/month/day and time in 24 H notation from the top.
Next, 100 fault errors are output in the order of log number, error code, and date and time of occurrence.
Next, 100 warning errors are output in the order of log number, error code, and date and time of occurrence.
When no error has occurred, No error is displayed in the error code area, and the date and time of occurrence is not displayed.

From the keyboard, type error and press enter.

```
2021/Sep./06,17:15,
Fault Error log,
No.,Code,Date,Time,
0,d60,2021/Sep./06, 16:53,
1,d21,2021/Sep./06, 15:54,
2,d20,2021/Sep./06, 15:54,
3,d21,2021/Sep./03, 18:19,
4,d20,2021/Sep./03, 18:19,
5,d21,2021/Sep./03, 17:42,
    ・
    ・(No. 6 to 95)
    ・
96,Noerror,
97,Noerror,
98,Noerror,
99,Noerror,
2021/Sep./06,17:15,
Warning Error log,
No.,Code,Date,Time,
0,F6d,2021/Sep./06, 16:56,
1,F6d,2021/Sep./06, 15:54,
2,A 9,2021/Sep./03, 17:42,
3,F6d,2021/Sep./03, 16:51,
4,F6d,2021/Sep./03, 16:47,
5,F6d,2021/Sep./03, 16:06,
    ・
    ・(No. 6 to 95)
    ・
96,Noerror,
97,Noerror,
98,Noerror,
99,Noerror,
```

6.4.7 apr : APR data

Data is output in year/month/day and time in 24 H notation from the top.
APR data for DR imaging are stored in APR number, set tube voltage, set tube current time product,
432 are output in the following order: setting focus (large or small), additional filter number, shooting mode (short or long).
Next, nine APR data in cassette mode are output.
Finally, the starting tube voltage of the warm-up mode is output.

From the keyboard, type apr and press enter.

```
2021/Sep./06,17:15,
DR Apr data,
1,75kV,8mAs,Small,Non-filter,Short,
2,75kV,8mAs,Small,Non-filter,Short,
3,80kV,8mAs,Small,Non-filter,Short,
4,75kV,8mAs,Small,Non-filter,Short,
5,75kV,8mAs,Small,Non-filter,Short,
    ・
    ・(No. 6 to 428)
    ・
429,40kV,0.32mAs,Small,Non-filter,Short,
430,40kV,0.32mAs,Small,Non-filter,Short,
431,40kV,0.32mAs,Small,Non-filter,Short,
432,40kV,0.32mAs,Small,Non-filter,Short,
Casette Apr data,
1,40kV,0.32mAs,Small,Non-filter,Short,
2,40kV,0.32mAs,Small,Non-filter,Short,
3,40kV,0.32mAs,Small,Non-filter,Short,
4,40kV,0.32mAs,Small,Non-filter,Short,
5,40kV,0.32mAs,Small,Non-filter,Short,
6,40kV,0.32mAs,Small,Non-filter,Short,
7,40kV,0.32mAs,Small,Non-filter,Short,
8,40kV,0.32mAs,Small,Non-filter,Short,
9,40kV,0.32mAs,Small,Non-filter,Short,
Warm-up Apr data,
50kV,
```

6.4.8 dap：Dap adjustment data(option)

Data is output in year/month/day and time in 24 H notation from the top.
Then, the state of DIPSW 3 -5 and data of the DAP adjustment factor are output.

From the keyboard, type dap and press enter.

```
2021/Sep./06,17:16,
DAP data,
DIPSW_3-5,OFF,
Adjust,100,
```

6.4.9 calcdap：Calculated DAP Adjustment Data

Data is output in year/month/day and time in 24 H notation from the top.
Subsequently, the ON/OFF state of the area dose calculation unit, the registered value of the horizontal leaf opening amount, and the registered value of the vertical leaf opening amount are output. Then, the All comp value is output in the order of no additional filter, additional filter 1 and 2, and ckV value according to each tube voltage.

From the keyboard, type calcdap and press enter.

```
2021/Sep./06,17:16,
Calc DAP data,
ON,
Horizontal,76,353,
Vertical,70,346,
Non-filter,
Comp,100,
40kV,84,
50kV,166,
60kV,270,
70kV,383,
80kV,508,
90kV,652,
100kV,807,
110kV,979,
120kV,1157,
130kV,1325,
133kV,1399,
Filter1,
40kV,72,
50kV,161,
    ・
    ・
    ・
130kV,1393,
133kV,1454,
Filter2,
Comp,100,
40kV,72,
50kV,161,
60kV,267,
70kV,383,
80kV,513,
90kV,663,
100kV,824,
110kV,995,
120kV,1187,
130kV,1393,
133kV,1454,
```

6.4.10 drset：DR Shooting Settings Data

Data is output in year/month/day and time in 24 H notation from the top.
It then outputs a time limit for short shooting, a time limit for long shooting, whether to enter a password when returning from Auto Power Off with a keyless option, and communication protocol settings.

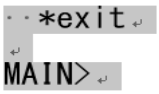
From the keyboard, type drset and press enter.

```
2021/Sep./06,17:16,
DR setting data,
Short,60,
Long,69,
Password,ON,
Protocol,01,
```

6.4.11 exit：End of dtout command

From the keyboard, type exit and press enter to exit the dtout command.

The dtout command can also be terminated by restarting the device.
MAIN > on the PC display.



6.5 End DR Maintenance

I. Turn off the unit.

II. Disconnect the cable from the maintenance PC to J4 on the XCONT board, and Connect the cable from the DR device control PC to the J4.

III. Restore DIPSW2 -4 and DIPSW2 -7 of the XCONT substrate.

7 How to respond to DR errors

7. How to respond to DR errors

7.1 Troubleshooting when DR-ID800 error occurs

Described on the next page.

7. How to respond to DR errors

Error Code	MC/CL	Query (Error)	Solution (Result)	Reference
10753	MC	System error [10753] is displayed on the monitor, and the FPD does not communicate with the main unit. (Error 10753: MP connection failed)	The F9 fuse of the MPC54A substrate was blown, so the substrate input voltage was checked. Since +24 V, +5 V, and +12 V are normal, the F9 fuse was replaced and the communication is returned.	
10753	MC	FPD cannot be connected. When restarting, error code [10753] system error detection is displayed at startup. RupcTool does not provide a wired or wireless connection to the FPD. The breakage of the fuse on the MPC54B substrate was not confirmed.	Generally speaking, in the case of 10753 error, the MPC54B substrate to PC is the suspect components.	
10753	MC	Unable to communicate with FPD due to system error 10753 Defect Event: FPD cannot be connected. The fuse of the MPC54B substrate was checked, but no abnormality was found. The connection cable to the MPC54B substrate was reseated, but no abnormality was found. Ping continuity to the MP was checked, but no abnormality was found. When the MP version was checked with RUPcTool, "Error" was displayed. When the MP substrate state was checked with RUPcTool, "Error" was displayed.	Communication with MP was not possible. If you haven't changed anything in particular and you're suddenly unable to connect, it's probably a hardware problem. Please check the LAN cable and MPC54B substrate.	
10753	MC	10753 MP-MC communication error occurs. A message appears prompting you to restart the DR device. Restart restores the normal operation. We don't know when the phenomenon will occur.	As a result of analysis by R&D, the 800IU and the panel (wired) are repeatedly reconnected, but the 800IU is not connected. In addition to the MPC54B substrate, please check the communication route such as the LAN cable.	
10753	MC	10753 Error (cannot connect to FPD) Error Details Main unit power OFF/DR ON/FPD wired connection 10753 error occurs on DR ON. No error occurs with wireless connection. MC Manager disappears when the error occurs. Starting MC Manager directly again works fine. Both 1211 SE and 1214 SE occurs. *When the error occurs, the IU unit repeatedly restarts, probably because it is trying to connect to the FPD (video upload).	I checked the video. The power supply to the MPC substrate seems to stop momentarily every 40 seconds. Please check the DC-AC power inverter output inside the mobile X-ray unit.	
10753	MC	A communication problem occurred between MC and 800IU.	MPC54B substrate and LAN cable are suspect parts.	
10753	MC	MP connection error	As a result of log analysis, it was found that not only the MP but also the SE (wireless connection) was not responding just before the MP disconnection occurred. The AP is connected to the LAN cable of the DR-ID 800IU. In addition, since the AP power supply is AC, it is supplied from the DC-AC inverter of the mobile X-ray unit (Same for 800IU AC power) Therefore, the probable causes are as follows. ・Power startup time of the AP unit is delayed. (Possible startup trouble) ・Poor contact of LAN cable for 800IU: AP and CSL and defective substrate. ・If there is a problem when the power supply starts, the DC-AC inverter of the mobile X-ray unit may be a candidate.	

7. How to respond to DR errors

10753	MC	MP connection error	Replaced with a new MPC54A substrate.	
11112	MC	Panel abnormality detection error occurred. The panel did not communicate, failing in radiography. (The panel used was 1212SE.)	Many 11112 errors occurred at that time. Since the detail code is 0x00000200, it is an error due to poor contact of the POW substrate or the cable connected to the POW board. However, this error has not occurred at all since then. After the error occurred, did you open the panel and check the cable contact failure? If so, I think the contact failure was improved by that. If not, there is a concern that it will happen again. If the problem recurs, reseat the cable connected to the POW substrate. If the problem is not resolved, replace the POW substrate.	
11112	MC	Since the error "11112" is displayed, the RMV substrate in the panel (1212SE) has been replaced. However, the error continues to appear.	The 11112 error detail code is 0x00000200. Please try replacing the POW substrate.	
1112	MC	DR-ID800 MC V16.6 FPD: 611SE During inspection, a message stating "Unable to capture due to FPD failure" appears, preventing the capture. Reconnecting the panel battery temporarily restores functionality, but the issue reoccurs after some time following the completion of the capture. Upon checking the FPD log, error 11112 has occurred multiple times. Disconnecting and reconnecting the cables related to RMV during panel disassembly did not improve the situation. The phenomenon is not limited to specific battery packs. Even after replacing the board, error 11112 (RMV board anomaly) was not resolved. Although the error does not occur every time, it seems to happen quite frequently. It does not appear to occur with the borrowed 612SE	If the issue is not resolved by replacing the "RMV board" and "POW board," we request that you replace the panel.	
1112	MC	1: Initialize the DR device 2: Error occurs after initialization (11112) 3: If capturing is started in this state, error (11247) occurs 4: Restores by reattaching the FPD battery pack Issue: Communication error between the console and FPD	I have reviewed the logs. It appears that communication errors are occurring involving the RMV board, with frequent records of communication errors on the RMV board. It seems that the RMV board is defective, so please replace the board.	
11228	MC	Panel crack determination error: panel crack of size 9 mm x 9 mm or larger	Since the data from the middle to the bottom is 0, there is a high possibility of TFT failure. As you probably know, there are quite a lot of drop impact records on GLG.log, so it is difficult to guarantee the performance. Panel replacement is recommended. However, if you want to check the possibility other than the failure of the TFT, we recommend that you check the effect by replacing the POW board, assuming the failure of the power supply supplied to the gate IC.	

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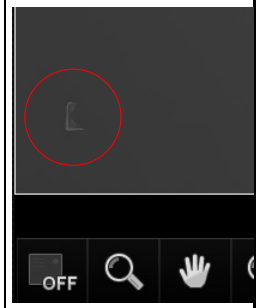
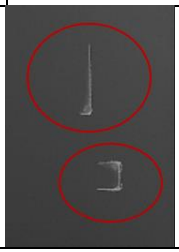
11228	MC	Panel abnormal detection error occurred (11228) Even after restarting the DR and panel, the issue persists. Panel communication error (11112) also occurs. Measures taken so far: Checked error logs and confirmed RMV board communication error. RMV board status: H/W status error}	Upon reviewing the logs you provided, communication errors with the RMV board were recorded. Please proceed with replacing the RMV board and the POW board. If this does not improve the situation, it is likely that the fault lies with the side sending data to the RMV board (the panel side). The [11228: panel fracture detection error] was recorded, indicating a possible panel fracture. Please replace the panel.	
11231	MC	I received a call from a customer asking for a repair because something like a "U shape" appeared on the bottom left of the screen. Since there was no change even when SID and FPD radiography position were changed, it was judged that the FPD side was the cause. FPD calibration was performed because there was no damage in appearance. The offset and gain calibrations are successful, but an error (error code 11231) is always output in the correction data generation in the defect calibration.	I checked the image. There is no doubt that the scintillator is defective. Please replace the panel.	
11231	MC	After frontal chest imaging, the corners of the image fell out in a fan-shaped white pattern, and then the same phenomenon occurred in other images. An attempt was made to perform panel gain calibration, but calibration failed with 12230 "Gain Correction Data Generation NG." Even after checking the error table and changing the irradiation field and conditions, it was not possible. I tried the defect correction calibration, but I could not perform it due to error 11231 "defect correction data generation error.	There seems to be a problem with TFT or scintillator. Please replace the panel.	
11231	MC	There was something like a triangle on the bottom. As for the situation at the time of calibration, offset and gain calibrations were successful, but no matter how many times the defect calibrations were performed, the error of 11231 occurred and data generation could not be performed.	It must be a defect of the TFT or scintillator inside the panel. Please replace the panel. There were several drop impact records on GLG. Please explain the handling precautions to the customer.	
11231	MC	When calibrated, Error 11231 occurs during defect correction calibration. We have since seen similar errors during normal startup.	The TFT is damaged by the impact and many defects are generated. The error has occurred because of this. Unfortunately, panel replacement is required.	
	MC	White out occurs at the bottom of the radiography image. It is not known when this occurred, but it occurred in all of the radiography images. The occurrence point is the same position.	I checked the image. Panel replacement required due to scintillator failure. In addition, there were many records of drop impact. Please tell the customer to handle it carefully.	

7. How to respond to DR errors

11239	MC	"Calibration collection failed" is displayed on the device.	At that time, the 11239 error (initialization calibration errors) occurred in the error log (ERRORB.LOG). It also occurs on the panel at a time other than the one indicated. Since this error did not occur only on certain panels, it is highly likely that this error is caused by external noise as described in the service manual. Calibration is performed regularly, so it is not a problem even if it fails once. However, if you are concerned about the error, check if there are any electronic devices near the panel that could cause noise.	
11239	MC	Calibration error at initialization	There is also a record of the drop impact in GLG, so it might have an effect, but there has been no error in the calibration during operation since the initialization. Check if there is any electronic device near the panel when the panel starts.	
11244	MC	It appears that an 11244 error occurred, making the FPD unusable (i.e., the device is unusable). We suspect a fault in the wireless module due to a 0x00000002 H/W error. The on-site report is as follows: 1: Power on the device 2: The LED on the side of the FPD lights up red 3: Connecting the wired SE cable triggers a <11244> panel communication error 4: Reattaching the battery pack and restarting the PC do not restore functionality Checking the error with the RU-PC Tool shows a 0x00000002 H/W error logged. A check of the board status indicates WLAN H/W Status: Error!, while other items are OK	Upon reviewing the logs you provided, abnormalities were recorded in communication with the wireless module. Please check the connection of the wireless module itself and the cables connected to it.	
11246	MC	"[11247] Radiography menu registration error" and "[11246] Panel communication error" are displayed at image acquisition. The phenomenon has occurred several times since May 10, and it has not occurred until then. Reseating the FPD battery resumes image acquisition.	Log analysis by R&D revealed that an error occurred due to a substrate (RMV substrate) failure inside the panel. Please replace the RMV substrate.	
11246	MC	[11246] "Panel communication error. Check the device status and restart the panel if there is any abnormality." Even if the FPD battery is replaced, the problem may not be resolved.	Log analysis by R&D revealed that the failure of the RMV substrate was confirmed. Please replace the substrate.	
11247	MC	"[11247] Radiography menu registration error" and "[11246] Panel communication error" are displayed at image acquisition. The phenomenon has occurred several times since May 10, and it has not occurred until then. Reseating the FPD battery resumes image acquisition.	Log analysis by R&D revealed that an error occurred due to a substrate (RMV substrate) failure inside the panel. Please replace the RMV substrate.	
11247	MC	1: Obtain patient information from MWM 2: Transition to the study window 3: Error 11247 4: Restore by reseating the FPD battery pack Problem: Error occurs during transition to the study window.	This phenomenon seems to be caused by the wireless environment. Since there seems to be a problem in the communication environment, measure the communication speed and checked the radio wave strength to check the radio environment. There is a communication failure in multiple panels, so please check the AP.	
11254	MC	Error when the panel does not support country codes or when a channel is configured in Edit Config	Reset the country code.	

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11254	MC	The customer pointed out that the wireless connection was delayed after the upgrade. When the FPD is wired on the device startup, an error screen appears after the console advance starts. After that, when the cable is disconnected to attempt the wireless connection, the FPD cannot be connected to the main unit. The connection is successful by reseating the battery of the FPD and restarting the FPD. When the FPD is connected wirelessly at the time of device startup, there is no error at startup, and there is no problem even if you switch from wireless-wired-wireless.	(1) I checked the log you sent me. There seems to be a 11254 error. S/W is checking the inside of the device when the panel starts up, but S/W cannot access the wireless module properly and it seems to cause an error. The suspect component is the wireless LAN module and the RMV substrate connected to it. Please take care of the part replacement. (2) It was not restored by (1), and was solved by the following implementation. There was a difference between the radio settings on the Cisco unit and the settings on the unit. The RU PC tool side was MG: 11N20&CH:2 and the Cisco was Channel40-5200Mhz. Problem was resolved after changing to MO:11N20 & CH:40.	
11270	MC	[11270] Memory code error appears at DR startup. [Memory card error] is displayed at the bottom right of the study registration window.	11270 error occurred. If this happens frequently, remove the SD card inside the panel and try cleaning the contacts. If it continues, please replace the SD card.	
11270	MC	Memory Error	There are some 11270 errors, but there is also a drop impact log of Q224494 in GLG. Due to the drop impact, the SD card installed in the RMV substrate may have contact defects sporadically. Please open the back cover, insert and remove the SD card, insert the SD securely and check the effect.	
11271	MC	1: Attach the battery pack to 1212SE 2: Error 11271 occurs 3: Removed and reinserted the battery pack, but it did not recover 4: After waiting for some time, reattached the battery pack and there was no abnormality 5: Monitored for a few days, but the above phenomenon recurred on February 12, 2024 Issue: An error 11271 occurred, preventing image capture	Upon reviewing the logs you provided, "RMV board communication failure" was frequently recorded. Has the RMV board already been replaced? Also, after replacing the board, have you installed the latest version of the panel software? Please confirm this. Additionally, there were many records of the MC software shutting down unexpectedly. This may have caused some files to be corrupted, leading to malfunctions. Please check if the shutdown method is correct (e.g., was it forcibly shut down?)	
12215	MC	Communication between FPD and PC is lost. Communication is restored by plugging in and unplugging the power pack. The above phenomenon has recently become frequent. (confirmed to have occurred on July 25 and 26, 2021) At the error occurrence, the battery is attached, at the bottom right on the PC, the antenna of the connection device status is displayed, but the battery is displayed as empty.	The 12215 battery access error occurred at this time. [12215] 2021/07/26 07:53:13 0019F3 @SE[Q120632] BATTERY ERROR Check the terminal of the panel or battery. Check the terminals for damage, deformation, or dirt. If the terminal is not a problem and the error only occurs on a unique panel, the BCN65A, REC65A, SPR65A, and RMV65A substrates and the harness between them are suspect parts.	
12215	MC	As soon as the battery pack is loaded into the FPD after charging the battery, the application displays a dead battery. Another battery pack has the same phenomenon.	I checked the battery capacity from the log and there was no problem. At that time on August 3, 12215 errors (battery access errors) occurred, so I think it is a problem on the panel side. For 12215 errors, the suspect parts are the BCN65A, REC65A, SPR65A, and RMV65A substrates and the harness between them.	

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12215	MC	Communication between FPD and PC is lost. Communication is restored by plugging in and unplugging the power pack. The above phenomenon has occurred for a long time, and this time, the same phenomenon occurred in the FPD borrowed from the factory due to the FPD failure.	The same error 12215 as you inquired has occurred. [12215] 2021/10/04 12:29:57 00C5D9 @SE[Q120524] BATTERY ERROR Check the terminal of the panel or battery. Check the terminals for damage, deformation, or dirt. If the terminal is not a problem and the error only occurs on a unique panel, the BCN65A, REC65A, SPR65A, and RMV65A substrates and the harness between them are suspect parts. Replace the part and identify the defective part.	
12215	MC	Battery access error	BCN, BTC, RMV, and the harness between the substrates are suspect parts. The electrode of the BCN substrate is particularly suspicious. Please check the parts. This problem is completed by replacing the BCN.	
12215	MC	Battery access error	If the suspect components of 12215 error are the CN65A, REC65A, SPR65A, and RMV65A substrates and the cables between them. The BCN65A substrate is most likely to be broken, but please check each substrate and cable. BCN65A replacement	
12230	MC	After frontal chest imaging, the corners of the image fell out in a fan-shaped white pattern, and then the same phenomenon occurred in other images. An attempt was made to perform panel gain calibration, but calibration failed with 12230 "Gain Correction Data Generation NG. Even after checking the error table and changing the irradiation field and conditions, it was not possible. I tried the defect correction calibration, but I could not perform it due to error 11231 "defect correction data generation error. There seems to be a problem with TFT or scintillator. Please replace the panel.	There seems to be a problem with TFT or scintillator. Please replace the panel.	
12231	MC	Defect correction calibration error	The following procedure explains how to handle the 12231/1121 error. ① Reinstall the unit data from the CD and restore the defect map to its initial state. ② Check if the error is resolved by re-calibration. If you want to check whether the scintillation has come off, check the image without shading. It can be checked by the presence of U-shape artifacts. The defect exceeded the specification only once, and the error appears to have occurred only then. Defects can come and go, so this time it seemed to be an instantaneous phenomenon. This error has since been resolved, and you can use it while it is resolved. If it happens continuously, please replace the panel.	

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12231	MC	When acquiring image with DR-ID 1212SE, horizontal lines appear in the lower half of the image.	When I checked the log, I found that there are streaks at the bottom of the image due to the GTD board. The GTD substrate was probably damaged by the impact of the drop. This substrate cannot be replaced on site, so please replace the panel.	
12600	MC	Correction data load read retry	This is probably because the correction file is corrupted. Please reinstall MC and perform MC registration and calibration.	
12601	MC	[12601] @MC has recovered from an unexpected shutdown. Some files might be corrupt is recorded multiple times.	Upon checking the logs, frequent system file load retries were recorded. It appears that the system files are corrupted, so please uninstall and reinstall the MC software.	
12601	MC	[【SM10005】Failed to connect to wireless LAN], "10753: Detected communication disconnection between MP (IU) and MC.	The following error was recorded near the time of the event in the log data you provided. [12601] 2024/10/03 13:02:44 @MC has recovered from an unexpected shutdown. Some files might be corrupt. This error indicates that an unexpected shutdown occurred, causing some files to become corrupt, and subsequently, the system failed to read from these files upon reboot. Such logs are not recorded when the power is turned off using the console's "Shutdown" menu. Could you please check if the power has been turned off using the "Key Switch" or "Main Circuit Breaker" before the console software has fully shut down? Additionally, since this error suggests that several files of the MC software might be corrupted, please reinstall the MC software to restore functionality.	
12700	MC	In the morning, when the device starts up, the message "Wireless is unavailable" or "Memory error" appears and images cannot be acquired.	There was a 12700 error in the Q225910 panel at that time. Since a drop impact was recorded on the Q225910 panel, it is possible that the SD card is about to be pulled out due to the impact. Open the back cover of the panel and check if the SD card is properly loaded.	
13005	CL	DICOM Storage fails when I try to store images via the wired connection after radiography. Also, a test radiography attempt fails. When the system restarts, "13005 Device status error (DDO file does not exist or is corrupted)" and "11246 SE communication error" are displayed.	Can you give me some information to identify this DICOM-Storage failed test? Information such as Accession No./StudyDate/StudyTime makes it easier to identify the study. Accession No./StudyDate/StudyTime are found in the Study list.	
E00135	CL/F-RIS	I was informed that an error [E00135] "An error occurred during network connection" occurred when acquiring RIS patients.	Contact INFOCOM.	
MD00001	CL	MD00001 (initialization failed) occurred at console startup.	There is a trace that Windows was forcibly terminated on 04/09, the file seems to have been corrupted because the power was forcibly turned off. Restoration from an already performed backup image was successful.	
MD10006	CL	10753 MP-MC communication error occurs. A message appears prompting you to restart the DR device. Restart will return to normal operation. We don't know when the phenomenon will occur.	As a result of analysis by design, the 800IU and the panel (wired) are repeatedly reconnected, but the 800IU is not connected. In addition to the MPC54B board, please check the communication route such as the LAN cable.	

7. How to respond to DR errors

MD11024	CL	Connection to radiography device has been disconnected.	TCP Retransmissions occur frequently, so it is probable that a failure occurred below the network layer. This may also be the cause of data transfer failures during initialization. Check the connection of the communication cable and the communication connector because the error has occurred in wired communication. If the problem continues to occur, replace the cable or HUB.	
RE 10035	CL	Failed to acquire order information. *Unable to retrieve information from RIS.	Error handling by RIS manufacturer. To check the DICOM connection: First of all, it is important to switch the DICOM logging mode to "Detailed recording mode" and collect the logs from the Acquisition Tool of Console's service utility. ・Scope of Local Service Although some knowledge about the DICOM standard is required, it is recommended to check the DICOM communication log (* 1) for data that violates the DICOM standard in the DICOM tag transmitted by RIS. At this time, switch the DICOM log mode to "Detailed recording mode". Otherwise, the tag information received from the RIS is not logged. ※C:¥Program Files¥FujiFilm¥IIP¥Log¥LogP¥Reception_BCRIS_MWLSCU.log"	
RE10122	CL	There is no problem with two-byte characters, but if you enter the patient's name in one-byte katakana, the study cannot start.	It is assumed that [RE10122] occurred because a one-byte katakana patient name was set in the field for setting a two-byte patient name.	
SM10010	CL	SM10010 In-House Radio Communication Error	1. Click windows10 settings. 2. Click Network and Internet. 3. Click Wi-Fi. 4. Managing Known Networks. 5. Check the SSID displayed. 6. Click the desired SSID. 7. Delete both "XXXX" and "XXXX2" profiles. 8. Connect only the "XXXX" access point. 9. Reboot the console and verify that the "SM10010" error does not occur.	
UU70026	CL	I try to set up for a miss shot analysis on DR-ID800. An error with the attached image occurs, and the summary button is not available.	[Solution] [UU70026] may occur if there is any inconsistency in the miss shot comment database. Open User Utility > Table Settings > Miss shot Comments, delete all miss shot categories, and then re-register the required miss shot comments. If [UU70026] still does not converge, please reproduce [UU70026] and provide the following 3 data. 1) Log data collected from Console service utility - Acquisition Tool 2) C:¥ProgramFiles¥FujiFilm¥IIP¥Data¥RetakeAnalysis folder 3)C:¥Program Files¥FujiFilm¥IIP¥Log¥RetakeAnalysis.log file	

	CL	An attempt was made to acquire an image, but it is fails and F68 was displayed. When the second or subsequent error occurred, the exposure was made after confirming that the Ready lamp in the lower right corner of the console was green, but an error occurred. User tried to shut it down, but the screen went gray. About an hour passed without shutting it down and the user contacted CSC. After that, I called back and user said it was running.	About long shutdown times After the user presses the System Exit button to exit the Console, WindowsOS termination processing is performed. A disk IO error occurred while the WindowsOS was exiting. This could be a hardware failure such as a physical hard drive failure, a bios failure, a SATA cable failure, a motherboard failure, or a power failure (power reduction). There was evidence that disk IO error occurred twice in 2021/3. The part cannot be identified, but the hardware seems to be having problems.	

7.2 How to get MobileDaRt Evolution Image Processing Options SW log

If an error occurs in MobileDaRt Evolution Image Processing Options SW, obtain the data in the following folder. All logs are saved in this folder.

C:¥ProgramData¥MobileDaRt Evolution Image Processing Options SW¥log